



USCGC WMEC 270 A Class
SPECIFICATION FOR DRYDOCK REPAIRS
FY2027

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REVISIONS RECORD

This page is used to record specification revisions, which may have occurred subsequent to a Revision 0 (Rev-0) package. Information listed is intended to provide contractors and field unit personnel a means to ensure all the current specification revision pages are present when reviewing or utilizing this specification package.

DATE	REV#	WORK ITEM#	CHANGES MADE

NOTE : All work item and paragraph numbers listed above for a given revision correspond to same numbers in the previous revision. This revised specification is self-contained with all of the above listed changes incorporated.

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CONSOLIDATED LIST OF REFERENCES

The below-listed documents form a part of this specification to the extent specified herein. Approval/publication dates or revision dates/numbers are also identified, to ensure that same document versions are used at the time of specification writing and during contract execution.

All Coast guard drawings, technical publications, and standard specifications will be provided to contractors by the Coast Guard at an appropriate time, or upon request, free of charge. Other Government documents may be accessed – free of charge – from links located on the SFLC website. Commercial sites provide access to their respective documents.

COAST GUARD DRAWINGS

Coast Guard Contract Guidance Drawing 901 WMEC 111-008, Rev -, Expanded Shell/Gen Arrg't Overlay for Shell Survey, 901-904

Coast Guard Drawing 618 WMEC 161-001, Rev B, Arrg't & Details Stern Tube & Strut

Coast Guard Drawing 618 WMEC 243-008, Rev A, Shafting Arrangement & Details

Coast Guard Drawing 618 WMEC 244-002, Rev -, Stern Tube & Strut Bearing Details

Coast Guard Drawing 627 WMEC 161-001, Rev B, Arrg't & Details Stern Tube & Strut

Coast Guard Drawing 627 WMEC 243-008, Rev K, Shaft Installation

Coast Guard Drawing 627 WMEC 244-005, Rev -, Stern Tube & Strut Bearing Details

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters and Rope Guards

Coast Guard Drawing 901 WMEC 111-004, Rev D, Transom Pl & Fr

Coast Guard Drawing 901 WMEC 111-001, Rev G, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-001, Rev H, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-002, Rev F, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-002, Rev G, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-003, Rev H, Shell Exp FR 101-182

Coast Guard Drawing 901 WMEC 111-003, Rev J, Shell Exp FR 101-182

Coast Guard Drawing 901 WMEC 114-001, Rev F, Skeg

Coast Guard Drawing 901 WMEC 114-002, Rev D, Bilge Keels

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters and Rope Guards

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters And Rope Guards

Coast Guard Drawing 901 WMEC 114-005, Rev -, Stern Flap Mods to Structure

Coast Guard Drawing 901 WMEC 123-002, Rev D, Chain Lkr Bhds & Details

Coast Guard Drawing 901 WMEC 162-001, Rev I, Stacks

Coast Guard Drawing 901 WMEC 171-001, Rev J, Main Mast

Coast Guard Drawing 901 WMEC 184-010, Rev F, XDRC Fdns and Shell Mods

Coast Guard Drawing 901 WMEC 184-024, Rev B, Stuct Mods Incidental to DS60 Doppler Install & ODOM DWDS Rip-Out

Coast Guard Drawing 901 WMEC 243-001, Rev K, Shafting Arrangement

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Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement
Coast Guard Drawing 901 WMEC 243-002, Rev G, S- Tubes, Strut Brg & Seal A&D
Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes Strut Brg and Seal Arr and Det
Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes, Strut Bearing & Seal Arrangement and Details
Coast Guard Drawing 901 WMEC 243-002, Rev G, S-Tubes, Strut Brg & Seal A&D
Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arr
Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arrangement
Coast Guard Drawing 901 WMEC 243-004, Rev E, Shafting Details
Coast Guard Drawing 901 WMEC 243-004, Rev E, Controllable Pitch Propeller Shafting Details
Coast Guard Drawing 901 WMEC 243-004, Rev E, CPP Shafting Details
Coast Guard Drawing 901 WMEC 243-004, Rev E, Shafting Details
Coast Guard Drawing 901 WMEC 243-004, Rev F, Controllable Pitch Propeller Shafting Details
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Coast Guard Drawing 901 WMEC 243-004, Rev G, CPP Shafting Details
Coast Guard Drawing 901 WMEC 243-004, Rev H, Shafting Details
Coast Guard Drawing 901 WMEC 243-005, Rev D, Sftg Align Anal and Procs
Coast Guard Drawing 901 WMEC 243-006, Rev C, Shaft Seal A&D
Coast Guard Drawing 901 WMEC 243-007, Rev D, Shaft Coupling Assemblies, CPP
Coast Guard Drawing 901 WMEC 243-007, Rev E, Controllable Pitch Propeller Shaft Coupling Assemblies
Coast Guard Drawing 901 WMEC 243-007, Rev E, Shaft Coupling Assemblies
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Coast Guard Drawing 901 WMEC 245-002, Rev F, CPCH Prop Hyd Sys A & D (901)
Coast Guard Drawing 901 WMEC 245-003, Rev E, Controllable Pitch Propeller Propeller Hub Assembly
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Coast Guard Drawing 901 WMEC 245-004, Rev E, Controllable Pitch Propeller Eshcer Wyss 270ft WMEC Blade and Palm
Coast Guard Drawing 901 WMEC 245-005, Rev -, C.P. Propeller
Coast Guard Drawing 901 WMEC 245-007, Rev B, Special Tools
Coast Guard Drawing 901 WMEC 245-007, Rev B, Special Tools
Coast Guard Drawing 901 WMEC 245-008, Rev D, Controllable Pitch Propeller Actuating Unit Type 300.3GN
Coast Guard Drawing 901 WMEC 245-008, Rev D, CPCH Prop Actuator Unit Assy
Coast Guard Drawing 901 WMEC 245-009, Rev L, CPCH Prop Hyd Sys Diag
Coast Guard Drawing 901 WMEC 245-010, Rev F, CPCH Prop Hyd Sys A & D (902-904)
Coast Guard Drawing 901 WMEC 245-013, Rev B, Pitch Actuator System Engagement Mechanisms, Links Shaft Clamps and Brackets
Coast Guard Drawing 901 WMEC 245-014, Rev A, CPP System Pitch Setters
Coast Guard Drawing 901 WMEC 245-015, Rev B, CPP System Pitch Setters Main Boards, Sub-Boards, Scales and Mechanical Stops

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Coast Guard Drawing 901 WMEC 256-002, Rev T, Mn and Aux SW Cooling Sys A&D
Coast Guard Drawing 901 WMEC 256-004, Rev A, Aux Salt Water Mods
Coast Guard Drawing 901 WMEC 256-004, Rev A, Aux. Salt Water Mods.
Coast Guard Drawing 901 WMEC 256-012, Rev -, Shaft Seal Vent Line Modification
Coast Guard Drawing 901 WMEC 256-013, Rev -, Mn and Aux Sw Cooling Sys Diag
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Coast Guard Drawing 901 WMEC 256-013, Rev A, Mn and Aux SW Cooling Sys Diag
Coast Guard Drawing 901 WMEC 259-002, Rev K, Diesel Exhaust System A & D
Coast Guard Drawing 901 WMEC 259-002, Rev K, Diesel Exhaust System A&D
Coast Guard Drawing 901 WMEC 424-004, Rev D, Doppler Speed Log System Replacement Ripout, Installation, Arrangement, Wiring and Details
Coast Guard Drawing 901 WMEC 451-010, Rev D, 1SG38-SINS2(V)45 Installation BWD & Details
Coast Guard Drawing 901 WMEC 524-003, Rev G, Sea Conn, Seachest Conn and Waster Piece A&D
Coast Guard Drawing 901 WMEC 529-001, Rev N, Ballasting & Emer Bilge Drainage Sys Diagram
Coast Guard Drawing 901 WMEC 562-003, Rev B, Rudderstock Seal
Coast Guard Drawing 901 WMEC 562-005, Rev K, A & B Class Rudder Stock And Steering Gear A & D
Coast Guard Drawing 901 WMEC 562-005, Rev K, Rudder Stock and Steering Gear A & D
Coast Guard Drawing 901 WMEC 565-005, Rev -, Fin Stabilizer Hydraulic System Diagram
Coast Guard Drawing 901 WMEC 565-007, Rev H, Fin Stabilizer Variable Assembly
Coast Guard Drawing 901 WMEC 565-020, Rev -, Fin Stabilizer Variable Assembly (Sperry 1890934)
Coast Guard Drawing 901 WMEC 565-031, Rev C, Fin Shaft Assembly, w/ Sleeve Fitted NSN 3040-01-104-5121
Coast Guard Drawing 901 WMEC 565-034, Rev -, Fin Stabilizer Clevis Pin
Coast Guard Drawing 901 WMEC 581-002, Rev C, Hawse & Chain Pipes
Coast Guard Drawing 901 WMEC 588-003, Rev M, Hlcptr Safety Net-A&D
Coast Guard Drawing 901 WMEC 602-002, Rev C, Distinguishing Figures
Coast Guard Drawing 901 WMEC 603-001, Rev A, Draft Marks
Coast Guard Drawing 901 WMEC 612-004, Rev A, NAVSEA Certified Flight Deck Net Installation.
Coast Guard Drawing 901 WMEC 633-001, Rev B, Cathodic Protection – A & D
Coast Guard Drawing 901 WMEC 801-009, Rev L, Outboard Profile
Coast Guard Drawing 901 WMEC 801-009, Rev M, Outboard Profile
Coast Guard Drawing 901 WMEC 801-012, Rev B, Curves of Form
Coast Guard Drawing 901 WMEC 801-022, Rev K, Docking Plan
Coast Guard Drawing 905 WMEC 114-003, Rev B, Shaft Fairwaters & Rope Guards
Coast Guard Drawing 905 WMEC 243-001, Rev B, Shafting Arrangement
Coast Guard Drawing 905 WMEC 243-002, Rev F, Stern Tubes Strut Brg and Seal Arr and Det
Coast Guard Drawing 905 WMEC 243-012, Rev B, Controllable Pitch Propeller Shaft Details
Coast Guard Drawing 905 WMEC 562-001, Rev G, Rudder
Coast Guard Drawing 905 WMEC 565-020, Rev -, Fin Stabilizer Tiller
Coast Guard Drawing FL 184-002, Rev A, SS60 Transducer Install Adapter Assy
Coast Guard Fleet Drawing FL 2804-12, Rev -, U.S.C.G. Emblem

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Coast Guard Fleet Drawing FL 2804-22, Rev A, Consolidated Visual ID for Cutters

DEVELOPER NOTE: Delete 901 WMEC 588-003 once Flight Deck Net Replacement TCTO is accomplished. If TCTO is not yet accomplished, delete 901 WMEC 612-004.

NAVSEA Drawing 803-2145807, Rev C, Propulsion Shafting and Components

NAVSEA Drawing 803-921865, Rev R, Cathodic Protection Hull Zinc Anodes

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Coast Guard Technical Publication (TP) 2699, SWBS 245, Section A, Feb 2018, Controllable Pitch Propeller System

Coast Guard Technical Publication (TP) 2718, Jan 2017, Manufacturer's Instruction Book - SWBS Group(s) 555-562, 270' WMEC "A" Class Cutters, Section A, SWBS 561, Steering Gear System, and SWBS 562, Rudder Stock Seal Assembly

Coast Guard Technical Publication (TP) 2719, Aug 2016, Manufacturer's Instruction Book SWBS Group(s) 565, 270' WMEC "A" Class Cutters, Fin Stabilizer Machinery

Coast Guard Technical Publication (TP) 5454, Nov 2013, Technical Manual for Type MA Mechanical Stern Tube Seal

Coast Guard Technical Publication (TP) 5490, Apr 2018, Fin Stabilizer System

Coast Guard Technical Publication (TP) 9063, SWBS 252, Aug 2016, Coast Guard Machinery Control System (CGMCS) – Pitch Control System & Throttle Control system – Vol 4

Coast Guard Technical Publication (TP) 9187, SWBS 243, May 2021, Propulsion Shafting and CPP System Service Manual

Fire Prevention and Response

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), Latest Version, General Requirements

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Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

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Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Inspect, Repair, and Test Auxiliary Machine Systems

Surface Forces Logistics Center Standard Specification 5550 (SFLC Std Spec 5550), Latest Version,

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2020, Requirements for Preservation of Ship Structures

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), Latest Version, Requirements for Preservation of Ship Structures

Surface Forces Logistics Center Standard Specification 8634 (SFLC Std Spec 8634), 2022, Drydocking

Surface Forces Logistics Center Standard Specification 8635 (SFLC Std Spec 8635), 2020, Temporary Services

Surface Forces Logistics Center Standard Specification 8636 (SFLC Std Spec 8636), 2022, Temporary Hull Accesses

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American Society for Nondestructive Testing (ASNT) SNT-TC-1A, 2020, Recommended Practice for Personal Qualification and Certification in Nondestructive Testing

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APM: If work is being done in DD, no cofferdam is needed; delete cofferdam protective measures paragraph 3.1.3 and above NAVSEA ref. Above NAVSEA ref is accessible on-line

ASTM D5162, 2021, Standard Practice for Discontinuity (Holiday) Testing of Nonconductive Protective Coating on Metallic Substrates

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ASTM International (ASTM) D5363, 2016, Standard Specification for Anaerobic Single-Component Adhesives (AN)

ASTM International (ASTM) F992, 2017, Standard Specification for Valve Label Plates

Code of Federal Regulations (CFR) Title 29, Part 1915, Occupational Safety and Health Standards for Shipyard Employment

Commercial Item Description (CID) A-A-59313, Nov 2003, Thread Compound; Anti-seize, Zinc Dust-Petrolatum

Federal Specification (Fed Spec) HH-P-151, Mar 1991, Packing; Rubber-Sheet, Cloth-Insert

International Standard Organization (ISO) 4406, 2017, Hydraulic Fluid Power – Fluids – Method for Coding the Level of Contamination by Solid Particles – Third Edition

Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) SP-61, 2013 Edition, Pressure Testing Of Steel Valves

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Manufacturers' Standardization Society of the Valve and Fittings Industry (MSS) SP-80, 2013 Edition, Bronze Gate, Globe, Angle and Check Valves

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MIL-A-18001, Dec 2015, Anodes, Sacrificial Zinc Alloy

MIL-A-22262, Mar 1996, Abrasive Blasting Media Ship Hull Blast Cleaning

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MIL-PRF-17672E, Oct 2012, Hydraulic Fluid, Petroleum, Inhibited

MIL-PRF-24647, Jan 2018, Paint System, Anticorrosive and Antifouling, Ship Hull

MIL-S-45180, 1998, Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant

MIL-S-45180, 1998; Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant

MIL-STD-1687, Sep 1994, Thermal Spray Process for Naval Ship Machinery Application

MIL-STD-2191, Nov 1992, Repair Welding, Weld Cladding, Straightening, and Cold Rolling of Main Propulsion Shafting

MIL-STD-2199A (SH), Jan 2020, Covering for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

MIL-STD-2199B (SH), Jan 2020, Coverings for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

MIL-STD-2199B, Jan 2020 Coverings for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

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Naval Sea Systems Command (NAVSEA) Underwater Ship Husbandry Manuals (UWSH) S0600-AA-PRO-160, Nov 2011, Chapter 16, Cofferdams

NAVSEA Technical Manual S9243-AW-TRS-011/SHAFT Rev 1, June 2015, Surface Ships Main Propulsion Shafting Refurbishment Procedures

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- The Society for Protective Coatings (SSPC) Surface Preparation Specification SSPC-SP 11, 2013, Power Tool Cleaning to Bare Metal
- The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP 10/NACE No.2, 2007, Near-White Blast Cleaning
- The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP 12/NACE No.5, 2002, Surface Preparation and Cleaning of Metals by Waterjetting Prior to Recoating
- The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP WJ-2, 2012, Waterjet Cleaning of Metals – Very Thorough Cleaning
- The Society for Protective Coatings (SSPC)/NACE International (NACE), 2007, Joint Surface Preparation Standard SSPC-SP 10/NACE No.2, Near-White Blast Cleaning
- The Society for Protective Coatings (SSPC)/NACE-International (NACE) Joint Surface Preparation Standard SSPC-SP 10/NACE No. 2, 2007, Near-White Metal Blast Cleaning

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CONSOLIDATED LIST OF CRITICAL INSPECTION ITEMS

The following is a list of work items, which contain Critical Inspection reports, which the Contractor must complete within the first 25% of the availability contract period (see SFLC Std Spec 0000, paragraph 3.2.6.5 (Inspection report particulars)):

Work Item	Title
1	Hull Plating, Side Scan, Ultrasonic Testing
2	Hull Plating, U/W Body, Ultrasonic Testing
3	U/W Body, Preserve, 100 percent
7	Appendages (U/W), Preserve (Internal) and Leak Test
9	Voids (Non-Accessible), Preserve (Internal) and Leak Test
13	Propulsion Propeller Shaft Coating, Inspect and Test
14	Propulsion Shafting; Remove, Inspect, and Reinstall
15	Propulsion Intermediate Shafts; Remove, Inspect, and Reinstall
18	Propulsion Shaft Seal (Mechanical And Inflatable) Assemblies, Inspect and Repair or Renew
21	Propulsion Shaft Bearings, External, Renew
23	Controllable Pitch Propellers, Clean and Inspect
24	Controllable Pitch Propeller (CPP) System, Inspect and Service
30	Sea Strainers, Duplex, All Sizes, Overhaul
31	Sea Strainers, Simplex, All Sizes, Clean and Inspect
32	Rudder Assemblies; Remove, Inspect And Reinstall
33	Fin Stabilizers (Bearings, Packing, Stock/Blade), Repair or Renew
38	Drydock

PRINCIPAL CHARACTERISTICS

270' WMEC (A-CLASS)	
PHYSICAL	
Length overall	270' 0"
Length between perpendiculars	255' 0"
Beam molded	38' 0"
Depth molded, main deck amidships	23' 6"
01 level amidships	31' 4"
Full load displacement	1,902.2 long tons
Draft, full load to baseline amidships	14' 3"
Highest projection above baseline *height approximated	≈ 110' 0"
Shore tie voltage requirements	2 cables, 400A / 450V, 3 phase
Frame spacing	1' 0"
MACHINERY	
Main propulsion	2 ALCO Model 18-251-F Diesel Engines, 3,600 BHP each
Ship's service generators	2 Caterpillar Model D-398-TA Diesel-driven KATO 6P6-1350 Generators. 475 KW each @ 0.8 power factor, 450VAC, 3 phase, 60 cycle
Emergency generators	Caterpillar Model D-348-TA Diesel-driven KATO 4P4-1900. 475KW @ 0.8 power factor, 450VAC, 3 phase, 60 cycle
Number of propellers	2
Propeller diameter	9' 0"
Number of blades, Each	4
Pitch	Controllable
Shaft RPM	260
Shaft diameter	10.55" at exit of hull
Anchor & chain	Two 4,000 lb Navy Stockless Anchors, 8 shots of chain each
TANK CAPACITIES	
Diesel fuel total (95%)	79,210 gal.
JP-5 total (95%)	25,890 gal.
Potable water total (100%)	8,553 gal.
Hydraulic & lube oil total (95%)	2,237 gal.

General Requirements

1. SCOPE

1.1 Intent. This standard specification invokes general requirements for conducting vessel repairs performed at a commercial contractor's facility for Coast Guard vessels.

1.2 Term interchangeability. The terms 'Contractor', 'CG Yard', 'NAVSTA EVERETT', 'shipyard', 'Base', and 'Coast Guard Industrial' are used interchangeably in this specification. Where the primary service provider is Coast Guard personnel, references to contractor and other noted descriptors within this specification or within drawings, publications, SFLC Standard Specifications or other commercial and military references are deemed the same as prime service provider.

2. REFERENCES

COAST GUARD DRAWINGS

None

COAST GUARD PUBLICATIONS

Coast Guard Commandant Instruction (COMDTINST) M10360.3 (series), Coatings and Color Manual
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), Latest Version,
General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), Latest Version,
Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), Latest Version,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

Code of Federal Regulations (CFR) Title 29, Part 1915, Occupational Safety and Health Standards for
Shipyard Employment

3. REQUIREMENTS

3.1 General. The Contractor must conform to all requirements specified in SFLC Std Spec 0000 and in this item, as applicable, during the performance of this availability. The requirements in this WI applies to all work under the scope of this contract, whether explicitly stated in all following work items or not, and to all other work subsequently authorized by changes, modifications, or extensions to the contract.

NOTE

NAVSEA drawings listed will be available FOR INSPECTION ONLY from the Coast Guard Port Engineer post-award. SFLC will not redistribute NAVSEA documents. Contractors can apply to NAVSEA headquarters directly for copies.

3.2 Fire watch requirements. The Contractor must refer to 3.3.1.3 (Fire watch requirements) of SFLC Std Spec 0000, in accomplishing the following task:

- Provide portable fire extinguishers for Coast Guard fire watch personnel. Coast Guard fire watch is in lieu of contractor personnel during the hours of 0800-1600, Monday through Friday, and limited to two Coast Guard fire watch personnel.
- Provide fire watch personnel and fire extinguishers for the duration of the availability period, during and beyond noted Coast Guard fire watch support.

3.3 Preservation requirements. The Contractor must accomplish all preservation tasks, including touch-ups, in accordance with SFLC Std Spec 6310.

3.3.1 Brand name approval. Ensure that all contractor-furnished coatings are in accordance with SFLC Std Spec 6310, Appendix C (Authorized Coatings for Use on Cutters and Boats).

3.3.2 Coating colors and system color schemes. The Contractor must obtain a written KO authorization to deviate from any coatings required in SFLC Std Spec 6310 Appendix C before work.

3.3.3 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”).

3.3.3.1 Quality control requirements. The Contractor must abide by the following when performing preservation related inspections. The following measurements must be randomly spaced throughout for the purposes of providing a representation of the entire prepared or coated surface.

3.3.3.1.1 Surface profile measurements. One surface profile measurement must be taken for every 200 square feet for the first 1000 square feet; for each additional 500 square feet or less, one profile measurement must be taken. Profile measurements must be taken in accordance with SFLC Std Spec 6310, paragraph 3.1.8.3 (Surface profile measurements). A “measurement” for surface profile is defined as follows:

- ASTM D4417, Method B: One profile measurement is the average (mean) of 10 individual readings.
- ASTM D4417, Method C: One profile measurement is the average (mean) of 2 individual readings.

3.3.3.1.2 Soluble salt conductivity measurements. 5 conductivity measurements must be taken every 1000 square feet. For submerged (immersed) applications conductivity measurements must not exceed 30 microsiemens/cm. For non-submerged (non-immersed) applications conductivity measurements must not exceed 70 microsiemens/cm. If a conductivity check fails, clean the surface in accordance with SFLC Std Spec 6310, paragraph 3.1.8.5 (Soluble salt removal).

3.3.3.1.3 Coating thickness measurements. Three area measurements must be taken for the first 1000 square feet; for each additional 1000 square feet, perform an additional area measurement. A “measurement” for coating thickness is defined as follows:

- SSPC-PA 2 defines an area measurement as 5 separate spot measurements randomly spaced throughout a 100 square foot area.
- SSPC-PA 2 defines a spot measurement as 3 gage readings and each new gage reading the probe must

be moved to a new location within a 1.5-inch diameter circle defining the spot.

3.4 Welding and brazing requirements. The Contractor must perform all welding and allied processes, and NDE in accordance with SFLC Std Spec 0740.

3.5 Environmental protection requirements. The Contractor must adhere to the following environmental protection requirements in accordance with the SFLC Stand Spec 0000:

3.5.1 Contractor operated (non USCG) facilities. The Contractor must provide and maintain environmental protection as defined in SFLC Std Spec 0000 Appendix A, Requirements for Environmental Protection at Contractor Operated (Non USCG) Facilities, as applicable, during the performance of this availability. Contractor must plan for and provide environmental protective measures to control pollution that develops during normal practice, as well as plan for and provide environmental protective measures required to correct conditions that develop during the project. Contractor must comply with applicable Federal, state, and local laws, codes, ordinances, and regulations in their entirety. Any reference to a specific portion of a Federal, state, or local law, code, ordinance, or regulation in this or any other item must not be construed to mean that relief is provided from any other sections of the law, code, ordinance, or regulation.

3.5.2 Test and procedures. The Contractor is required to promptly conduct tests and procedures for the purpose of assessing whether operations are in compliance with applicable Environmental Laws. Analytical work must be done by qualified laboratories; and where required by law, the laboratories must be certified.

3.5.3 Regulatory notifications. The Contractor is responsible for all regulatory notification requirements in accordance with Federal, State and local regulations. In cases where the Coast Guard must also provide public notification, such as storm water permitting, the Contractor must coordinate with the Contracting Officer or COR, and if work is being performed at a USCG Facility, the local Facility Engineer or Engineering Officer. The Contractor must submit copies of all regulatory notifications to the Contracting Officer and the local Facility Engineer or Engineering Officer prior to commencement of work activities. Regulatory notifications must be provided for including but not limited to demolition, renovation, National Pollutant Discharge Elimination System (NPDES) defined site work, and remediation of controlled substances such as asbestos, hazardous waste, and lead paint.

3.5.4 Environmental manager. The Contractor must appoint in writing an Environmental Manager for the project, who is responsible for coordinating Contractor compliance with Federal, State, local, and station environmental requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements, including hazardous waste handling, storage, manifesting, and disposal; implement the Contractors' Environmental Management Plan; ensure that all environmental permits are obtained, maintained, and closed out; ensure compliance with Storm Water Program Management requirements; ensure compliance with Hazardous Materials including storage, handling, and reporting requirements; as well as coordinate any remediation of regulated substances such as lead, asbestos, and polychlorinated biphenyl (PCB). This may be a collateral position; however the individual must be trained to accomplish the following duties; ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure all Contractor personnel are trained in 40 CFR requirements and individual position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out.

3.5.5 HW disposal. Contractor must comply with SFLC Std Spec 0000 Appendix A, Requirements For Environmental Protection At Contractor Operated (Non USCG) Facilities for HW disposal.

3.6 Local Policy. The Contractor must refer to site (e.g., Base) Regulations and Instructions for details regarding local policies (e.g., crane services, parking, or facility usage).

3.7 SFLC Standard Specification approved changes. Approved updates to the 2022 SFLC Standard Specifications will appear here.

none

4. NOTES

4.1 QA inspection forms. QA inspection forms (QA-1 thru QA-5), required in SFLC Std Spec 6310 to be completed and submitted during preservation of “critical-coated surfaces”, are provided at the end of this document.

DRAFT

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**QA-1 - QUALITY ASSURANCE INSPECTION FORM
(PRESERVATION CHECKLIST)**

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE
LOCATION OF WORK (INCL. FRAME #'S)			AREA (SQFT)

CHECKPOINT 1 – COATING SYSTEM COMPLIANCE				
	Ensure all coatings are in compliance with SFLC Std Spec 6310, Appendix C.			
CHECKPOINT 2 - PAINT STORAGE				
	Ensure all coatings are kept at a temperature of 65 to 85°F at all times, unless otherwise specified by the coating mfg.			
CHECKPOINT 3 - AMBIENT CONDITIONS				
	Ensure surface and surrounding temperatures are each between 50 and 90°F for water-containing coatings, and 35 and 95°F for other coatings, unless otherwise specified by the coating manufacturer(s).			
	Ensure maximum relative humidity (RH) is as follows, from surface preparations through final curing of topcoat: 50% for tanks, voids, and vent plenum; and 85% for all other areas, unless otherwise specified by manufacturer(s).			
	Ensure surface temperature is at least 5°F above the dew point, unless otherwise specified by the coating mfg.			
CHECKPOINT 4 - PRE-SURFACE PREPARATION				
	Remove surface contaminants (soluble salts, loose rust, mud, and marine growth) with low pressure fresh water wash down (maximum 5,000 psi). If oil and grease are present, perform solvent cleaning, as per SSPC SP-1.			
	Verify equipment setup, blast media, and surface preparation methods match designated test coupon.			
CHECKPOINT 5 - SURFACE PREPARATION				
	Verify environmental conditions (see CHECKPOINT 3).			
	Ensure cleanliness of prepared surface is as per specification (i.e.: SSPC SP-11, SP-10, SP WJ-2...).			
	Verify surface anchor profile using ASTM D4417-Methods B or C against SFLC Std Spec 6310. Conduct profile readings at a minimum of 5 locations for the first 1000-sqft area, and 2 locations for each succeeding 1000-sqft area.			
	Measure soluble salt conductivity in accordance with SSPC-Guide 15. Conduct 5 measurements per each 1000-sqft area (max. threshold: 70 microsiemens/cm for non-submerged surfaces, 30 microsiemens/cm for submerged surfaces).			
CHECKPOINT 6 - PRIMER COAT APPLICATION				
	Verify environmental conditions (see CHECKPOINT 3).			
	Verify proper mixing and stand-in (induction) times.			
	Ensure no paint is applied when the temperature is expected to drop to freezing before the paint has dried.			
	Ensure surfaces are completely dry, unless otherwise allowed by the coating manufacturer(s).			
	Verify wet film thickness (WFT) at random, to prevent under or over application. Verify final DFT.			
	Brush out all runs, sags, drips, and puddles.			
	Perform visual inspection for holidays and other defects.			
CHECKPOINT 7 – STRIPE COAT APPLICATION				
	Verify environmental conditions (see CHECKPOINT 3).			
	Ensure overcoating window is as per manufacturer's instructions.			
	After primer coat (mist coat after inorganic zinc), brush-apply un-thinned coat of same primer paint over edges, weld seams, cut-outs, and areas of complex geometries @ 3-4 mils wet film thickness (WFT).			
CHECKPOINT 8 – TOP COAT APPLICATION				
	Verify environmental conditions (see CHECKPOINT 3).			
	Ensure overcoating window is as per manufacturer's instructions.			
	Verify proper mixing and stand-in (induction) times, as applicable.			
	Verify wet film thickness at random, to prevent under or over application.			
	Brush out all runs, sags, drips, and puddles.			
CHECKPOINT 9 – FINAL INSPECTION				
	Verify final system dry film thickness. Conduct 5 sets of 3 readings for each of the first 3 100-sqft areas, followed by 5 sets of 3 readings for each succeeding 1000-sqft area.			
	Ensure that system cure is in accordance with manufacturer's recommendation for intended service.			
	Ensure potable water tank exhaust ventilation is maintained continuously from and during coating application through final system cure, to exhaust all solvent to the atmosphere and to prevent solvent entrapment.			
	For immersion coatings (including tank U/W body), record date and time of the following events: Final coat application: / ; Return to service or removal from environment controls: /			
CHECKPOINT 10 – RECORD KEEPING				
	Complete, sign, and submit all provided QA Inspection Forms.			
NAME OF QP-1/AMPP INSPECTOR		SIGNATURE	CERT. #	DATE / TIME
NAME OF CG REPRESENTATIVE		SIGNATURE	UNIT	DATE/TIME

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QA-2 - QUALITY ASSURANCE INSPECTION FORM
(ENVIRONMENTAL READINGS)

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE

Use one sheet for each activity. Record conditions every four hours from before surface preparation to application of final coating system coat.							
DATE & TIME	ACTIVITY (SURFACE PREPARATION, PRIMER COAT, BARRIER COAT, TOP COAT, ETC...)	LOCATION (FRAME & DECK, RELATION TO EQUIPMENT, ETC.)	TEMPERATURE				% REL. HUMIDITY
			DEW PT.	SURFACE	AMBIENT	ΔT DP - SURFACE	
NAME OF QP-1/AMPP INSPECTOR		SIGNATURE			CERT. #		DATE / TIME
NAME OF CG REPRESENTATIVE		SIGNATURE			UNIT		DATE/TIME

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**QA-3a - QUALITY ASSURANCE INSPECTION FORM
(SURFACE PROFILE LOG FOR PROFILE MEASUREMENTS IAW ASTM D4417-METHOD-C)**

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE
LOCATION OF WORK (FRAME REFERENCES)			AREA (SQFT)

SURFACE PREPARATION METHOD		PROFILE ACHIEVED (MILS)		
		MIN	MAX	MEAN
SSPC-SP-10/AMPP No. 2	☐			
SSPC-SP WJ-2/AMPP WJ-2	☐			
SSPC-SP-3	☐			
SSPC-SP-11	☐			
SSPC-SP-11 (inaccessible area)	☐			
Brush-blasting (non-metallic substrate)	☐			
ABRASIVE MANUFACTURER:		ABRASIVE SIEVE SIZE:		

PLACE SURFACE PROFILE REPLICA TAPES IN THE SPACES PROVIDED BELOW, TO SERVE AS PERMANENT QA RECORD. MAINTAIN A SEPARATE LOG FOR EACH LOCATION. WHEN AN AREA IS DIVIDED INTO SEPARATE SECTIONS, MAINTAIN A SEPARATE LOG FOR EACH SECTION.					
Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here	
Reading (mils):		Reading (mils):		Reading (mils):	
Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here	
Reading (mils):		Reading (mils):		Reading (mils):	
Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here	
Reading (mils):		Reading (mils):		Reading (mils):	
Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here	
Reading (mils):		Reading (mils):		Reading (mils):	
Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here		Place Surface Profile Replica Tape Here	
Reading (mils):		Reading (mils):		Reading (mils):	
MEAN MIL READING (IAW ASTM D4417-METHOD C) FOR ABOVE 15 READINGS:					

NAME OF QP-1/AMPP INSPECTOR	SIGNATURE	CERT. #	DATE / TIME
NAME OF CG REPRESENTATIVE	SIGNATURE	UNIT	DATE/TIME

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QA-3b - QUALITY ASSURANCE INSPECTION FORM
(SURFACE PROFILE LOG FOR PROFILE MEASUREMENTS IAW ASTM D4417-METHOD-B)

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE
LOCATION OF WORK (INCL. FRAME #'S)		AREA (SQFT)	

SURFACE PREPARATION METHOD		PROFILE ACHIEVED (MILS)		
		MIN	MAX	MEAN
SSPC-SP-10/AMPP No. 2	☐			
SSPC-SP WJ-2/AMPP WJ-2	☐			
SSPC-SP-3	☐			
SSPC-SP-11	☐			
SSPC-SP-11 (inaccessible area)	☐			
Brush-blasting (non-metallic substrate)	☐			
ABRASIVE MANUFACTURER:		ABRASIVE SIEVE SIZE:		

RECORD MEASUREMENTS TAKEN IN THE SPACES PROVIDED BELOW, TO SERVE AS PERMANENT QA RECORD. MAINTAIN SEPARATE LOG FOR EACH LOCATION. WHEN AN AREA IS DIVIDED INTO SEPARATE SECTIONS, MAINTAIN A SEPARATE LOG FOR EACH SECTION.					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Reading (mils):					
Mean Mils Reading (IAW ASTM D4417-Method B for above 10 readings (by column):					
Mean Reading (mils)					

NAME OF QP-1/AMPP INSPECTOR	SIGNATURE	CERT. #	DATE / TIME
NAME OF CG REPRESENTATIVE	SIGNATURE	UNIT	DATE/TIME

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**QA-4 - QUALITY ASSURANCE INSPECTION FORM
(SURFACE SOLUBLE SALT CONDUCTIVITY LOG)**

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE
LOCATION OF WORK (INCL. FRAME #'S)			AREA (SQFT)

[illegible]

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QA-5 - QUALITY ASSURANCE DATA FORM
(COATING THICKNESS)

(Use one sheet for each sequence)

VESSEL NAME	HULL #	WORK ITEM #	WORK ITEM TITLE

COATING MFG	PRODUCT NAME	BATCH #	INDUCTION TIME	COATING SYSTEM SEQUENCE (PRIMER/TOUCHUP/3RD COAT, ETC.)

DRY FILM THICKNESS (DFT) MEASUREMENTS IAW SSPC-PA 2.

SPOT	1	2	3	4	5	AVERAGE VALUE
*BASE METAL READING (BMR)						
*Required, If Magnetic Pull-Off (Type I/Banana) Gauge Is Used.						

LOCATION (FRAME REFERENCE):						
SPOT	1	2	3	4	5	OVERALL AVG. DFT
1						
2						
3						
AVG.						
LOCATION (FRAME REFERENCE):						
SPOT	1	2	3	4	5	OVERALL AVG. DFT
1						
2						
3						
AVG.						
LOCATION (FRAME REFERENCE):						
SPOT	1	2	3	4	5	OVERALL AVG. DFT
1						
2						
3						
AVG.						

APPLICATION METHOD (AIRLESS, CONVENTIONAL SPRAY, ROLLED)	AVERAGE DFT

NAME OF QP-1/AMPP INSPECTOR	SIGNATURE	CERT. #	DATE / TIME
NAME OF CG REPRESENTATIVE	SIGNATURE	UNIT	DATE/TIME

4.2 Tank and Void Assessment form.

SFLC-ESD-25		TANK AND VOID ASSESSMENT FORM	
PRINT	RESET		
GENERAL DATA Note: Use SFLC-ESD-29 for Compartments other than Tanks & Voids.			
Inspector's Name:		Organization:	Contact Info:
Cutter Name:		Cutter Class:	Hull:
Tank:		Service:	Tank Area: Gallons:
Solid Balast:	Access Compt:	Date:	Assessment Reason:
ACCESS DATA			
Manhole and cover condition:		Tank Penetration Condition:	
VENT OVERFLOW DATA			
Present: ☐ Yes ☐ No Check Valve Installed: ☐ Yes ☐ No Check Valve Operates Properly: ☐ Yes ☐ No			
LADDER DATA			
NR of Ladder(s) Present: ☐ Yes ☐ No Ladder Damaged: ☐ Yes ☐ No Ladder Material:			
TANK LEVEL INDICATOR (TLI) DATA			
TLI Present in Tank: ☐ Yes ☐ No TLI Damaged: ☐ Yes ☐ No TLI Type:			
SOUNDING TUBE DATA			
Sounding Tube Present in Tank: ☐ Yes ☐ No Sounding Tube Damaged: ☐ Yes ☐ No			
Striker Plate Damaged (>50%): ☐ Yes ☐ No			
CATHODIC PROTECTION DATA			
Cathodic Protection in Tank: ☐ Yes ☐ No Total Zincs: Number of Zincs > 50% Depleted:			
1-6 Cleanliness & Housekeeping			
Clean to light layer or residue		1-2 (G)	Comments:
Loose accumulation scale		3-4 (Y)	
Impending residue and sediments		5-6 (R)	
% 1-6 Coating Systems			
All Painted Surfaces		1-2 (G)	Comments:
		3-4 (Y)	
		5-6 (R)	
% 1-6 Structural			
Corrosion		1-2 (G)	Comments:
		3-4 (Y)	
		5-6 (R)	
Pitting & Grooving		1-2 (G)	Comments:
		3-4 (Y)	
		5-6 (R)	

SFLC-ESD-25	TANK AND VOID ASSESSMENT SHEET
PRINT	

Structural Integrity Data			
Estimated Total Linear Feet of Structure Requiring Repair:		Estimated Total Square Feet of Plating Requiring Repair:	
Cracks/ Fractures Present:	☐ Yes ☐ No	Buckling/Deflections/ Distortions Present:	☐ Yes ☐ No
Holes Present:	☐ Yes ☐ No	Material Wastage Present:	☐ Yes ☐ No
All Welds Intact:	☐ Yes ☐ No	Structural Evaluation Recommended:	☐ Yes ☐ No

PHOTOGRAPHS
Pictures Taken (enter quantity):
Note: To add pictures to this form, Work Station must have Adobe Acrobat (not Reader) installed. Add all photos and photo comments to a word document and save file. Open this form and click "Combine Files". Add the document with photos you've just saved and save as a new combined .pdf files.
Additional Comments: DRAFT

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WORK ITEM 1: Hull Plating, Side Scan, Ultrasonic Testing

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to perform thickness measurements of underwater (u/w) hull plating by ultrasonic side scan method.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 111-001, Rev H, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-002, Rev G, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-003, Rev J, Shell Exp FR 101-182

Coast Guard Drawing 901 WMEC 111-004, Rev D, Transom Pl & Fr

Coast Guard Contract Guidance Drawing 901 WMEC 111-008, Rev -, Expanded Shell/Gen Arrg't
Overlay for Shell Survey, 901-904

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General
Requirements

OTHER REFERENCES

American Society for Nondestructive Testing (ASNT) SNT-TC-1A, 2020, Recommended Practice for
Personal Qualification and Certification in Nondestructive Testing

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.3 Side scan requirements and procedures.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Hull zinc anode system
- Docking blocks

3.2 Work coordination and preparation. The Contractor must plan all specified vessel work detailed in this specification package to ensure the side scan survey does not necessitate a work stoppage. The Contractor must abide by the following guidance, to facilitate the ultrasonic survey:

- Ensure that all u/w body surface preparation and primer coat application tasks are completed within ten days of lifting the vessel.
- Provide a continuous supply of fresh water at .50 gpm within 50 feet of the vessel with standard hose bib fitting.
- Provide 110 VAC electrical power, and scaffolding or other suitable means of access to bring inspection personnel within one foot of all the body surface inspection areas that are five feet above ground level.
- After completion of the hull survey, and authorized hull repairs, resume/complete u/w body preservation process by applying the second coat of the anti-corrosive coating system, and all anti-fouling paint coats.
- Pay close attention to manufacturer's recommended practices in regards to paint system overcoat time limitations and required degree of surface cleanliness prior to resuming u/w body coating application.

3.3 Side scan requirements and procedures. The Contractor must ensure the following referring to the Coast Guard drawings listed in Section 2 (References) herein as guidance during the performance of the tasks specified below and submit a CIR.

3.3.1 Testing company and personnel. Provide the services of a qualified ultrasonic testing company, with suitable side scan equipment, to conduct a "Scanning Crawler Automated Test" of the vessel's u/w hull plating for the presence of metal corrosion and deterioration. Ensure the following:

- All scanning operations are completed by personnel meeting the personnel qualification and certification, as required by contractor process. This contractor-developed personnel qualification and certification process must meet, at a minimum, ASNT Recommended Practice No. SNT-TC-1A.
- An ASNT Level III Analyst is on site, to oversee the project and evaluate the collected data.
- An ASNT Level III or II Analyst is on site, to verify with a hand held gauge all areas on the hull identified as thinner than the specified criteria, in order to expedite all repair procedures.
- Qualified personnel are provided to complete all on-site corrosion inspections.

3.3.2 Coast Guard notification.

3.3.2.1 Provide a daily update to the COR on all areas inspected and potential issues or schedule changes.

3.3.2.2 Notify the COR at least 3 days prior to completion for an onsite review of readings and repair areas.

3.3.3 Scanning parameters. Use the following parameters during the course of this inspection:

- Automated scanning resolution must be set to 0.5" x 0.5"
- Plates where scan data indicating wastage greater than or equal to 25% nominal plate thickness must be highlighted in "RED" and designated "repair areas".
- Plates where scan data indicating wastage greater than or equal to 15% to 24% of nominal plate thickness must be highlighted in "YELLOW".
- Plates where scan data indicating wastage less than 15% of nominal plate thickness must be highlighted in DARK GREEN.
- Hand held UT readings must be taken to verify bad metal areas found with the side scan equipment. All readings must be provided to the COR and the lowest reading marked on the hull at the bad metal location.
- Areas that are accessible but cannot be reached by the automated scanner must be scanned with a hand-held meter at a resolution of 2" x 2". Any repair areas found during the hand scan must be marked on the ship's hull and their approximate positions recorded on the plate drawings and report summary.

3.3.3.1 Any scanning parameter or scope of work changes, including deviations to the color scheme specified above, must be approved by the KO.

3.3.3.2 The "echo-to-echo" technique, or use of multiple backwall echoes to exclude coating thickness, is not authorized due to unreliability in this application.

3.3.4 Ensure that the side scan inspection area includes the Port, Starboard, and Transom exterior shell plate below the water line, but excludes the following list.

- Areas under and within 6-inches of blocks and side blocks (areas that are accessible 6 inches from blocks and side blocks must be hand scanned.)
- Areas up to 6-inches around any obstacle on the hull including sea suction, discharges, rudderposts, knuckle, bow stem, stabilizer tubes, and stern tubes, as applicable.
- Any areas blocked by immovable scaffolding or shipyard activity.

3.4 Side scan reports. The Contractor must submit to the COR a final report within 3 working days after the side scan inspection is completed. A copy of all repair areas must be clearly marked on the provided shell plate drawing(s), and provide a summary document of same areas. Submit two hard copies of the report, and electronically submit the report and raw data via email to the COR, as part of the CIR submission prior to departing the job site. Ensure the reports include:

- Inspection overview.
- Data sheets for each plate with designated repair areas outlined in blue. Each data sheet must contain at least one longitudinal reference (e.g. frame or bulkhead number) and one athwartships reference (e.g. hull stringer number).
- Mosaic of scanned data overlaid semi-transparently on Coast Guard Drawings 901 WMEC 111-001, 901 WMEC 111-002, and Coast Guard Drawing 901 WMEC 111-03; and additionally on Coast Guard Contract Guidance Drawing 901 WMEC 111-008.

3.4.1 Photos. Provide digital photos of all designated repair areas.

3.4.2 Electronic files. Ensure electronic submissions contain a complete copy of the report, interactive data files, and digital pictures. All electronic files must be compatible with standard USCG workstation operating software.

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Electronic files format must be such that the CG has the capability to change the percent of wastage (format conditioning) per plate. Electronic files must be capable of being opened using MS Word, MS Excel, or PDF software. Digital pictures must be recorded in JPEG format.

3.5 **Repair plan.** The Contractor must submit a plan to accomplish repairs based on the final side scan report. The report must include all applicable external and internal interferences in way of structural repairs. Submit CFR.

3.5 Repair plan. The Contractor must submit a plan to accomplish repairs based on the final side scan report. The report must include all applicable external and internal interferences in way of structural repairs. Submit CFR.

3.6 Side shell plating renewal. The Contractor must crop and renew 400 square feet of steel hull plating, as designated by the Coast Guard Inspector and using Coast Guard Drawings 901 WMEC 111-001, 901 WMEC 111-002, and Coast Guard Drawing 901 WMEC 111-03; and additionally on Coast Guard Contract Guidance Drawing 901 WMEC 111-008 as guidance.

3.6.1 Touch-up preservation, general. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

3.6.2 Boundary test, generic. The Contractor must verify the integrity of all boundaries affected by this work item using one of the methods described in SFLC Std Spec 0740, Appendix C. Submit a CFR.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 2: Hull Plating, U/W Body, Ultrasonic Testing

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to conduct ultrasonic testing (UT) of U/W hull plating.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 111-001, Rev H, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-002, Rev G, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-003, Rev J, Shell Exp FR 101-182

Coast Guard Drawing 901 WMEC 801-022, Rev K, Docking Plan

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022,
Welding and Allied Processes

OTHER REFERENCES

None.

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.2 (UT measurements).

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 UT measurements. The Contractor must perform a total of 200 UT measurements of U/W hull surfaces, in locations designated by the Coast Guard Inspector, using Coast Guard Drawings 901 WMEC 111-001, 901 WMEC 111-002, 901 WMEC 111-003, and 901 WMEC 801-022 for guidance, in accordance with SFLC Std Spec 0740, Appendix C. Mark up a copy of Coast Guard Drawings 901 WMEC 111-001, 901 WMEC 111-002, and 901 WMEC 111-003 to show all locations selected for measurement and record all measurements in Table I (UT Measurements Summary Sheet - Port/Stbd Shell) provided at the end of this item. Submit a CIR, to also include the marked up drawing(s) and completed Table I.

3.2.1 In addition to the UT measurements, take a total of 100 pit-depth measurements, using a suitable pit depth gauge.

4. NOTES

4.1 UT Table. Table 1 provided below is for the recording of all UT measurements.

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TABLE 1 - UT MEASUREMENTS SUMMARY SHEET PORT/STBD SHELL

[illegible]

WORK ITEM 3: U/W Body, Preserve, 100 percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to preserve U/W body surfaces.

NOTES

1. The purpose of the preservation tasks covered by this item is to completely remove all existing coatings and recoat the U/W body surfaces with a new coating system.

2. U/W body surfaces are as defined in 4.1 (Definitions) and exclude areas covered by docking blocks unless the vessel is fleeted for the express purpose of preserving blocked areas.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 801-009, Rev M, Outboard Profile

Coast Guard Drawing 901 WMEC 602-002, Rev C, Distinguishing Figures

Coast Guard Drawing 901 WMEC 603-001, Rev A, Draft Marks

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

The Society for Protective Coatings (SSPC) Surface Preparation Specification No.1 (SSPC-SP 1),
2016, Solvent Cleaning

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

-
- 3.2.3 Post-surface preparation cleaning and inspection

3.1.2 Tech Rep. The Contractor must refer to SFLC Std Spec 0000, paragraphs 3.2.4.2.1 (Painting contractor certification program requirement) and 3.2.4.2.2 (Coating Tech Rep), for applicable requirements.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection) to the following components, spaces and equipment:

- All scuppers and overboard discharges.
- Waterline area where U/W body coating system interfaces with freeboard coating system (unless freeboard surfaces are also being preserved).
- Fathometer transducer surfaces.
- Sea valve openings.
- Propeller shaft bearings and seals.
- Rudder bearings.
- Propellers.
- Hydraulic systems (CPP, rudders, and fin stabilizers).
- Zinc anodes (unless anodes are being renewed).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Sea chest grating.
-

NOTE

The hull fairings are not to be removed simply as an interference for 100% near white blasting unless found to be failing.

3.1.5 Initial inspection. Prior to removing the existing coating system, the Contractor must inspect and verify whether all draft marks have permanent markings (weld beads or impressions), showing their location on the hull. Submit a CFR.

3.1.5.1 Fairing inspection. Prior to preservation preparation, inspect hull fairing condition. Submit a CFR.

3.1.5.2 Weld bead inspection. Prior to preservation preparation, using Coast Guard Drawings 901-WMEC 602-002 and 901-WMEC 603-001 as guidance, inspect all weld bead markings for draft marks and boot top. If weld beads are missing, make note of the existing boot top and submit a CFR.

3.1.6 Water used in preparation and washing procedures. The Contractor must ensure that water used in all surface preparation tasks, including pre-surface preparation wash and water jetting, is of sufficient purity and quality that it does not prevent the surface being cleaned from achieving the required degree of surface cleanliness or non-visible contamination criteria.

3.1.6.1 Ensure that surface preparation water does not contain sediments or other impurities that are destructive to the proper functioning of the cleaning equipment.

3.1.6.2 Ensure that all water used in any surface preparation or cleaning procedures is captured, contained, and all spent water disposed of in accordance with all Federal, state and local regulations.

3.1.7 Surface preparation optional methods. The Contractor has the option of using either high/ultrahigh pressure water jetting or abrasive blasting to achieve the required surface preparation, prior to application of the coating system specified in 3.2 (Preservation requirements). The Contractor may add abrasives to the water jet stream, for one or both of the following reasons:

- Achieving greater productivity.
- Achieving the required surface profile.

NOTES

1. Existing coating system on the U/W body surfaces may have a nominal thickness in excess of that which was originally installed, because of “patch-coats” applied during past drydock availabilities.

2. Water jetting without abrasive addition does not provide any additional anchor profile to the surface, beyond what was present after the previous surface preparation.

3. Whenever fleetling is required for any reason, if the U/W Body has not yet been fully preserved, the areas adjacent to all docking blocks must be fully preserved at least 2 feet in each direction prior to refloating the vessel. If the vessel will be placed onto a different blocking position, the areas in way of the new blocking position must be fully preserved.

3.2 Preservation requirements. The Contractor must accomplish the following tasks:

3.2.1 Pre-surface preparation wash. Accomplish low-pressure (less than 5,000 psi) freshwater wash of all affected surfaces, to remove soluble chlorides and other surface contaminants. Refer to SSPC-SP 1, for guidance.

3.2.2 Surface preparation and coating application.

The Contractor must accomplish the following tasks:

3.2.2.1 Prepare and coat the U/W hull surfaces with the system specified for “Underwater (U/W) Body and Boot-Top (U/W Body and Boot-Top, Steel Hulls (Up to 7 Years), in Salt Water”, in SFLC Std Spec 6310, Appendix A (Cutter and Boat Exterior Painting Systems). Ensure that the anticorrosive (AC) is tacky prior to applying the first antifouling (AF) top-coat.

3.2.2.2 Ensure that the first AF coat is applied over the AC undercoating, while it is still tacky.

3.2.3 Post-surface preparation cleaning and inspection. After completion of surface preparation and prior to coating application, accomplish the following tasks and submit a CIR.

3.2.3.1 Perform a visual inspection of the prepared U/W body steel substrate.

3.2.3.2 Perform solvent cleaning of all prepared surfaces, in accordance with SSPC-SP 1. Capture, contain, and dispose of all wastes from solvent cleaning, in accordance with all Federal, state and local regulations..

NOTE

Lists of all authorized coating materials and suppliers are listed in SFLC Std Spec 6310, Appendix C (Cutter and Boat Authorized Coatings).

3.2.3.2 Ensure the alternating color coats must be as follows: Red/Grey/Black/Red/Black. The first two coats are anti-corrosive and the final three coats are anti-foulant in accordance with SFLC Std Spec 6310. The first four coats are applied to both U/W body and boot-top. The final black AF coat is for boot-top only..

3.3 Draft mark painting. The Contractor must paint all draft marks with two coats of a “Polysiloxane” or “Silicone Alkyd Enamel” coating, at 2.0-3.0 mils DFT, white (17925).

3.4 In-process quality control. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”). Surfaces being preserved are considered “critical-coated surfaces”.

3.5 AF system protective measures – in the event of prolonged atmospheric exposure. If the vessel will not be refloated within the immersion time as recommended by the coating system manufacturer, the Contractor must employ suitable measures - including but not limited to keeping the coating system wet and away from direct sun light, to avoid film damage that may reduce or impact the final performance of the AF coating system when placed into service.

4. NOTES

4.1 Definitions.

4.1.1 U/W Body. The underwater body is defined as the areas from the bottom of the keel to the upper edge of the boot-topping, as shown on Coast Guard Drawing 901 WMEC 801-009, including rudders, transducer hull rings, sea chests (including the sea chest inlet up to the final connection with the flange face at the sea chest isolation valve(s)), bilge keels, struts, skegs, gratings, rope guards, and all other appendages.

WORK ITEM 4: Hull Plating Freeboard, Preserve, 100 Percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to preserve 100% of the freeboard surfaces defined in 4.1 Definition of freeboard surfaces.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 588-003, Rev M, Helicopter Safety Net-A&D

Coast Guard Drawing 901 WMEC 612-004, Rev A, NAVSEA Certified Flight Deck Net Installation.

DEVELOPER NOTE: Delete 901 WMEC 588-003 once Flight Deck Net Replacement TCTO is accomplished. If TCTO is not yet accomplished, delete 901 WMEC 612-004.

Coast Guard Drawing 901 WMEC 801-009, Rev L, Outboard Profile

Coast Guard Fleet Drawing FL 2804-12, Rev -, U.S.C.G. Emblem

Coast Guard Fleet Drawing FL 2804-22, Rev A, Consolidated Visual ID for Cutters

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

OTHER REFERENCES

None.

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection) and to the following specific components, spaces and equipment:

- Areas where underwater body coating system interfaces with freeboard coating system (unless u/w body surfaces are also being preserved).
- Adjacent deck surfaces and deck fittings.
- Liferails
- Vents
- Bulkheads
- Vent intakes
- Deck equipment.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Anchors and anchor chain
- Flight deck nets.
- Developer Note: Fix formatting for paragraph 3.1.4.1 and 3.1.4.2 then delete this bullet. Select dwg 612-004 or 588-003 in paragraph 3.1.4.2 depending on whether Flight Deck Net TCTO has been accomplished.
- 3.1.4.1 Flight deck nets initial inspection. Prior to removing flight deck nets, the Contractor must inspect, in the presence of the Coast Guard Inspector, the flight deck net installation to document initial condition. Submit a CFR.
- 3.1.4.2 Flight deck net restoration. The Contractor must reinstall flight deck nets in accordance with Coast Guard Drawing 901 WMEC 588-003 or 901 WMEC 612-004. After reinstallation, the Contractor must accomplish a visual inspection, in the presence of the Coast Guard Inspector to ensure that the flight deck net installation is satisfactory.

3.2 Initial inspection. Prior to removing the existing coating system, the Contractor must inspect and verify whether or not all hull identification numbers and letters have permanent markings (weld beads or impressions), showing their location on the hull. Submit a CFR.

3.3 Pre-surface preparation wash. The Contractor must accomplish low-pressure (less than 5,000 psi) fresh water wash of all affected surfaces, to remove soluble chlorides and other surface contaminants. Capture, contain, and dispose of wash water for proper disposal in accordance with all Federal, state and local regulations.

3.4 Surface preparation optional methods. The Contractor must have the option of using either high/ultrahigh pressure water Jetting or abrasive blasting to achieve the required surface preparation, prior

to application of the coating system specified in 3.6 (Preservation requirements). The Contractor may add abrasives to the waterjet stream, for one or both of the following reasons:

- Achieving greater productivity.
- Achieving the required surface profile.

NOTE

Waterjetting without abrasive addition does not provide any additional anchor profile to the surface, beyond what was present after the previous surface preparation.

3.5 Substrate inspection. After completing surface preparation and before coating application, the Contractor must perform a visual inspection of the prepared substrate, and submit a CFR.

3.6 Preservation requirements. The Contractor must accomplish the following tasks:

3.6.1 System particulars. Prepare and coat 100% of the freeboard surfaces as designated in paragraph 1.1 (Intent) with the system specified for “Freeboard/Superstructure/Mast (Freeboard/Superstructure)” in SFLC Std Spec 6310. Select the following:

- Select “Option I” system, for the applicable metal substrate.
- White (17925), as the top/finish coat color..

3.6.2 Visual identification markings. The Contractor must paint all distinctive visual safety and identification markings as follows:

3.6.2.1 Decals. Coast Guard Emblem decals may be substituted for painted emblems. Vinyl adhesive letters and numbers may also be substituted for painted letters and numbers - at the Contractor’s discretion.

3.6.2.2 Painted markings. Paint all distinctive visual identification markings, including the Coast Guard diagonal stripes, "U.S. COAST GUARD" legends, hull numbers, and draft marks in accordance with SFLC Std Spec 6310, and Coast Guard Drawings FL 2804-12 and FL 2804-22 as applicable.

NOTES

Surfaces being preserved are considered “critical-coated surfaces”.

Although there may be several possible procurement sources for the Coast Guard Emblem decals and vinyl adhesive letters and numbers, one known source is:

**Brace Enterprises
10250 SE 138 Terrace
Dunnellon, FL 34431**

**352-489-4442 / Fax: 352-489-4476
www.braceenterprise.com**

3.6.3 Tear drop. Not applicable.

3.7 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”).

4. NOTES

4.1 Definition of freeboard surfaces. For the purposes of this item the freeboard is defined as continuous shell plating, from the upper limit of the boot-topping to the 01 Deck Level, as shown on Coast Guard Drawing 901 WMEC 801-009, included are the following:

Transom, port, and starboard steel surfaces, including anchor pockets and rail stanchions, up to the steel portion of the Detacouple on the 01 Level.

DRAFT

WORK ITEM 5: Hull and Structural Plating, Steel Plate, Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew steel shell plating.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 111-001, Rev G, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-002, Rev F, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-003, Rev H, Shell Exp FR 101-182

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

NOTE

This item does not address interferences in way of repair work. Cost of interference removal and reinstallation must be negotiated via the Change Request process.

3.2 Renewal. The Contractor must renew steel plating, as designated by the Coast Guard Inspector, up to 300 square feet, using Coast Guard Drawings 901 WMEC 111-001, 901 WMEC 111-002, and 901 WMEC 111-003 as guidance.

3.3 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, paragraph 3.1.13 (Touch-ups and minor coating repairs.)

4. NOTES

This section is not applicable to this work item.

WORK ITEM 6: Propulsion Shaft Fairwaters and Rope Guards, Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew all fairwaters and rope guards on external shafting for a designated shaft.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters And Rope Guards

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2020,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

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3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

3.1.5 Work coordination. The Contractor must coordinate the inspection tasks specified herein with all propeller shaft-related and U/W body hull preservation tasks, as specified in other work items in this specification package.

3.2 Removals. The Contractor must accomplish the following tasks:

3.2.1 Remove and dispose of existing fairwaters and rope guards in accordance with all Federal, state and local regulations.

3.2.2 Grind smooth any welding left on struts after the removal of rope guards.

3.3 Requirements. The Contractor must renew the designated fairwater and rope guard and associated fasteners. Accomplish the following tasks, using the referenced drawings as guidance:

3.3.1 Prepare and coat the fairwater and rope guard surfaces in accordance with SFLC Std Spec 6310.

3.3.2 Prior to preservation, visually inspect all fairwater and rope guard surfaces, including hardware mounting brackets/surfaces for corrosion, cracks, and all other defects. The Contractor must clean and chase all threads to be reused. Submit a CFR.

3.3.3 Provide and install new rope guards, after coating system has cured sufficiently to prevent being damaged, using 901 WMEC 114-003 as guidance. Provide and install new fasteners.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 7: Appendages (U/W), Preserve (Internal) and Leak Test

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to perform leak test and preserve internal surfaces of the designated U/W appendages.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 111-001, Rev H, Shell Exp Fwd 101
Coast Guard Drawing 901 WMEC 111-002, Rev G, Shell Exp Aft 182
Coast Guard Drawing 901 WMEC 111-003, Rev J, Shell Exp FR 101-182
Coast Guard Drawing 901 WMEC 114-001, Rev F, Skeg
Coast Guard Drawing 901 WMEC 114-002, Rev D, Bilge Keels
Coast Guard Drawing 901 WMEC 114-005, Rev -, Stern Flap Mods to Structure

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements
Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020,
Welding and Allied Processes

OTHER REFERENCES

ASTM International (ASTM) D5363, 2016, Standard Specification for Anaerobic Single-
Component Adhesives (AN)
MIL-S-45180, 1998, Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant
MIL-PRF-16173E, Oct 2017, Corrosion Prevention Compound, Solvent Cutback, Cold-
Application

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.2 Leak test.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

3.2 Leak test. In the presence of the Coast Guard Inspector, the Contractor must accomplish the following tasks for all appendages as designated in Table 1 below, using the Coast Guard drawings listed in Section 2 (References) as guidance:

- Remove and dispose of all existing plugs (e.g. drain plugs, fill/vent plugs), as applicable.
- Submit a CIR noting any liquids draining from any of the appendages – indicating the presence of a leak.
- If a Change Request has been authorized and released based on the CIR, the Contractor must repair leaks.
- Dispose of any drained liquids in accordance with all Federal, state, and local laws and regulations.

NOTE

Some liquids that are drained may be classified as hazardous materials or hazardous waste depending on state and local regulations. The vessel environmental coordinator may assist with determination of waste category.

TABLE 1 – U/W APPENDAGES

APPENDAGE	FRAME	SIDE
Bilge Keel	118 to 170	P/S
Skeg	150 to 230	CL
Fin Stabilizers	96 to 103	P/S

3.3 Preservation of void internal surfaces. The Contractor must accomplish the following tasks for each appendage listed in Table 1, using the Coast Guard Drawings listed in Section 2 (References) herein as guidance:

- Remove any existing rust preventive compound using steam, solvent, vapor degreaser, or other suitable method.

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- Fill all appendage interior surfaces with a rust preventive compound conforming to MIL-PRF-16173, Class II, Grade 3, to coat all surfaces.
- Drain, collect, and dispose of remaining compound in accordance with all applicable Federal, state, and local regulations.
- Ensure that the coated surfaces are left exposed to the atmosphere for 24 hours to allow for adequate drying. Submit CFR.

3.4 Plug renewal. Upon completion of all work on appendages, the Contractor must accomplish the following tasks:

- Chase the hull insert threads of all plugs.
- Renew all stainless steel, Type 316 and Monel plugs (as applicable) in place of those removed.
- Coat new plugs prior to installation with a sealing and locking compound conforming to ASTM D5363-AN0123 or a flexible joint compound conforming to MIL-S-45180, Type II.

3.5 Appendage air test. The Contractor must accomplish the following tasks for any appendage from the above list designated by the Coast Guard Inspector, with an approved Change Request resulting from the CIR from paragraph 3.2:

- Perform an air test of the appendage, in accordance with SFLC Std Spec 0740, Appendix C.
- Submit a CFR.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 8: Propulsion Shaft Fairwaters And Rope Guards, Inspect

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to inspect all fairwaters and all rope guards on external shafting.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters and Rope Guards

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2020, Requirements for Preservation of Ship Structures

OTHER REFERENCES

The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP 10/NACE No.2, 2007, Near-White Blast Cleaning

The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP WJ-2, 2012, Waterjet Cleaning of Metals – Very Thorough Cleaning

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Inspection - initial. Prior to commencement of work, the Contractor must visually inspect the shaft fairwater and rope guard installations using the references listed in Section 2. Submit a CFR. If a Change Request is authorized, the Contractor must renew a designated fairwater or rope guard in lieu of repairing existing.

3.3 Removals. The Contractor must accomplish the following tasks:

3.3.1 Remove and retain existing fairwaters and rope guards. Discard all mounting hardware.

3.3.2 Grind smooth any welding left on struts after the removal of rope guards, and clean all surfaces thoroughly.

NOTE

This may require the removal of fairing material before fairwater socket hex-head bolts can be extracted.

3.3.3 Remove and install new fairing material as required to access bolts.

3.4 Inspection particulars. The Contractor must accomplish the following tasks, using the referenced drawings as guidance:

3.4.1 Prepare the fairwater and rope guard surfaces in accordance with SFLC Std Spec 6310 to the equivalent of a "SSPC-SP 10/NACE No. 2, Near-White Blast Cleaning", with a 1.5 to 2.5 mil anchor profile; or to "SSPC-SP WJ-2/L".

3.4.2 Visually inspect all fairwater and rope guard surfaces, including hardware mounting brackets/surfaces for corrosion, cracks, and all other defects. Submit a CFR.

3.4.3 Coat all fairwater and rope guard surfaces, using the coating system specified for "Underwater (U/W) Body and Boot-Top, Steel Hulls (Up to 7 Years), in Salt Water", in SFLC STD 6310, Appendix A (Cutter and Boat Exterior Painting Systems), and using the drawings listed in Section 2 as guidance.

3.5 Reinstallation. After coating system has cured sufficiently to prevent being damaged, the Contractor must reinstall fairwaters and rope guards. Renew all fasteners.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 9: Voids (Non-Accessible), Preserve (Internal) and Leak Test

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to perform leak test and preserve internal surfaces of non-accessible voids.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 111-001, Rev H, Shell Exp Fwd 101

Coast Guard Drawing 901 WMEC 111-002, Rev G, Shell Exp Aft 182

Coast Guard Drawing 901 WMEC 111-003, Rev J, Shell Exp FR 101-182

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

OTHER REFERENCES

ASTM International (ASTM) D5363, 2016, Standard Specification for Anaerobic Single-Component Adhesives (AN)

MIL PRF-16173, Oct 2017, Corrosion Preventive Compound, Solvent Cutback, Cold-Application

MIL-S-45180, 1998; Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.2 Leak test.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

3.2 Leak test. In the presence of the Coast Guard Inspector, the Contractor must accomplish the following tasks for all designated non-accessible voids (see Table 1 for list) and submit a CIR, using the Coast Guard drawings listed in Section 2 (References) as guidance:

- Remove and dispose of existing plugs (e.g. drain plugs, fill/vent plugs), as applicable.
- Submit a CIR noting any liquids draining from any of the voids - indicating the presence of a leak.
- Dispose of any drained liquids in accordance with all Federal, state, and local laws and regulations.

CAUTION

Some liquids that are drained may be classified as hazardous materials or hazardous waste depending on state and local regulations. The vessel environmental coordinator may be able to assist with determination of waste category.

3.3 Preservation of void internal surfaces. The Contractor must accomplish the following tasks for each void designated in Table 1, using the Coast Guard Drawings listed in Section 2 (References) as guidance:

- Remove any existing rust preventive compound using steam, solvent, vapor degreaser, or other suitable method.
- Fill all void interior surfaces with new rust preventive compound conforming to MIL-PRF-16173, Class II, Grade 3, to coat all surfaces.
- Drain, collect, and dispose of remaining compound in accordance with all applicable Federal, state, and local regulations.
- Ensure that the coated surfaces are left exposed to the atmosphere for 24 hours to allow for adequate drying.

TABLE 1 - NON-ACCESSIBLE VOIDS

VOID	FRAME	SIDE
Shallow Water Depth Sounder Transducer Fairing	80	P
Speed Log Transducer Fairing	76	P
Bow Stem Void 4-F-0-V	F	S

3.4 Plug renewal. Upon completion of all work on the designated voids, the Contractor must accomplish the following tasks:

- Chase the hull insert threads of all plugs.
- Renew all stainless steel, Type 316, or Monel plugs, in place of original plugs.
- Coat new plugs prior to installation with a sealing and locking compound conforming to ASTM D5363-AN0123 or a flexible joint compound conforming to MIL-S-45180, Type II.

3.5 Void air test. If a Change Request is authorized and released, the Contractor must accomplish the following tasks for any designated non-accessible void from the above list:

- Apply a vacuum or pressure of 2 psig to the void, in accordance with SFLC Std Spec 0740, Appendix C.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 10: Voids, Accessible, Clean and Inspect**1. SCOPE**

1.1 Intent. This work item describes the requirements for the Contractor to clean and inspect the following void(s):

TABLE 1 - VOIDS

TYPE OF STRUCTURE	LOCATION	CAPACITY - 95% (GALLONS)	LOW SUCTION (GALLONS)
Transducer Well	4-75-0-Q	1,200	
Void	4-12-0-W	7,330	

1.2 Government-furnished property.

None.

2. REFERENCES**COAST GUARD DRAWINGS**

Coast Guard Drawing 901 WMEC 529-001, Rev N, Ballasting & Emer Bilge Drainage Sys Diagram

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

OTHER REFERENCES

ASTM International (ASTM) D1330, 2015, Standard Specification for Rubber Sheet Gaskets

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.2 Operational test, initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of the TLI's for tanks listed in paragraph 1.1 (Intent), to demonstrate existing operational condition. Submit a CFR.

3.3 Cleaning and inspection requirements. The Contractor must refer to Coast Guard Drawing 901 WMEC 529-001 for guidance. to accomplish the following tasks:

3.3.1 Content removal. The Contractor must remove and dispose of all fluids and/or residues in accordance with all applicable Federal, state, and local regulations.

3.3.2 Cleaning. The Contractor must remove access cover(s). Clean the designated structure's interior surfaces free of all foreign materials, such as sediment, sludge and fungal growth. Remove all persistent residues, taking care not to damage the tank coating system. Remove cleaning media and residues continuously from the compartment during the washing process. Remove any residual wash media and wipe up residual moisture with clean lint-free cloths. Collect, contain, and dispose of all wash media, residues, and cleaning materials in accordance with all Federal, state, and local regulations.

3.3.3 Inspection. The Contractor must visually inspect all interior surfaces, including, but not limited to bulkheads, floor and overhead plating, structural members, manhole cover surfaces, fasteners and gasket seating surfaces. Complete the "Tank and Void Assessment Form" (found in the Work Item titled "General Requirements") and submit via CFR. Electronic PDF version of this form will be furnished by the COR upon request. Include the following, as applicable:

- Structural condition.
- Inaccessible areas, if any.
- Condition of coating, including measurements taken, percentage, location, and type of coating failure.
- Tank level indicator (TLI) and/or float switch condition.
- Sounding tube and striker plate condition.

- Suction and discharge piping condition.
- Fastener material and condition.

3.4 Closing. The Contractor must ensure that the compartment(s) remain open for approximately 24 hours after completion of the tasks specified above. Notify the COR at least 24 hours prior to closing the compartment(s) and ensure tank closure is accomplished in the presence of the Coast Guard Inspector. After satisfactory inspection by the Coast Guard Inspector, and completion of all authorized repairs, close the manhole cover(s) with new gasket material conforming to ASTM D1330 and cotton grommets on the studs (as applicable). The Contractor must renew up to 10% of missing or damaged nuts and washers.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.5 Operational test, post repairs. After completion of work and in the presence of the Coast Guard inspector, the Contractor must thoroughly test and demonstrate the TLI's for tanks listed in paragraph 1.1 (Intent), to be in satisfactory operating condition. Submit a CFR.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 11: Mast, Preserve, 100 Percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to preserve the aluminum Mast (see paragraph 4.2 for definition of “mast” surfaces).

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 171-001, Rev J, Main Mast

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 3042 (SFLC Std Spec 3042), 2022,
Shipboard Electrical Cable Removal, Relocation, Splice, Repair, and Installation

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 The Contractor must, once the equipment has been removed from the Mast, ensure it is stored in a protective environment. Areas/equipment/components to be protected include, but are not limited to the following (as applicable):

- Cable runs
- Label plates
- Connectors
- Rubber gaskets
- Yardarm blinkers
- Wind speed air vanes
- Navigation lights
- Ship's bell
- Adjacent bulkhead and deck surfaces
- Contact points
- Insulators
- Rubber shock mounts
- Safety climb.

3.1.3.2 The Contractor must coil up electrical cabling down to associated deck/bulkhead penetrations. Mark, seal, and cover all coiled cabling. Ensure the cable is protected from abrasion and damage.

CAUTION

Do not bend any electrical cable to a radius less than 20 times their diameter.

WARNING

Do not hoist a radar pedestal using its array.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following.

- Radar gear, including associated pedestals
- Navigation lights
- Electrical fittings
- Rigging
- Antennas
- Wind direction indicator
-

3.1.4.1 The Contractor must notify the COR 72 hours in advance of beginning interference removals.

3.2 Personnel qualification. The Contractor must ensure that all Contractor personnel performing removal and installation of electronics equipment must have the training of an electronics journeyman and be

knowledgeable in the areas of electrical and physical standards for equipment such as radar, waveguides and antennas.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.3 Operational test - initial. Prior to commencement of work, the Contractor must witness Coast Guard Personnel perform an initial operational test of all mast electrical and electronic equipment to demonstrate existing operational condition. Submit a CFR.

3.4 Mast removal. The Contractor must be aware that removal of the masts is not practical for this class of cutters.

3.4.1 Surface preparation optional methods. The Contractor has the option of using either high/ultrahigh pressure water jetting or abrasive blasting to achieve the required surface preparation, prior to application of the coating system specified in 3.3 (Preservation requirements). The Contractor may add abrasives to the water jet stream, for one or both of the following reasons:

- Achieving greater productivity.
- Achieving the required surface profile.

NOTE

Water jetting without abrasive addition does not provide any additional anchor profile to the surface, beyond what was present after the previous surface preparation.

3.4.2 Substrate inspection. After surface preparation and before the coating application, the Contractor must perform a visual inspection of all cleaned mast surfaces, including electrical hangers, hardware, and safety rail hardware. Submit a CFR

3.5 Preservation requirements. The Contractor must prepare and coat the mast surfaces using the system specified for “Freeboard/Superstructure/Mast (Freeboard/Superstructure)” in Appendix A (Cutter and Boat Exterior Painting Systems) of SFLC Std Spec 6310. Select the following:

- “Option I” system, for aluminum surfaces
- Black (17038), as the top/finish color.

3.6 Weather-proofing requirements. The Contractor must weatherproof all connectors, cables and fasteners in accordance with SFLC Std Spec 3042.

3.7 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for critical-coated surfaces).

NOTES

1. Unless a containment system is used for surface preparation dust and debris and coating application overspray during pier side/dockside preservation, the following must be adhered to:

a. All surface preparation tools/equipment must be vacuum-shrouded.

- b. Coatings must be applied by brushing or rolling.**
- 2. Coating application by brushing or rolling may require additional coats to obtain required dry film thickness.**
- 3. Surfaces being preserved are considered “critical-coated surfaces”.**

3.8 Repair of un-intended damages. The Contractor must repair all damages (including overspray) incurred to vessel surfaces (including those not covered by the intended scope of this work item), during surface preparation and paint application procedures.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.9 Operational test – post repairs. After completion of work, the Contractor must witness an operational test (by Coast Guard personnel, including C4IT personnel) of all mast electrical and electronic equipment, to prove satisfactory operating condition. Submit a CFR.

4. NOTES

4.1 Coast Guard Inspector. CG Inspector will ensure that antennas are mounted and weather-proofed before final operational test.

4.2 Definition of mast. For the purposes of this work item, “mast” is defined as all vertical and horizontal structural surfaces as detailed on Coast Guard Drawing 901 WMEC 171-001 from the 03 deck level up, which include, but are not limited to the main mast, all equipment platforms, stub mast, yardarms, life rails, ladder rungs, access openings, and access covers. Stub mast climbing safety rails are excluded and must not be painted.

WORK ITEM 12: Propulsion Shafting, Strain Gauge Alignment, Drydock, Check

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to perform strain gauge alignment checks prior to vessel dry-docking, during dry-docking, and after refloating.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-005, Rev D, Sftg Align Anal and Procs

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

OTHER REFERENCES

MIL-STD-2199B (SH), Jan 2020, Coverings for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep. The Contractor must provide the services of a Qualified Technical Representative who is familiar with the strain gauge alignment method to accomplish the following on site:

- Advise on manufacturer's proprietary system information.
- Assist with proper alignment, clearances, and repair methods associated with stern tube modification and propulsion shaft system removal and installation.

- Ensure compliance with manufacturer's procedures and standards during system disassembly, inspection, and reassembly as applicable.

3.1.2.1 Ensure the Tech Rep has experience with the system/equipment stated above and demonstrated on their résumé.

3.1.2.2 Submit the name and résumé of the Tech Rep to the COR at the Arrival Conference.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Mechanical seal
- Bulkhead seal
- Shaft coating

3.2 Definitions and standard units. For the purposes of this specification and any reports generated, the following definitions apply.

3.2.1 Optimal alignment. Defined as a state of shaft alignment (through all phases of bearing wear down) where:

- Bearing loads are within a satisfactory range
- Bending stresses are minimal
- Gear box bearing loads (Fwd and Aft) always remain positive and are near equal at midlife wear
- Shaft slope through long bearings matches the bearing slope within a reasonable range
- Horizontal loads are minimized

NOTE

“Optimal alignment” is the goal when components are being moved or relocated e.g. foundations altered, chockfast poured, new strut being installed, new stern tube installed, etc.

3.2.2 Acceptable alignment. Defined as a state of achievable shaft alignment where components are constrained (movement/relocations not within work scope) but:

- Current bearing loads are within a satisfactory range and if possible remain satisfactory though continued bearing wear down
- Horizontal loads are minimized

NOTE

“Acceptable alignment” is the goal whenever “Optimal alignment” cannot be achieved because the scope of associated work does not provide for relocation of components necessary.

3.2.3 Standard units. All reports, correspondences, and discussions must be in Standard/US Customary Units and not in SI/Metric Units.

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3.3 Strain gauge alignment check. The Contractor must perform the specified actions in Table 1 below using Coast Guard Drawings 901 WMEC 243-001 and 905 WMEC 243-005 as guidance.

TABLE 1 - TASKS

#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	REQUIREMENTS	
				PARAGRAPH	OTHER
1a	Strain Gauge Alignment Check, Pre-Drydock	1	Port Shaftline	3.3.1 3.3.1.2 3.3.1.3 3.3.2 3.3.4	Accomplish prior to drydocking.
1b	Strain Gauge Alignment Check, Pre-Drydock	1	Stbd Shaftline	3.3.1 3.3.1.2 3.3.1.3 3.3.2 3.3.4	Accomplish prior to drydocking.
2a	Strain Gauge Alignment Check, Drydock	1	Port Shaftline	3.3.1 3.3.1.1 3.3.1.3 3.3.3 3.3.4	Accomplish after drydocking the vessel and before performing any hot work and/or structural modifications.
2b	Strain Gauge Alignment Check, Drydock	1	Stbd Shaftline	3.3.1 3.3.1.1 3.3.1.3 3.3.3 3.3.4	Accomplish after drydocking the vessel and before performing any hot work and/or structural modifications.
3a	Strain Gauge Alignment Check, Post Repair	1	Port Shaftline	3.3.1 3.3.1.1 3.3.1.3 3.3.3 3.3.4 3.3.5.1 3.3.6	Accomplish in drydock after all structural and other shafting repairs and modifications have been completed, and before re-floating the vessel.
3b	Strain Gauge Alignment Check, Post Repair	1	Stbd Shaftline	3.3.1 3.3.1.1 3.3.1.3 3.3.3 3.3.4 3.3.5.1 3.3.6	Accomplish in drydock after all structural and other shafting repairs and modifications have been completed, and before re-floating the vessel.
4a	Strain Gauge Alignment Check, Post Drydock	1	Port Shaftline	3.3.1 3.3.1.2 3.3.1.3 3.3.2 3.3.4 3.3.5.2	Accomplish after re-floating the vessel.
4b	Strain Gauge Alignment Check, Post Drydock	1	Stbd Shaftline	3.3.1 3.3.1.2 3.3.1.3 3.3.2 3.3.4 3.3.5.2	Accomplish after re-floating the vessel.

3.3.1 Shaft alignment check by strain gage method. The Contractor must furnish all necessary equipment and install sufficient strain gages, measure strains, and calculate vertical and horizontal reactions using the strain gage measurement method. Perform the strain gage measurements under the direct supervision of personnel experienced in and regularly engaged in shaft alignment by the strain gage method.

3.3.1.1 When specified in Table 1, install strain gages internally and externally to measure all internal and external bearing reactions.

3.3.1.2 When specified in Table 1, install strain gages internally to measure all internal bearing reactions.

3.3.1.3 All strain gages must remain at the same locations throughout all required inspections. Do not duplicate cost of installation if strain gauges are already installed from previous evolutions.

3.3.2 Alignment check prerequisites, afloat. The Contractor must ensure the following conditions are met for the afloat alignment check:

- All shaft components are installed.
- Vessel is in the water for at least 24 hours before alignment inspection.
- All permanent ballast are in place.
- Fuel, water, and temporary ballast tanks are filled to normal operating levels, generally half to 75% capacity.
- Vessel is on an even keel.

NOTE

It is highly recommended that the Cutter get the pre-dock strain gauge reading done in advance instead of shortly prior to the dry-docking.

- All major machinery weighing over 500 pounds is either installed or simulated by equivalent weights appropriately located
- The shafts must be free to rotate and the stern tube face seal shall not be any tighter than necessary

NOTE

Small boats are not considered major machinery and therefore do not need to be simulated by equivalent weights in their absence.

3.3.3 Alignment check prerequisites, drydock. The Contractor must ensure the following conditions are met for the in-dock alignment check:

- All shaft components are installed.
- The shafts must be free to rotate and the stern tube face seal shall not be any tighter than necessary

3.3.4 Strain gauge report. The Contractor must submit a written report of results via CFR with the following information to the COR within 24 hours after alignment check:

- All specified (i.e. external and/or internal) bearing reactions
- Deviation from specified bearing reactions in Figure A5.4 of Coast Guard Drawing 901 WMEC 243-005
- A description of the alignment procedures used to obtain the bearing loads

- Strain gauge locations
- Bearing locations and reactions in shaft modeling
- Detail dimensions and configuration of the shaft model
- All mathematics values used in the shaft model such as material properties, spring constant, thermal growth effects, bearing offsets, all concentrated weights, etc
- All assumptions applied in the shaft model
- All constant values used in the shaft model
- Any indication of misalignment
- Any recommended corrective procedures. The recommended corrective procedures must include, but not be limited to:
 - New bearing off-sets to achieve an Optimal alignment (as defined in 3.2.1)
 - Other alternative to achieve an Optimal alignment
 - Detailed procedures to achieve shaft alignment
 - Table of calculated bearing wear down analysis based on prior strain gauge reports (furnished upon request by COR)

3.3.5 Strain gauge report analysis. The Contractor must provide comparative analysis of strain gauge reports as follows. This analysis may be furnished with the Strain Gauge Report described in paragraph 3.3.4.

3.3.5.1 When specified in Table 1, compare results of the “Strain Gauge Alignment Check, Post Repair” with results of “Strain Gauge Alignment Check, Drydock” and provide analysis, comments and recommendations.

3.3.5.2 When specified in Table 1, compare the findings of the inboard bearings of the “Strain Gauge Alignment Check, Post Drydock” with the “Strain Gauge Alignment Check, Pre-Drydock” conditions and provide analysis, comments and recommendations.

3.3.6 Shaft coating repair. Prior to re-floating vessel, and after all other shafting repairs, the Contractor must repair areas of the shaft GRP coating damaged in performance of strain gauge mounting and removal in accordance with Appendix B, paragraph B.5.3.2 “Repair of in-depth defects in GRP” of MIL-STD-2199.

- After curing, accomplish inspection and spark test in accordance with Appendix B, paragraph B.5.2 of MIL-STD-2199. Submit CFR with results of the spark test.

4. NOTES

4.1 Results submittal. The COR will forward a copy of each strain gauge report to the MECPL Engineering Branch.

WORK ITEM 13: Propulsion Propeller Shaft Coating, Inspect and Test

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to inspect the port and starboard propeller shaft coating.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes Strut Brg and Seal Arr and Det

Coast Guard Drawing 901 WMEC 243-004, Rev E, Controllable Pitch Propeller Shafting Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements

OTHER REFERENCES

None.

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.2 (Inspection particulars).

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance

with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Rope guard
- Fairwaters

3.2 Inspection particulars. Using the Coast Guard drawings listed in Section 2 (References) as guidance, the Contractor must clean the shaft free of all marine growth and accomplish the following inspections. Submit CIR.

3.2.1 Visual inspection. Visually inspect the following components for excessive wear and other mechanical defects such as nicks, cracks, and corrosion:

- Shaft surfaces
- All shaft and propeller associated fasteners
- Couplings (as applicable)
- Stern tube/shaft sealing systems (as applicable)
- Packing ring and glands (as applicable).

3.2.2 Shaft coating inspection. Inspect the shaft fiberglass covering for porosity as follows:

3.2.2.1 Visually inspect the plastic covering for evidence of lack of adhesion, and other defects.

3.2.2.2 Conduct a high voltage spark test of the covering at 250 to 300 volts per mil of glass reinforcement (approximately 15,000 volts for a four-ply covering).

NOTE

Pay particular attention to the joining line between sleeves, couplings, and shafts.

Coast Guard personnel will operate all shipboard machinery and equipment.

3.3 Operational test, post repairs. After completion of work, the Contractor must witness an operational test (by Coast Guard personnel) of the shafting system, for a minimum of 4 hours, to prove satisfactory operating condition. Submit a CFR.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 14: Propulsion Shafting; Remove, Inspect, and Reinstall

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to remove, inspect, and reinstall port and starboard propeller shafts.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	**Propeller shaft	NSN: 2010-01-135-7065	2 ea.	158,558.75

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arr
 Coast Guard Drawing 901 WMEC 243-002, Rev G, S- Tubes, Strut Brg & Seal A&D
 Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arr
 Coast Guard Drawing 901 WMEC 243-004, Rev E, Shafting Details
 Coast Guard Drawing 901 WMEC 243-007, Rev D, Shaft Coupling Assemblies, CPP

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements
 Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Inspect, Repair, and Test Auxiliary Machine Systems
 Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes
 Coast Guard Technical Publication (TP) 2699, Feb 2020, Technical manual for the Controllable Pitch Propeller

OTHER REFERENCES

NAVSEA Technical Manual S9243-AW-TRS-011/SHAFT Rev 1, June 2015, Surface Ships Main Propulsion Shafting Refurbishment Procedures

MIL-PRF-16173E, Oct 2017, Corrosion Prevention Compound, Solvent Cutback, Cold-Application

MIL-STD-2199B, Jan 2020 Coverings for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

ASTM D5162, 2021, Standard Practice for Discontinuity (Holiday) Testing of Nonconductive Protective Coating on Metallic Substrates

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2.2 Inspections
- 3.5 Government's right for change out

3.1.2 Tech Rep.

Not Applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 Protection of fine surfaces. The Contractor must protect all fine surfaces in accordance with SFLC Std. Spec 5000. If equipment will be stored outside, apply preservative coating in accordance with MIL-PRF-16173 Grade 1 to unprotected steel surfaces including flanges, tapers, threads, spigots, keyways, and o-ring grooves.

3.1.3.2 Hydraulic system contamination protection. The Contractor must maintain hydraulic cleanliness of the shaft bores. Cleanliness Grade A as defined in SFLC Std Spec 5000 is required.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Rope guard.
- Fairwaters.
- CPP hub.
- Shaft seal.

3.2 Work to be accomplished. Using Coast Guard Drawings 901 WMEC 243-001, 901 WMEC 243-002, 901 WMEC 243-003, 901 WMEC 243-00, 901 WMEC 243-007 and TP 2699 as guidance, as applicable, the Contractor must accomplish the following:

3.2.1 Removals.

CAUTION

Be aware historically couplings have been difficult to remove, often requiring added heat to loosen.

Ensure shaft is protected and supported at all bearing journals (minimum), at all times, until reinstallation.

When transporting/moving/lifting, ensure the shaft is supported at all times at a minimum of three locations along the shaft.

- Remove and clean all CPP oil tubes free of all oil and grease.
- Mark or measure existing coupling draw on each shaft for reference, and match-mark shaft coupling bolts to their respective holes, to facilitate reinstallation procedures.
- Remove the shaft and attached components (as applicable) to a suitable repair facility.

3.2.2 Inspections. Clean the shaft free of all marine growth and surface contaminants, and accomplish the following inspections, as applicable (submit a CIR):

3.2.2.1 Visual inspections. Accomplish a visual inspection of each internal and external surface of each flange, bore, sleeve, and shaft body in accordance with Sections 3-1.4.1, 3-1.4.2, 3-2.2.1, 3-2.3.4, of S9243-AW-TRS-011/SHAFT.

3.2.2.2 If a Change Request has been authorized and released, the Contractor must accomplish borescopic inspection of each shaft bore in accordance with Section 3-2.2.2 of S9243-AW-TRS-011/SHAFT.

3.2.2.3 Visually inspect the following components, as applicable, for excessive wear and other mechanical defects such as nicks, cracks, and corrosion, etc. Determine the extent of repairs required for each stern tube assembly, if any.

- Bearings, bearing carriers, epoxy-resin chocking and associated fasteners (as applicable).
- All shaft and propeller-associated fasteners.
- Couplings.

3.2.2.4 Dimensional Inspections. Complete the following measurements. Fill out and submit the enclosed Shaft Inspection worksheet in Paragraph 4.1 with CIR submission:

3.2.2.4.1 Measure and record all keys and keyways in accordance with section 3-1.2 of S9243-AW-TRS-011/SHAFT.

NOTE

Before taking shaft runout readings, slowly spin the shaft on a lathe for approximately two hours to eliminate any sagging that may have occurred during handling and storage.

3.2.2.4.2 Measure and record shaft concentricity and outer diameter in accordance with sections 3-3.6 and 3-3.7 of S9243-AW-TRS-011/SHAFT except as follows:

- In lieu of sleeve fit areas on the shaft, record runout and outer diameter readings on finished surface of installed sleeves. Readings for sleeve runout must be provided every 90 degrees around the shaft circumference, and outer diameter must be recorded in two perpendicular planes, for each location.
- Runout and outer diameter readings are not required for areas of shafting covered by fiberglass/GRP protective coating.

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3.2.2.4.3 Accomplish a dimensional inspection of each flange, spigot, and bore in accordance with sections 3-1.4.3, 3-1.4.4, 3-1.4.5, and 3-2.2.4. of S9243-AW-TRS-011/SHAFT.

3.2.2.5 NDE. Perform NDE of all bearing sleeves, shaft tapers, coupling bolts, keyways, and weld repaired surfaces in accordance with SFLC Std Spec 0740, Appendix C.

3.2.2.6 Protective coating inspection. Inspect the shaft fiberglass/GRP covering for porosity as follows:

3.2.2.6.1 Conduct, at close range, a careful visual inspection of the full length of all accessible/exposed GRP for evidence of deterioration, physical damage, lack of adhesion, and/or any other defects. Particular attention is to be paid to the GRP-to-shaft sleeve interfaces for indication of seawater penetration, as evidenced by rust color staining bleeding through at subject interfaces. If damaged or suspect conditions are found, mark all areas with chalk (do not use crayon or "grease" pencil as they have paraffin and waxes in their makeup).

3.2.2.6.2 Conduct a high voltage spark test on the GRP surface of each shaft in accordance with ASTM D5162 and Section B.5.2 of MIL-DTL-2199B.

3.2.2.6.3 If a Change Request is released and approved by the KO, the Contractor must renew the shaft coating/covering. See separate Contract work item for guidance.

3.2.3. Oil tube inspection. Perform a visual and dimensional inspection of removed inner and outer oil tubes for damage or deterioration, paying particular attention to all threads, spacers, couplings, welds, and o-ring grooves. For each tube, measure and record total indicated runout between tube assembly end and nearest spacer and between each pair of spacers. Measure and record spacing between spacers and tube ends to the nearest 1/8".

3.3 Reinstallation. The Contractor must reinstall the propeller shaft to its original configuration and accomplish the following, using Coast Guard Drawings 901 WMEC 243-001, 901 WMEC 243-002, 901 WMEC 243-003, 901 WMEC 243-00, 901 WMEC 243-007 and TP 2699, as applicable.

3.3.1 Reassemble the CPP oil tube assembly into the associated shaft with new seals and fasteners in accordance with TP 2699.

3.3.2 Take care not to damage the sleeves, fiberglass coating, the taper and threads during installation.

3.4 Final shaft coupling alignment check. The Contractor must accomplish the following:

3.4.1 Verify the shaft coupling is aligned to its original position using the reference marks established prior to its removal.

3.5 Government's right for change out. The Contractor must be aware that the Government reserves the right to furnish a new shaft for installation in place of the existing shaft. If the Government exercises this right, the Contractor must transport the spare shaft to a suitable inspection facility, and perform the inspections specified in paragraph 3.2.2 (Inspections) and submit a CIR. The Contractor must return the removed shaft to the Coast Guard PA as an MTI item.

3.6 Operational test, post repairs. After completion of work, the Contractor must witness an operational test (by Coast Guard personnel) of the shafting system, for a minimum of 4 hours, to prove satisfactory operating condition. Submit a CFR.

4. NOTES

4.1 Shaft OD and concentricity inspection worksheet.

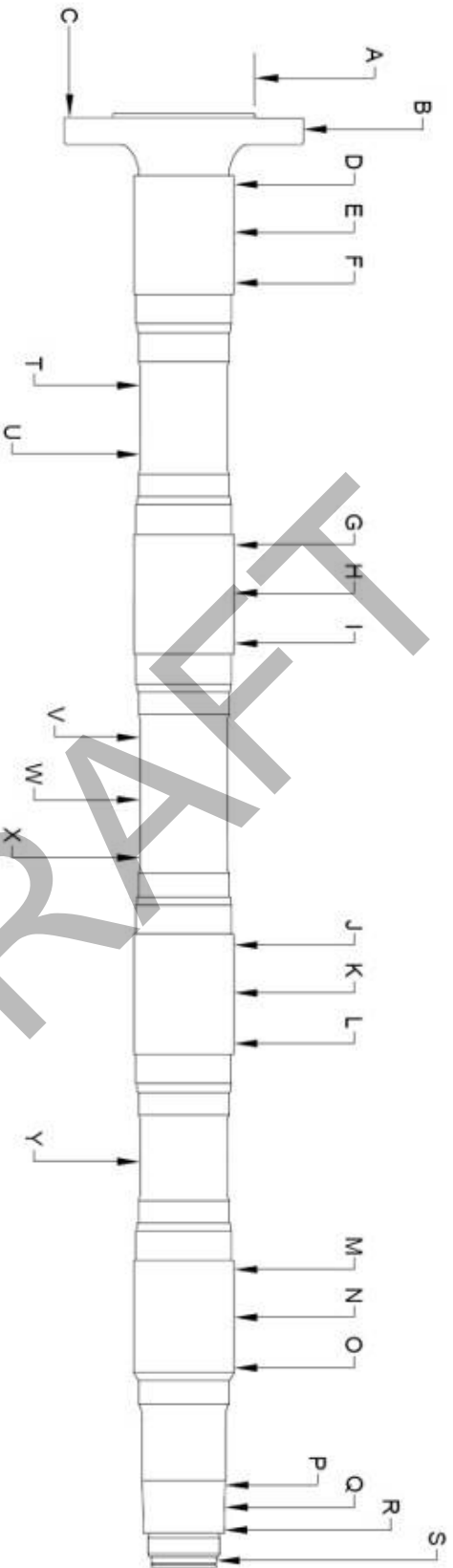
DRAFT

270' A/B CLASS PROPELLER SHAFT INSPECTION

Date: _____
Hull: _____

Shaft Serial #: _____
Port/Stbd: Starboard _____

Inspected By: _____
Organization: _____



(NOT TO SCALE)

VISUAL INSPECTION COMMENTS/NOTES:

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270' A/B CLASS PROPELLER SHAFT INSPECTION

(PORT/STBD)

(Draft-00)

STATION	SHAFT CONCENTRICITY MEASUREMENTS **					OD MEASUREMENTS***	
	0 Deg	90 Deg	180 Deg	270 Deg	MAX TIR	0-180 Deg	90-270 Deg
A						N/A	N/A
B							
C****						N/A	N/A
D							
E							
F							
G							
H							
I							
J							
K							
L							
M							
N							
O							
P						N/A	N/A
Q						N/A	N/A
R						N/A	N/A
S						N/A	N/A
T*							
U*							
V*							
W*							
X*							
Y*							

* Readings to be taken only if fiberglass coating is designated for removal/renewal by contract work item or CR.

** Concentricity measurements to be in accordance with S9243-AW-TRS-011 Para. 3-3.6

*** Outer Diameter (OD) measurements to be in accordance with S9243-AW-TRS-011 Para. 3-3.7

**** This reading to be the face runout of the flange at the outer periphery.

(Sheet 2 of 2)

WORK ITEM 15: Propulsion Intermediate Shafts; Remove, Inspect, and Reinstall

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to remove, inspect, and reinstall the port and starboard intermediate shafts.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	**Intermediate Shaft	NSN: 2010-01-364-7026	2 ea.	9,745.00

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arr
 Coast Guard Drawing 901 WMEC 243-002, Rev G, S-Tubes, Strut Brg & Seal A&D
 Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arr
 Coast Guard Drawing 901 WMEC 243-004, Rev H, Shafting Details
 Coast Guard Drawing 901 WMEC 243-007, Rev E, Shaft Coupling Assemblies
 Coast Guard Drawing 901 WMEC 245-008, Rev D, CPCH Prop Actuator Unit Assy

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 2697, Mar 2024, Main Reduction Gear
 Coast Guard Technical Publication (TP) 2698, Oct 2019, Manufacturer's Instruction Book-SWBS Group(s) 242-245, SWBS 243, Section B – Stuffin g Box
 Coast Guard Technical Publication (TP) 2699, Jan 2023, Manufacturer's Instruction Book-SWBS Group(s) 245, Controllable-Pitch Propeller System
 Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures
 Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022,
Welding and Allied Processes

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.3 (Inspections).

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Bulkhead stuffing box
- Propeller shaft
- Thrust shaft
- Thrust bearing
- CPP OD box
- Shaft counter
- CPP oil tubes

3.2 Removal. The Contractor must accomplish the following for the port and stbd intermediate shafts, using TP 2699, TP 2697 and Coast Guard Drawing 901 WMEC 243-003 for guidance:

3.2.1 Mark existing shaft position, and match-mark shaft coupling bolts to their respective holes, to facilitate reinstallation procedures.

3.2.1.1 Prior to disassembly measure existing position of reduction gear.

3.2.1.1.1 Remove OD box and adaptor flange.

3.2.1.1.2 Fabricate and install .500 inch special tool on front of reduction gear to aid in maintaining .500 inch standoff between reduction gear flange and control shaft mounting face shown on Coast Guard Drawing 901 WMEC 245-008.

3.2.1.2 Measure overall length of intermediate shaft from flange face to coupling face and record.

3.2.1.3 Measure coupling number III face to shaft face distance and record.

3.2.2 Remove the intermediate shaft and attached components to a suitable repair facility. Provide support to the reduction gear, to prevent it from sagging.

3.3 Inspections. The Contractor must clean each shaft and associated components, including the inside shaft bore, free of all surface contaminants, and accomplish the following inspections (Submit a CIR):

3.3.1 Visual inspection. The Contractor must visually inspect all cleaned surfaces, including bearing components, shaft surfaces, and shaft bore, for excessive wear and other defects.

NOTE

Before taking shaft runout readings, slowly spin the shaft on a lathe for approximately two hours to eliminate any sagging that may have occurred during handling and storage.

3.3.2 Measurements.

3.3.2.1 Measure the diameter and concentricity in three locations on each shaft journal.

3.3.2.2 Measure the diameter and concentricity of the non-journal surfaces every 3 feet of shaft length.

3.3.2.3 Measure the flange rim and face runouts on both shaft ends.

3.3.2.4 Measure all coupling bolts.

3.3.3. Oil tube inspection. Perform a visual and dimensional inspection of removed inner and outer oil tubes for damage or deterioration, paying particular attention to all threads, spacers, couplings, welds, and o-ring grooves. For each tube, measure and record total indicated runout between tube assembly end and nearest spacer and between each pair of spacers. Measure and record spacing between spacers and tube ends to the nearest 1/8".

3.4 Installation. Using Coast Guard Drawings and TPs listed under Section 2 (References) as guidance, as applicable, the Contractor must reinstall the intermediate shaft to original configuration.

3.4.1 Take care not to damage machined surfaces during installation.

3.4.2 Ensure the intermediate shaft is aligned with the reduction gear output coupling flange to original configuration, using the reference marks established prior to its removal.

3.4.2.1 Using the .500 inch special tool on front of reduction gear, ensure position of reduction gear relative to control shaft mounting face has not changed and is in accordance with Coast Guard Drawing 901 WMEC 245-008

3.4.2.2 After assembly, measure overall length of intermediate shaft from coupling face to flange face to ensure it is equal to the measurement taken in step 3.2.1.2 above.

3.4.2.3 After assembly, measure coupling number III face to shaft face distance to ensure it is equal to the measurement taken in step 3.2.1.3 above.

3.4.3 Renew the bulkhead stuffing box packing materials in accordance with TP 2699.

3.5 Shaft tunnel preservation. After shaft removal, the Contractor must accomplish the following:

3.5.1 Prepare and coat the shaft tunnel surfaces (17" diameter between bulkhead 165 and 169), using the system specified for "Bilges, Cofferdams, and Forepeaks, Steel" in SFLC Std Spec 6310 Appendix B (Cutters and Boats Interior Painting Systems). Select Red as the top coat color.

3.5.2 After completion of surface preparation and before application of primer coat, perform a visual inspection of the prepared shaft tunnel surfaces, and submit a CFR.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.6 Operational test, post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate the shafting system to be in satisfactory operating condition. Submit a CFR.

3.7 Government's right for change out. The Contractor must be aware that the Government reserves the right to furnish a new intermediate shaft for installation in place of the existing intermediate shaft. If the Government exercises this right, the Contractor must return the removed shaft to the Coast Guard PA as an MTI item. After a Change Request has been authorized and released, the Contractor must transport the new shaft to a suitable inspection facility, and perform the inspections specified in paragraph 3.3 (Inspections). Submit a CFR.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 16: Propulsion Shaft Coating, Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew the fiberglass reinforced plastic (FRP) covering for the port and stbd propeller shafts.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-004, Rev F, Controllable Pitch Propeller Shafting Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2020, Requirements for Preservation of Ship Structures

OTHER REFERENCES

MIL-STD-2199A (SH), Jan 2020, Covering for Waterborne Main Propulsion Shafting on U.S. Naval Surface Ships and Submarines

The Society for Protective Coatings (SSPC)/NACE International (NACE), 2007, Joint Surface Preparation Standard SSPC-SP 10/NACE No.2, Near-White Blast Cleaning

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 The Contractor shall maintain existing CPP hydraulic system cleanliness; take all necessary precautions to prevent the introduction of contaminants into the inner shaft's sensitive hydraulic area. Whenever disconnecting or removing components from the hydraulic system, completely seal all openings to the rest of the system, immediately, using either caps (for externally threaded connection points), bolt-on blanks, or taped-on discs/covers (durable plastic or sheet-metal no less than 1/16-inch thick).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Propellers
- Propeller Shafts

3.2 Surface preparation. The Contractor must remove and dispose of all existing FRP covering over the shaft surfaces.

3.2.1 The Contractor must abrasive-blast the bare shaft surfaces to a "Near-White Blast" standard, in accordance with SSPC SP-10/NACE No. 2, and obtain a surface anchor profile of 3.0-4.5 mils.

NOTE

Exercise extreme care not to damage shaft and shaft sleeve surfaces during covering removal and surface preparation.

3.3 Shaft surface inspection. The Contractor must perform a visual inspection of the cleaned shaft surfaces for signs of corrosion, wear, and other defects; submit a CFR.

NOTE

The shaft coating system required by MIL-STD-2199 has been modified. The new coating system involves applying a four coating shaft covering system which includes in sequential order:

- Primer
- Elastomeric Coating
- Sealing/Fairing compound
- Glass Reinforced Plastic.

3.4 New covering installation. The Contractor must perform the FRP covering renewal for the the port and stbd propeller shafts, shown on Coast Guard Drawings 901 WMEC 243-001 and 901 WMEC 243-004, in accordance with MIL-STD-2199A (SH). Before the cleaned shaft surfaces begin to flash-rust, accomplish the following:

- Prepare resin in accordance with manufacturer's recommendations, and apply a coat to the cleaned shaft surfaces, not to exceed a total of 60 mils thickness.

- Apply a ply of glass tape wound spirally onto the coated surfaces, butting the edges.
- Repeat the resin and glass tape applications, reversing the direction of the spiral warp for each subsequent application, until a four-ply covering has been formed.
- Ensure that all entrapped air is worked out, and the glass tape is wet-out during each ply installation.

3.4.1 Covering inspection. The Contractor must conduct a high voltage spark test in accordance with Section 5.6.1 of MIL-STD-2199 of the covering, after it has hardened, at 250 to 300 volts per mil of glass reinforcement (approximately 15,000 volts for the four-ply covering), paying particular attention to the joining line between sleeves, couplings, and shafts. Any defects in the coating are unsatisfactory. Repair any leaks and repeat the testing procedures, as necessary. Submit a CFR.

3.4.2 Preservation of covering. Upon completion of covering inspection, and after all repairs, if any, the Contractor must prepare and coat the FRP covering, using the system specified for “U/W Body and Boot-Top, Steel Hulls (Up to 7 Years), in Salt Water” in SFLC Std Spec 6310, Appendix A (Cutters and Boats Exterior Coating Systems).

NOTE

Exercise extreme care not to damage FRP covering during surface preparation.

Some versions of SFLC Std Spec 6310 direct not to coat the FRP covering. This specification takes precedence over SFLC Std Spec 6310 for instance(s) of disagreement.

3.4.3 The Contractor must, during shafting handling, shaft reinstallation, and reinstallation of any fairwaters or rope guards take the necessary precautions to prevent damage to the newly installed shaft coating.

3.4.4 In the related Definite Item “Propulsion Shafting Removal, Inspection, and Maintenance” the Contractor is required, after each shaft and coupling have been installed on the cutter, to renew the shaft covering over the coupling and the exposed shaft ends. The Contractor must not repeat that cost in the bid of this Item.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 17: Propulsion Shafting, Straighten

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to straighten a designated shaft.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev K, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-004, Rev E, CPP Shafting Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020,
Welding and Allied Processes

OTHER REFERENCES

MIL-STD-2191, Nov 1992, Repair Welding, Weld Cladding, Straightening, and Cold Rolling of
Main Propulsion Shafting

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Shaft straightening requirements. Based on inspection results obtained from a related shaft work item in this specification package, or provided by the Coast Guard Inspector, the Contractor must straighten the designated shaft, using Coast Guard Drawings 901 WMEC 243-001, 901 WMEC 243-004, and MIL-STD-2191 for guidance and as specified herein.

3.3 Pre-straightening inspection. Prior to initiating straightening processes, the Contractor must determine the location, magnitude and type of bend in the shaft, with the entire shaft supported on centers in a lathe, as follows:

- Before taking shaft runout readings, spin the shaft on the lathe for approximately two hours to eliminate any sagging that may have occurred during handling and storage.
- Conduct a dial indicator survey of shaft runout at stations spaced every two feet of shaft length to determine the magnitude and location of the bend.
- At each station, mark the position of maximum runout on the periphery of the shaft.

NOTE

Be aware that the marks on the shaft periphery will be in a straight line if the shaft contains a simple bend, and will spiral part way around the shaft, if the shaft is both bent and twisted.

- Plot the maximum runout to an exaggerated scale, in order to locate the center of the bend, and to determine the extent of straightening required.
- Submit a CFR including locations and magnitudes of the bend.

3.4 Straightening procedures. The Contractor must straighten the shaft to equal or less than 0.005" total indicator runout, using the localized heating (hot spotting method for carbon steel shafts, in accordance with MIL-STD-2191, Section 11.2.2.2 and sub-paragraphs).

3.5 Post-straightening inspections. After completion of straightening procedures, the Contractor must accomplish inspections, and submit a CFR:

- Perform NDE of all straightened areas in accordance with Std Spec 0740, Appendix C.
- Recheck the shaft surfaces, as specified in paragraph 3.3 (Pre-straightening inspection).

4. NOTES

This section is not applicable to this work item.

WORK ITEM 18: Propulsion Shaft Seal (Mechanical And Inflatable) Assemblies, Inspect and Repair or Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to overhaul the stern tube seal assemblies for the port and starboard propulsion shafts, located in the Reefer & Dry Stores (4-169-0-A). The stern tube seal assemblies consist of both mechanical and inflatable components.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	Stern tube seal major overhaul kit	NSN: 5330-01-527-0959	2 ea.	34,607.00
N	**Stern tube seal split seat	NSN: 5330-01-359-5555	2 ea.	12,708.00
N	**Stern tube seal face change out kit	NSN: 5330-01-527-0696	2 kt.	8,954.00
N	Automatic Air Eliminator	PN: AE36 NSN: 4820-01-690-3058	2 ea.	543.00

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 256-012, Rev -, Shaft Seal Vent Line Modification
Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement
Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes Strut Brg and Seal Arr and Det
Coast Guard Drawing 901 WMEC 243-006, Rev C, Shaft Seal A&D

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 5454, Nov 2013, Technical Manual for Type MA Mechanical Stern Tube Seal
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements
Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2020, Requirements for Preservation of Ship Structures

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.2 Inspection and repair requirements

3.1.2 Tech Rep. The Contractor must provide the services of a Qualified Technical Representative who is familiar with the John Crane MA stern tube seal system/equipment to accomplish the following on site:

- Advise on manufacturer's proprietary system information.
- Assist with installation and repair method(s).
- Ensure compliance with manufacturer's procedures and standards during system disassembly, inspection, and reassembly as applicable.

3.1.2.1 Ensure the Tech Rep has experience with the system/equipment stated above and demonstrated on their résumé.

3.1.2.2 Submit the name and résumé of the Tech Rep to the COR at the Arrival Conference.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Deck plating and framing
- Shelves/supply storage
- Safety grating
- Propulsion shaft
- Cooling water piping
- Air line

3.2 Inspection and repair requirements. The Contractor must overhaul the port and starboard stern tube seal assemblies, in accordance with the referenced TP 5454, Chapter 4, Paragraph 4.9 (Complete Overhaul of Seal) and Chapter 8 (Installation), and as specified below. Measure seat wear in accordance with TP 5454, Chapter 4, Section 4.7. Visually inspect the shaft seal mounting faces (including bulkhead mounting flange), and exposed adjacent surfaces for evidence of corrosion or deterioration. Submit a CIR, including recommended repairs, if any.

3.2.1 Seat machining. Machine the seal seats in accordance with TP 5454, Chapter 6, Section 6.4. If a Change Request is released and approved by the KO, build-up/repair the shaft seal bulkhead mounting flange using a commercially available metallic polymer (e.g. Belzona or equivalent).

3.2.2 Seal compression setting. Set the stern tube seal compression during the installation in accordance with TP 5454, Chapter 6, Section 6.2 (Setting Seal Compression).

3.2.3 Alignment of seat. Align each seat to run true with its associated shaft in accordance with TP 5454, Chapter 6, Paragraph 6.3, and as follows:

3.2.3.1 Attach a dial indicator so as to indicate either the face of the seat, aft side of the face carrier, or the flat side of the forward main clamp ring near the butt joint, to measure relative misalignment of the seat in relation to the stationary face insert, as the shaft rotates.

3.2.3.2 Engage turning gear to rotate the shaft.

3.2.3.3 Adjust the alignment bolts and drive bolts to have a total indicator reading of 0.002" or less while the shaft is rotating. Take readings for three complete revolutions of the shaft.

3.2.3.4 Once alignment is complete, lock the alignment bolts with their locking nuts while the shaft is rotating and the indicator is still installed. Recheck alignment, after locking, in the presence of the Coast Guard Inspector. Submit CFR.

3.2.3.5 Disengage turning gear and install splash guard onto the face carrier.

3.2.4 Automatic Air Eliminator, Renew. The Contractor must rip out and dispose of the port and starboard shaft seal air eliminators and renew with government furnished Automatic Air Eliminators, as shown on Coast Guard Drawing 901 WMEC 256-012. Renew all disturbed seals, gaskets and fasteners.

3.3 Tests. The Contractor must perform the following tests to the port and starboard stern tube seal assemblies:

3.3.1 Stern tube seal leak test. Pressurize the stern tube seal with clean fresh water through the flanged water connections at the bottom of the mounting ring to 20 psig. Inspect and correct for any leakage. Submit CFR.

3.3.2 Inflatable seal air test. Pressurize the inflatable seal to 60 psi using Nitrogen, in lieu of Carbon Dioxide.

3.3.2.1 Visually inspect to ensure all pipe joints are tight.

3.3.2.2 With seal inflated to 75-90 psi, shut off all supply valves, and monitor air leakage. Pressure must not drop more than 5 psi in 15 minutes. Submit CFR. Correct any discrepancy noted.

WARNING

Do not rotate shaft while testing inflatable seal!

3.3.3 Inflatable seal water leakage test. After the vessel is waterborne, operate inflatable seal at normal pressure in accordance with TP 5454, Chapter 2, Section 2.3 (use Nitrogen, in lieu of Carbon Dioxide).

3.3.3.1 Slowly open the seal drain valve and drain the seal cavity. Once seal cavity is drained, measure the seal leakage which must not exceed one pint per minute. Submit CFR. Take corrective action if the leakage is exceeding one pint per minute.

3.3.3.2 Align the seal water flush lines for normal operation.

3.4 Stern tube split seal change-out. The Contractor must be aware that the Government reserves the right to furnish a new stern tube split seal for installation in-place of the one removed. When the Government exercises this right, the Contractor must dispose of the removed stern tube split seals as designated by the Coast Guard Inspector.

WARNING

Do not drain any fluids including fresh water, into any space, bilge, or exterior location.

3.5 Pipe flushing. The Contractor must flush all new and disturbed piping with clean fresh water until all debris is removed but no longer than five minutes. Ensure flushing fluid is directed to move scale and foreign debris away from installed machinery to prevent possible damage upon operational testing. Submit a CFR documenting date and time of flushing process and level of pipe cleanliness.

3.5.1 Dispose of flushing fluid in accordance with all applicable Federal, state, and local regulations.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.6 Operational test, post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate all items or shipboard devices that have been disturbed, used, repaired, altered, or installed to be in satisfactory operating condition. Submit a CFR.

3.7 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

4. NOTES

4.1 Face seal only overhaul kit is available if CIR results of paragraph 3.2 indicate that a complete overhaul of the seal assembly is not required. Only one of the kits (major overhaul kit or face change out kit) should be used and the unused kit returned to stock.

DRAFT

WORK ITEM 19: Propeller Shaft Sleeves, Renew**1. SCOPE**

1.1 Intent. This work item describes the requirements for the Contractor to renew the sleeve(s) on a designated propeller shaft.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	Stern tube sleeve (aft)	NSN: 2010-01-345-2315	1 ea.	10,792.56
N	Propeller Strut sleeve (aft)	NSN: 2010-01-340-5362	1 ea.	12,725.00
N	Propeller Strut sleeve (fwd)	NSN: 2010-01-345-2314	1 ea.	13,180.16
N	Stern tube sleeve (fwd)	NSN: 2010-01-340-6083	1 ea.	8,865.14

2. REFERENCES**COAST GUARD DRAWINGS**

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arr

Coast Guard Drawing 901 WMEC 243-002, Rev G, S-Tubes, Strut Brg & Seal A&D

NAVSEA Drawing 803-2145807, Rev C, Propulsion Shafting and Components

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Using Coast Guard Drawings 901 WMEC 243-001 and 901 WMEC 243-002 as guidance, the Contractor must accomplish the following, for a designated propeller shaft:

3.2.1 Removal. The Contractor must remove and dispose of the sleeve(s).

3.2.2 Inspections. The Contractor must clean all shaft journal surfaces, and accomplish the following:

3.2.2.1 Perform a visual inspection of the shaft journal surfaces for corrosion, deterioration, and other signs of failures.

3.2.2.2 Measure and record the shaft journal outside diameter and the shaft bearing inside diameter in the following positions:

- Forward.
- Middle.
- Aft.

3.2.2.3 Conduct nondestructive testing of all shaft surfaces in the way of sleeve fit areas, in accordance with SFLC Std Spec 0740, Appendix C.

3.3 Inspection report. The Contractor must submit a CFR for the completed inspection and measurements.

3.4 New sleeve installation requirements. After the Coast Guard Inspector has reviewed all measurements, the Contractor must accomplish the following tasks for the Government-furnished shaft sleeves:

3.4.1 Machining of inside diameter (ID). Bore new sleeves' ID to the dimensions based on the recorded journal OD readings, to facilitate an interference fit, as specified below.

3.4.2 Installation. Install the new sleeves onto the shaft with a shrink-fit equal to 0.00075-inch per inch of shaft diameter, in accordance with NAVSEA Drawing 803-2145807, Table 8-1 (Shaft Sleeve Shrink Fit).

3.4.3 Machining of outside diameter. After installation, machine the sleeves OD based on the shaft bearing's ID reading; including the required bearing clearance.

3.5 Concentricity measurement. The Contractor must check the outside diameter of each shaft sleeve to ensure it is concentric with the axis of rotation within 0.0005 inch (0.001 inch runout). Submit CFR.

3.6 Polishing. The Contractor must polish the sleeve surfaces to a 32 micro-inch RMS finish, upon completion of installation and all required machining.

3.7 Final documentation. The Contractor must provide a complete set of the new sleeve dimensional readings to the COR within 48 hours after completion of sleeve installation. At a minimum, the set of readings must include the new sleeve length, diameters (inside and outside) at the bearing journals, concentricity, and polish. Submit CFR.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 20: Propulsion Shafting, Optical or Laser Alignment Check

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to survey propulsion system alignment in drydock using optical or laser instrumentation.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arr

Coast Guard Drawing 901 WMEC 243-002, Rev G, S- Tubes, Strut Brg & Seal A&D

Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arr

Coast Guard Drawing 901 WMEC 243-004, Rev E, Shafting Details

Coast Guard Drawing 901 WMEC 243-007, Rev D, Shaft Coupling Assemblies, CPP

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Alignment survey personnel. The Contractor must provide the services of an Alignment Engineer who is an expert in propulsion shafting alignment using optical sighting or laser alignment method to perform the propulsion shafting alignment check on all bearings between the propeller and reduction gear.

3.3 Work coordination and preparation. The Contractor must coordinate surveys with all other work items detailed in the specification, to include work items for propulsion shaft removal and bearing renewals, to ensure the alignment survey does not necessitate a work stoppage. Abide by the following:

3.3.1 Conduct survey between the hours of 2200 and 0700 to minimize hull deflection that may occur during the day due to heating of cutter from the sun.

3.3.2 Erect any staging required to provide access to bearing locations.

3.4. Survey particulars. Using Coast Guard Drawings 901 WMEC 243-001, 901 WMEC 243-002, 901 WMEC 243-003, 901 WMEC 243-00, 901 WMEC 243-007 and TP 2699 as guidance, the Contractor must conduct two alignment surveys for each propulsion shaft line. The first survey must be accomplished prior to performing hot work or structural repairs and the final survey must be accomplished after all hot work has been completed.

3.4.1 Establish a reference line or line of sight (LOS) perpendicular to the reduction gear output flange and concentric with the centerline.

3.4.2 Install two targets, at a minimum, at each propulsion shaft bearing location between the reduction gear output flange and propeller, including line shaft bearings where feasible. Install targets at forward and aft ends of the bearing housing. Record horizontal and vertical offsets of each target location from the established LOS, as well as longitudinal or axial distance of the target from the center of the reduction gear output flange along the LOS.

3.4.2.1 Targets must be installed with old bearings removed, so that the measurements obtained represent the center of the housing or chocking/bedding compound. Alternatively, survey may be conducted with new, unworn bearings installed. Identify the surface actually measured in the report submission (e.g. bearing housing, chocking bed, or new bearing).

3.5 Final Report. Provide a CFR of the findings and the Alignment Engineer's recommended solutions to Coast Guard Inspector. This report must describe procedures used and an analysis of the obtained data including any identified movement of bearing positions, as well as recommendations for realignment (if necessary).

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 21: Propulsion Shaft Bearings, External, Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew the external, water-lubricated, propulsion bearings for port and starboard propeller shafts.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	Stern tube bearing	NSN: 3120-01-189-1949	2 ea.	24,320.00
N	Intermediate Strut Bearing	NSN: 3120-01-189-1087	2 ea.	9,647.53
N	Propeller Strut Bearing	NSN: 3120-01-189-1088	2 ea.	12,752.00

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 114-003, Rev C, Shaft Fairwaters and Rope Guards

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes, Strut Bearing & Seal Arrangement and Details

Coast Guard Drawing 901 WMEC 243-004, Rev G, CPP Shafting Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

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- 3.2.1 Removal and inspection of existing bearings.
- 3.2.2 Shaft and housing inspections.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Shafts
- Fairwaters
- Rope guards.

3.2 Bearing renewal particulars. The Contractor must accomplish the following, Coast Guard Drawings 901 WMEC 114-003, 901 WMEC 243-001, 901 WMEC 243-002, and 901 WMEC 243-004 as guidance. Submit forms in electronic format. Electronic copies of forms may be obtained from the COR upon request.

3.2.1 Removal and inspection of existing bearings. Remove all existing external bearings identified in paragraph 1.2 and perform the following inspections. Fill out "USCG Drydock Bearing Measurements Form (1 of 2)" provided in Paragraph 4.3 of this work item. Submit CIR.

3.2.1.1 Measure and record the inner diameter (ID) of each bearing at a minimum of two locations 90 degrees apart for both the forward and aft ends of each bearing.

3.2.1.2 Perform visual inspection of the internal bearing surfaces for any defects such as uneven wear, scoring, dis-bonding, missing material, or other damage and record findings in the "Remarks" section.

3.2.2 Shaft and housing inspections. Accomplish the following inspections. Submit a CIR.

3.2.2.1 Perform visual inspection of the existing chocking material, if installed in the bearing housing. Measure the ID at all steps for the purpose of machining the new bearing.

3.2.2.2 Perform visual inspection of the shaft sleeve in way of removed bearings. Measure and record outer diameter (OD) of each sleeve at two locations 90 degrees apart at the forward, center, and aft positions.

3.2.3 New bearing installation. The Contractor must accomplish the following using new Government-furnished external propulsion shaft bearings.

3.2.3.1 Machine the new Government-furnished bearings to fit the bearing housing in accordance with the manufacturer's installation recommendations. Drill and tap new bolt holes on the new bearings to match the bearing housings for proper fit.

3.2.3.2 Install the new bearings in accordance with the bearing manufacturer's recommendations, using new Monel or type 316 stainless steel fasteners.

3.2.4 New bearing measurements. Perform the following inspections after bearing installation. Fill out “USCG Drydock Bearing Measurements Form (2 of 2)” provided in Paragraph 4.3 of this work item. Submit CFR.

3.2.4.1 Measure and record the inner diameter (ID) of each bearing at a minimum of two locations 90 degrees apart for both the forward and aft ends of each bearing.

3.2.4.2 If matching shaft sleeve has been renewed, measure and record the outer diameter of the new sleeve at two locations 90 degrees apart at the forward, center, and aft positions.

NOTE:

See Optional related work item “Propeller Shaft Sleeves, Renew” for details on sleeve renewal.

3.2.5 Clearance inspection. After shaft reinstallation, measure and record all accessible external bearing clearances. Measurements must be taken 4-6” into the bearing from the forward and aft faces for all accessible locations. Fill out USCG Drydock Bearing Measurements Form in Paragraph 4.3 of this work item.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.3. Operational test, post repairs. After completion of work and in the presence of the Coast Guard Inspector, the Contractor must thoroughly test and demonstrate all item that have been used, repaired, altered, or installed to be in satisfactory operating condition. Submit a CFR.

4. NOTES

4.1 Port Engineer Responsibilities. The Port Engineer will forward the completed USCG Drydock Bearing Measurements Forms to the following cognizant engineering departments for review and documentation.

- Product Line Systems and Equipment Section (SES)
- Engineering Services Division (ESD) - Naval Architecture and Marine Engineering (NAME) – Main Propulsion (MAIN-PROP) at SFLC-ESD-NAME-MainProp-PropulsionReports-SMB-Access@uscg.mil.

4.2 General guidance for minimum and maximum operating bearing clearances. If no other bearing clearance limit criteria exists (e.g. Drawing notes or other Coast Guard standards), the following numbers may be used as a guide for acceptable range of clearances for newly installed bearings.

TABLE 1 – STANDARD BEARING CLEARANCES

DIAMETER OF JOURNAL (INCHES)	MINIMUM OPERATING CLEARANCE OF WATER-LUBRICATED BEARINGS (INCHES)	MAXIMUM OPERATING CLEARANCE OF WATER-LUBRICATED BEARINGS (INCHES)
2 to 2.499	0.017	0.091
2.5 to 2.99	0.018	0.100
3 to 3.99	0.020	0.107

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4 to 4.99	0.022	0.120
5 to 5.995	0.025	0.131
6 to 6.99	0.027	0.142
7 to 7.99	0.030	0.154
8 to 8.99	0.032	0.165
9 to 9.99	0.035	0.177
10 to 10.99	0.037	0.187
11 to 11.99	0.040	0.197
12 to 12.99	0.042	0.207
13 to 13.99	0.045	0.217
14 to 14.99	0.047	0.225
15 to 15.99	0.050	0.235

4.3 Bearing inspection forms. See next page for inspection forms for old and new bearings.

DRAFT

USCG Dry Dock Bearing Inspection Form (1 of 2)

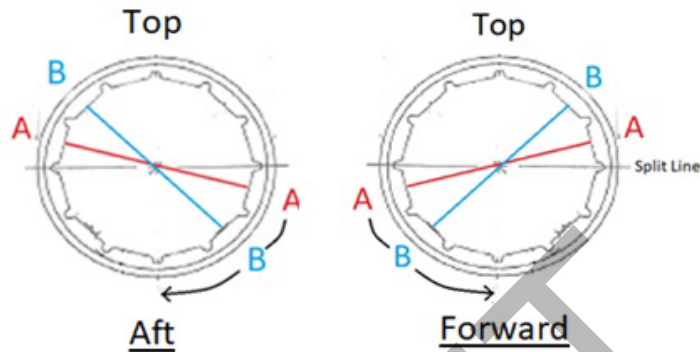
Bearing Position:

Cutter:

Date:

Old Bearing Inside Diameter Inspection

After shaft removal record bearing condition and ID measurement and shaft sleeve OD measurement to the nearest 0.001" Record "N/A" for any location not applicable or not required.

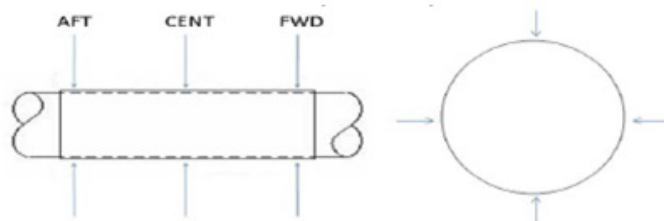


LOC	AFT	FWD
A		
B		
C		
D		
E		
F		
G		
H		

Bearing Condition Remarks (Cracks, De-bonding, Wear, other Damage):

Shaft Sleeve Outside Diameter Measurements

Measure the outer diameter of the shaft sleeve corresponding to the bearing in three location along the bearing length: Aft, Center, and Fwd. At each location, take two measurements of the shaft sleeve at 90 deg increments (vertical and horizontal).



	AFT	CENT	FWD
Vertical			
Horizontal			

USCG Dry Dock Bearing Measurements Form (2 of 2)

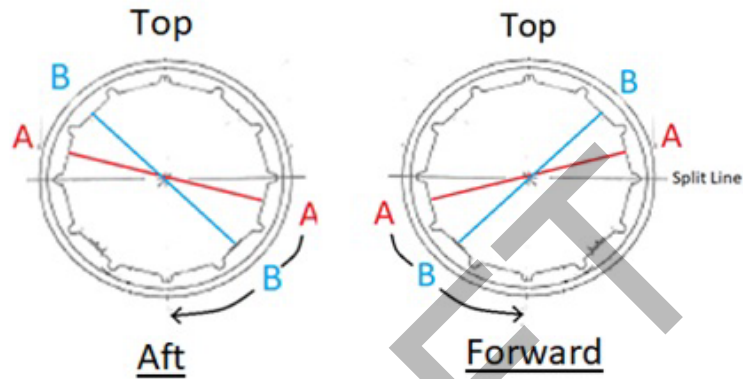
Bearing Position:

Cutter:

Date:

New Bearing Inside Diameter Measurements

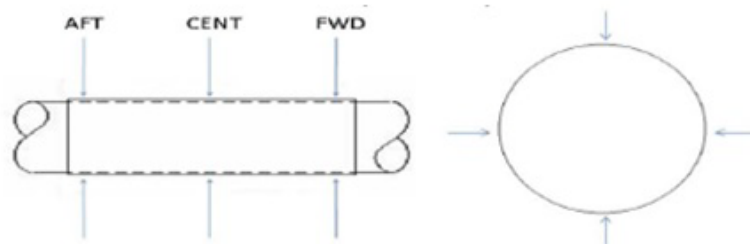
Measure inner diameter of each new bearing after machining as depicted on the sketch below to the nearest 0.001". Record "N/A" for any location not applicable or not required by the work item.



LOC	AFT	FWD
A		
B		
C		
D		
E		
F		
G		
H		

Shaft Sleeve Outside Diameter Measurements

Measure the outer diameter of the shaft sleeve corresponding to the renewed bearing in three location along the bearing length: Aft, Center, and Fwd. At each location, take two measurements of the shaft sleeve at 90 deg increments (vertical and horizontal)



	AFT	CENTER	FWD
Vertical			
Horizontal			

USCG Dry Dock Bearing Clearance Measurement Form

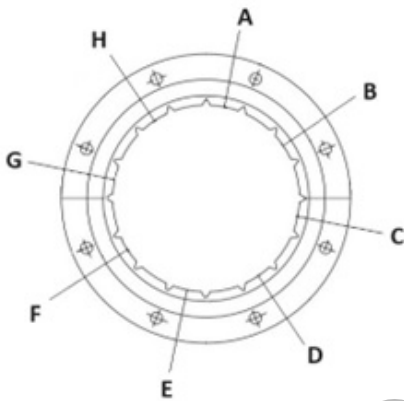
Port/STBD:

Cutter:

Date:

INSTRUCTIONS:

Use this form to record existing propulsion shaft bearing clearances. Clearance is defined as the gap between the inner diameter of the bearing material surface and the outer diameter of the shaft sleeve. Measurements must be taken approximately 4-6" into the bearing from the forward and/or aft faces for each bearing. Record all measurements to the nearest 0.001" in the Bearing Clearance Table. For all measurements not accessible or not required by the work item, record "N/A".



*Image orientation: Facing forward for all bearings
(Position "C" to STBD, Position "G" to PORT)*

BEARING CLEARANCE TABLE (nearest 0.001")

Position	Main Strut		Intermediate Strut		Stern Tube	
	FWD	AFT	FWD	AFT	FWD	AFT
A						
B						
C						
D						
E						
F						
G						
H						

REMARKS:

WORK ITEM 22: Stern Tube Interior Surfaces, Preserve, 100 Percent

WORK ITEM 1: SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to preserve stern tube interior surfaces.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement

Coast Guard Drawing 901 WMEC 243-002, Rev G, Stern Tubes, Strut Bearing & Seal Arrangement and Details

Coast Guard Drawing 901 WMEC 243-003, Rev F, Shaft Removal Arrangement

Coast Guard Drawing 901 WMEC 243-004, Rev F, CPP Shafting Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

OTHER REFERENCES

MIL-A-22262, Mar 1996, Abrasive Blasting Media Ship Hull Blast Cleaning

MIL-PRF-24647, Jan 2018, Paint System, Anticorrosive and Antifouling, Ship Hull

The Society for Protective Coatings (SSPC)/NACE-International (NACE) Joint Surface Preparation Standard SSPC-SP 10/NACE No. 2, 2007, Near-White Metal Blast Cleaning

The Society for Protective Coatings (SSPC) Surface Preparation Specification No.11 (SSPC-SP 11), 2012, Power-Tool Cleaning to Bare Metal

The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP 12/NACE No.5, 2002, Surface Preparation and Cleaning of Metals by Waterjetting Prior to Recoating

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection) and to the following specific components, spaces and equipment:

- Stern tube flange surfaces.
- Engine Room.
- Bearings and bearing housings.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Rope guard.
- Fairwaters.

3.2 Preservation requirements. Using Coast Guard drawings 901 WMEC 243-001, 901 WMEC 243-002, 901 WMEC 243-003, and 901 WMEC 243-004 as guidance, the Contractor must preserve 100% of the interior tube surfaces as follows:

3.2.1 The Contractor must not prepare or coat waterborne bearing contact areas inside the stern tube. This includes bearing lands, areas in contact with bearing shell, areas in contact with Cutless type bearings, etc. These areas must remain uncoated to provide required fits.

3.2.2 Surface preparation. The Contractor must prepare surfaces by waterjetting to a SSPC-SP 12/NACE No. 5, WJ-2/L standard, or by one or a combination of the surface preparation methods described in Table 1:

TABLE 1 – SURFACE PREPARATION

ABRASIVE BLASTING	MECHANICAL CLEANING
Abrasive-blasting to SSPC-SP10/NACE No. 2, using grit conforming to MIL-A-22262 (1.5 to 2.5 mil anchor profile). Used reference MIL-A-22262, Mar 1996, Abrasive Blasting Media Ship Hull Blast Cleaning for guidance	Power tool cleaning to a SSPC-SP 11 (1.0 mil anchor profile).

3.3 Inspection. After surface preparation and before coating application, the Contractor must perform a visual inspection of the prepared stern tube surfaces, and submit a CFR.

3.4 Surface coating. The Contractor must prime and coat the prepared surfaces with two coats of a “High Turbulence Coating”, at 5.0-6.0 mils DFT each. Used reference MIL-PRF-24647, Jan 2018, Paint System, Anticorrosive and Antifouling, Ship Hull for guidance.

NOTE

List of authorized suppliers for “High Turbulence Coatings” is provided in SFLC Std Spec 6310, Appendix C (Authorized Coatings for Use on Cutters and Boats).

3.5 In-process quality control. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”).

NOTE

Surfaces being preserved are considered “critical-coated surfaces”.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 23: Controllable Pitch Propellers, Clean and Inspect

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to clean and inspect the propeller immediately after drydocking.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 245-004, Rev E, Controllable Pitch Propeller Eshcer Wyss
270ft WMEC Blade and Palm

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

OTHER REFERENCES

NAVSEA Technical Manual S9245-AR-TSM-010, December 2012, Marine Propeller Inspection,
Repair, and Certification

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.3 (CPP inspection).

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.1.5 Competent personnel. Only competent, trained, qualified, and experienced personnel who are familiar with naval propeller critical geometries and characteristics must perform or supervise CPP cleaning operations and conduct visual technical inspection. Propeller Visual Technical Inspections must be performed by a NAVSEA qualified inspector in accordance with NAVSEA S9245-AR-TSM-010.

3.2 CPP Cleaning. The Contractor must thoroughly clean the Cutter's controllable pitch propeller or propellers depicted in Coast Guard Drawing 901 WMEC 245-004 to remove all marine growth and fouling.

3.2.1 The Contractor must ensure cleaning and inspection must be performed as soon after dry-docking as practical (not to exceed 24 hours) and prior to any other related Controllable Pitch Propeller work is performed. Cleaning is not to be confused with polishing which is not the subject of this specification.

3.2.2 The Contractor must ensure CPP cleaning must not result in damage to or the removal of blade or hub material. The following methods are acceptable for cleaning installed CPPs and fairwater caps in drydock and are listed in order of precedence.

3.2.2.1 Hydroblasting (high pressure water jet cleaning) may be used on all CPP and fairwater surfaces. For best results, perform hydroblasting immediately after drydocking and before marine growth has dried. At NO TIME must hydroblasting on CPP or fairwater cap surfaces exceed 10,000 pounds per square inch. Do not direct the water jet directly into blade seal cavities since this may cause undesirable water and/or contaminant intrusion into the CPP's internals.

3.2.2.2 Hand cleaning, including scraping and wire brushing is more effective if performed prior to drying of marine growth. Use hard plastic or wood scraper or soft bronze or brass wire brushes to clean all CPP surfaces. DO NOT use metallic scrapers or scrapers impregnated with abrasive material. Scraping and wire brushing must be performed by hand and not powered. The final cleaning operation of CPP edges (within 3 inches of the edges and tips) must be cleaned by hand with plastic surface conditioning material (e.g. Scotch-Brite "greenies" or equivalent).

3.2.2.3 Powered cleaning methods with surface conditioning discs must only be used on CPP surfaces of minimal curvature and must not be used within 6 inches of blade edges, tips, cusps, fillets (excluding hub to blade interface) or other sensitive areas. Powered surface conditioning discs must only be utilized by experienced personnel who are familiar with navy propeller critical geometries and characteristics. The use of powered nylon or plastic brushes is acceptable for cleaning edges and tips, provided proper care is taken to avoid removal of metal. When cleaning the outer periphery of the CPP blades, the brushes and disks must be kept flat on the surface. Do not attempt to round the edges of the propeller blade with brushes and disks.

3.3 CPP Inspection. After cleaning, the Contractor must perform a visual technical inspection of all external CPP components. Visual Technical Inspection of propeller blades must be performed in accordance with NAVSEA S9245-AR-TSM-010. Submit inspection form 9245/2 (See Paragraph 4.1 for hyperlink). Submit A CIR.

3.3.1 The Contractor must visually inspect and record the condition of all CPP surfaces including blades (suction and pressure sides) , fillet areas, edges and tips, blade palms and hub noting defects and damage such as: nicks, dents, bends, cracks, curling, flat spots, ridges, surface finish, grooves, gouges, pitting, porosity, cavitation, erosion, waviness, and roughness.

4. NOTES

4.1 Form 9245/2 for visual technical inspection can be found using the following hyperlink:
<http://www.dema.mil/NPP/forms.aspx>.

DRAFT

WORK ITEM 24: Controllable Pitch Propeller (CPP) System, Inspect and Service

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to inspect and service the Controllable Pitch Propeller (CPP) system.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	O.D. Box Inspection Kit	NSN: 3010-01-470-3519	2 kt.	1,380.00
N	Filter	NSN: 4330-01-177-4069	12 ea.	169.00
N	**Oil Feed Bushing	NSN: 3120-12-329-5287	2 ea.	14,760.00
N	**Ball Bearing	NSN: 3110-00-156-6745	2 pr.	121.00
N	**Bearing Bushing	NSN: 2010-01-469-8639	2 ea.	1,198.00
N	**Bearing Bushing	NSN: 3120-12-198-7607	8 ea.	1,255.00
N	**Spool valve assembly	NSN: 4820-01-513-2509	2 ay.	25,348.00
N	Cartridge check valves	NSN: 4820-01-228-6988 PN: 2201C-1	6 ea.	14.00
N	Strainer seal kit	NSN: 5330-01-266-6513 PN: 909156	2 at.	31.00
N	Selector valve seal kit	NSN: 4820-01-358-6162 PN: 7350-103	6 ea.	47.00
N	Magnets	NSN: 4330-01-258-2746 PN: 905904 WV	12 ea.	140.00
N	Check Valve	NSN: 4820-01-198-7138 PN: Compact Controls CP103-1-B-0-05, Or Parker CVH201P	12 ea.	61.00
N	**Relief Valve Assy	NSN: 4820-01-194-5761	6 ea.	101.00
N	Strainer Element, Sediment	NSN: 4730-01-456-6689	4 ea.	184.00
N	Filter Assembly Seal Kit	NSN: 5330-01-358-1291 PN: 1300014	12 kt.	64.00
Y	**Propeller Blade-RH (Port)	NSN: 2010-01-119-3972	1 se.	40,000.00
Y	**Propeller Blade-LH (Stbd)	NSN: 2010-01-120-2871	1 se.	40,000.00
N	O-Ring	NSN: 5330-01-120-5229	8 ea.	9.00
N	O-Ring	NSN: 5330-01-118-8529	8 ea.	817.00
N	Blade Washer	NSN: 5310-01-335-4361	48 ea.	213.00
N	**Shoulder bolt	NSN: 5306-01-118-8337	8 ea.	2,139.00
N	**Shoulder blade bolt	NSN: 5306-01-119-3607	16 ea.	1,523.00

****New or refurbished equipment that the Government may provide for installation in place of existing equipment.**

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-001, Rev L, Shafting Arrangement
Coast Guard Drawing 901 WMEC 245-001, Rev E, CPCH Prop Hyd Sys Diag
Coast Guard Drawing 901 WMEC 245-002, Rev F, CPCH Prop Hyd Sys A & D (901)
Coast Guard Drawing 901 WMEC 245-004, Rev E, Controllable Pitch Propeller Escher Wyss
270 ft WMEC Blade and Palm
Coast Guard Drawing 901 WMEC 245-005, Rev -, C.P. Propeller
Coast Guard Drawing 901 WMEC 245-007, Rev B, Special Tools
Coast Guard Drawing 901 WMEC 245-008, Rev D, Controllable Pitch Propeller Actuating Unit
Type 300.3GN
Coast Guard Drawing 901 WMEC 245-009, Rev L, CPCH Prop Hyd Sys Diag
Coast Guard Drawing 901 WMEC 245-010, Rev F, CPCH Prop Hyd Sys A & D (902-904)
Coast Guard Drawing 901 WMEC 245-013, Rev B, Pitch Actuator System Engagement
Mechanisms, Links Shaft Clamps and Brackets
Coast Guard Drawing 901 WMEC 245-014, Rev A, CPP System Pitch Setters
Coast Guard Drawing 901 WMEC 245-015, Rev B, CPP System Pitch Setters Main Boards, Sub-
Boards, Scales and Mechanical Stops

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 2699, SWBS 245, Section A, Feb 2018, Controllable
Pitch Propeller System
Coast Guard Technical Publication (TP) 9187, SWBS 243, May 2021, Propulsion Shafting and
CPP System Service Manual
Coast Guard Technical Publication (TP) 9063, SWBS 252, Aug 2016, Coast Guard Machinery
Control System (CGMCS) – Pitch Control System & Throttle Control system – Vol 4
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements
Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022,
Welding and Allied Processes
Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022,
Auxiliary Machine Systems

OTHER REFERENCES

International Standard Organization (ISO) 4406, 2017, Hydraulic Fluid Power – Fluids – Method
for Coding the Level of Contamination by Solid Particles – Third Edition
MIL-PRF-17672E, Oct 2012, Hydraulic Fluid, Petroleum, Inhibited
Society of Automotive Engineers (SAE) Aerospace Material Specification AMS-C-6183, 2019,
Cork and Rubber Composition Sheet; For Aromatic Fuel and Oil Resistant Gaskets

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2.1 Operational test, initial
- 3.2.1.1 Pitch checks
- 3.2.5 CPP hub inspection (and all subparagraphs)

3.1.2 Tech Rep. The Government will provide a Tech Rep to facilitate CPP inspections and overhauls. The Contractor must provide labor, material and tools to accomplish the below tasks marked with (*), as directed by the Government Tech Rep.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.3.1 Protection of fine surfaces. The Contractor must protect all fine surfaces in accordance with SFLC Std. Spec 5000.

3.1.3.2 Follow General Maintenance Precautions of TP 2699 Section 4-2.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000 (Interferences).

3.1.5 Hydraulic system cleanliness. The Contractor must continuously monitor hydraulic system cleanliness for the duration of this work item in accordance with SFLC Std Spec 5000 paragraph 3.1.4. Cleanliness Grade A as defined in SFLC Std Spec 5000 is required.

3.2 Maintenance particulars. The Contractor must use Coast Guard Technical Publications (TP) 2699 and 9187, and Coast Guard Drawings 901 WMEC 245-002 and/or 901 WMEC 245-010 as guidance to accomplish the following tasks in Table 1 below.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

TABLE 1 – TASK REQUIREMENTS

#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	APPENDIX/PARA. FROM SFLC STD SPEC 5000	ADDITIONAL REQUIREMENTS (WORK ITEM PARA.)
1	*Operate and Inspect, Initial	2	Port and Stbd CPP	3.2.1 <u>Operate and inspect</u> C2.4.2.2 <u>Testing electrical sensors</u>	3.2.1 <u>Operational test, initial</u> 3.2.1.1 <u>Pitch checks</u> 3.2.1.2 <u>Pitch markings</u> 3.2.1.3 <u>Gauges</u> 3.2.1.4 <u>Float level switches, pressure switches, pressure</u>

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#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	APPENDIX/PARA. FROM SFLC STD SPEC 5000	ADDITIONAL REQUIREMENTS (WORK ITEM PARA.)
					transducers and RTDs 3.2.1.5 <u>Flex hoses</u>
2	Hydraulic Fluid, Sample, Pre-Repair	2	Port and Stbd CPP Hydraulic System	C2.1.3 <u>Sample and test</u>	3.2.2 <u>Hydraulic fluid sampling and testing</u>
3	Hydraulic Fluid, Drain	2	Port and Stbd CPP Hydraulic System	C2.1 <u>Fluids</u>	3.2.3 <u>System fluid drainage and disposal</u> 3.2.3.1 <u>CPP Hub fluid sample</u>
4	*Remove	2 sets	Port and Stbd Propeller Blades	3.2.5 <u>NDE</u>	3.2.4 <u>Propeller blades removal</u> 3.2.4.1 NDT 3.2.4.2 <u>Blade palm to hub body dimensional check</u> Propeller blade cleaning and inspections covered in definite Work Item "Controllable Pitch Propellers, Clean and Inspect".
5	*Inspect	2	Port and Stbd CPP Hub	N/A	3.2.5 <u>CPP hub inspection</u> 3.2.5.1 <u>Thrust readings</u> 3.2.5.2 <u>Trunnion bushing clearance readings</u> 3.2.5.3 <u>Borescope inspection</u>
6	Clean and Inspect	2	Port and Stbd Hydraulic System Tanks	N/A	3.2.6 <u>Clean and inspect hydraulic fluid tanks</u> All subparagraphs
7	*Overhaul	2	Port and Stbd Actuating Units	3.1.7 <u>Cleaning of hydraulic system components at reassembly</u> 3.2.3 <u>Disassemble and inspect</u> A2.4 <u>Fine surface inspections</u>	3.2.7 <u>Actuating unit overhaul</u> 3.2.7.1 Inspection and measurements 3.2.7.2 Seals, gaskets, retainer ring renewals 3.2.7.3 Inspect Teleflex cables and bushings
8	*Overhaul	2	Port and Stbd Hydraulic Control and Pressure Filter Manifolds	3.1.7 <u>Cleaning of hydraulic system components at reassembly</u> 3.2.3 <u>Disassemble and inspect</u> A2.4 <u>Fine surface inspections</u> C2.4 <u>Valves and manifolds</u>	3.2.8 <u>Hydraulic control and pressure filter manifolds</u> 3.2.8.1 Disposal 3.2.8.2 Pressure filter manifold disassemble clean and inspect 3.2.8.3 Pressure filter manifold reassembly, component

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#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	APPENDIX/PARA. FROM SFLC STD SPEC 5000	ADDITIONAL REQUIREMENTS (WORK ITEM PARA.)
				C2.4.2 <u>Disassembly</u> C2.4.2.1 <u>Testing solenoid assemblies</u> C2.4.2.2 <u>Testing electrical sensors</u>	renewals 3.2.8.4 Hydraulic control manifold disassemble clean and inspect 3.2.8.5 Hydraulic control manifold reassembly, component renewals 3.2.8.6 Strainer disassemble, clean and inspect; reassemble renew parts.
9	*Reinstall	2 sets	Port and Stbd Propeller Blades	N/A	3.2.9 <u>Blade reassembly/renewals</u>
10	Renew	1,500 gallons	Port and Stbd Hydraulic Fluid	C2.1.1 <u>Renewal</u> 3.1.5 <u>Mandatory renewals</u>	3.2.10 <u>System fluid renewal</u> 3.2.10.1 Storage tank
11	Flush	2	Port and Stbd CPP Hydraulic System	C2.1.2 <u>Filtering</u>	3.2.11 <u>System flushing</u> All subparagraphs
12	*Set Relief Valves	12	Port and Stbd Hydraulic System Relief Valves	C2.4.4 <u>Setting and tagging pressure control valves in place</u>	3.2.12 <u>CPP system hydraulic pressure relief valves</u> 3.2.12.1 Set in place 3.2.12.2 Tag
13	*Op Test	2	Port and Stbd CPP System	N/A	3.2.13 <u>Operational test, post repairs</u> 3.2.13.1 Pressure test
14	Hydraulic Fluid, Sample, Post-Repair	2	Port and Stbd CPP Hydraulic System	C2.1.3 <u>Sample and test</u>	3.2.2 <u>Hydraulic fluid sampling and testing</u>
15	Groom	2	Port and Stbd Control Console	3.2.6 <u>Groom and lubricate</u>	3.2.14 <u>Control console groom</u>
16	Sea Trial	1	Port and Stbd CPP System	N/A	3.2.15 <u>Sea trial</u>

NOTE

Tasks and paragraphs marked with asterisk (*) require specific knowledge and OEM specifications and procedures available to the Tech Rep. Contractor to provide labor, material and tools to accomplish the work as directed by the Tech Rep.

*3.2.1 Operational test, initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of all items or shipboard devices to be disturbed, used, repaired, or altered, to demonstrate existing operational condition. Submit a CIR.

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- Operate CPP system in accordance with TP 2699 Chapter 2. Note operating pressure.
- While system is operating, record any leaks, unusual vibration or noise. Noise is defined as loud grinding caused by metal-to-metal contact. When standing beside a hub in drydock, the normal pitching of the blades should be soundless.

*3.2.1.1 Pitch checks. Prior to commencement of work, the Contractor must check pitch of blades against indicated pitch scale in accordance with TP 2699 Section 4-8 Paragraph B.1 and Section 4-9 Paragraph B.3, and Coast Guard Drawings 901 WMEC 245-013, 901 WMEC 245-014 and 901 WMEC 245-015 to ensure synchronization. Operate system using two hydraulic pumps and record time for pitch to advance from zero pitch to full ahead pitch, zero pitch to full astern pitch, and design ahead pitch to full astern pitch. Submit a CIR.

*3.2.1.2 Pitch markings. Verify reference marks for zero pitch, full ahead and full astern are present on the blades and hub, and readable in accordance with Coast Guard Drawing 901 WMEC 245-004 General Note 7.h. Submit a CFR to restore markings if missing or unreadable.

*3.2.1.3 Gauges. Verify all system gauges have valid calibration stickers. See TP 2699 Section 4-8 Paragraph F and Coast Guard Drawing 901 WMEC 245-001 or 901 WMEC 245-009 as applicable for guidance. Submit CFR for any gauges that do not have current or valid calibration stickers in accordance with SFLC Std Spec 5000 Appendix C Paragraph C2.5.2.

*3.2.1.4 Float level switches, pressure switches, pressure transducers and RTDs. Verify all float level switches, pressure switches, pressure transducers, sump heaters and RTDs function correctly in accordance with SFLC Std Spec 5000 Appendix C Paragraph C2.4.2.2 using TP 2699 Section 4-8 Paragraph E.1 through E.4 as guidance. Submit a CFR.

*3.2.1.5 Flex hoses. Inspect all flex hoses and verify Service Life Dates are current. Submit a CFR for any flex hoses that are damaged, missing ID tags or whose Service Life Dates are expired.

3.2.2 Hydraulic fluid sampling and testing. Prior to and immediately following the work herein, the Contractor must sample and test the system hydraulic fluid in accordance with SFLC Std Spec 5000, Appendix C, Paragraph C2.1.3. Submit a CFR.

3.2.3 System fluid drainage and disposal. The Contractor must drain all hydraulic oil from the Port and Stbd CPP systems including tanks (approximately 1,500 gallons) into a suitable container using TP 2699 Section 4-10 Paragraph A.1 as guidance. Dispose of removed fluids in accordance with all applicable Federal, state and local regulations.

3.2.3.1 CPP Hub fluid sample. While CPP Hubs are being drained, obtain fluid sample and test in accordance with SFLC Std Spec 5000, Appendix C, Paragraph C2.1.3 (this sample is separate from the samples required in paragraph 3.2.2). Submit a CFR.

*3.2.4 Propeller blades removal. The Contractor must remove the propeller blades from CPP hub assemblies under the direction of the Government Tech Rep. Before blade removal, ensure that the propeller shaft is restrained from turning. See TP 2699 Section 8-3 Paragraphs A.1 through A.11 for guidance. Ensure protective measures are applied to removed blades.

3.2.4.1 Perform non-destructive testing of all blade bolts, blade bolt holes, dowel pins and dowel pin holes in accordance with SFLC Std Spec 0740, Appendix C. Submit a CFR.

*3.2.4.2 Blade palm to hub body dimensional check: For each blade, measure and record the OD of the blade palm in the area of the o-ring sealing surface and the ID of the corresponding sealing surface in the blade port counterbore. See Figure 1. Record dimensions in Data Sheet 1.

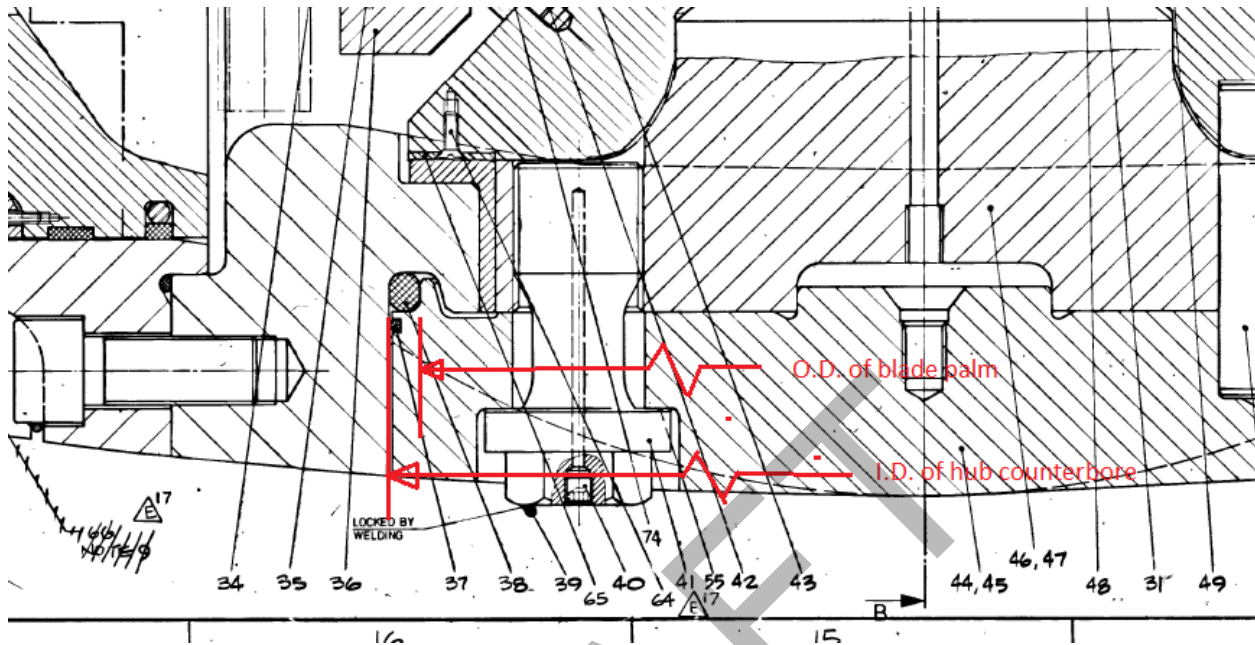


FIGURE 1: CPP BLADE PALM DIMENSIONAL CHECK

*3.2.5 CPP hub inspection. The Contractor must accomplish Port and Stbd CPP Hub inspections below under the direction of the Tech Rep. Tech Rep must evaluate data collected and make a recommendation to the COR if hub will need further corrective action using Table 2 as guidance. Submit CIR containing results of all inspections in subparagraphs below.

TABLE 2: CPP HUB CORRECTIVE ACTIONS

INSPECTION RESULT AS DEEMED BY GOVT TECH REP (PARAGRAPH 3.2.5)	ACTION
Satisfactory	No further corrective action.
Inconclusive and further Hub Inspections are necessary	Activate optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Inspect”.
Unsatisfactory	Activate optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Renew”.

If a Change Request is released and approved by the KO, activate optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Inspect”.

If a Change Request is released and approved by the KO, activate optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Renew”.

NOTE

Thrust dimension and trunnion face dimension measured below are not toleranced dimensions, but combined with bushing clearance readings and bore scope inspection can give a general idea of the hub condition.

*3.2.5.1 Thrust readings. For each blade trunnion, measure axial thrust in accordance with Tech Rep instructions. Record thrust measurements in Data Sheet 2.

*3.2.5.2 Trunnion bushing clearance readings. For each blade, measure clearance between blade trunnion and hub collar bushing using feeler gauges in 4 places, 90 deg apart. See Figure 2. Record clearance measurements in Data Sheet 3.

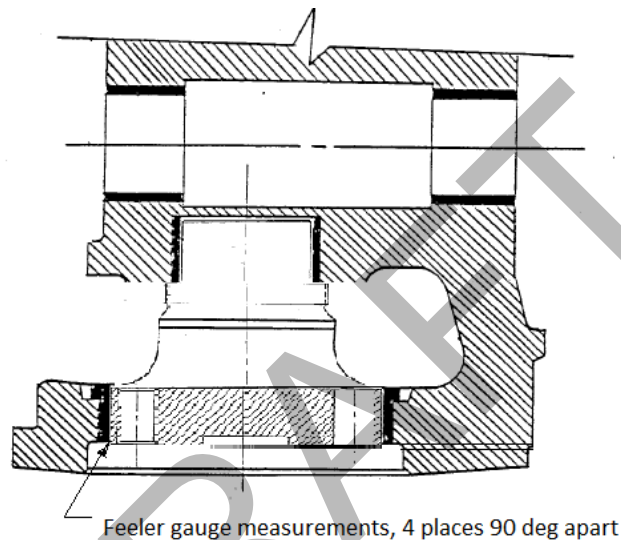


FIGURE 2: CPP BLADE TRUNNION TO HUB COLLAR BUSHING MEASUREMENTS

*3.2.5.3 Borescope inspection. For each hub, accomplish borescope inspection of the hub interior to inspect for signs of excessive wear or damage.

3.2.6 Clean and inspect hydraulic fluid tanks. The Contractor must clean and inspect the System Sump Tanks (4-162-1-F and 4-162-2-F), Head Tanks, and Storage Tank (4-162-0-F). Accomplish the following:

3.2.6.1 Plug log. The Contractor must keep a written record of all plugs put in any tank vents. A separate list must be kept for each tank being entered.

3.2.6.1.1 Ensure that all plugs are removed from each tank (as applicable) upon completion of work in the tank.

3.2.6.1.2 Ensure the plug log is available to the Coast Guard inspector when the inspector is performing his close-out inspection on each tank.

3.2.6.2 Cleaning requirements. The Contractor must remove tank cover(s) and clean tank interior surfaces free of all foreign materials, such as sediment or sludge, taking care not to damage the coating system (if applicable). Remove cleaning media and residues continuously during the washing process.

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Remove any residual wash media and wipe up residual moisture with clean lint-free cloths. Extreme care must be exercised to prevent damage to the tank level indicator sensors, during the cleaning process.

3.2.6.3 Tank content and waste disposal. The Contractor must dispose of residual tank contents and all cleaning fluids in compliance with all applicable Federal, state, and local laws, ordinances and regulations. Document a complete chain of custody record of the removed tank contents and generated wastes, from the vessel to the point of final destination or delivery. Submit document to the COR upon completion of work.

3.2.6.4 Inspection. The Contractor must visually inspect all tank interior surfaces, including, but not limited to bulkheads, floor and overhead plating, structural members, manhole cover surfaces, fasteners and gasket seating surfaces. Complete the “Tank and Void Assessment Form” (found in the Work Item titled “General Requirements”) and submit via CFR. Electronic PDF version of this form will be furnished by the COR upon request. Include the following, as applicable:

- Tank structural condition.
- Inaccessible areas.
- Condition of tank coating (if applicable), including measurements taken, percentage, location, and type of coating failure.
- Tank level indicator (TLI) and/or float switch condition (as applicable).
- Sounding tube and striker plate condition.
- Suction and discharge piping condition.
- Fastener material (stainless steel) and condition.

3.2.6.5 Tank closing. The Contractor must ensure that the tank(s) remain open for at least 24 hours after completion of all KO-authorized repair and preservation procedures. Notify the COR at least 24 hours prior to closing the tank(s) and ensure tank closure is accomplished in the presence of the Coast Guard Inspector. After satisfactory inspection by the Coast Guard Inspector and completion of all authorized repairs, close tank manhole cover(s) with new gasket material conforming to AMS-C-6183, and renew any damaged or missing fasteners. For purpose of bid, assume 10% of existing fasteners will require renewal.

*3.2.7 Actuating unit overhaul. The Contractor must overhaul Port and Stbd CPP Actuating Unit under the direction of the Tech Rep using TP 2699, Section 8-2 Paragraphs A.1 through A.46, B.1 through B.13 and C.1 through C.81, TP 9187, and Coast Guard Drawings 901 WMEC 245-001 or 901 WMEC 245-009 and 901 WMEC 245-008 as guidance:

*3.2.7.1 Accomplish inspection and measurements of components in Data Sheets 4 and 5. Submit a CFR for any out of spec components.

*3.2.7.2 Renew all seals, gaskets and retainer rings using items provided in the Government Furnished overhaul kit.

3.2.7.3 Inspect the Teleflex E-MAN engagement and control cables for binding and misalignment. Submit a CFR. Clean and lubricate the pitch setter mechanism using TP 9063 and Coast Guard Drawings 901 WMEC 245-013, 901 WMEC 245-014 and 901 WMEC 245-015 as guidance.

*3.2.8 Hydraulic control and pressure filter manifolds. The Contractor must overhaul Port and Stbd CPP hydraulic control manifolds and pressure filter manifolds under the direction of the Tech Rep, using TP 2699, Appendix "D" (Pressure Filter Manifold) and TP 2699, Appendix "E" (Control Manifold) as guidance:

3.2.8.1 Dispose of all renewed components.

3.2.8.2 Disassemble the pressure filter manifold, and clean and inspect all components and fasteners. Submit a CFR.

*3.2.8.3 Reassemble and reinstall the pressure filter manifolds, and accomplish the following with Government furnished material:

- Renew all Check Valves (NSN 4820-01-198-7138)
- Renew all Filter Elements (NSN 4330-01-177-4069)

NOTE

Cartridge relief valves NSN 4820-01-194-5761 are obsolete with no replacement identified. Re-use whenever possible.

- Clean and inspect all Relief Valves (NSN 4820-01-194-5761). Submit a CFR.
- Renew all Filter Assembly seals using Filter Seal Kit (NSN 5330-01-358-1291)
- Renew all Cartridge check valves (NSN 4820-01-228-6988)
- Renew all Selector Valve seals using Selector valve seal kits (NSN 4820-01-358-6162)

*3.2.8.4 Disassemble the hydraulic control manifolds, and clean and inspect all components and fasteners. Submit a CFR

*3.2.8.5 Renew all o-rings, gaskets and seals (Contractor furnished) and reassemble and reinstall hydraulic control manifolds.

*3.2.8.6 Disassemble the strainers, and clean and inspect all components and fasteners. Submit a CFR.

- Renew seals using seal kit (Seals, gasket and o-rings, NSN 5330-01-266-6513)
- Renew all Strainer Elements (NSN 4730-01-456-6689)
- Renew all Magnets (NSN 4330-01-258-2746)

*3.2.9 Blade reassembly/renewals. After all components have been inspected and allowable dimensional clearances have been verified, reassemble the blades. Renew all blade seals. See TP 2699 Section 8-3 Paragraphs C.1 through C.9 for guidance.

3.2.10 System fluid renewal. After all authorized repairs, the Contractor must renew 100% of the system hydraulic fluid, sumps, head tanks and system contents in accordance with TP 2699 Section 4-10 Paragraph A.3 and using Coast Guard Drawing 901 WMEC 245-001 or 901 WMEC 245-009 as guidance. See Table 3 for approved lubricant. Replenish any system fluid lost during repair and/or flush to obtain 100% capacity. Accomplish the following:

TABLE 3 – AUTHORIZED LUBRICANTS

OEM SPEC OIL
MIL-PRF-17672 2135TH

NOTE

Each system should be completely void of air after cycling; oil level must be monitored after the shaft is rotated. If turning the shaft releases more air, more oil must be added to the system.

3.2.10.1 Storage tank. Top off with hydraulic fluid filtered through a 10-micron absolute, non-bypass type filter designed specifically for filtering oils and other lubricants.

3.2.11 System flushing. The Contractor must flush the hydraulic system with the system fully assembled and vessel re-floated (to ensure blade seals are properly lubricated) using the system's operating fluid in accordance with SFLC Std Spec 5000 Appendix C Paragraph C2.1.2 (Filtering) and the following:

3.2.11.1 Install a 3 micron filter on the return line directly off port "A" of the O.D. Box. The Contractor must furnish all necessary fittings and filtering rig. Equipment tie-in shall minimize loss of fluid when connecting/disconnecting equipment (installation of temporary isolation valves for example). Use this filter as an inspection point to determine if fragments are still coming out of the return piping. If necessary, to maintain system recommended temperature, install an external in-line heater at this location.

NOTE

Carefully monitor the heater to ensure that they do not char the oil around the elements.

3.2.11.2 While operating the system normally, using system pumps, cycle pitch full astern/ahead repeatedly, while periodically rotating the shafting. Continue to monitor return pressure, ensuring that it remains within system guidelines.

3.2.11.3 After 15 cycles, shut down and inspect the return filter. Continue to repeat 3.2.11.2 until target particulate count of 18/16/13 or better per ISO 4406 is achieved.

3.2.11.4 Submit a detailed report to the COR within 24 hours. Report shall include:

- Method used to obtain the particulate contamination reading.
- Method used to determine particle size.
- Final contamination level readings.

3.2.11.5 After flushing, remove and dispose of temporary filters and mesh screens. Renew all system hydraulic filters.

*3.2.12 CPP system hydraulic pressure relief valves. The Contractor must clean, inspect and test the below-listed pressure relief valves.

*3.2.12.1 After completion of inspection and testing, reset the relief valves to their set pressure in the presence of the Tech Rep in accordance with SFLC Std Spec 5000 Appendix C Paragraph C2.4.4 using

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Coast Guard Drawings 901 WMEC 245-001 or 901 WMEC 245-009 and TP 2699 Section 4-8 Paragraph A.1 through A.5 as guidance.

3.2.12.2 Furnish and tag each relief valve with data plates that include the test pressure, test date, and the name of test facility.

TABLE 3 - PRESSURE RELIEF VALVES

NUMBER	QTY	SIZE	SET PSI	SERVICE
RV-2	6	1/2	750 ¹	HP Filter Manifold
RV-3	2	1/8	150	Pilot Relief (Controls RV-6)
RV-4	2	3/4	150	Head Tank Pump Discharge
RV-5	2	1-1/2	15	Cooler Discharge to Sump
RV-6	2	1-1/2	500	Main Pump Discharge
RV-7	1	1-1/2	28	Hydraulic oil transfer pump

¹ TP and Dwg 901 WMEC 245-009 indicate that this setting should be 600 psig but MEC-ENG-SES SME has discovered that this setting is too close to RV-6 and 750 psig is a more appropriate setting.

***3.2.13 Operational test, post repairs.** After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and Ship's Force and demonstrate all items or shipboard devices that have been disturbed, used, repaired, altered, or installed to be in satisfactory operating condition and then repeat the requirements of paragraph 3.2.1.1 (Pitch checks) while the vessel is still in drydock. Submit a CFR.

- Pitch change rate: Elapsed time between pitch stop limits must be 20 seconds +/- 3 seconds
- Hydraulic oil temp: Temperature must be between 105°F and 135°F. This reading is taken from the thermometer at the inlet side of the oil cooler.
- Pitch drift: Set pitch to Full ahead. Once pitch is met, pitch must not drift astern.
- Check pointer attached to command input shaft for actuating unit. Ensure pointer lines up with "0" mark on the scale while the blades are at "0" pitch marks on the hub

***3.2.13.1 Pressure test.** Perform pressure test of fully assembled system in accordance with TP 2699 Section 8-3 Paragraph C.8 prior to installing/repairing shaft coating. Verify no leaks in the shaft line.

3.2.14 Control console groom. The Contractor must plan for and allow Ship's Force up to 2 days to conduct a CPP control console system groom prior to placing the vessel back in the water.

3.2.15 Sea trial. The Contractor must conduct a four-hour sea trial to test the CPP system with the hub operating in all pitch positions. Submit a CFR.

4. NOTES

4.1 Coast Guard Inspector responsibilities. The Coast Guard Inspector will be responsible for the following:

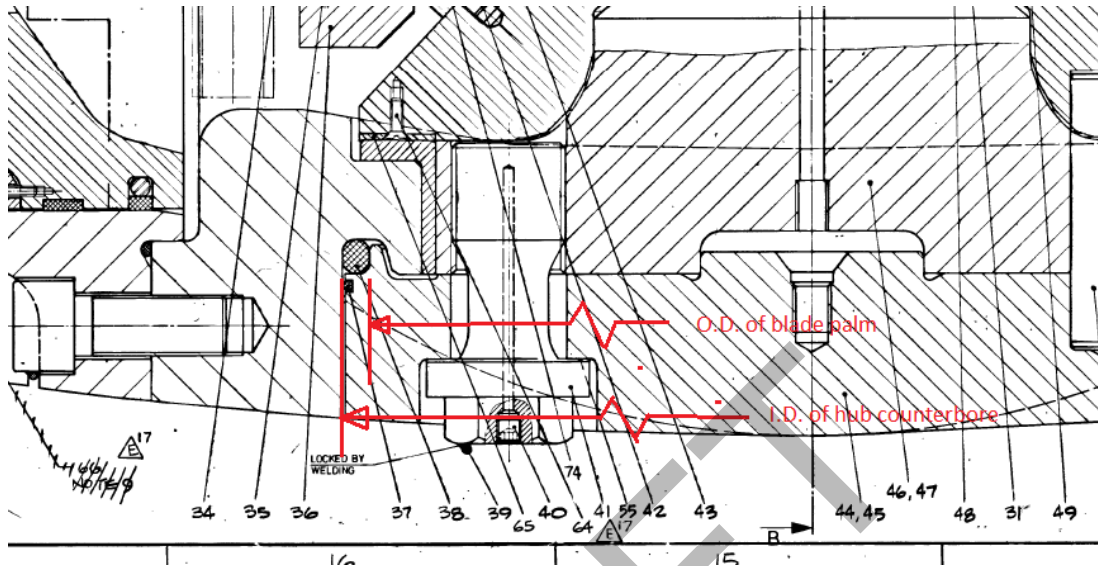
4.1.1 Visually verify the presence of protective coverings each and every time the hydraulic system is opened and exposed to potential contamination. The Coast Guard Inspector will initial and date as verified the CFR provided by the Contractor, detailing protective measures taken.

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- 4.1.2 Visually verify tank cleanliness prior to closure.
- 4.1.3 Visually verify cleanliness of all hoses and containers used for removal and installation of fluids.
- 4.1.4 Confirm all above verifications in writing on CFR provided by Contractor.

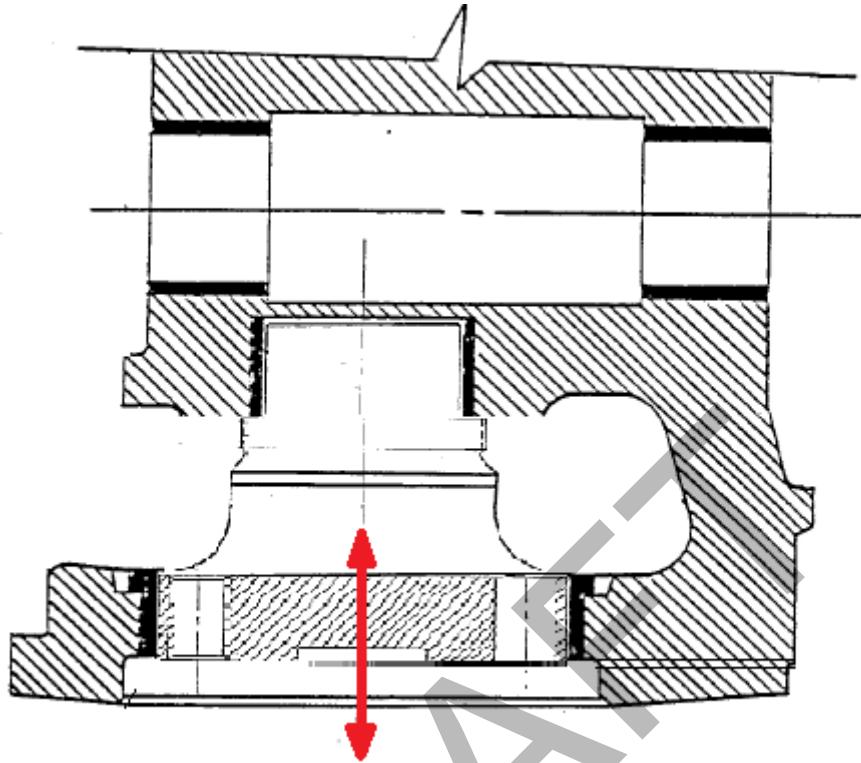
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DATA SHEET 1 - 270 BLADE PALM AND HUB MEASUREMENTS



PORT		STBD	
BLADE #1	MEASUREMENT	BLADE #1	MEASUREMENT
ID of Hub Counterbore		ID of Hub Counterbore	
OD of Blade Palm		OD of Blade Palm	
BLADE #2	MEASUREMENT	BLADE #2	MEASUREMENT
ID of Hub Counterbore		ID of Hub Counterbore	
OD of Blade Palm		OD of Blade Palm	
BLADE #3	MEASUREMENT	BLADE #3	MEASUREMENT
ID of Hub Counterbore		ID of Hub Counterbore	
OD of Blade Palm		OD of Blade Palm	
BLADE #4	MEASUREMENT	BLADE #4	MEASUREMENT
ID of Hub Counterbore		ID of Hub Counterbore	
OD of Blade Palm		OD of Blade Palm	

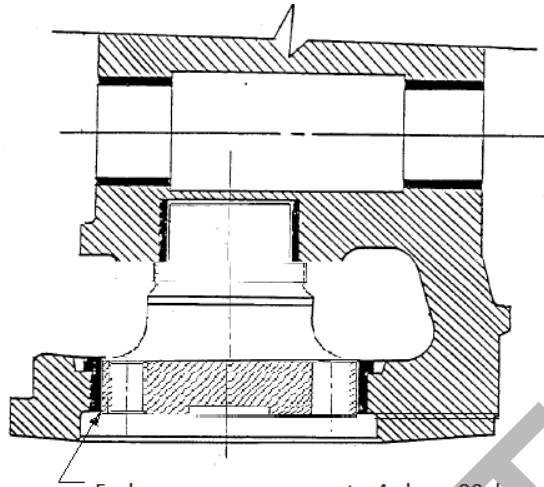
DATA SHEET 2 - 270 TRUNNION THRUST MEASUREMENTS



PORT		STBD	
HUB PORT #1	MEASUREMENT	HUB PORT #1	MEASUREMENT
Trunnion Axial Thrust (total axial travel of trunnion within blade port)		Trunnion Axial Thrust (total axial travel of trunnion within blade port)	
HUB PORT #2	MEASUREMENT	HUB PORT #2	MEASUREMENT
Trunnion Axial Thrust (total axial travel of trunnion within blade port)		Trunnion Axial Thrust (total axial travel of trunnion within blade port)	
HUB PORT #3	MEASUREMENT	HUB PORT #3	MEASUREMENT
Trunnion Axial Thrust (total axial travel of trunnion within blade port)		Trunnion Axial Thrust (total axial travel of trunnion within blade port)	
HUB PORT #4	MEASUREMENT	HUB PORT #4	MEASUREMENT
Trunnion Axial Thrust (total axial travel of trunnion within blade port)		Trunnion Axial Thrust (total axial travel of trunnion within blade port)	

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DATA SHEET 3 - 270 BLADE AND HUB MEASUREMENTS



Feeler gauge measurements, 4 places 90 deg apart

PORT		STBD	
HUB PORT #1	MEASUREMENT	HUB PORT #1	MEASUREMENT
Clearance between trunnion and collar bushing	Fwd _____	Clearance between trunnion and collar bushing	Fwd _____
	Aft _____		Aft _____
	Port _____		Port _____
	Stbd _____		Stbd _____
HUB PORT #2	MEASUREMENT	HUB PORT #2	MEASUREMENT
Clearance between trunnion and collar bushing	Fwd _____	Clearance between trunnion and collar bushing	Fwd _____
	Aft _____		Aft _____
	Port _____		Port _____
	Stbd _____		Stbd _____
HUB PORT #3	MEASUREMENT	HUB PORT #3	MEASUREMENT
Clearance between trunnion and collar bushing	Fwd _____	Clearance between trunnion and collar bushing	Fwd _____
	Aft _____		Aft _____
	Port _____		Port _____
	Stbd _____		Stbd _____
HUB PORT #4	MEASUREMENT	HUB PORT #4	MEASUREMENT
Clearance between trunnion and collar bushing	Fwd _____	Clearance between trunnion and collar bushing	Fwd _____
	Aft _____		Aft _____
	Port _____		Port _____
	Stbd _____		Stbd _____

**DATA SHEET 4 – OD BOX SPOOL VALVE ASSY (ACCOMPLISH DURING TP 2699 SECTION 8-2
PARAGRAPH B.13)**

DRAFT

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PORT _____ STBD _____

Control Bushing I.D. P/N 8-350-170 Item 14

1st land _____

2nd land _____

3rd land _____

Compare with control piston O.D.

Control Bushing housing I.D. _____

Control bushing O.D. _____

Clearance _____

Control Piston O.D. P/N 8-350-168 item 19

Clearance _____

1st land _____

2nd land _____

3rd land _____

pilot piston O.D. P/N 8-376-932 Item 17

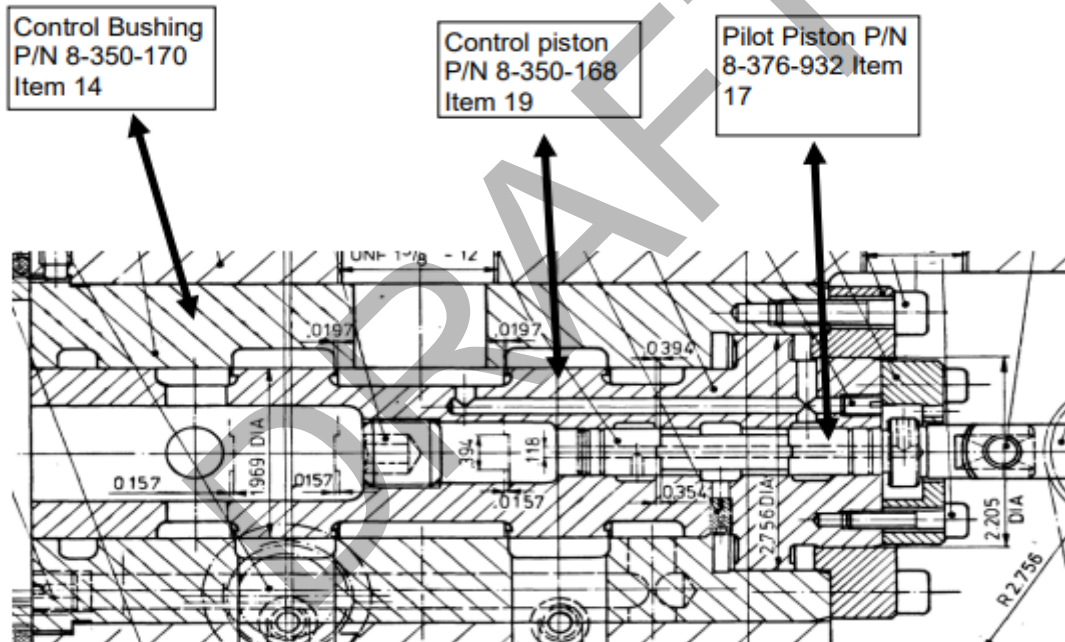
Clearance _____

Pilot Piston Bore I.D. _____

1st land _____

2nd land _____

Note item: 14,19 and 17 come as a matched set NSN 4820-01-513-2509

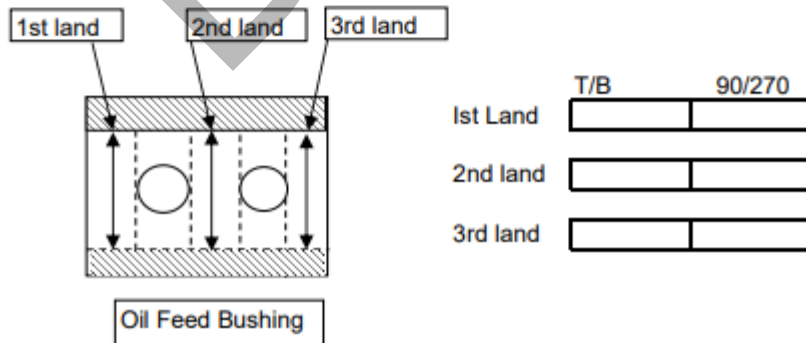
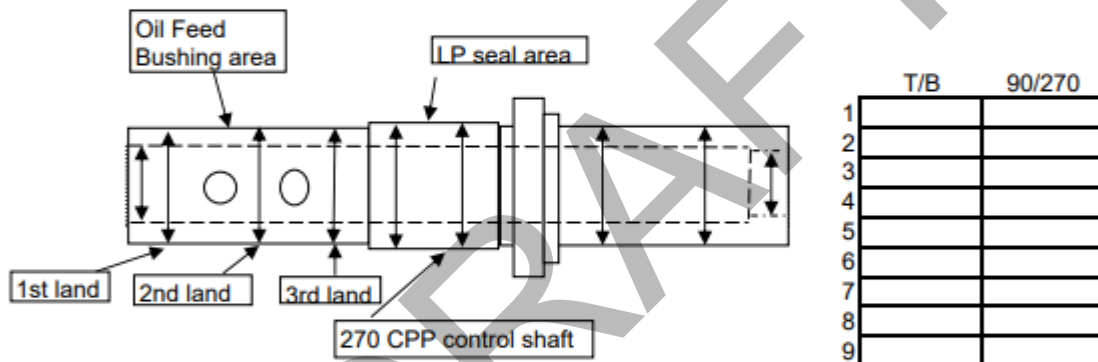
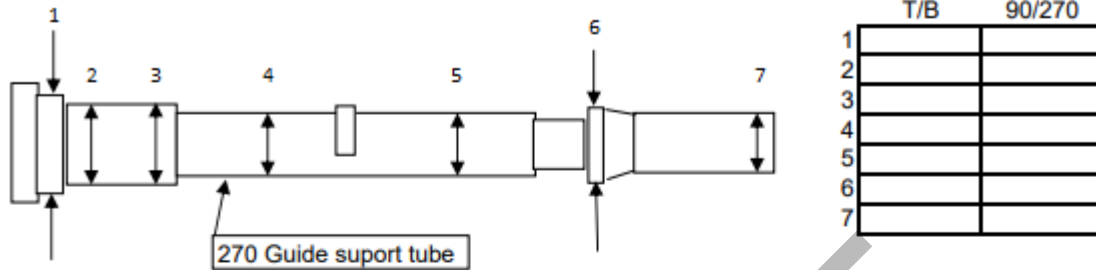


NOTES:

DATA SHEET 5 – OD BOX CONTROL SHAFT AND OIL FEED BUSHING ASSY (ACCOMPLISH DURING TP 2699 SECTION 8-2 PARAGRAPH A.13 AND A.16)

PORT _____
STBD _____

ALL MICROMETER READINGS WILL BE TAKEN FROM LEFT TO RIGHT



WORK ITEM 25: Controllable Pitch Propeller (CPP) Hubs, Renew (Optional)

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew a designated controllable pitch propeller hub assembly.

NOTE

This Work Item is written to renew one hub only (port or stbd as designated by the Change Request). Contractor's pricing should reflect the price of one hub only. If both port and stbd hubs require renewal (as designated via Change Request), the WI will be activated twice.

1.2 Government-furnished property.

NOTE

Although both hubs are listed in the GFP Table below, Government will furnish a Left Hand OR a Right Hand Hub Assembly for this instance of this Work Item as designated in the Change Request. If the other Hub requires renewal also, it will be furnished when a 2nd instance of this Work Item is activated.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	**Hub installation kit	NSN: 2010-01-470-3258	1 kt.	2,565.00
Y	**Left hand hub assembly	NSN: 2010-01-119-5469	1 ea.	897,576.00
Y	**Right hand hub assembly	NSN: 2010-01-118-8538	1 ea.	920,016.00
N	*Special tools (see 4.2 (Government-loaned installation tools))	N/A	1 se.	1,000.00

*Government-loaned property, which shall be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-007, Rev E, Controllable Pitch Propeller Shaft Coupling Assemblies

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Coast Guard Drawing 901 WMEC 245-003, Rev E, Controllable Pitch Propeller Propeller Hub Assembly

Coast Guard Drawing 901 WMEC 245-007, Rev B, Special Tools

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 2699, SWBS 245, Section A, Feb 2018, Controllable Pitch Propeller System

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Inspect, Repair, and Test Auxiliary Machine Systems

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep. The Government will provide a Tech Rep to facilitate CPP inspections and overhauls. The Contractor must provide labor, material and tools to accomplish the below tasks marked with (*), as directed by the Government Tech Rep.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.3.1 Protection of fine surfaces. The Contractor must protect all fine surfaces in accordance with SFLC Std. Spec 5000.

3.1.3.2 Hydraulic system contamination protection. The Contractor must maintain existing hydraulic cleanliness in accordance with SFLC Std. Spec 5000.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Shafting
- Rudders (if Hub is not removed with the tailshaft as an assembly)
- Hydraulic fluid (drained in Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”)
- Blades (removed in Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”)

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- Oil tubes

3.1.5 Hydraulic system cleanliness. The Contractor must continuously monitor hydraulic system cleanliness for the duration of this work item in accordance with SFLC Std Spec 5000 paragraph 3.1.4. Cleanliness Grade A as defined in SFLC Std Spec 5000 is required.

3.2 Maintenance particulars. The Contractor must use Coast Guard Technical Publication (TP) 2699 and Coast Guard Drawing 901 WMEC 245-003 as guidance to accomplish the following tasks in Table 1 below.

TABLE 1 – TASK REQUIREMENTS

#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	APPENDIX/PARA. FROM SFLC STD SPEC 5000	ADDITIONAL REQUIREMENTS (WORK ITEM PARA.)
1	*Renew	1	Designated CPP Hub	3.1.7 <u>Cleaning of hydraulic system components at reassembly</u> A2.4 <u>Fine surface inspections</u>	3.2.1 <u>Removal and reinstallation</u> 3.2.1.1 <u>Hub assembly removal</u> This paragraph must be descoped if hub is removed from tail shaft in optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Inspect” 3.2.1.2 <u>Old hub preparation for shipping</u> 3.2.1.2.1 3.2.1.2.2 3.2.1.2.3 3.2.1.2.4 3.2.1.3 <u>New hub preparation</u> 3.2.1.3.1 3.2.1.3.2 3.2.1.3.3 3.2.1.3.4 3.2.1.4 <u>Hub assembly installation</u> This paragraph must be descoped if hub will be reinstalled onto tail shaft in optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Inspect”

NOTE

Tasks and paragraphs marked with asterisk (*) require specific knowledge and OEM specifications and procedures available to the Tech Rep. Contractor to provide labor, material and tools to accomplish the work as directed by the Tech Rep.

*3.2.1 Removal and reinstallation. The Contractor must remove and install the designated hub assembly under the direction of the Tech Rep. Removal equipment, lifting devices and securing devices shall be in a safe operating condition and adequate for the work to be accomplished.

NOTE

The Government will furnish the special tools shown on Coast Guard Drawing 901 WMEC 245-007.

NOTE

Although the blades, hub, and shaft can be removed separately, TP 2699 recommends the blades be removed first and the hub and shaft assemblies removed as a unit. If there is no separate tasking for removing tail shaft, CPP hub may be removed from tail shaft separately.

*3.2.1.1 Hub assembly removal. The Contractor must remove the designated CPP hub assembly from the tailshaft under the direction of the Tech Rep using TP 2699 Section 8-5 Paragraphs B.1 through B.6 and Coast Guard Drawings 901 WMEC 243-007 and 901 WMEC 245-003 as guidance.

3.2.1.2 Old hub preparation for shipping. The Contractor must package the old hub, re-using the shipping crate and packing materials in which the new hub was shipped, for Mandatory Turn-In.

3.2.1.2.1 Ensure the shipping steel blanks (with seals) are installed onto the blade ports.

3.2.1.2.2 Connect the hub oil head tank to the hub and top off the head tank with CPP hydraulic fluid.

3.2.1.2.3 Check and correct any oil leaks.

3.2.1.2.4 Turn over the hub to the Coast Guard Property Administrator (PA) as an MTI item.

3.2.1.3 New hub preparation. The Contractor must prepare the new hub for installation as follows:

3.2.1.3.1 Transport the new hub to a suitable repair facility.

3.2.1.3.2 Open the shipping crate and disconnect the hub oil head tank from the hub while in the crate; retain the crate and shipping materials to use to ship the old hub.

3.2.1.3.3 Remove the new hub; drain and dispose of all oil in accordance with all applicable Federal, state, and local regulations.

*3.2.1.3.4 Perform a visual inspection of the hub under the supervision of the Tech Rep. Submit a CFR.

*3.2.1.4 Hub assembly installation. The Contractor must install designated CPP hub assembly onto tailshaft under the direction of the Tech Rep using TP 2699 Section 8-5 Paragraphs D.28 through D.39

and Coast Guard Drawing 901 WMEC 245-003 as guidance using the Government-furnished installation kit.

NOTE

Post installation pressure test, operational test and pitch checks to be accomplished on Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”.

4. NOTES

4.1 Coast Guard Inspector responsibilities. The Coast Guard Inspector will be responsible for the following:

4.1.1 Visually verify the presence of protective coverings each and every time the hydraulic system is opened and exposed to potential contamination. The Coast Guard Inspector will initial and date as verified the CFR provided by the Contractor, detailing protective measures taken.

4.1.2 Visually verify tank cleanliness prior to closure.

4.1.3 Visually verify cleanliness of all hoses and containers used for removal and installation of fluids.

4.1.4 Confirm all above verifications in writing on CFR provided by Contractor.

4.2 Government-loaned installation tools. The Government-loaned “special tools” are shown on Coast Guard Drawing 901 WMEC 245-007 for hub disassembly.

DRAFT

WORK ITEM 26: Controllable Pitch Propeller (CPP) Hubs, Inspect (Optional)

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to disassemble and inspect a designated controllable pitch propeller hub assembly.

NOTE

This Work Item is written to inspect one hub only (port or stbd as designated by the Change Request). Contractor's pricing should reflect the price of one hub only. If both port and stbd hubs require inspection (as designated via Change Request), the WI will be activated twice.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	**Hub Installation kit	NSN: 2010-01-470-3258	1 kt.	2,565.00

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 243-007, Rev E, Controllable Pitch Propeller Shaft Coupling Assemblies

Coast Guard Drawing 901 WMEC 245-003, Rev E, Controllable Pitch Propeller Propeller Hub Assembly

Coast Guard Drawing 901 WMEC 245-007, Rev B, Special Tools

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 2699, SWBS 245, Section A, Feb 2018, Controllable Pitch Propeller System

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Auxiliary Machine Systems

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep. The Government will provide a Tech Rep to facilitate CPP inspections and overhauls. The Contractor must provide labor, material and tools to accomplish the below tasks marked with (*), as directed by the Government Tech Rep.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 Protection of fine surfaces. The Contractor must protect all fine surfaces in accordance with SFLC Std. Spec 5000.

3.1.3.2 Hydraulic system contamination protection. The Contractor must maintain existing hydraulic cleanliness in accordance with SFLC Std. Spec 5000.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Shafting
- Rudders (if Hub is not removed with the tailshaft as an assembly)
- Hydraulic fluid (drained in Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”)
- Blades (removed in Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”)
- Oil tubes

3.1.5 Hydraulic system cleanliness. The Contractor must continuously monitor hydraulic system cleanliness for the duration of this work item in accordance with SFLC Std Spec 5000 paragraph 3.1.4. Cleanliness Grade A as defined in SFLC Std Spec 5000 is required.

3.2 Maintenance particulars. The Contractor must use Coast Guard Technical Publication (TP) 2699 and Coast Guard Drawing 901 WMEC 245-003 as guidance to accomplish the following tasks in Table 1 below.

NOTE

Tasks and paragraphs marked with asterisk (*) require specific knowledge and OEM specifications and procedures available to the Tech Rep. Contractor to provide labor, material and tools to accomplish the work as directed by the Tech Rep.

TABLE 1 – TASK REQUIREMENTS

#	TASK TYPE	QTY	COMPONENT OR ASSEMBLY	APPENDIX/PARA. FROM SFLC STD SPEC 5000	ADDITIONAL REQUIREMENTS (WORK ITEM PARA.)
1	*Remove	1	Designated CPP Hub	N/A	3.2.1 <u>Hub assembly removal</u>
2	*Disassemble and Inspect	1	Designated CPP Hub	3.2.3 <u>Disassemble and inspect</u> 3.1.7 <u>Cleaning of hydraulic system components at reassembly</u> A2.4 <u>Fine surface inspections</u>	3.2.2 <u>Hub inspection</u> 3.2.2.1 Hub disassembly and trunnion removal 3.2.2.1.1 Trunnion bushing ID measurements 3.2.2.1.2 Trunnion OD measurements 3.2.2.1.3 Component inspections 3.2.2.2 Assessment 3.2.3 <u>Hub reassembly</u> 3.2.3.1 Hub component renewals (if approved by Change Request)
3	*Reinstall	1	Designated CPP Hub	N/A	3.2.4 <u>Hub assembly installation</u>

NOTE

The Government will furnish the special tools shown on Coast Guard Drawing 901 WMEC 245-007.

NOTE

Although the blades, hub, and shaft can be removed separately, TP 2699 recommends the blades be removed first and the hub and shaft assemblies removed as a unit. If there is no separate tasking for removing tail shaft, CPP hub may be removed from tail shaft separately.

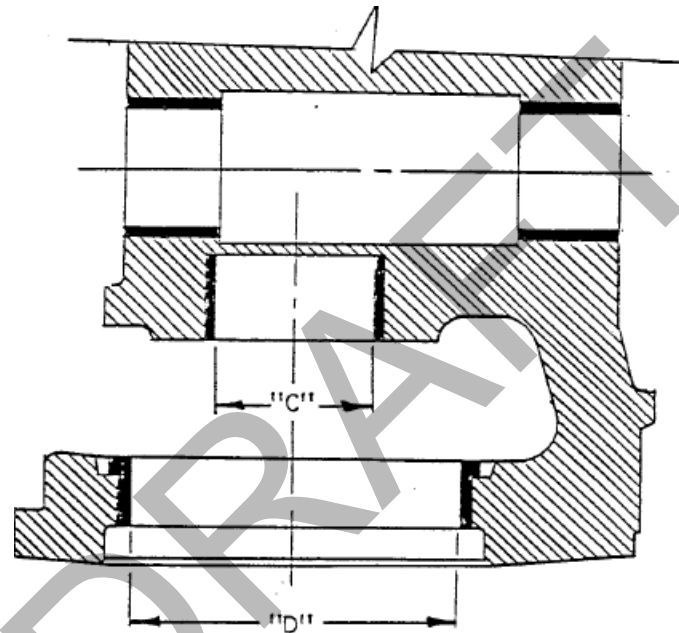
*3.2.1 Hub assembly removal. The Contractor must remove the designated CPP hub assembly from the tailshaft under the direction of the Tech Rep using TP 2699 Section 8-5 Paragraphs B.1 through B.6 and Coast Guard Drawings 901 WMEC 243-007 and 901 WMEC 245-003 as guidance. Transport to suitable location for hub disassembly where Cleanliness Grade A in accordance with SFLC Std Spec 5000 can be maintained.

*3.2.2 Hub inspection. The Contractor must perform the following under the direction of the Tech Rep for a designated CPP hub, using TP 2699 and Coast Guard Drawing 901 WMEC 245-003 as guidance. Submit a CFR.

*3.2.2.1 Disassemble CPP hub in accordance with TP 2699 Section 8-5 Paragraphs B.7 through B.27.

*3.2.2.1.1 At each CPP blade port, measure and record trunnion bushing ID as shown in Figure 1 to nearest 0.001" (2 places, 90 degrees apart). Submit a CFR.

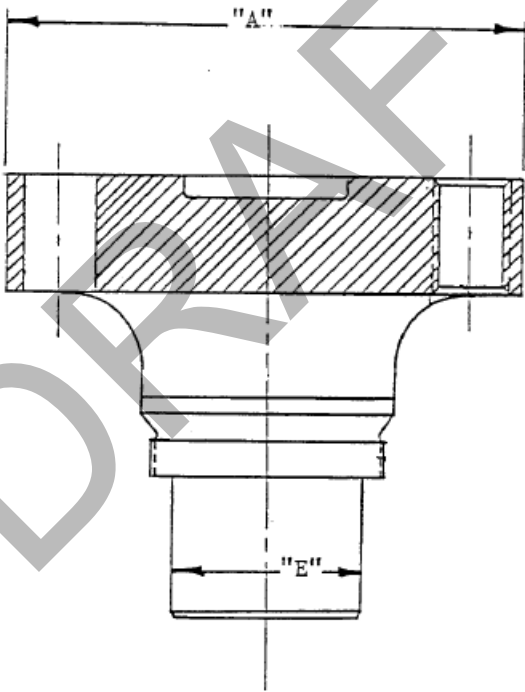
*3.2.2.1.2 For each CPP blade trunnion, measure and record dimensions shown in Figure 2, to nearest 0.001" (2 places, 90 degrees apart). Submit a CFR.



POS	MEASURED (2 PLACES, 90 DEGREES APART)
"C" Blade 1	
"C" Blade 2	
"C" Blade 3	
"C" Blade 4	

POS	MEASURED (2 PLACES, 90 DEGREES APART)
“D” Blade 1	
“D” Blade 2	
“D” Blade 3	
“D” Blade 4	

FIGURE 1: HUB DIMENSIONS



POS	MEASURED (2 PLACES, 90 DEGREES APART)
“A” Blade 1	
“A” Blade 2	

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POS	MEASURED (2 PLACES, 90 DEGREES APART)
“A” Blade 3	
“A” Blade 4	
“E” Blade 1	
“E” Blade 2	
“E” Blade 3	
“E” Blade 4	

FIGURE 2: BLADE TRUNNION DIMENSIONS

*3.2.2.1.3 Inspect remaining components in accordance with TP 2699 Section 8-5 Paragraphs C.1 through C.6 except do not accomplish any repair or replacement specified therein. Submit a CFR.

*3.2.2.2 Tech Rep must evaluate data collected and make a recommendation to the COR if hub will need further corrective action using Table 2 as guidance.

TABLE 2: CPP HUB CORRECTIVE ACTIONS

INSPECTION RESULT AS DEEMED BY GOVT TECH REP (PARAGRAPHS 3.2.2.1.1 AND 3.2.2.1.2)	ACTION
Satisfactory	No further corrective action. Reassemble hub and install.
Unsatisfactory	Activate optional Work Item “Controllable Pitch Propeller (CPP) Hubs, Renew”. Reassemble hub (so that it may be prepared for shipping).

*3.2.3 Hub reassembly. The Contractor must reassemble designated CPP hub under the direction of the Tech Rep in accordance with TP 2699 Section 8-5 Paragraphs D.1 through D.28 and using Coast Guard Drawing 901 WMEC 245-003 as guidance.

*3.2.3.1 If a Chane Request is approved, the Contractor must renew any components determined to be out-of-specification under the direction of the Tech Rep using Government Furnished material.

*3.2.4 Hub assembly installation. The Contractor must install designated CPP hub assembly onto tailshaft under the direction of the Tech Rep using TP 2699 Section 8-5 Paragraphs D.29 through D.39 and Coast Guard Drawing 901 WMEC 245-003 as guidance using the Government-furnished installation kit.

NOTE

Post installation pressure test, operational test and pitch checks to be accomplished on Work Item “Controllable Pitch Propeller (CPP) System, Inspect and Service”.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 27: Fathometer Transducer, Renew**1. SCOPE**

1.1 Intent. This work item describes the requirements for the Contractor to renew the transducer(s).

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	***Fathometer Transducer, Shallow Water	NSN: 5845-01-668-2181 PN: SS60	1 ea.	1,884.26

***Government-furnished property, which is to be supplied by either the vessel or the C5I Service Center.

2. REFERENCES**COAST GUARD DRAWINGS**

Coast Guard Drawing 901 WMEC 184-010, Rev F, XDCR Fdns and Shell Mods
 Coast Guard Drawing 901 WMEC 451-010, Rev D, 1SG38-SINS2(V)45 Installation BWD & Details
 Coast Guard Drawing FL 184-002, Rev A, SS60 Transducer Install Adapter Assy

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements
 Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

OTHER REFERENCES

Federal Specification (Fed Spec) HH-P-151, Mar 1991, Packing; Rubber-Sheet, Cloth-Insert

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 Transducer and lines. Protect all exposed ends of transmission lines, control lines, connectors, and cables from weather, moisture, and physical damage while they are disconnected from the transducer. Take all necessary measures to protect transducer from damage during the performance of work. Inform the COR, in writing, of all damage, if any, that is incurred by the transducer.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Transducer renewal particulars. The Contractor must accomplish the following tasks using Coast Guard Drawings FL 184-002, 901 WMEC 184-010 and 901 WMEC 451-010 for guidance:

3.2.1 Transducer removal. Disconnect and remove the existing fathometer transducer. Retain the mounting hardware for reinstallation. Dispose of the removed transducer.

APM NOTE: Determine whether the existing transducer is the SS60 or the older SS505. If it is an SS505, the existing hull ring adapter will require modification. Add the following paragraph below the Note.3.2.1.1 Adapter modification. Modify the existing hull ring adapter using Coast Guard Drawing FL 184-002 as guidance.

NOTE

Do not discard any hull ring adapter(s). They are to be reused with the new transducer(s).

3.2.2 New transducer installation. Install new Government-furnished transducer(s), in place of the removed. Reinstall each hull ring adapter and transducer in accordance with the applicable drawing listed in Section 2 (References) using new gaskets, mold release coating, and adhesive sealants. Apply mold release aerosol coating to both adaptor orifice and transducer to facilitate future removal of the transducer from the adaptor ring.

NOTE

A piece of neoprene or a washer may need to be placed at the bottom of the hull ring adaptor to facilitate transducer and bottom of hull ring flushness.

3.2.2.1 Using chocking compound as transducer bedding material, install transducer into hull ring adaptor. After the chocking compound has cured, paint the bottom half of the hull ring adaptor and transducer in accordance with drawing notes.

3.2.2.2 Renew the transducer head to hull ring gasket with new gasket conforming to HH-P-151, Class 4. Using appropriate thickness gasket, place gasket on adaptor plate of transducer; apply ½" bead of marine adhesive on adaptor plate along the inside edge of the gasket. Install transducer flush with hull.

3.2.2.3 Reconnect each transducer cable to its terminal board in the adjacent connection box. Renew all stuffing tube packing and seals. Inspect the connection in the presence of the Coast Guard Inspector. Submit a CFR. If cable renewal is authorized via approved Change Request, cut the transducer cable to length leaving 2-foot service loop in the cable. Ensure cables are installed in a manner to prevent any chaffing which may require new cable way clamps and stuffing tubes.

3.2.2.4 Following completion of transducer installation, fabricate a ¾" × 3" × ¼" aluminum label plate, engraved with ¼-inch high numbers stating the installation date. Affix a label plate, using suitable epoxy cement. Ensure that the label plate is visible and legible, without the need to move equipment or cables.

3.2.2.5 Thoroughly clean the transducer radiating surfaces with a strong solution of household detergent to remove any oil, dirt, and all other surface contaminants.

3.3 Transducer seal. If not waterborne, the Contractor must inspect the transducer to ensure that its outer circumference is installed flush to the hull ring and the transducer cover plate forms a tight seal with the hull ring. Submit CFR.

3.4 Post installation tests. After completion of work, the Contractor must accomplish the following tasks, in the presence of the Coast Guard Inspector:

3.4.1 Preliminary leak test. If not waterborne, inspect and perform a water hose test of all affected boundaries in accordance with SFLC Std Spec 0740, Appendix C. If waterborne, slowly back off the emergency cover plate bolts and allow seawater to flood the transducer well. No leakage allowed. Submit a CFR.

3.4.2 Not waterborne. Replace the hull ring set screws with new Monel fasteners, coating the threads with anti-seize compound. Recess the set screws ⅛" and cover with beeswax tallow.

3.4.3 Waterborne. Using a diver, remove and retain the hull ring set screws. Install an emergency cover plate and gasket over the transducer to be renewed. If available, the vessel's emergency cover plate may be used; otherwise a new cover plate must be fabricated and turned over to the vessel at the completion of the work.

3.4.3.1 Slowly back off the fasteners that normally maintain the transducer pressure boundary to sea. Ensure that any leakage through the emergency cover plate pressure boundary is minor prior to proceeding with transducer removal.

3.4.3.2 Upon completion of work and preliminary leak testing, completely remove the emergency cover plate. Clean all vacated tapped holes of anti-seize compound and beeswax tallow. Chase all threads. Install new Monel set screws in each hull ring, coating the threads with anti-seize compound. Recess the set screws ⅛" and cover with beeswax tallow.

3.5 Leak repairs. The Contractor must observe each transducer for leaks during the vessel re-floating process (if dry docked) or emergency cover plate removal (if waterborne). Repair all leaks detected. Submit a CFR.

3.6 Operational test – post repairs. After completion of work, the Contractor must thoroughly test and prove the transducer(s) to be in satisfactory operating condition. Submit a CFR.

4. NOTES

4.1 Operation of equipment. Coast Guard personnel will operate all shipboard machinery and equipment.

DRAFT

WORK ITEM 28: Doppler Speed Log Transducer, Clean and Inspect

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to clean, inspect and reinstall the doppler speed log transducer.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	**FURUNO DS-630-30 Doppler Speed Log Transducer with 30M Cable	NSN: 5845-01-678-5056	1 ea.	7,097.35

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 184-024, Rev B, Stuct Mods Incidental to DS60 Doppler Install & ODOM DWDS Rip-Out

Coast Guard Drawing 901 WMEC 424-004, Rev D, Doppler Speed Log System Replacement Ripout, Installation, Arrangement, Wiring and Details

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 10500, December 2019, Doppler Sonar, Model DS-60
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

OTHER REFERENCES

The Society for Protective Coatings (SSPC)/NACE International (NACE) Joint Surface Preparation Standard SSPC-SP 10/NACE No.2, 2007, Near-White Blast Cleaning

The Society for Protective Coatings (SSPC) Surface Preparation Specification SSPC-SP 11, 2013, Power Tool Cleaning to Bare Metal

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 Protect all exposed ends of transmission lines, control lines, connectors, and cables from weather, moisture, and physical damage while they are disconnected from the transducer.

3.1.3.2 Take all necessary measures to protect transducer from damage during the performance of work specified herein. Inform the COR, in writing, of all damage, if any, that is incurred by the transducer.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Insulation

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.2 Operational test, initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of all items or shipboard devices to be disturbed, used, repaired, or altered, to demonstrate existing operational condition. Submit a CFR.

3.3 Transducer maintenance and inspection. The Contractor must accomplish the following tasks while the vessel is not waterborne using TP 10500 and Coast Guard Drawings 901 WMEC 184-024 and 901 WMEC 424-004 for disassembly and reassembly guidance.

3.3.1 Transducer removal. Disconnect and remove the existing doppler speed log transducer, along with the associated transducer hull adapter(s) and gasket(s), if so installed. Retain all mounting hardware for reinstallation. Remove the emergency cover plate setscrews from the transducer hull ring.

3.3.2 Cleaning.

3.3.2.1 Transducer. Clean the removed doppler speed log transducer. To remove marine growth the use of mild detergent, soft bristled brush, putty knife, or wet sandpaper (#220 grade or finer) to restore a smooth surface to the transducer face are authorized. Special attention must be made to not damage the end of the transducer.

3.3.2.2 Hull. Clean each transducer hull ring and associated adapter (if so equipped) in accordance with either SSPC SP-10 or SP-11. Clean all vacated tapped holes of anti-seize compound and beeswax tallow. Chase all threads. Protect all tapped holes with plugs or temporary bolts to exclude debris.

3.3.3 Inspection. Perform all inspections in the presence of the Coast Guard Inspector. Visually inspect transducer for cracks. Inspect all cleaned surfaces and surrounding welds for corrosion. Submit CFR.

3.3.4 Preservation. Prepare, prime and paint the hull opening and surrounding areas under the work item for underwater body preservation. Prepare, prime and paint the surfaces of the transducer hull ring adapter in accordance with the Coast Guard drawing(s) listed in Section 2 (References).

3.3.5 Reinstallation. After the last coat of paint has cured for at least 24 hours (cure time is dependent upon specific paint system used and environmental conditions), reinstall hull ring adapter and transducer in accordance with the applicable drawing using new gaskets, mold release coating, and adhesive sealants. Inspect the speed doppler to ensure the outer circumference of the doppler is installed flush to the hull ring.

3.3.5.1 Be aware that the Government, at no additional consideration, may elect to furnish new transducers for reinstallation instead of reusing the existing.

3.3.5.2 Ensure that the horizontal major axis of each transducer is oriented with respect to the vessel's centerline. Reuse retained mounting hardware, coating all bolt threads with anti-seize compound.

3.3.5.3 Reconnect each transducer cable to its terminal board in the adjacent connection box. Renew all stuffing tube packing and seals. Inspect the connection in the presence of the Coast Guard Inspector.

3.3.5.4 Each transducer must be protected from damage with a metal cover plate bolted to the hull until just prior to undocking.

3.4 Boundary test, water hose. The Contractor must inspect and perform a water hose test of all affected boundaries in accordance with SFLC Std Spec 0740, Appendix C. Submit a CFR.

3.5 Leak test. After completing all authorized repairs, the Contractor must test the Doppler system's operation and observe the Doppler for leaks during the vessel refloating process. Ensure zero visible leakage from or deformation of mechanical parts by repairing all leaks and discrepancies. Submit a CFR.

3.6 Operational test, post repairs. After completion of work and in the presence of the Coast Guard Inspector, the Contractor must thoroughly test and demonstrate the equipment listed below to be in satisfactory operating condition. Submit a CFR.

- Doppler Speed Log

4. NOTES

This section is not applicable to this work item.

WORK ITEM 29: Sea Valves and Waster Pieces, Overhaul Or Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to overhaul or renew sea water (hull) valves and Waster Pieces.

1.1.1 Valves designated for overhaul.

TYPE	SIZE (INCHES)	DESCRIPTION	FRAME NO. VALVE NO.	PRESSURE (PSIG)
Remote BFLY	5	Bronze- #2 MSW OVBD	2-MSW-5	
Remote BFLY	6	Steel -ENG RM Eductor OVBD	3-152-2	200
Remote BFLY	5	Bronze #1 MSW OVBD	1-MSW-5	
Globe	8	Bronze #1MDE Seachest	1-MSW-1	150
Globe	8	Bronze #2 MDE Seachest	2-MSW-2	150
Butterfly	10	Bronze Strainer COV	3-91-2	150
Butterfly	10	Bronze Strainer X-Connect	3-90-4	150
Butterfly	10	Bronze Strainer COV	3-91-4	150
Butterfly	8	ASW Supply COV	2-92-2	150
Butterfly	8	Bronze ASW OVBD	3-95-6	200
Globe 90° Angle	10	Bronze Seachest	3-95-4	150
Globe 90° Angle	10	Bronze Seachest	3-95-2	150
Globe 90°	6	Bronze ASW SW Supply	3-190-3	150
Globe 90°	6	Bronze ASW SW Supply	3-190-1	150
Gate	6	I/P ASW X-Conn	2-115-2	150
Gate	6	Doppler Speed Log		50

The Contractor must visually inspect the 3-inch diameter Aft Grey Water overboard discharge line connected to the overboard discharge valve (located at Frame 150, starboard Engine Room), and submit a CFR.

1.1.2 Valves designated for renewal.

TYPE	SIZE (INCHES)	DESCRIPTION	FRAME NO. VALVE NO.	PRESSURE (PSIG)
Butterfly	4	Bronze BEB OVHD DISCH	3-118-2	400
Butterfly	4	Bronze #2 GEN AFT Winding Cooler OVBD	3-116-2	200
Butterfly	4	Bronze #1 GEN AFT Winding Cooler OVBD	3-116-7	200
Check	4	Bronze Drain Check STBD Forward	2-103-1	150
Check	4	Bronze DRAIN Check PORT Forward	2-103-2	150
Remote BFLY Gear	4	Bronze ASW From FSS/CHW Unit OVBD	3-81-3-1	150
Butterfly	4	Bronze RLF VLV OVBD COV	3-90-1	150
Check Valve	4	Inport ASW	4-115-2	150
Scupper	2	Bronze AFT Waste Water OVBD	3-148-1	150
Gate	1-1/4	Bronze OWS OVBD DISCH	3-141-2	150
Gate	1-1/2	Bronze Seachest Blow Down	3-116-5	150
Angle Globe	4	Bronze BEB Suction	3-117-1	150
Gate	1-1/2	Bronze Seachest Vent	3-118-1	150
Gate	3	Bronze #2 GEN J/W Cooler OVBD	3-105-4	150
Gate	3	Bronze #1 GEN J/W Cooler OVBD	3-105-3	150
Navy Gate	3	Bronze #1 FIRE Pump OVBD	3-87-4	150
Scupper	2	Bronze FWD Wastewater DSHCH	3-88-2	150
Scupper	2	Bronze SEW Hold Tank OVBD	3-96-3	150
Gate	3	Bronze GAG, SEW Hold TK EMER OVRFL	3-99-1	150
Scupper	3	Bronze Wastewater Drain OVBD PORT	2-85-2	150
Scupper	3	Bronze Wastewater Drain OVBD STBD	2-90-1	150
Gate	2	Bronze EDG DISCH	2-244-1	150
Butterfly	2	EDG ASW Overboard	2-249-1	150
Navy Gate	3	Reefer ASW OVBD	3-187-2	150

Gate	1	Reefer ASW OVBD	3-187-1	150
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1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 256-013, Rev A, Mn and Aux SW Cooling Sys Diag
Coast Guard Drawing 901 WMEC 256-002, Rev T, Mn and Aux SW Cooling Sys A&D
Coast Guard Drawing 901 WMEC 256-004, Rev A, Aux. Salt Water Mods.
Coast Guard Drawing 901 WMEC 259-002, Rev K, Diesel Exhaust System A&D
Coast Guard Drawing 901 WMEC 524-003, Rev G, Sea Conn, Seachest Conn and Waster Piece A&D

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020, General Requirements
Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

OTHER REFERENCES

American Society of Mechanical Engineers (ASME) B16.34, 2017, Valves-Flanged, Threaded, and Welding End
ASTM International (ASTM) F992, 2017, Standard Specification for Valve Label Plates
Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) SP-61, 2013 Edition, Pressure Testing Of Steel Valves
Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) SP-67, 2017 Edition, Butterfly Valves
Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) SP-72, 2018 Edition, Ball Valves with Flanged or Butt-Welding Ends for General Service
Manufacturers' Standardization Society of the Valve and Fittings Industry (MSS) SP-80, 2013 Edition, Bronze Gate, Globe, Angle and Check Valves

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Deck plates
- Piping
- Insulation
- Pipe contents
- Associated system tanks and filters
- Valve flanges.

3.2 Valve material. The Contractor must assume all above-listed valves (paragraphs 1.1.1 and 1.1.2) have a bronze housing and flanged connections for the purposes of bidding. The Contractor must verify required valve list against referenced drawings and by ship-check prior to purchasing any materials. If the actual required valve list differs from what is listed in this work item, submit a CFR.

3.3 Fluid handling. The Contractor must drain and dispose of all residual fluids in the piping system in accordance with all Federal, state, and local rules and regulations.

3.4 Remove. The Contractor must remove all designated valves in Section 1.1 (Intent). Immediately after valve removal, install blank flanges and gaskets over all openings and secure each flange with at least two bolts, 180 degrees apart. Visually inspect associated flanges and piping; submit a CFR.

3.5 Contractor's option for valve renewal. The Contractor may, at no additional cost to the Government, opt to renew valves designated for overhaul if preferable for the Contractor. If the Contractor elects to renew valves designated for overhaul, ensure all new valves are commercial-standard type valves, conforming to the applicable standard listed in Table 1 (Valve Standards). New valves must be equivalent (including identical material) to the valve being renewed. Visually inspect the piping, flange and mounting arrangements; and submit a CFR detailing any required modifications to accommodate the new valve(s). If the Contractor elects to renew a valve after attempting overhaul, it will be at no additional cost to the Government.

NOTE

This work item requires the Contractor to provide to the Government installed valves meeting the designated test requirements. This work item includes the cost of EITHER overhaul OR renewal of each designated valve.

3.6 Overhaul. The Contractor must accomplish the following as required for each valve designated for overhaul (not including valves the Contractor has opted to renew) to meet the specified valve testing standard:

3.6.1 Disassemble. The Contractor must disassemble the valve to the extent necessary to perform the required work.

NOTE

Complete disassembly of some valves may not be necessary to accomplish overhaul.

3.6.2 Clean. The Contractor must clean all internal surfaces and visually inspect for defects in body and structural material. Inspect the surface finish and condition of seats, disks, parting faces, plugs, and sealing surfaces.

3.6.3 Machine. The Contractor must conduct all machining necessary, including but not limited to grind, lap and spot-in seat-to-disk, in order to obtain an acceptable leakage rate at or below valve testing standards (see Table I below).

3.6.4 Reassemble. The Contractor must reassemble the valve using new hardware and software (packing, O-rings, gaskets, seal rings, non-metallic seats, pins, washers, inserts, etc.).

3.6.5 Test. The Contractor must test the overhauled valves in accordance with the applicable standards listed in Table 1 (Valve Standards). Submit a CFR.

TABLE 1 - VALVE STANDARDS

VALVE TYPE	INDUSTRY STANDARD
Steel Valves	MSS SP-61
Butterfly Valve	MSS SP-67
Ball Valves, Flanged or Butt-Welded Ends	MSS SP-72
Bronze Gate, Globe, Angle and Check Valves	MSS SP-80
All others	ASME B16.34

3.7 Renewal.

3.7.1 Valve renewal. The Contractor must renew all designated with commercial-standard type valves, conforming to the applicable standard listed in Table 1 (Valve Standards). The Contractor must replace any Mil-Std valves listed for renewal with equivalent commercial standard valves. The Contractor must be aware substitution of body material or trim set is not authorized.

3.7.2 Waster piece inspection. The Contractor must visually inspect all waster pieces associated with the valves specified for overhaul and/or renewal listed in paragraph 1.1 (Intent), as applicable. Evaluate the percentage of remaining material for each waster piece and submit a CFR. Renewal of waster pieces may be the subject of a CR.

3.8 Valve reinstallation/installation. Upon completion of all authorized work, the Contractor must accomplish the following:

- Remove and dispose of all blank flanges and associated gaskets.
- Reinstall/install all overhauled and new valves with new gaskets.
- Renew all missing or damaged valve label plates.
- Renew all bolting hardware.
- Prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in

accordance with SFLC Std Spec 6310, paragraph 3.1.13 (Touch-ups and minor coating repairs).

3.9 Valve labeling. The Contractor must install valve label plates on all new valves in accordance with ASTM F992.

3.10 Leak test. After completing all authorized mechanical (i.e. threaded, bolted, etc.) joint repairs, the Contractor must test the effected seawater system's operation using the system fluid at normal operating pressure. Ensure zero visible leakage from or deformation of mechanical parts by repairing all leaks and discrepancies. Submit a CFR.

3.11 Hydrostatic test. After all authorized repairs, the Contractor must hydrostatically test all new and disturbed piping and components of the effected seawater system in accordance with SFLC Std Spec 0740, Appendix C, Hydrostatic Test. Ensure zero leakage from or permanent deformation of pressure-containing parts by repairing all leaks, deformations, and discrepancies. Submit a CFR.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 30: Sea Strainers, Duplex, All Sizes, Overhaul

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to overhaul the following Duplex Sea Strainer(s).

SERVICE	SIZE (INCHES)	QTY	MANUFACTURER
Shaft Cooling	1	2	Kraissl
Refrigeration System	1	1	Kraissl
SSDG	3	4	Kraissl
Reverse Osmosis	1.5	2	Kraissl

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 256-013, Rev -, Mn and Aux Sw Cooling Sys Diag

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

OTHER REFERENCES

ASTM International (ASTM) D1330, 2015, Standard Specification for Rubber Sheet Gaskets

Society of Automotive Engineers (SAE) Aerospace Material Specification (AMS) G-6032, 2020 Grease, Plug Valve, Gasoline and Oil Resistant, NATO Code Number G-363, Metric

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2.2 Cleaning and inspection

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Deck plates and grating
- Piping and hoses.

3.2 Requirements. The Contractor must overhaul and test the designated strainers (see paragraph 1.1 (Intent)), shown on Coast Guard Drawing 901 WMEC 256-013. The Contractor must accomplish the following for each sea strainer in accordance with manufacturer's recommendations:

3.2.1 Remove. Secure, isolate, and remove to a suitable repair facility the designated sea strainers. Install blank flanges and gaskets over the piping system openings and secure with at least two bolts 180-degrees apart to provide a watertight seal. Ensure that the blank flanges are installed immediately after the sea strainers are removed and remain in-place until strainers are reinstalled, removing only for sea valve visual inspection.

3.2.2 Clean and inspect. After removing the sea strainers, visually inspect in-place the exposed sea valves and piping/hoses located adjacent to the strainers. Completely disassemble and thoroughly clean all strainer components to bright metal without damage or deformation to any component. Visually inspect all disassembled components. Submit A CIR.

3.2.3 Renewal. Any strainer baskets, covers, and packing gland fasteners that require renewal as determined by the CIR must be addressed on a Change Request.

3.2.4 Reassemble. Reassemble each strainer. Lubricate the valves with plug valve grease conforming to AMS-G-6032.

3.2.5 Plug valve test. Where applicable, lubricate the valves with plug valve grease conforming to AMS-G-6032. Hydrostatically test the plug valves of all duplex strainers for each strainer chamber, at a test pressure of 15 psig. To determine leakage rate the strainer must have the outlet blanked and the unpressurized side top cover removed (two tests per duplex strainer). While maintaining the test pressure at the inlet side, the non-pressurized side must not fill within twenty minutes. Provide a CFR which must include, but is not limited to the following information:

- Grease type, including batch number and manufacturer.
- Time elapsed for chamber to fill, if less than twenty minutes.

CAUTION

Ensure that the drain valve for each chamber is closed.

3.3 Reinstall. After completion and acceptance by the Coast Guard Inspector; the Contractor must reinstall all overhauled, cleaned and new strainers to their original locations, as shown on Coast Guard Drawing 901 WMEC 256-013. Renew all gaskets with material conforming to ASTM D1330 of the same size and configuration as those gaskets removed. Renew all hardware with nickel-copper alloy fasteners.

3.4 Notification. The Contractor must notify the Coast Guard Inspector at least 24 hours prior to reinstalling sea strainers.

3.5 Government's right for changing out strainers. The Contractor must be aware that the Government reserves the right to furnish new sea strainer(s) for installation in place of the existing strainer(s). If the Government exercises this right, the Contractor must dispose of the old strainer(s) in accordance with all applicable Federal, state, and local regulations.

3.6 Contractor's option for strainer renewal. The Contractor may opt to renew strainers designated for overhaul if preferable for the Contractor, and is at no additional cost to the Government. New strainers must be equivalent (including identical material) to the strainer being renewed. Visually inspect the piping and mounting arrangements, and submit a CFR detailing any required modifications to accommodate the new strainer(s).

3.7 Leak test. After completing all authorized mechanical (i.e. threaded, bolted, etc.) joint repairs, the Contractor must test the seawater system's operation using the system fluid at normal operating pressure. Ensure zero visible leakage from or deformation of mechanical parts by repairing all leaks and discrepancies. Submit a CFR.

3.8 Surface preservation, 100% interior. The Contractor must perform 100% preservation in accordance with SFLC Std Spec 6310, Appendix B (Cutter and Boat Interior Painting Systems) on the following interior surfaces:

- Sea strainer surfaces
- All adjacent structural members.

3.8.1 Preparation method. The Contractor must prepare interior surfaces using system specified for "Bilges, Cofferdams and Forepeaks, Steel, Option II".

3.8.2 Coating method. The Contractor must apply finish/top coat color to interior surfaces as follows: match existing adjacent surfaces. Do not paint packing glands, valve stems, threads, and similar working surfaces.

4. NOTES

4.1 Definition. The terms 'duplex' as used in this work item refer to the generic strainer type, not a manufacturer's brand name.

WORK ITEM 31: Sea Strainers, Simplex, All Sizes, Clean and Inspect

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to clean and inspect the following simplex sea strainers:

TABLE 1 – STRAINERS

SERVICE	SIZE (INCHES)	QTY	MANUFACTURER
ASW Cooling	10	2	Staples & Pfeiffer
MSW Cooling	8	2	Staples & Pfeiffer
BEB Suction	4	1	Staples & Pfeiffer
ASW/EDG Cooling	6	2	Staples & Pfeiffer

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 256-013, Rev -, Mn and Aux SW Cooling Sys Diag
Coast Guard Drawing 901 WMEC 256-002, Rev T, Mn and Aux SW Cooling Sys A&D
Coast Guard Drawing 901 WMEC 256-004, Rev A, Aux Salt Water Mods
Coast Guard Drawing 901 WMEC 259-002, Rev K, Diesel Exhaust System A & D

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements
Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

Society of Automotive Engineers (SAE) AMS-G-6032, August 2012, Grease, Plug Valve, Gasoline and Oil Resistant, NATO Code Number G-363, Metric

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2.2 Cleaning and inspection

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Deck plates and grating
- Piping and hoses.

3.2 Requirements. The Contractor must clean and inspect the designated strainers (see paragraph 1.1 (Intent)), shown on Coast Guard Drawings 901 WMEC 256-013, 901 WMEC 256-002, 901 WMEC 256-004, and 901 WMEC 259-002. The Contractor must accomplish the following for each sea strainer in accordance with manufacturer's recommendations:

3.2.1 Disassembly. Disconnect and disassemble the strainer to the extent possible to allow for the inspection tasks specified herein. Do not unship the simplex strainers. Install blank flanges, including suitable gasket materials over the resultant openings, each secured with at least two bolts 180-degrees apart, to provide a watertight seal. Ensure that the blank flanges are installed immediately after the strainers has been removed, and remain in-place until the strainers are about to be reinstalled.

3.2.2 Cleaning and inspection. Clean all disassembled strainer components to bright metal without damage or deformation. As applicable, inspect the components, including packing glands. Submit a CIR.

3.2.3 Reassembly. After completion and acceptance by the Coast Guard Inspector, reassemble all strainers. Renew all hardware with nickel-copper alloy (Monel) fasteners. Renew all gaskets, o-rings and seals. As applicable, lubricate the valves with plug valve grease conforming to SAE AMS-G-6032.

3.3 Notification. The Contractor must notify the Coast Guard Inspector at least 24 hours prior to reinstalling sea strainers.

3.4 Government's right for changing out strainers. The Contractor must be aware that the Government reserves the right to furnish new sea strainer(s) for installation in place of the existing strainer(s). If the

Government exercises this right, the Contractor must dispose of the removed strainer(s) in accordance with all applicable Federal, state, and local regulations.

3.5 Leak test. After completing all authorized mechanical (i.e. threaded, bolted, etc.) joint repairs, the Contractor must test the seawater system's operation using the system fluid at normal operating pressure. Ensure zero visible leakage from or deformation of mechanical parts by repairing all leaks and discrepancies. Submit a CFR.

3.6 Surface preservation. The Contractor must perform 100% preservation of strainer surfaces in accordance with SFLC Std Spec 6310, Appendix B (Cutter and Boat Interior Painting Systems). Prepare and coat interior surfaces using system specified for “Bilges, Cofferdams and Forepeaks, Steel, Option II”. Apply finish/top coat color to interior surfaces to match existing adjacent surfaces. Do not paint packing glands, valve stems, threads, and similar working surfaces.

4. NOTES

4.1 Definition. The term ‘simplex’ as used in this work item refers to the generic strainer type, not a manufacturer’s brand name.

DRAFT

WORK ITEM 32: Rudder Assemblies; Remove, Inspect And Reinstall

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to remove, inspect and reinstall the port and starboard rudder assemblies, located in the Steering Gear Room (3-228-0-E).

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	**Rudder Assembly	NSN: 2040-01-486-0550	2 ea.	182,000.00
N	**Upper Bearing	NSN: 3110-01-121-8032	2 ea.	3,910.00
N	**Rudder Stock Sleeve	NSN: 2040-01-F91-0187	2 ea.	4,269.00
N	John Crane Seal kit	NSN: 5330-01-527-0778	2 ea.	5,955.00
N	**Thordon Lower Bearing (SXL 20.7" X 17.5" X 14)	PN: F546150102	2 ea.	1,000.00

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 562-003, Rev B, Rudderstock Seal

Coast Guard Drawing 901 WMEC 562-005, Rev K, Rudder Stock and Steering Gear A & D

Coast Guard Drawing 905 WMEC 562-001, Rev G, Rudder

Coast Guard Drawing 901 WMEC 243-008, Rev A, Hub & Rudder Removal Padeye Arr. & Details

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

Coast Guard Technical Publication (TP) 2718, Jan 2017, Manufacturer's Instruction Book - SWBS Group(s) 555-562, 270' WMEC "A" Class Cutters, Section A, SWBS 561, Steering Gear System, and SWBS 562, Rudder Stock Seal Assembly

OTHER REFERENCES

MIL-STD-1687, Sep 1994, Thermal Spray Process for Naval Ship Machinery Application

MIL-PRF-16173, Sep 2006, Corrosion Preventive Compound, Solvent Cutback, Cold-Application

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2 Carrier bearing clearance measurement.
- 3.3.2 Inspections.

3.1.2 Tech Rep. The Contractor must provide the services of a qualified Tech Rep, who is familiar with the Thordon bearing and John Crane seals as installed on the 270-ft Coast Guard Medium Endurance Cutters, to accomplish the following, on site:

- Advise on manufacturer's proprietary system information.
- Assist with proper repair methods.
- Ensure compliance with manufacturer's procedures and standards during disassembly, inspection, repair, modification, calibration, and reassembly of the equipment/system; and be present when work is being performed on the seal assembly and sub-assemblies.

3.1.2.1 Ensure the Tech Rep has experience with the system/equipment stated above and demonstrated on their résumé.

3.1.2.2 Submit the name and résumé of the Tech Rep to the COR at the Arrival Conference.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to, the following:

- Flammable fluid contents.

3.1.5 Operational test, initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of the steering gear and rudder to demonstrate existing operational condition. Submit a CFR.

3.1.5.1 Record the time it takes to run the rudders hard over from 35 deg right rudder to 35 deg left rudder, then repeat in the other direction. Verify that the rudder hard over time is not greater than 20 seconds for single HPU operation, or 10 seconds when operating both HPU's.

3.2 Carrier bearing clearance measurement. Before maintenance work is begun, the Contractor must measure the rudder carrier bearing clearances of both rudders as shown on Coast Guard Drawing 901 WMEC 562-005. Submit a CIR.

3.3 Rudder maintenance requirements. The Contractor must perform the following tasks for each rudder assembly, using TP 2718; and Coast Guard drawings 901 WMEC 243-008, 901 WMEC 562-003, 905 WMEC 562-001, and 901 WMEC 562-005 as guidance:

3.3.1 Removals.

3.3.1.1 Crop and retain the existing rudder stock nut cover plate.

3.3.1.2 Disconnect and remove the tiller, upper roller bearing, bearing housing, rudder shaft seal assembly, retainer plates and other associated mounting components. Disconnect and transport the rudder assembly to a suitable repair facility.

3.3.1.2.1 Scribe a mark at existing rudder nut position on the rudder stock, and on the tiller and rudder stock assembly, to facilitate later reinstallation.

3.3.1.2.2 If heat is necessary to facilitate the removal of the roller bearing inner race from the stock, do the following, prior to applying heat:

- Remove all grease from upper bearing surfaces.
- Remove rollers and outer race.

3.3.2 Inspections. Clean all removed and disassembled components free of all surface contaminants, and perform the following inspections. Submit a CIR for each inspection.

3.3.2.1 Rudder blade. Accomplish the following for the both port and starboard rudder blades:

3.3.2.1.1 Visually inspect all rudder blade welds.

3.3.2.1.2 In the presence of the Coast Guard Inspector, remove all fill and drain plugs from rudder and check for the presence of water. Submit a CFR.

3.3.2.1.3 Air test the rudder in the presence of the Coast Guard Inspector in accordance with SFLC Std Spec 0740, Appendix C. Test pressure must be 2 psig. There must be no pressure drop allowed over a test period of ten minutes. Identify location of all leaks. Submit a CFR.

3.3.2.1.4 Following completion of work, chase the hull insert threads of all drain and vent plugs. Install new stainless steel, Type 316, or Monel drain plugs. Coat the new plugs prior to installation with a sealing and locking compound conforming to ASTM D5363-AN0123 or a flexible joint compound conforming to MIL-S-45180, Type II.

3.3.2.2 Rudder stock. Set the rudder stock in a lathe and accomplish the following:

- Visually inspect the entire rudder stock surfaces for wear and corrosion.
- Measure the stock runout; perform and document all other measurements specified on attached "Rudder Stock Measurements" sheets.
- Examine the condition of metal spraying coating.
- Measure the outside diameter of each stock journal in way of seal area and bearing sleeve at

the top, bottom and midpoint.

- Conduct NDI of the tapers, keys, and keyways in accordance with Std Spec 0740, Appendix C.

NOTE

The stock diameter in way of “John Crane” seal is a nominal size 11.100" + 0.017" -0.016".

3.3.2.3 Upper bearing.

- Clean and visually inspect the removed roller bearing.
- Measure the inner diameter of the roller bearing and outside diameter of rudder stock.
- Measure axial clearance (end-play). Clearance must not equal or exceed 0.031 inch greater than the original installed axial clearance.

NOTE

The design axial clearances for new carrier bearing are minimum 0.0274 inch and maximum 0.0417 inch. See Note 29 of Coast Guard Drawing 901 WMEC 562-005.

3.3.2.4 Lower bearing.

- Visually inspect the existing “Thordon” lower bearing for any abnormal or excessive wear.
- Measure the inside diameter at the top, middle and bottom positions.
- Complete and submit the attached Bearing Clearance Report.

3.3.3 Component renewals.

3.3.3.1 If a Change Request has been authorized and released, renew the bearing liner with Government-furnished sleeve/liner, as shown on Coast Guard Drawing 901 WMEC 562-005.

3.3.3.2 If a Change Request has been authorized and released, renew the thermal spray coating on the rudder stock, in accordance with MIL-STD-1687. Use thermal spray high velocity oxygen fuel (HVOF) thermal coating process for coating application, and accomplish the following tasks:

- Machine the rudder stock to remove existing thermal coating or chrome plating; grinding or machining to excavate all defects noted.
- Clean and prepare the surfaces to be HVOF coated in accordance with MIL-STD-1687.
- Coat the prepared surfaces with HVOF thermal sprayed tungsten carbide/nickel to the required thickness specified as shown on Coast Guard Drawing 901 WMEC 562-005.
- Machine and grind the coated areas to match the design dimension or a dimension specified by the COR.
- Check the stock runout to ensure no heat distortion on the straightness of the rudder stock, and compare the runout reading with the reading taken in paragraph 3.3.2.2 (Rudder stock).
- Polish the coated areas to a minimum of 125 RMS or a finish specified in the rudder drawing.

3.3.3.3 If a Change Request has been authorized and released, renew the lower bearing as follows:

3.3.3.3.1 Remove and dispose of the lower bearing.

3.3.3.3.2 After authorized repairs, if any, take all measurements of the rudder stock and lower housing required to facilitate the machining of the new Government-furnished “Thordon” bearing. Submit CFR (see 4.1 (Thordon bearing particulars)).

3.3.3.3.3 Install the new bearing into the bearing housing, in place of the removed.

3.3.4 Rudder reinstallation. Upon completion of all inspections and authorized repairs, if any, reassemble and reinstall each rudder assembly, as shown on Coast Guard Drawings 905 WMEC 562-001 and 901 WMEC 562-005. Apply all necessary sealant and fill cavities, as specified on Coast Guard Drawing 901 WMEC 562-005.

3.3.4.1 During the final assembly, clean the bore of the rudder and the stock taper area free from dirt and other foreign matter, including lubricants.

3.3.4.2 In the presence of the Coast Guard Inspector, check the fit of the rudder stock taper and the associated rudder by bluing. Obtain a minimum of 75% uniform fit between the rudder stock taper and rudder. Do not use heat to compensate for a poor fit on the rudder stock taper. If the fit is less than 75 percent, grind the rudder bore with fine emery cloth until the 75 percent contact is achieved. Submit a CFR.

3.3.4.3 Reinstall and snug the rudder stock nut, ensuring the nut is torqued to the value specified in Coast Guard Drawing 901 WMEC 562-005 General Note 10. Submit the final torque value to the COR, after the nut has been properly secured. After the nut has been secured, fabricate and install two nut locking bars in accordance with Coast Guard Drawing 901 WMEC 562-005 General Note 31 and using Coast Guard Drawing 901 WMEC 562-005 Detail 28-A as guidance, ensuring that each bar is perpendicular to the other.

3.3.4.4 Reinstall the rudder stock nut cover plate, as shown on Coast Guard drawings 905 WMEC 562-001 and 901 WMEC 562-005. Prior to reinstallation of the rudder stock nut cover plate, perform a visual inspection of the rudder stock nut and locking bar welds, in the presence of the Coast Guard Inspector. Submit a CFR stating that the lock bar installation is complete.

3.3.4.4.1 After the rudder stock nut cover plate has been reinstalled, air test the associated nut access void in accordance with Std Spec 0740, Appendix C. Submit a CFR.

3.3.4.4.2 Coat the nut access void surfaces with corrosion preventive compound conforming to MIL-PRF-16173, Class II, Grade 3.

3.3.4.4.3 Renew the stock nut cover plate plug.

3.3.4.5 Install the upper bearing into bearing housing. If the outer race of the upper bearing cannot be tightly fitted into the bearing housing, apply bearing manufacturer's recommended tightening compound to secure the upper bearing into the bearing housing to prevent loosening. Renew the seal and packing material in the bearing housings, as shown on Coast Guard Drawing 901 WMEC 562-005.

3.3.5 Mechanical seal assembly installation. Accomplish the following for each “John Crane” mechanical seal assembly:

3.3.5.1 Measure the amount of wear on the seat surface in accordance with TP 2718. If still within allowable specification, machine the stationary seal faces to remove grooves and wear at 25 RPM with

0.005 inch per revolution feed rate to achieve a 63 micro inch finish. Polish with a block and 600 grit wet and dry emery paper. Submit a CFR.

3.3.5.2 Reassemble and reinstall the seal assembly to original configuration, using the Government-furnished seal kits. Fabricate and install new Contractor-furnished cork-rubber gasket in accordance with Coast Guard Drawing 901 WMEC 562-003 for the mounting flange.

3.3.6 Tiller reinstallation.

3.3.6.1 Reinstall tiller arm in accordance with prescribed marking, ensuring proper orientation.

3.3.6.2 Reinstall feedback rod mechanism and ensure proper adjustment.

3.3.6.3 Tack weld the locking bars to the tiller arm nuts.

3.3.6.4 Check the rudder alignment with the pilot house rudder angle indicator and adjust as necessary to obtain correct alignment within 1/2 degree port or starboard.

3.3.7 Bearing axial clearance check. Check the carrier bearing axial clearance (end play) of each installed rudder assembly.

NOTE

Total allowable axial travel of the installed assembly is 0.036" – 0.072" in accordance with Detail 83-C of 901 WMEC 562-005.

3.4 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

3.5 Government's right for changing out rudder assembly. The Contractor must be aware that the Government reserves the right to furnish a new rudder assembly for installation in place of the existing rudder assembly. If the Government exercises this right, the Contractor must turn over the removed rudder assembly to the Coast Guard PA as a MTI item, with the completed "Rudder Stock Measurements" sheets.

3.6 Government's right for changing out the upper bearing. The Contractor must be aware that the Government reserves the right to furnish a new upper bearing for installation in place of the existing upper bearing. If the Government exercises this right, the Contractor must dispose of the old bearing in accordance with all applicable Federal, state, and local regulations.

3.7 Operational test, post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate rudder system to be in satisfactory operating condition. Submit a CFR.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.7.1 Dock trial. Conduct a test of the steering gear in accordance with the Section 6-11 (Scheduled Test procedure) through Section 6-15 (Gyro Mode Test) of TP 2718. The rudder must travel a full 70 degree

arc (35 degrees to port and to starboard) without chattering or binding. Ensure no leaks are present. Provide a written testing report to the COR within 24 hours.

3.7.1.1 Record the time it takes to run the rudders hard over from 35 deg right rudder to 35 deg left rudder, then repeat in the other direction. Verify that the rudder hard over time is not greater than 20 seconds for single HPU operation, or 10 seconds when operating both HPU's.

3.7.2 Sea trial. Upon satisfactory completion of dock trials, conduct a sea trial of the steering gear system. The system must be operated in all possible modes of operation and demonstrate satisfactory operation in accordance with the manufacturer's specifications.

4. NOTES

4.1 Thordon bearing particulars. Government-furnished Thordon bearings are reserved and stored at the bearing manufacturer's facilities. Upon request, the bearing manufacturer will machine new bearing to the final dimensions based on the measurements of the existing bearings. The new bearings will then be shipped to the Contractor's facility within 2 to 3 days, after being machined, for installation. See Detail 61-C of Coast Guard Drawing 901 WMEC 562-005 in regards to effect of temperature on dimensions.

4.2 Rudder stock runout. The runout should be no more than 0.005 inch.

4.3 Coast Guard personnel. Coast Guard personnel will operate all shipboard machinery and equipment.

4.3 Seal Manufacturer. Original seal manufacturer (John Crane) has been acquired by Wartsila Lips, Inc.

Vessel: _____

Date: _____

Preliminary/Final: _____

Machinist: _____

RUDDER STOCK MEASUREMENTS SHEET

INSPECTION POINTS (SEE FIGURE 1)		DESIGN DIMENSIONS	ACTUAL DIMENSION	M & TE ID#	QC / QA INSP.
aa	Overall Length	143.97" Reference			
bb	Tiller keyway depth For / Aft	0.820" to .870"	/		
cc	Tiller keyway width	1.500" +.004" -.000"			
dd	Carrier ring keyway width	2" +.002" -.000"			
ee	Torque keyway width	1.75" +.003" -.001"			
ff	Small end taper to bottom of keyway	27.87" reference			
gg	Length of Keyway	9.750" +.030" -.000"			
hh	Torque keyway depth	0.875" +/- .005			
ii	End of thread to small end taper	6.50" +/- .010"			
jj	Taper length	45.00" reference			
kk	Sleeve journal length	28.40" +/- .010"			

ll	Rudder seal journal	21.114" +/- .005"			
mm	Roller bearing journal length	6.860" +.002" -.001"			
nn	Bearing housing seal journal length	11.495" +/- .005"			
oo	Tiller arm journal	19.00" +/- .010"	/		
pp	shaft sleeve location	41.75"			
qq	sleeve OA length	31"			
rr	Sleeve small end length	7"			
A1	Tiller journal diameter	8.250" +.002" -.003"	/		
A2	Tiller journal diameter	8.250" +.002" -.003"	/		
A3	Tiller journal diameter	8.250" +.002" -.003"	/		
B	Journal diameter	9.937" +.002" -.000"	/		
C	Carrier ring keyway diameter	8.937" +.003" -.000"	/		
D	Roller bearing journal diameter	10.235" +.000" -.004"	/		
E1	Seal area diameter	11.10" +.010" -.010"	/		
E2	Seal area diameter	11.10" +.010" -.010"	/		
E3	Seal area diameter	11.10" +.010" -.010"	/		
F1	Sleeve journal diameter	14.50" +.002" -.000"	/		
F2	Sleeve journal diameter	14.50" +.002" -.000"	/		
F3	Sleeve journal diameter	14.50" +.002" -.000"	/		
F4	Sleeve diameter	17.625" +.000" -.002"	/		
F5	Sleeve diameter	17.625" +.000" -.002"	/		
F6	Sleeve diameter	16"	/		
G	Radius	0.375"			
H	Small end taper diameter	8.25" +.000" -.002"	/		
	Shaft taper = 1.667 inch/ft				
1) All diameters must be taken along the same plane at two locations 90 degrees apart. 2) When viewed from the tiller end, initial reading must be taken 45 degrees clockwise from the forward tiller. 3) F1, 2 and 3 must be recoded where a sleeve does not exist. 4) Note any imperfections here:					

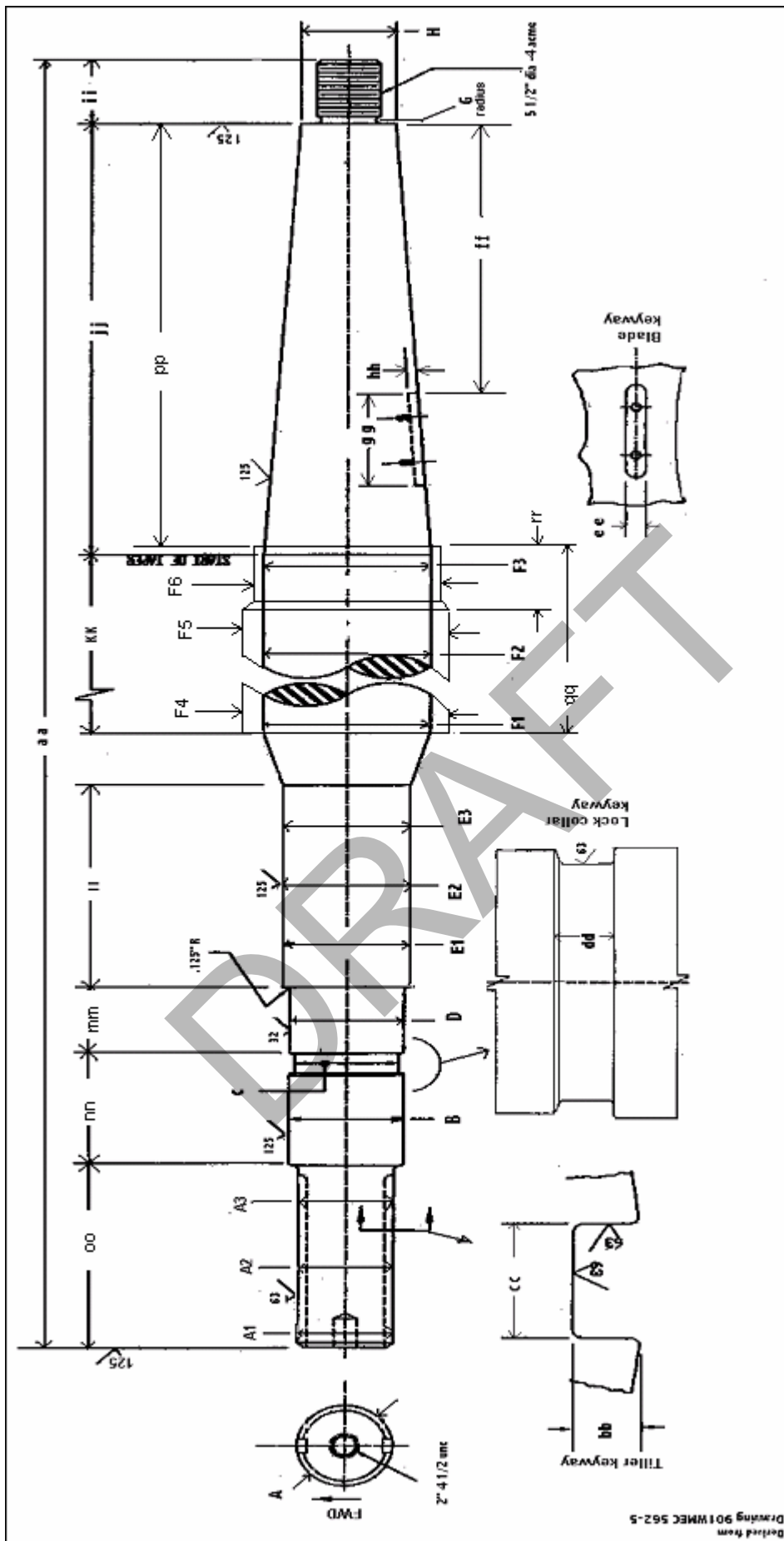
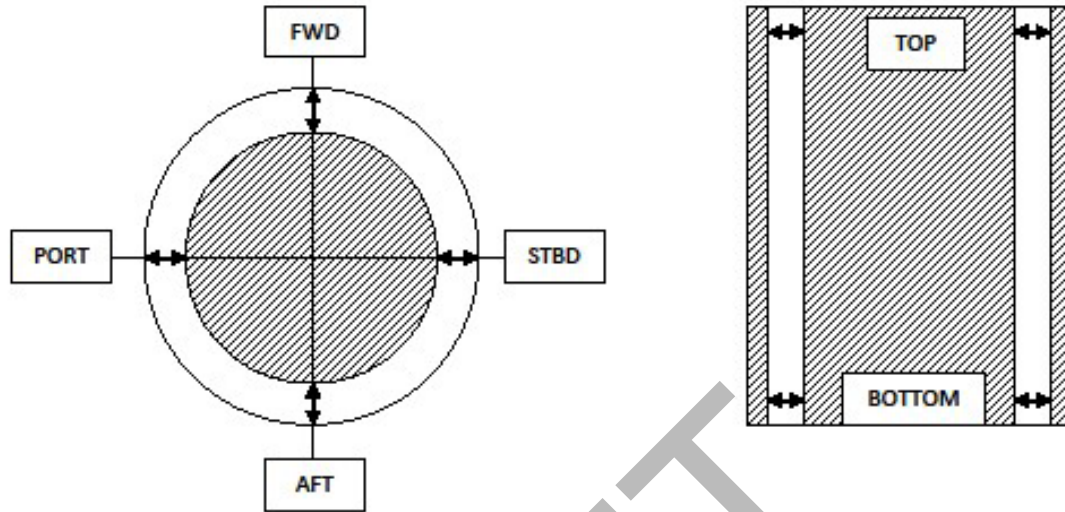


FIGURE 1 - 270 RUDDER STOCK

(Draft-00)

RUDDER BEARING CLEARANCE REPORT



PORT RUDDER

TOP

F-A Clearance = FWD _____ (nearest .001 in) + AFT _____ (nearest .001 in) = _____ in

P-S Clearance = PORT _____ (nearest .001 in) + STBD _____ (nearest .001 in) = _____ in

BOTTOM

F-A Clearance = FWD _____ (nearest .001 in) + AFT _____ (nearest .001 in) = _____ in

P-S Clearance = PORT _____ (nearest .001 in) + STBD _____ (nearest .001 in) = _____ in

STBD RUDDER

TOP

F-A Clearance = FWD _____ (nearest .001 in) + AFT _____ (nearest .001 in) = _____ in

P-S Clearance = PORT _____ (nearest .001 in) + STBD _____ (nearest .001 in) = _____ in

BOTTOM

F-A Clearance = FWD _____ (nearest .001 in) + AFT _____ (nearest .001 in) = _____ in

P-S Clearance = PORT _____ (nearest .001 in) + STBD _____ (nearest .001 in) = _____ in

WORK ITEM 33: Fin Stabilizers (Bearings, Packing, Stock/Blade), Repair or Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to overhaul both the port and starboard fin stabilizer systems.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	Seal, Plain	PN: 1856959-3 (Sperry Marine) = 17855H1L5 NSN: 5330-01-625-6617	4 ea.	383.45
N	O-Rings	PN: MS9021-381 NSN: 5331-00-763-2637	4 ea.	9.21
N	Packing Material	NSN: 5330-00-902-5505	2 Roll	885.14
N	Packing Material	NSN: 5330-00-527-7496	16 ft.	16.30
N	Thordon Bearing Set Assembly	PN: TBFINSTABTG13886BRG NSN: 3110-01-599-8155	2 ea.	11,916.00
N	Inflatable Seal	NSN: 5330-01-103-6518	2 ea.	1,031.62
Y	**Roller Bearing	NSN: 3110-00-155-9065	2 ea.	897.92
N	O-ring	NSN: 5331-01-306-4879	2 ea.	9.02
N	O-ring	PN: MS9021-359 NSN: 5331-00-435-0431	2 ea.	5.60
N	O-ring	PN: MS9021-377 NSN: 5331-01-129-0629	2 ea.	9.58
N	Seal, Plain	PN: 1856959-2 (Sperry Marine) = 17444H1L5 NSN: 5331-01-625-4197	4 ea.	113.73
N	**Seal Retainer	NSN: 5330-01-382-7103	2 ea.	228.00
N	Lantern Rings	NSN: 4320-01-377-2106	2 ea.	720.33
N	**Seal Housing	NSN: 5330-01-382-9160	2 ea.	1,622.00
N	**Tiller Ram Pin	NSN: 5315-01-419-1523	4 ea.	829.00
N	**Tiller Ram Clevis Pin	NSN: 5315-01-685-7187	4 ea.	600.00
N	**Hydraulic Cylinder	NSN: 3040-01-103-6565	4 ea.	5,652.21
Y	**Fin Stock with sleeves	NSN: 3040-01-104-5121	2 ea.	17,849.79
Y	*Fin Removal Tools	NSN: 5180-01-F10-1826	1 set	15,000.00

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 565-005, Rev -, Fin Stabilizer Hydraulic System Diagram
Coast Guard Drawing 901 WMEC 565-007, Rev H, Fin Stabilizer Variable Assembly
Coast Guard Drawing 901 WMEC 565-020, Rev -, Fin Stabilizer Variable Assembly (Sperry 1890934)
Coast Guard Drawing 901 WMEC 565-031, Rev C, Fin Shaft Assembly, w/ Sleeve Fitted NSN 3040-01-104-5121
Coast Guard Drawing 901 WMEC 565-034, Rev -, Fin Stabilizer Clevis Pin
Coast Guard Drawing 905 WMEC 565-020, Rev -, Fin Stabilizer Tiller

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 2719, Aug 2016, Manufacturer's Instruction Book SWBS Group(s) 565, 270' WMEC "A" Class Cutters, Fin Stabilizer Machinery
Coast Guard Technical Publication (TP) 5490, Apr 2018, Fin Stabilizer System
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements
Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes
Surface Forces Logistics Center Standard Specification 5000 (SFLC Std Spec 5000), 2022, Auxiliary Machine Systems

OTHER REFERENCES

MIL-PRF-16173, Oct 2017, Corrosion Preventive Compound, Solvent Cutback, Cold-Application

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph:

- 3.3.2.3 (Fin and shaft assemblies).

3.1.2 Tech Rep. The Contractor must provide the services of a Qualified Technical Representative who is familiar with the Thordon bearing system, to accomplish the following on site:

- Advise on manufacturer's proprietary system information.
- Assist with installation and repair methods.
- Ensure compliance with manufacturer's procedures and standards during system disassembly, inspection, and reassembly as applicable.

3.1.2.1 Ensure the Tech Rep has experience with the system/equipment stated above and demonstrated on their résumé.

3.1.2.2 Submit the name and resume of the Tech Rep to the COR at the Arrival Conference.

3.1.3 Protective measures, general. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.3.1 Protection of fine surfaces. The Contractor must protect all fine surfaces in accordance with SFLC Std. Spec 5000.

3.1.3.2 Hydraulic system contamination protection. The Contractor must maintain existing hydraulic cleanliness in accordance with SFLC Std. Spec 5000.

NOTE

Be aware that plastic bags may be used only when arrangement or configuration prevents the use of the other sealing methods specified above.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Flammable liquid contents.

3.1.5 Fin removal tools. The Contractor must be aware that the Government will provide a set of special tools for use in removing the fin stabilizers. Inspect the tools, upon receipt, and return the tools to the Coast Guard Inspector, upon completion of work. The Contractor must be responsible for repairing or renewing the tools, if damaged due to improper use or loss, as applicable.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.1.6 Operational test, initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of the fin stabilizer system to demonstrate existing operational condition. Submit a CFR.

3.2 Docking interference. Prior to beginning work, the Contractor must set up blocks for docking in such manner to avoid any interference with the operational test or repair work of the fin stabilizer system.

3.3 Maintenance requirements. The Contractor must accomplish the following for both port and starboard fin stabilizer systems.

CAUTION

Fins may be free to rotate. Prior to disassembly of fin machinery, the Contractor should use suitable means to secure the fin shafts and blades from uncontrolled movement or rotation during tiller disassembly.

3.3.1 Tiller Assembly. The Contractor must perform the following tasks, using TP 2719, Section 6-64 and, Figure 7-1 for guidance:

3.3.1.1 Disconnect and remove, and disassemble the tiller assembly.

3.3.1.2 Perform NDI of the tiller pins, clevis pins, tiller, and clevis for cracks, in accordance with SFLC Std Spec 0740, Appendix C. Submit a CFR.

NOTE

It has been noted on several 270 installations that the tiller has not been fabricated correctly, resulting in loose fit of tiller on fin shaft. There should be a gap (approximately .060") between tiller halves to ensure sufficient clamping force around the shaft.

3.3.1.3 Check that there is a gap between tiller half faces in accordance with Coast Guard Drawing 905 WMEC 565-020. Submit a CFR.

3.3.1.3.1. Tiller repairs. If there is no discernible gap present between tiller halves or there is evidence that the tiller has slipped down the shaft, and a Change Request has been approved and released, accomplish the following.

3.3.1.3.1.1 Machine .030" off each tiller half mating face (4 places per tiller assembly, 8 places total). See Coast Guard Drawing 905 WMEC 565-020 as guidance.

3.3.1.3.1.2 Check the fit of the tiller to the shaft by bluing. Obtain a minimum of 75% uniform fit between the tiller and the shaft.

NOTE

Weld repair is limited to bringing the bore back to original tolerances. Defects affecting the structural integrity of the casting require replacement of the casting.

3.3.1.3.1.3 If necessary to restore 75% contact, weld up tiller bore to the extent required to re-bore tiller to specified ID in accordance with Coast Guard Drawing 905 WMEC 565-020.

3.3.1.3.1.4 If tiller requires re-boring, prior to machining final tiller bore, install .060" thick shims between tiller halves and bolt the halves together. Remove shims prior to reinstalling tiller on shaft. Recheck fit to ensure minimum 75% uniform fit between the tiller and the shaft. Submit a CFR documenting final percentage of contact.

3.3.1.4 The Contractor must be aware that the Government reserves the right to furnish new tiller pins for installation in place of the existing pins. If the Government exercises this right, the Contractor must dispose of the old tiller pins in accordance with all applicable Federal, state, and local regulations.

3.3.1.5 After completion of all other authorized work, reassemble and reconnect the tiller assembly and actuators. Torque the 1-1/2"-8 UN Grade 8 nylon-locking tiller clamp nuts to achieve a stretch of 0.006 ± 0.0003 " in accordance with Coast Guard Dwg 901 WMEC 565-020 Section B-B. Where it is not possible to measure stretch at assembly, torque the studs to 1000 ft-lbs using Molykote 1000 after bench-testing to verify that this torque value will result in a stretch of 0.006" with the tiller assembled. Torque value must be adjusted if other lubricants are used.

3.3.2 Fin and shaft assemblies. The Contractor must accomplish the following task for the fin and shaft assemblies, using TP 2719, Sections 6-78, 6-82, 6-83, Figure 7-1, and Coast Guard Drawing 901 WMEC 565-031 as guidance:

3.3.2.1 Remove removable access plates, to facilitate shaft removal.

3.3.2.2 Disconnect and remove the fin and shaft assemblies to a suitable repair facility. Disassemble fin and shaft assemblies.

3.3.2.3 Clean all components free of marine growth and all other surface contaminants, and perform the following inspections (Submit a CIR):

3.3.2.3.1 Inspect the fin void for the presence of water contamination.

3.3.2.3.2 Visually inspect all shaft and fin components for signs of corrosion, damage, and other defects.

3.3.2.3.3 Set each shaft between lathe centers, providing supports at each bearing surface, and measure the runouts along the shaft to check for straightness. Measure the shaft taper diameters and runouts. Check the shaft sleeves for wear. Measure the shaft sleeve outside diameter at the forward, middle and aft positions. Take a skim cut of the nickel-copper-silicon alloy sleeve surfaces to remove any minor grooves.

3.3.2.3.4 Perform NDI of shaft tapers, keys and keyways of the shafts, in accordance with SFLC Std Spec 0740, Appendix C.

3.3.2.3.5 Check to ensure that there is proper interference fit between each fin shaft keyway and associated key.

3.3.2.3.6 Fit the shaft with its associated fin, and inspect contact areas by bluing.

3.3.2.4 Upon completion of inspections and authorized repairs, if any, complete the following:

3.3.2.4.1 Reassemble the fin and shaft assemblies. If shaft and fin contact is observed to be less than 75 percent, hand grind the fin hub with fine emery cloth until the 75 percent contact pattern is achieved.

3.3.2.4.2 After all authorized repairs, if any, fill the fin stabilizer voids with a rust preventive compound conforming to MIL-PRF-16173, Grade 3, to coat all surfaces. Drain, collect, and dispose of remaining compound in accordance with all applicable Federal, state, and local regulations. Ensure that the coated surfaces are left exposed to the atmosphere for 24 hours to allow for adequate drying.

3.3.2.4.3 Plug holes in fins with epoxy using TP 2719 as guidance.

3.3.2.4.4 Replace the below listed components with Government-furnished components. Dispose of all replaced components in accordance with all applicable Federal, state, and local regulations.

TABLE 1 – REPLACEMENT COMPONENTS

COMPONENTS	QTY	INDEX NUMBER SHOWN ON FIG 7.1 & TABLE 7-3 OF TP 2719
Seal retainers	2 ea.	130
Lantern Rings	2 ea.	68 thru 70

NOTE

Each lantern ring consists of two halves and are secured by two socket head cap screws 0.375-16UNC-3A x 1.00 LG.

3.3.3 Threaded-end rod. The Contractor must accomplish the following tasks for the threaded-end rod, using Figure 7.1 and Table 7-3, Index number 151, in TP 2719 for guidance:

3.3.3.1 Remove and inspect the mechanical lock threaded rod for defects. Submit a CFR.

3.3.3.2 Reinstall the threaded-end rod to original configuration.

3.3.4 Outer seals. The Contractor must replace the port and starboard fin stabilizer outer seals in accordance with TP 2719, Section 4-22, with Government-furnished outer seals (Part No. 1856959-3) and O-rings (Part No. MS9021-381), as shown on Figure 7.1 and Table 7-3 of TP 2719.

3.3.5 Packing. The Contractor must replace the existing port and starboard fin stabilizer packing materials (Items 89 and 90), shown on Coast Guard Drawing 901 WMEC 565-007, with new Government-furnished packing materials, in accordance with TP 2719, Section 6-74.

3.3.6 Bearings. The Contractor must replace the existing bearings with Government-furnished Thordon bearings.

3.3.6.1 A Thordon Bearing Technical Representative must be consulted to establish the correct installation dimensions of the bearing depending on ambient temperatures, water swell, and thermal expansion, liner's OD, and housing's ID. Thordon uses a program that determines initial sizing that will result in the correct installed/operating running clearance. Provide Coast Guard Inspector with programs output and calculated initial (dry) bore diameter as well as final wet/swollen running diameter.

3.3.6.2 Following Thordon's technical guidance perform all required machining of bearings to facilitate installation.

3.3.6.3 Measure the inside diameter of the bearing housing; machine the outside diameter of the Thordon Bearing assembly to fit in accordance with Coast Guard Drawing 901 WMEC 565-007. Drill grease passage into bearing to facilitate lubrication.

3.3.7 Inflatable seals. The Contractor must replace the existing inflatable seals (Item 25), as shown on Coast Guard Drawing 901 WMEC 565-007, with Government-furnished inflatable seals, in accordance with TP 2719, Section 6-83, paragraph "d".

3.3.8 Spherical roller bearings. The Contractor must accomplish the following for the spherical roller bearings:

3.3.8.1 Inspect the spherical roller bearings for wear and/or damage.

3.3.8.2 Renew seals (Item 26 and 27) as shown on Coast Guard Drawing 901 WMEC 565-007.

3.3.8.3 The Contractor must be aware that the Government reserves the right to furnish a new roller bearing(s) for installation in place of the existing roller bearings. If the Government exercises this right, the Contractor must turn over the removed roller bearings to the Coast Guard PA as an MTI item.

3.3.9 System reassembly and reinstallation. After completing all authorized repairs, the Contractor must accomplish the following, using TP 2719, Sections 6-79, 6-84, and 6-85 as guidance.

3.3.9.1 Reassemble and reinstall each fin and shaft assembly, using the following Government-furnished overhaul components:

TABLE 2 – OVERHAUL COMPONENTS

COMPONENTS	QTY
Preformed packing	2 ea.
O-rings	2 ea.
O-rings	2 ea.
Seals, Plain	4 ea.
Seals, Plain	4 ea.

3.3.9.2 Reinstall the removed access plates.

3.3.9.3 The Contractor must be aware that the Government reserves the right to furnish a new shaft or fin for installation in place of the existing shaft or fin. If the Government exercises this right, the Contractor must turn over the removed shaft or fin to the Coast Guard PA as an MTI item.

3.3.10 Post installation tasks. After completion of fin system reinstallation, the Contractor must accomplish the following:

3.3.10.1 Perform air test to each fin in accordance with SFLC Std Spec 0740, Appendix C. Submit a CFR.

3.3.10.2 Renew the system hydraulic fluid in accordance with TP 5490 and SFLC Std Spec 5000, Appendix C, Paragraph C2.1 (Fluids).

3.3.10.3 Filter the system hydraulic fluid in accordance with SFLC Std Spec 5000, Appendix C, Paragraph C2.1 (Fluids).

3.3.10.4 Lubricate the following components:

- Thordon bearing, in accordance with TP 2719.
- Repeat back linkage arm spherical bushing, in accordance with TP 2719, Section 4-4.
- Spherical roller bearing, in accordance with TP 2719, Section 4-17.

3.3.10.5 Measure baseline values for Thordon bearings on the port and starboard fins using the procedures outlined in TP 2719, Chapter 4, paragraph 21. Submit a CFR.

3.4 Government's right for change out. The Contractor must be aware that the Government reserves the right to furnish new components (listed in Table 3 below) for installation in place of the existing components. If the Government exercises this right, the Contractor must dispose of the removed seal housing and seal retainer in accordance with all applicable Federal, state, and local regulations, and turn over the removed fin stock to the Coast Guard PA as an MTI item.

TABLE 3 – COMPONENTS POSSIBLE FOR CHANGING OUT

COMPONENTS	QTY
Seal Housing	2 ea.
Seal Retainer	2 ea.

Fin Stock with sleeves	2 ea.
------------------------	-------

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.5 Operational test, post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate all items or shipboard devices that have been disturbed, used, repaired, altered, or installed to be in satisfactory operating condition. Submit a CFR.

3.5.1 Test the fin stabilizer systems in accordance with TP 5490. Operate the fins through the complete range of travel utilizing the manual directional valves.

3.5.2 Conduct a minimum of four-hour sea trial.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 34: Tanks, Ballast, Preserve, 100 Percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to prepare and coat 100% of the surfaces of the following tank(s):

TABLE 1 - TANKS

TYPE OF STRUCTURE	LOCATION	CAPACITY - 95% (GALLONS)	LOW SUCTION (GALLONS)
Ballast	3-B-0-W	4,679	
Ballast	4-75-1-W	1,327	
Ballast	4-75-2-W	1,811	
Ballast	4-103-1-W	7,960	
Ballast	4-103-2-W	7,960	
Ballast	4-169-0-W	3,239	
Ballast	4-228-1-W	3,521	
Ballast	4-228-2-W	3,521	
Ballast	4-242-0-W	2,989	

Note: Tanks also used for storing grey water – considered immediately dangerous to life and health (IDLH).

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 529-001, Rev N, Ballasting & Emer Bilge Drainage Sys
Diagram

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

ASTM International (ASTM) D1330, 2010, Standard Specification for Rubber Sheet Gaskets

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Temporary access openings. Due to limited access to work areas, the Contractor may, with express permission of the KO (via submission of a CFR), cut access holes to facilitate accomplishment of the work specified herein. Perform all work required to open and close the access openings in accordance with SFLC Std Spec 8636.

NOTE

Any ballast tanks being listed for preservation will have been pumped down to low suction by ship's force prior to arrival at the Contractor facilities.

3.3 Content removal. The Contractor must remove and dispose of all fluids and/or residues in accordance with all applicable Federal, state, and local regulations. Plug all inlet and outlet piping in the tank to prevent contaminants from entering the tank. Use plugs with an attached lanyard, ring or other system that will ensure plugs are not lost in the pipe openings. Maintain a plug accountability log outside the tank(s) to prevent any of the installed temporary plugs from being lost inside the tank or forgotten inside at tank closure.

«HEALY_NOTE»

3.4 Surface preservation. The Contractor must accomplish the following tasks for the tanks listed in paragraph 1.1 (Intent):

3.4.1 The Contractor must remove and retain the tank manhole cover(s).

3.4.2 The Contractor must prepare and coat all tank interior surfaces (including internal surfaces of manhole cover(s), manhole cover hull ring(s) extending outward to the weld line that ties the hull ring into the tank plating on the tank exterior) using the system specified for "Tanks and Voids, Grey Water, Sewage, and CHT Tanks; Option II " in SFLC Std Spec 6310, Appendix B (Cutter and Boat Interior Painting Systems). Select finish/top coat color to match existing.

3.4.3 The Contractor must prepare and coat all manhole cover external surfaces to match existing adjacent surfaces, using the system specified for "Decks, Metal Interior and Non-Skid Areas (Metal Decks – No application of deck coverings)", in SFLC Std Spec 6310, Appendix B (Cutter and Boat Interior Painting Systems). Select finish/top coat color to match existing adjacent surfaces.

3.5 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for "critical-coated surfaces").

NOTE

Surfaces being preserved are considered "critical-coated surfaces".

3.6 Inspection. After surface preparation and before coating application, the Contractor must visually inspect all tank interior and manhole cover surfaces; including, but not limited to bulkheads, floor and overhead plating, structural members, fasteners and gasket seating surfaces. Submit a CFR including the following, as applicable:

- Structural condition.
- Inaccessible areas.
- Tank level indicator (TLI) and/or float switch condition.
- Sounding tube and striker plate condition.
- Suction and discharge piping.
- Fastener condition.

3.7 Tank closing. The Contractor must ensure that the tank(s) remain open for at least 24 hours after completion of the tasks specified above. Notify the COR at least 24 hours prior to closing the tank(s).

3.7.1 After satisfactory inspection by the Coast Guard Inspector and after all authorized repairs, accomplish the following:

- Reinspect all TLIs, as applicable, to verify proper operation. Submit CFR.
- Close tank manhole cover(s) with new gasket material conforming to ASTM D1330 and new cotton grommets on all studs (as applicable).

3.7.2 The Contractor must renew up to 10% of missing or damaged nuts and washers.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 35: Chain Locker, Preserve, 100 Percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to prepare and preserve 100% of the Chain Locker surfaces.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 123-002, Rev D, Chain Lkr Bhds & Details

Coast Guard Drawing 901 WMEC 581-002, Rev C, Hawse & Chain Pipes

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

Surface Forces Logistics Center Standard Specification 8636 (SFLC Std Spec 8636), 2022,
Temporary Hull Accesses

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Eductor strainer.
- Anchors and chains.

3.2 Temporary access openings. With express permission of the KO via submission of a CFR and in accordance with SFLC Std Spec 8636, the Contractor may perform all work required to cut open and close temporary access openings to facilitate accomplishment of the work specified herein.

3.3 Surface preservation. The Contractor must notify the Coast Guard Inspector 48 hours prior to beginning surface preparation. Remove access cover(s); prepare and coat all the port and starboard chain locker surfaces, including the sump, bellmouths, hawse pipes, chain pipes, bottom grate and all structural members, and including the internal surfaces of the access cover(s), using the system specified for “Chain Lockers”, in SFLC Std Spec 6310, Appendix B (Cutter and Boat Interior Painting Systems). Select finish/top coat color to match existing adjacent surfaces.

NOTE

High pressure waterjetting is not acceptable for surface preparation on surfaces to be coated with inorganic zinc.

3.4 Inspection. After surface preparation and before paint application, the Contractor must perform a visual inspection of all prepared surfaces and sounding tubes in accordance with Coast Guard Drawing 901 WMEC 123-002 and 901 WMEC 581-002. Submit a CFR.

3.5 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”). Surfaces being preserved are considered “critical-coated surfaces”.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 36: Superstructure, Preserve, 100 Percent

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to preserve the steel and aluminum superstructure surfaces.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 162-001, Rev I, Stacks

Coast Guard Drawing 901 WMEC 602-002, Rev C, Distinguishing Figures

Coast Guard Drawing 901 WMEC 801-009, Rev L, Outboard Profile

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

Specific equipment and components to be protected include, but are not limited to:

- Life lines.
- Bulkheads.
- Port lights.
- Ventilation system components.
- Deck surfaces.
- Cableways and cable runs.
- Mast and staffs.
- Windows.
- Stuffing tube gaskets.
- JP-5 hoses.
- MSB davit fall guides.
- Deck fittings.
- Helicopter safety nets.
- Hoses.
- Exterior doors and hardware (dogs, wedges, handles, latches, and hinges).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the below-listed:

- Glide Slope Indicator.
- Expansion Joint at Frame 119.
- SRBOC Launchers.
- Boat Davits.
- King Post.
- Lights.
- J-bar davits.
- Electrical/electronic equipment.
- Junction boxes.
- Flight deck cameras.
- 1MC speakers.
- Ribbons.
- Awards.
- Fire station hoses.

NOTE

Existing coating system on the Superstructure surfaces may have a nominal thickness in excess of that which was originally installed, as a result of “patch-coats” applied during past availabilities.

3.2 Initial inspection. Prior to removing the existing coating system, the Contractor must inspect and verify whether or not all hull identification numbers and letters have permanent markings (weld beads or impressions), showing their location on the hull. Submit a CFR.

3.3 Pre-surface preparation wash. The Contractor must accomplish low-pressure (less than 5,000 psi) fresh water wash of all affected surfaces, to remove soluble chlorides and other surface contaminants. Capture, contain, and dispose of wash water for proper disposal in accordance with all Federal, state and local regulations.

3.4 Surface preparation optional methods. The Contractor has the option of using either high/ultrahigh pressure water jetting or abrasive blasting to achieve the required surface preparation, prior to application of the coating system specified in 3.6 (Preservation requirements). The Contractor may add abrasives to the waterjet stream, for one or both of the following reasons:

- Achieving greater productivity.
- Achieving the required surface profile.

NOTE

Waterjetting without abrasive addition does not provide any additional anchor profile to the surface, beyond what was present after the previous surface preparation.

3.5 Substrate inspection. After completion surface preparation and before coating application, the Contractor must perform a visual inspection of the prepared substrate, and submit a CFR.

3.6 Preservation requirements. The Contractor must prepare and coat the superstructure surfaces, using the system specified for “Freeboard/Superstructure/Mast (Freeboard/Superstructure), in SFLC Std Spec 6310, Appendix A (Cutter and Boat Painting Systems) .

3.6.1 The Contractor must select “Option I” system, for the applicable metal substrate.

3.6.2 Stack Surfaces. Prepare and coat exterior stack surfaces using the system specified for “Smoke Stack Casing, Steel, Exterior” in Appendix A (Cutter and Boat Exterior Painting Systems) of SFLC Std Spec 6310. Install all associated emblems supplied by the cutter over preserved areas.

3.7 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000 In-process QC measures for critical-coated surfaces. Surfaces being preserved are considered “critical-coated surfaces.”

3.7.1 Coast Guard Inspector. The Contractor must notify the Coast Guard Inspector, a minimum of 4 hours prior to any quality assurance inspection related to the preservation of the critical coated surface. This includes the pre-surface preparation, surface preparation, and all coating application check points outlined on the QA-1 Quality Assurance Inspection Form.

3.8 Abrasive grit compliance conformance. If abrasive blasting is selected by the Contractor as the surface preparation method, the Contractor must ensure that the abrasive material used on all aluminum surfaces is an aluminum oxide, garnet or equivalent inert material conforming CID A-A-59316, Type I & IV.

4. NOTES

4.1 Definition of superstructure. For the purposes of this item, superstructure surfaces are defined as vertical and horizontal surfaces (excluding deck and mast surfaces) from the 01 Deck Level up; including, but not limited to the exterior shell, bulkheads and all attachments secured or bracketed to bulkheads, fire stations, stacks exterior, interior vertical surface of the stack tops, hangar (fixed and telescoping sections), doors, ladders, metallic rails, stanchions, stand-offs, brackets, piping, stuffing tubes, vent ducting, structural tees, angles, and gussets. Also included are the fantail bulkhead on the Main Deck, bulwarks (inboard side), the inside of the fashion plates aft, all weather deck vents, deck fittings, scuttles, hatches, risers, deck piping, overhangs, undersides of overhangs, and overheads.

4.2 Unit's responsibility. Unit crew will install new superstructure damage control (DC) decals, compartment check-off holders, and name plates, after the completion of work.

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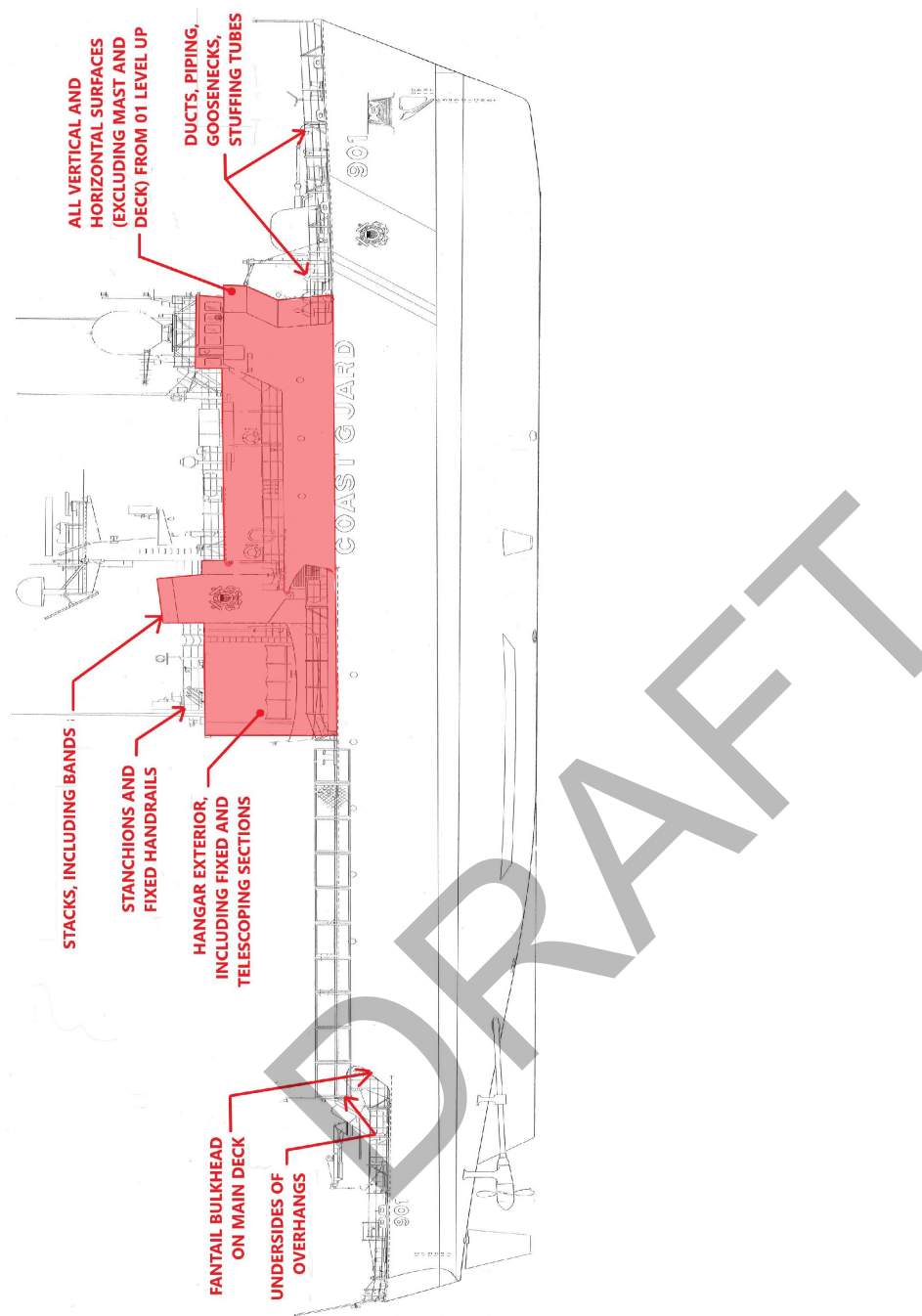


FIGURE 4.1 – SUPERSTRUCTURE SURFACES

WORK ITEM 37: Cathodic Protection, Zinc Anodes, Renew,

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to renew the following cathodic protection anodes:

QTY	TYPE/DESCRIPTION	SIZE (INCHES)	LOCATION
3	ZHS/23	6 x 12 x 1-1/4	Sea chest
108	ZHC/23	6 x 12 x 1-1/4	Hull
120	ZEP	2 diameter	Hull/Stern tube

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 633-001, Rev B, Cathodic Protection – A & D
NAVSEA Drawing 803-921865, Rev R, Cathodic Protection Hull Zinc Anodes

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements
Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022,
Requirements for Preservation of Ship Structures
«REF_A»

OTHER REFERENCES

Commercial Item Description (CID) A-A-59313, Nov 2003, Thread Compound; Anti-seize, Zinc
Dust-Petrolatum
MIL-A-18001, Dec 2015, Anodes, Sacrificial Zinc Alloy

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

3.2 Renewal particulars. The Contractor must accomplish the following tasks. Use Coast Guard Drawing 901 WMEC 633-001 and NAVSEA Drawing 803-921865 as guidance.

3.2.1 Removals. The Contractor must remove and dispose of all designated anodes (see Table under paragraph 1.1 (Intent)). Accomplish the following additional tasks, as applicable:

3.2.1.1 Retain all anode mounting hardware (studs, nuts, and washers).

3.2.1.2 Grind flush all residual roughness with the hull or mounting surfaces.

3.2.1.3 Visually inspect all mounting hardware, including studs, washers, and nuts. Verify the quality of the zinc anode stud weldment to the hull by tapping perpendicular on the nutted stud with a rubber hammer and submit a CFR.

3.2.2 New installations. Furnish and install new anodes conforming to MIL-A-18001, in place of the removed.

- Drill holes in each mounting strap, to facilitate stud attachment.

NOTE

All bolt studs may not have the exact same spacing.

- Renew up to 100% of mounting hardware.
- Ensure two star type lockwashers are installed: one between the nut and anode strap, and one between the anode strap and ship structure, to ensure electrical continuity between the anode and the attachment point.
- Apply antisieze compound, conforming to (CID) A-A-59313, to studs before each anode installation.
- Chip, grind, or wire-brush smooth all straps, welds, and mounting studs, as applicable. In addition, ensure that all installed anode surfaces are free of paint.

3.3 Electrical resistance testing. The Contractor must measure the electrical resistance between each anode surface and an adjacent metal ship structure, utilizing an OHM meter and a scale, to ensure that it is less than 0.1 ohm. Submit a CFR.

3.4 Touch-up preservation. The Contractor must prepare and coat the below listed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, paragraph 3.1.13 (Touch-ups and minor coating repairs.)

- External portions of studs, nuts, washers and straps, if applicable.
- Hull surfaces in way of removed anodes.

3.4.1 Wire brush all anodes within 24 hours before refloating the cutter. Take care not to disturb the U/W body coating system.

3.4.2 Do not paint new anodes.

4. NOTES

This section is not applicable to this work item.

DRAFT

WORK ITEM 38: Drydock

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to drydock the vessel, undock the vessel, and perform various drydocking-related tasks.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	*Transducer Cover Plate	PN: FL-184-002, DET 15-E	1 ea.	50.00

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

***Government-furnished property, which is to be supplied by either the vessel or the C4IT Service Center.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC 562-005, Rev K, A & B Class Rudder Stock And Steering Gear A & D

Coast Guard Drawing 901 WMEC 801-012, Rev B, Curves of Form

Coast Guard Drawing 901 WMEC 801-022, Rev K, Docking Plan

Coast Guard Drawing 905 WMEC 562-001, Rev G, Rudder

Coast Guard Drawing FL 184-002, Rev A, SS60 Transducer Install Adapter Assy

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 8634 (SFLC Std Spec 8634), 2022, Drydocking

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2020, Welding and Allied Processes

OTHER REFERENCES

MIL-PRF-16173, Oct 2017, Corrosion Preventive Compound, Solvent Cutback, Cold Application

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences). Known interferences include, but are not limited to the following:

- Anchor assembly
- Load conditions
- U/W body appendages

3.2 Key personnel. The Contractor must determine key personnel and require their presence during all drydocking phases as required by SFLC Standard Spec 8634. Submit list of key personnel to the COR.

3.3 Docking and undocking. The Contractor must perform drydock and undock in accordance with SFLC Std Spec 8634, using Coast Guard Drawing 901 WMEC 801-022 for guidance.

3.4 Cutter conditions. The Contractor must use the Full Load Condition values listed below all in inches, for purposes of performing Pre-Award calculations as described in Appendix A (Requirements for Calculations) of SFLC Std Spec 8634.

- Displacement (LT FW): 1890
- VCG (FT ABL): 16.62
- LCG (FT AFT MS): 4.62
- LCF (FT AFT MS): 14.14
- Trim (FT by stern): 0.734
- MT1” (LT-FT/Inch): 272.76

3.4.1 Blocking. The Contractor must facilitate docking the vessel in designated block position by arranging keel and side blocks and ensuring minimum block heights as shown on the vessel's docking plan.

TABLE 1 – BLOCKING

POS. #	BLOCK HEIGHT ABOVE DOCKING FACILITY DECK	STEEL PLATE	SITUATION AWARENESS	FIN STABILIZERS	ADDITIONAL
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	MIN.	RUDDER REMOVAL	PROPELLER REMOVAL	SHAFT REMOVAL				
TBD	48"	NA	30' 0" minimum from stern for shaft removal		NA	NA	If the drydock has a sliding block arrangement, the Dockmaster must exercise special care and/or set side blocks between frames 96 and 104 (fin stabilizer area) by a different method which clears the fin stabilizers.	None

3.5 Pre-award calculations. The Contractor must provide to the KO a set of pre-award calculations, as described in SFLC Std Spec 8634 Appendix A.

3.5.1 Provide vertical side/bilge block offsets for any side blocks placed in a location where vertical offsets are not already defined by the docking plan.

3.5.2 Submit an alternate blocking arrangement, as part of the pre-award calculation submission, to compensate for any changes from the docking plan.

3.5.2.1 If the alternate blocking arrangement interferes with U/W hull plate inspection or 100% preservation of U/W body surfaces required under separate work items in this specification package, the Contractor must include a plan of how inspection/preservation will be accomplished. Plan must include any modifications necessary to the prescribed docking plan including removing, shifting, repositioning blocks, or fleeing the vessel at no additional cost to the Government.

NOTE

1. The USCG has established several approved alternate docking plans for each vessel-class, to facilitate complete access to the entire U/W body structure, for periodic comprehensive inspection and/or 100% preservation. This inspection and preservation pattern and periodicity is a configuration management concern that is a vital requirement of the contract. Intention to deviate from these plans (fewer blocks, block spacing, additional blocks, etc) is to be identified on pre-award calculations (Paragraph 3.5.2); an alternate docking plan and mitigation strategy will be provided to maintain preservation and inspection configuration.

2. Pre-award calculations may be deemed unsatisfactory - and may adversely affect contract award if they are submitted with the following detrimental factors:

a. No methods have been proposed that meet the requirements for the specified docking plan.

b. Proposed block shifting or fleeting risk-mitigating plan may result in delays in period of performance.

c. Proposed alternate docking plan violates USCG configuration management policies.

3.6 Planned availability, immediate work and routine inspections. The Contractor must perform designated routine drydocking work, in accordance with SFLC Std Spec 8634, paragraph 3.5.4 (within twenty four hours after docking).

3.6.1 Upon the COR convening the Coast Guard Underwater Hull Inspection Board (UWHIB), the Contractor must facilitate and participate in the UWHIB inspections of the underwater hull. The Contractor must provide a designated hull repair supervisor to accompany the UWHIB and mark on the hull proposed repairs areas, as necessary.

NOTES

1. The COR will convene the UWHIB as soon as possible after the vessel has been dry-docked and the hull has been cleaned. No other work must take place until the UWHIB completes their inspections.

2. The UWHIB will recommend the extent of underwater body coating system preservation required based on the conditions found during the underwater hull survey.

3.7 Fuel offloading. The Contractor must be aware that fuel offloading is not mandatory to drydock the vessel.

3.8 Fleeting. Not required.

3.9 Additional requirements. In addition to all other specified routine work, the Contractor must accomplish the following tasks, for each rudder assembly:

3.9.1 Crop and retain the existing rudder stock nut cover plate, shown on Coast Guard Drawings 905 WMEC 562-001 and 901 WMEC 562-005.

3.9.2 Perform a visual inspection of the rudder stock nut and locking bar welds, in the presence of the Coast Guard Inspector. Submit a CFR stating that the stock nut and locking installation is secured.

3.9.3 Reinstall the rudder stock nut cover plate, as shown on Coast Guard Drawings 905 WMEC 562-004 and 901 WMEC 561-005. After the rudder stock nut cover plate has been reinstalled, do the following:

3.9.3.1 Air test the associated nut access void in accordance with SFLC Std Spec 0740, Appendix C. Submit a CFR.

3.9.3.2 Coat the nut access void surfaces with corrosion preventive compound conforming to MIL-PRF-16173, Class II, Grade 3.

4. NOTES

4.1 Arrival conditions. The COR will advise the Contractor of the actual tank and draft readings when the vessel arrives, and will discuss with the Contractor any liquid loading changes necessary.

DRAFT

WORK ITEM 39: Temporary Services, Provide - Cutter

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to provide temporary services to the Cutter, during the performance of this availability.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

None

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2020,
General Requirements

Surface Forces Logistics Center Standard Specification 8635 (SFLC Std Spec 8635), 2020,
Temporary Services

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

None.

3.1.2 Tech Rep.

Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, paragraph 3.3.5 (Interferences).

APM:

- 1. Select the desired Temporary Logistics from the list provided. Select “Y” or “N”, as applicable in Table. Provide quantities, wherever required in Tables 1, 2, and 3.**
- 2. If vessel is going to the CG Yard, DO NOT INCLUDE Parking. Cutter should request parking spaces in their Arrival LOGREQ.**
- 3. Determine if Duty Section berthing is needed, for how many personnel, and whether it is for the duration of the availability. This is a separate requirement and should not be confused with “Provide Temporary Berthing”, which may be a separate item in the spec package, or under a separate contract outside of this package.**
- 4. Verify specific storage needs (e.g. Life rafts stored in Dry Stores, etc.).**
- 5. Verify Electrical Capacities as defined in SFLC Std Spec 8635 Table III. For cutters with electrical service capability for 2x400A, 3x400A and/or 4x400A service, add verbiage to Table I - Service Selection below on sub-paragraph line ‘3.3.5’ to describe the exact current requirement for the availability (e.g., "For this availability, Contractor must provide 450 V AC, 3 x 400 A service").**

3.2 Temporary service particulars. The Contractor must provide the below listed temporary services, in accordance with SFLC Std Spec 8635.

TABLE 1 - SERVICE SELECTION

*SUB-PARAGRAPH	TITLE	Y/N
3.3.1	Office space	
3.3.2	Telephone and internet access	
3.3.3	Parking	
3.3.4	Duty section berthing: __ male, __ female. Duty section berthing must be provided for {Note: Choose one and delete the rest: (1) The entire duration of the availability, (2) during disruption of berthing areas or (3) days}	
3.3.5	Electrical power (including all requirements in associated sub-paragraphs)	
3.3.6	Hull grounding straps (not applicable when cutter is waterborne)	
3.3.7	Compressed air (including all requirements in associated sub-paragraphs)	
3.3.8	Hazardous material/hazardous waste disposal (see Tables 2 and 3 below)	
3.3.9	Heavy lift equipment: { day(s)/or hour(s)}	
3.3.10	Water supply	
3.3.10.1	Potable water: { gallons per day, at psig. } { or gallons, bulk }	
3.3.10.2	Hot-circulating water	
3.3.10.3	Cooling water	
3.3.10.4	Firemain system (including all requirements in associated sub-paragraphs)	
3.3.11	Steam (including all requirements in associated sub-paragraphs)	
3.3.12	Refuse disposal	
3.3.13	Sewage and grey water disposal (including all requirements in associated sub-paragraphs)	

*SUB-PARAGRAPH	TITLE	Y/N
3.3.14	Storage – General (including all requirements in associated sub-paragraphs):	
3.3.14	Dry stores.	
3.3.14	Paint and flammable stores.	
3.3.14	Refrigerated stores.	
3.3.15	Small boat storage (including all requirements in associated sub-paragraphs)	

*Each sub-paragraph number relates directly to the identical sub-paragraph number in SFLC Std Spec 8635.

TABLE 2 - HAZARDOUS WASTE DISPOSAL – LIQUIDS (GALLONS)

PAINT THINNERS	ENGINE COOLANT	BILGE WATER
Xx	xx	xx

TABLE 3 - HAZARDOUS WASTE DISPOSAL – SOLIDS

OILY FILTERS	OILY RAGS (LBS)	EMPTY 1-GAL CONTAINER*	EMPTY 5-GAL CONTAINER*	EMPTY 55-GAL CONTAINER*
xx	xx	xx	xx	xx

*Previously housed hazardous materials.

3.2 Extended temporary services. If the performance period of the contract is extended by the KO, the contractor must continue to provide all temporary services as specified herein for the extension period.

4. NOTES

This section is not applicable to this work item.

WORK ITEM 40: Auxiliary Sea Water Piping, Renew

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to replace the auxiliary seawater cooling pumps and renew designated auxiliary sea water piping.

1.2 Government-furnished property.

None.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 905 WMEC-185-021, SLEP FOUNDATIONS TO SUIT ASW PUMP REPLACEMENT

Coast Guard Drawing 905 WMEC-256-002, MN AND AUX SW COOLING SYS A & D

Coast Guard Drawing 905 WMEC-256-004, AMS SW COOLING SYS A&D

Coast Guard Drawing 905 WMEC-256-005, MISC EQPT SW COOLING A&D

Coast Guard Drawing 905 WMEC-256-011, MN AND AUX SW COOLING SYS DIAG

Coast Guard Drawing 905 WMEC-256-13, SLEP EDG SEA WATER COOLING PUMP REPLACEMENT

Coast Guard Drawing 905 WMEC-342-002, SLEP CAT C18 SSDG & EDG ASW MODIFICATIONS

Coast Guard Drawing 905 WMEC-256-015, SLEP ASW PUMP REPLACEMENT & MISC MODS

Coast Guard Drawing 905 WMEC-507-1, VALVE LABEL PLATE LIST

Coast Guard Drawing YD-505-001, ORIFICE PLATE STANDARD DETAIL

COAST GUARD PUBLICATIONS

Coast Guard Commandant Instruction (COMDTINST) M10360.3C, Coatings and Color Manual

Coast Guard Commandant Instruction (COMDTINST) M10360.3, Appendix B (Cutter and Boats Interior Painting System)

Coast Guard Commandant Instruction (COMDTINST) M9000.6F, Naval Engineering Manual

Coast Guard Commandant Instruction (COMDTINST) M5100.47 (series), Safety and Environmental Health Manual

Coast Guard Commandant Instruction (COMDTINST) 9077.1 (series), Equipment Tag-Out Procedure Manual

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

Surface Forces Logistics Center Standard Specification 0740 (SFLC Std Spec 0740), 2022, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 6310 (SFLC Std Spec 6310), 2022, Requirements for Preservation of Ship Structures

OTHER REFERENCES

ANSI/ASQC 9001-2008, Quality Management Systems – Model for Quality Assurance in Design, Development, Production, Installation, and Servicing Requirements

ASTM B127, Standard Specification for Nickel-Copper Alloy (UNS N04400) Plate, Sheet, and Strip

ASTM B466, Standard Specification for Seamless Copper-Nickel Pipe and Tube

ASTM International (ASTM) F992, 2017, Standard Specification for Valve Label Plates

MIL-DTL-15024, General Specification for Plates, Tags, and Bands for Identification of Equipment

MIL-DTL-24696, Detail Specification Gasket, Sheet, Non-Asbestos

MIL-STD-777, Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

MIL-T-16420, Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).

Manufacturers Standardization Society of the Valve and Fittings Industry (MSS), SP-58, 2018, Pipe Hangers and Supports – Materials, Design, Manufacture, Selection, Application and Installation

NAVSEA Drawing 803-1385866, Penetration Bulkhead and Decks

Society of Naval Architects and Marine Engineers (SNAME), T&R Bulletin 3-39, Guide for Shop and Installation Tests

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- None

3.1.2 Tech Rep. Not applicable.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Seawater
- Piping System

- Electrical connections

3.1.5 Structural. All fabrication, welding, sil-brazing, and inspection shall be in accordance with SFLC Standard Specification 0740_STD, "Welding and Allied Processes".

3.1.5.1 Conduct 250 UT Shots of 82 tank top. Submit CFR.

3.1.6 Inspection. A visual inspection is required of exposed areas after interferences are removed. Submit a Condition Found Report (CFR) if damage or excessive corrosion is found.

3.1.7 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces, in accordance with the following:

- SFLC Std Spec 6310, Requirements for Preservation of Ship Structures.
- COMDTINST M10360.3C, Coatings and Color Manual
- COMDTINST M10360.3, Appendix B (Cutter and Boats Interior Painting System)

3.1.8 Orifice plates. Orifice plates shall be manufactured in accordance with USCG Drawing No. YD-505-001. Orifice plates in systems exposed to seawater shall be manufactured from ASTM B127 Alloy 400.

3.1.8.1 Orifice plates shall be installed in straight runs of pipe a minimum of 5 pipe diameters in length upstream and 10 pipe diameters downstream, unobstructed by any bend, fitting, valve, etc.

3.1.8.2 Orifice plates shall be installed such that the label tabs are visible and the plates can easily be accessed for service or replacement.

3.1.9 Material. Pipe, valves, and fittings shall be in accordance with MIL-STD-777 for the service and conditions of the system unless otherwise noted. Gaskets and flanges shall be Garlock style 3200 or equivalent meeting MIL-DTL-24696 Type II.

3.2 Fluid handling. The Contractor must remove and dispose of removed fluids from the affected piping system, in accordance with in accordance with SFLC Std Spec 0000, Appendix A, Requirements for Environmental Protection and all applicable Federal, state, and local regulations.

WARNING

Do not drain ANY fluids, including fresh water, into any space, bilge, or exterior location.

3.2.1 Pipe hangers. Furnish, fit, and install new pipe hangers on all disturbed piping in accordance with MSS SP-58.

3.2.2 Pipe labeling. The Contractor must label disturbed piping as follows:

3.2.2.1 Stencil the following onto the pipe surfaces:

- Name of the piping system service.
- Destination, where feasible.
- Direction of flow, indicated by an arrow three inches long pointing away from the lettering (for reversible flow, point an arrow away from each end of the lettering).

3.2.2.2 Ensure all lettering and arrow(s) are as follows:

- In general, black color except white for dark-colored piping.
- Applied in conspicuous locations and preferably near control valves.

3.2.3 Valve labeling. The Contractor must renew all missing and damaged valve label plates on or adjacent to disturbed pipe runs, and install new valve label plates on new valves, in accordance with ASTM F992. Use Coast Guard Drawing 905-WMEC-507-1 and applicable system diagram as guidance.

3.3 Operational test - initial. Prior to commencement of work, the Contractor must witness Coast Guard personnel perform an initial operational test of all items or shipboard devices to be disturbed, used, repaired, or altered, to demonstrate existing operational condition. Submit a CFR.

3.4 System verification. The Contractor must verify piping sizes and fittings used to calculate orifice sizes. Use Coast Guard Drawing 905-WMEC-256-015 and 905-WMEC-256-011 as guidance. Submit a CFR with drawing redlines.

3.5 ASW pump foundation replacement. The Contractor must replace the pump foundation located in Auxiliary Machinery Space #2 in accordance with Coast Guard Drawing 905-WMEC-185-021. Contractor must provide and renew all fasteners, hardware, seals and gaskets.

3.5.1 Removal. The Contractor must remove the existing auxiliary seawater pumps. Use 905-WMEC-256-004 as guidance. Submit a CFR.

3.5.2 Foundation. The Contractor must replace foundation and brackets in accordance with 905-WMEC-185-021. Submit a CFR.

3.5.3 Foundation preservation. The contractor must prepare and coat foundation and brackets in accordance with SFLC Std Spec. 6310, Appendix B.

3.5.4 Installation. The Contractor must install the pumps in accordance with CG Dwg. 270-WMEC-256-015 and 905-WMEC-185-021.

3.6 ASW piping. The Contractor must renew the piping and components in accordance with Coast Guard Drawing 905-WMEC-256-015.

3.6.1 Replace fin cooler orifice. The Contractor must replace the orifice plate in the combined fin stabilizer hydraulic oil cooler seawater discharge piping in accordance with Coast Guard Drawing 905-WMEC-256-015.

3.6.2 Replace HVAC condenser orifice. The Contractor must replace the orifice in the discharge piping for the HVAC condensers in accordance with Coast Guard Drawing 905-WMEC-256-015.

3.6.3 Replace piping. The Contractor must replace the 1-1/4 in piping between each ASW pump bleed orifice and the swing check valve located in lower auxiliary machine space 3-82-0-E with 70/30 Cu-Ni Class 200 piping. Use Coast Guard Drawing 905-WMEC-256-015, 905-WMEC-256-002 and 905-WMEC-256-004 as guidance.

3.6.4 Pipe flushing. The Contractor must flush all new and disturbed piping with clean fresh water until all debris is removed but no longer than five minutes. Ensure flushing fluid is directed to move scale and

foreign debris away from installed machinery to prevent possible damage upon operational testing. Submit a CFR documenting date and time of flushing process and level of pipe cleanliness.

3.6.4.1 Dispose of flushing fluid in accordance with all applicable Federal, state, and local regulations.

WARNING

Do not drain ANY Fluids (including fresh water) into any space, bilge, or exterior location.

3.7 Testing. After all work has been completed, the Contractor must, in the presence of the Coast Guard Inspector, accomplish the following tasks, and submit required documentation:

3.7.1 Hydrostatic test. After all authorized repairs, the Contractor must hydrostatically test all new and disturbed piping and components of the affected piping system in accordance with SFLC Std Spec 0740, Appendix C, Hydrostatic Test. Ensure zero leakage from or permanent deformation of pressure-containing parts by repairing all leaks, deformations, and discrepancies. Submit a CFR.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.7.2 Operational test - post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate all items or shipboard devices that have been disturbed, used, repaired, altered, or installed to be in satisfactory operating condition. Submit a CFR.

3.9 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

4. NOTES

WORK ITEM 41: Ballast, Installation

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to install ballast to correct a list that exceeds one degree at full load, min-op and/or the critical stability point, as a result of the weight changes associated with other SLEP work items.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
	None			

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

***Government-furnished property, which is to be supplied by either the vessel or the C4IT Service Center.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 905 WMEC-191-004, SLEP FIXED BALLAST INSTALLATION (SENECA)

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022, General Requirements

OTHER REFERENCES

None

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph.

- 3.2 Ballast Installation.

3.1.2 Tech Rep. None

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

3.2 Ballast Installation. The contractor must install ballast if the total list after all SLEP work items are completed is calculated to exceed one degree at full load, min-op, or critical stability point(s), as indicated by the weight report. The contractor must submit a CIR prior to installing ballast to obtain the quantity and location of the ballast. The quantity and location of the ballast shall be calculated based on the arrival condition of the cutter combined with the weight report. The quantity and location of ballast will not be known prior to the start of the availability. All ballast shall be fully encapsulated lead ingots with a rubber lining attached between the ingots and any structure. The ballast is predicted to not exceed 5000 lbs. and the expected location is along the shell between frames 103 and 124 outboard of longitudinal L-10 port/starboard. Use Coast Guard Drawing 905-WMEC-191-004 as guidance.

4. NOTES

4.1 This section not applicable to this work item.

WORK ITEM 42: MK 38 MOD 3, Install

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to remove the MK 75 76mm Gun Weapon System (GWS), MK 92 RADAR, and associated equipment; and install the MK 38 MOD 3 Machine Gun System (MGS), a maintenance platform around the MK 38, an ammunition hoist between the magazine and weather deck adjacent to the maintenance platform, new ammunition storage solution, and all ancillary equipment associated with the MK 38.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	MK 38 MOD 3 GWS	NSN: Needed	1	
N	MAIN COMPUTER UNIT (MCU)	NSN: 8566096	1	
N	MK 1 MOD 1 MULTI-FUNCTION DISPLAY (MFD)	NSN: 8566093	1	
N	MK 1 MOD 1 MAIN CONTROL PANEL (MCP)	NSN: 8566092	1	
N	COMMANDER SAFETY PANEL (CSP)	NSN: 8566094	1	
N	BATTERY CHARGER, I/P 115VAC, 60HZ, 1PH, O/P 24-28VDC	NSN: 8566095	1	
N	REMOTE OPERATION CONSOLE (ROC)	NSN: N/A	1	

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

***Government-furnished property, which is to be supplied by either the vessel or the C4IT Service Center.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 905-WMEC-179-001, Rev 1, SLEP MKD 38 MAINTENANCE PLATFORM A&D

Coast Guard Drawing 905-WMEC-184-015, Rev 1, SLEP MK 38 MGS CSC 02-63-0-C MISC FNDS

Coast Guard Drawing 905-WMEC-184-016, Rev 1, SLEP MK 92 FOUNDATION REMOVAL

Coast Guard Drawing 905-WMEC-187-003, Rev 1, SLEP MK 38 MGS AMMO HOIST FDN & DETAILS

Coast Guard Drawing 905-WMEC-441-006, Rev 1, SLEP MK38 MGS WAVEGUIDE DRY AIR SYSTEM REMOVALS

Coast Guard Drawing 905-WMEC-512-018, Rev 1, SLEP MK 38 MGS MAGAZINE 1-26-0-M VENTILATION MODS

Coast Guard Drawing 905-WMEC -512-019 Rev 1, SLEP MK 38 MGS MAGAZINE 1-26-0-M VENTILATION POWER MODS

Coast Guard Drawing 905-WMEC-522-004, Rev 1, SLEP MK38 MGS MAGAZINE 1-26-0-M SPRINKLER MODS

Coast Guard Drawing 905-WMEC-533-010, Rev 1, SLEP MK38 MGS FRESH WATER REMOVALS

Coast Guard Drawing 905-WMEC-551-009, Rev 1, SLEP MK38 MGS AMMO HOIST MODS

Coast Guard Drawing 905-WMEC-153-002, Rev 2, SLEP MK38 MGS TEMP ACCESS CUT

Coast Guard Drawing 905-WMEC-256-014, Rev 1, SLEP MK38 MGS ASW REMOVALS

Coast Guard Drawing 905-WMEC-320-013, Rev AT, ELECTRONIC POWER DISTRIBUTION

Coast Guard Drawing 905-WMEC-320-014, Rev AN, ELECTRICAL ONE LINE DIAGRAM

Coast Guard Drawing 905-WMEC-321-006, Rev B, WIREWAY 02 LEVEL AND ABOVE

Coast Guard Drawing 905-WMEC-331-005, Rev J, LIGHTING SYSTEM – MAIN DECK

Coast Guard Drawing 905-WMEC-331-007, Rev F, LIGHTING SYSTEM 02 LEVEL AND ABOVE

Coast Guard Drawing 905-WMEC-436-003, Rev P, FIRE & SMOKE DETECTION & QUARTERDECK MONITORING SYSTEM BLK & IWD

Coast Guard Drawing 905-WMEC-400-008, Rev P, COMMAND & SURVEILLANCE WIRING DATA COED, VOL 4, IC EQUIPMENT

Coast Guard Drawing 905-WMEC-400-009, Rev AA, COMMAND & SURVEILLANCE WIRING DATA COED, VOL 5, NAVIGATION SYSTEM CABLING

Coast Guard Drawing 905-WMEC-400-010, Rev J, COMMAND & SURVEILLANCE WIRING DATA COED, VOL 6, WEAPONS & FIRE CONTROL CABLING

Coast Guard Drawing 905-WMEC-400-011, Rev AJ, COMMAND & SURVEILLANCE WIRING DATA COED, VOL 7, POWER CABLING

Coast Guard Drawing 905-WMEC-401-001, Rev AT, ARRANGEMENT PILOT HOUSE AND BRIDGE WINGS

Coast Guard Drawing 905-WMEC-401-002, Rev BL, ARRANGEMENT SENSOR ROOM & COMMAND SUPPORT CENTER

Coast Guard Drawing 905-WMEC-401-006, Rev AM, ARRANGEMENT ANTENNA AND WAVE GUIDE

Coast Guard Drawing 901-WMEC-401-007, Rev A, ARRANGEMENT MODIFICATIONS – MISC SPACE

Coast Guard Drawing 905-WMEC-406-003, Rev A, TOPSIDE GROUNDING, BONDING & RADHAZ, WARNINGS, NOTES & DETAILS

Coast Guard Drawing 901-WMEC-410-002, Rev E, COMMAND & CONTROL SYS BLK WRG DIAG & DETAILS

Coast Guard Drawing 901-WMEC-410-009, Rev N, SEAWATCH BLOCK DIAGRAM & INSTALLATION DETAILS

Coast Guard Drawing 901-WMEC-410-010, Rev J, SEAWATCH WIRING DATA COED

Coast Guard Drawing 901-WMEC-424-004, Rev E, DOPPLER SPEED LOG SYSTEM REPLACEMENT RIP-OUT, INSTALLATION, ARRANGEMENT, WIRING AND DETAILS

Coast Guard Drawing 905-WMEC-426-001, Rev AA, GYRO SYSTEM BLOCK & ISOMETRIC WIRING DIAGRAM

Coast Guard Drawing 901-WMEC-431-003, Rev J, MAIN IC SWITCHBOARD ARRANGEMENTS, ELEMENTARY WIRING DIAGRAM & DETAILS

Coast Guard Drawing 901-WMEC-431-005, Rev G, REPLACEMENT MAIN IC SWITCHBOARD WIRING DATA COED

Coast Guard Drawing 905-WMEC-432-001, Rev L, SOUND POWERED TELEPHONE SYSTEM BLOCK & IWD

Coast Guard Drawing 905-WMEC-433-001, Rev N, ANNOUNCING SYSTEM BLK & IWD

Coast Guard Drawing 905-WMEC-440-003, Rev AW, RADIO COMMUNICATION BWD & DETAILS

Coast Guard Drawing 905-WMEC-441-001, Rev F, WG DRY AIR SYSTEM DIAGRAM

Coast Guard Drawing 901-WMEC-441-008, Rev C, SEA TEL 4010W-53 KU BAND SYSTEM INSTALLATION

Coast Guard Drawing 905-WMEC-455-001, Rev J, IFF SYSTEM BLOCK & IWD

Coast Guard Drawing 905-WMEC-472-001, Rev L, ESM SYSTEM CKTS CI & CT BLOCK & IWD

Coast Guard Drawing 905-WMEC-480-001, Rev P, MK 92 FCS BLOCK & ISO WIRING DIAGRAM

Coast Guard Drawing 905-WMEC-480-002, Rev 2, SLEP MK-75 GUN / MK-92 FCS RIP-OUT BWD, ARRANGEMENT & WAVEGUIDE DETAILS

Coast Guard Drawing 905-WMEC-481-001, Rev 3, SLEP MK-38 MGS INSTALLATION BWD & DETAILS

Coast Guard Drawing 905-WMEC-481-003, Rev 3, SLEP MK-38 MGS WIRING DATA (COED)

Coast Guard Drawing 905-WMEC-514-006, Rev D, HVAC SENSOR RM & CSC 02 LVL A & D

Coast Guard Drawing 905-WMEC-514-017, Rev B, HVAC SNSR RM & CSC 02 LVL A & D

Coast Guard Drawing 905-WMEC-533-010, SLEP MK38 MGS FRESH WATER REMOVALS

Coast Guard Drawing 905-WMEC-713-004, Rev -, SLEP MAGAZINE (1-26-0-M) ARR & DETAILS

Coast Guard Drawing 905-WMEC-187-002, Rev 1, SLEP MK 38 GWS FDN ADAPTER

Coast Guard Drawing 905-WMEC-514-033, Rev 1, SLEP MK38 MGS MAGAZINE 1-26-0-M CW MODS

Coast Guard Drawing FL-711-037, Rev 3, SHIPBOARD INSTALLATION, MK 38 MOD 3 MACHINE GUN SYSTEM (MGS)

Coast Guard Drawing FL-713-002, Rev B, MK38 MGS READY SERVICE LOCKER FAB'N

COAST GUARD PUBLICATIONS

Surface Forces Logistics Center Standard Specification 0000, (SFLC Std Spec 0000) 2022 Edition, General Requirements
Surface Forces Logistics Center Standard Specification 0740, (SFLC Std Spec 0740) 2022 Edition, Welding and Allied Processes
Surface Forces Logistics Center Standard Specification 3041, (SFLC Std Spec 3041) 2022 Edition, Shipboard Electrical Cable Test
Surface Forces Logistics Center Standard Specification 3042, (SFLC Std Spec 3042) 2022 Edition, Shipboard Electrical Cable Removal, Relocation, Splice, Repair and Installation
Surface Forces Logistics Center Standard Specification 6310, (SFLC Std Spec 6310) 2022 Edition, Requirements for Preservation of Ship Structures
Coast Guard Commandant Instruction (COMDTINST) M10360.3C, Coatings and Color Manual
Coast Guard Commandant Instruction (COMDTINST) M10360.3, Appendix B (Cutter and Boats Interior Painting System)
Coast Guard Commandant Instruction (COMDTINST) M9000.6F, Naval Engineering Manual
Coast Guard Commandant Instruction (COMDTINST) M4500.5 (series), Property Management Manual
Coast Guard Commandant Instruction (COMDTINST) M5100.47 (series), Safety and Environmental Health Manual
Coast Guard Commandant Instruction (COMDTINST) 9077.1 (series), Equipment Tag-Out Procedure Manual

OTHER REFERENCES

Military Standard, MIL-STD-1310(SERIES), Shipboard Bonding, Grounding, & Other Techniques for Electromagnetic Compatibility and Safety
Institute of Electrical and Electronics Engineers Standard, IEEE-45, Recommended Practice for Electrical Installations on Shipboard
MIL-DTL-15024, General Specification for Plates, Tags, and Bands for Identification of Equipment
MIL-DTL-24643, General Specification for Low Smoke Electric Cables and Cords for Shipboard Use
NAVSEA Drawing 803-1385866, Rev E, Penetration, Bulkhead and Deck
NAVSEA Drawing 804-5773931, Rev A, Acoustic & Thermal Insulation for Compartments Installation Details

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.10.2 Temporary Access Cuts

- 3.13.10 MK 75 gun foundation

3.1.2 Tech Rep. Thirty days prior to tech representative services the Contractor must contact the PRO via CFR.

NOTE

The Government will provide the services of a Technical Representative. The Technical Representative will provide onsite technical support for system testing, commissioning, certification and onboard training of the ship's personnel for the new system.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.3.1 The Contractor shall furnish and install suitable covering to seal off and protect all non-affected surfaces/equipment and spaces in the vicinity of the work area against contamination during the performance of work. Upon completion of work, remove protective material and inspect for the presence of contamination. Clean all equipment and spaces, contaminated due to improper protection, to original state of cleanliness.

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Piping:
 - Seawater
 - Compressed Air
 - Sewage (VCHT)
- Sheathing and Lagging piping
- Deck plating and structural framework
- Insulation
- Duct work
- Lighting
- Fire/Smoke detectors
- Wiring and Wireways
- VHF Speaker #5 starboard overhead @ Frame 72 CSC 02-63-0-C
- Audio Amplifier starboard overhead @ Frame 72 CSC 02-63-0-C
- IFF Alarm starboard overhead @ Frame 72 CSC 02-63-0-C
- Television starboard overhead @ Frame 69 CSC 02-63-0-C

3.2 Tag-out. The Contractor must secure and tag out all equipment affected by this work item in accordance with SFLC Std Spec 0000, General Requirements, paragraph 2.

3.3 Gas freeing. When entering a tank, compartment, confined space, or conducting hot work, the Contractor must refer to and comply with SFLC Std Spec 0000, General Requirements, paragraph 3.3.1.2.1.

3.4 Welding and brazing. The Contractor must perform all welding, brazing, and non-destructive testing (NDT) in accordance with SFLC Std Spec 0740, Welding and Allied Processes.

3.5 Inspection. The Contractor must perform a visual inspection of all sensors, switches, and exposed areas upon arrival after interferences are removed. The Contractor must perform a visual inspection of exposed areas after interferences are removed. Submit a Condition Found Report (CFR) if damage or excessive corrosion is found.

3.6 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces, in accordance with SFLC Std Spec 6310, Requirements for Preservation of Ship Structures.

3.7 Environmental compliance. Contractor must dispose of all system fluids, cleaning and flushing fluids and wastes in accordance with SFLC Std Spec 0000, Appendix A, Requirements for Environmental Protection.

3.8 Electrical requirements. The Contractor must accomplish all electrical removals work identified in Tables 1 and 2 in accordance with SFLC Std Spec 3042 and test cables in accordance with SFLC Std Spec 3041.

TABLE 1 – MK 92 REMOVALS

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
Y	COMBINED ANTENNA ASSEMBLY W/ JUNCTION BOX	MK 53 MOD 0	1	
Y	SAFETY SWITCHES	MK 142 MOD 0	1	
Y	RADAR RECEIVER-TRANSMITTER	MK 69 MOD 0	1	
Y	WAVEGUIDE AIR DRYER	MK 13 MOD 0	1	
Y	CAS DSOT TEST SET	MK 602 MOD 0	1	
Y	SERVO CONTROL CABINET	MK 160 MOD 0	1	
Y	COMPUTER SET	AN/UYK-7(V)	1	
Y	INPUT/OUTPUT CONSOLE PERIPHERAL	SPRS	1	
Y	WEAPON CONTROL CONSOLE	MK 106 MOD 0	1	
Y	POWER MONITOR COUPLER ASSEMBLY (TRACK) (SEARCH)	6076597-1	1	
N	FIRE CONTROL SWITCHBOARD	2887849	1	
Y	WAVEGUIDE			

TABLE 2 - MK 75 REMOVALS

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
MTI	ITEM DESCRIPTION	NSN/PN	QTY	
Y	GUN 76MM, 62 CAL	MK 75 MOD 0	1	
Y	SLIP RING	MK 10 MOD 0	1	
Y	GUN CONTROL PANEL	MK 378 MOD 0	1	
Y	ANTI-ICING BOX ASSY	MK 56 MOD 0	1	
Y	WATER CONTROL PANEL	MK 390 MOD 0	1	
Y	TRAIN THERMOSTAT		1	
Y	TRAIN HEATING ELEMENT		1	
Y	KLAXON	IC-B3S4	2	

3.8.1 Disposition. Damage and contamination of all MTI equipment must be avoided. The condition code for equipment removed is "M" until it reaches the maintenance facility to be inspected and coded.

3.8.1.1 All MTI equipment in Table 1 - MK 92 REMOVALS must be crated and shipped to Lockheed Martin for disposition. The Fire Control Switchboard is not MTI; upon removal, scrap unit. The MK 92 FCS equipment shipped to the GFE warehouse at Lockheed Martin at the following address:

Lockheed Martin – Rotary and Mission Systems
 405 Heron Drive
 Pureland Warehouse
 Bridgeport, NJ 08014
 M/F: MK 92
 POC: Colleen Mason ((856) 975-4254) / Laura Oliver ((856) 975-4256)

3.8.1.2 All MTI equipment in Table 2 - MK 75 REMOVALS must adhere to the following disposition instructions:

3.8.1.2.1 CG Yard Ordnance Repair Facility (ORF) will perform depot-level maintenance and repair on the MK 75 Gun Mount including removal, disassembly, cleaning, and inspection of all major MK 75 Gun Mount components including: (1) hydraulic system, (2) lower gun loading system, (3) upper gun loading system, (4) carriage assembly, (5) equilibrators and compensator assemblies, (6) slide assembly, (7) weathershield assembly and gun barrel, (8) train system, (9) elevation system, and (10) Gun Mount control system. CG Yard ORF will classify all reclaimed materiel (principal end items) in terms of

readiness for issue and use, or to identify action(s) required to change the status of materiel. All serviceable component parts from the MK 75 Gun Mount's major assemblies shall be classified Supply Condition Code (SCC) "A" (Serviceable) per Department of Defense (DoD) regulations.

3.8.1.2.2 All unserviceable, but repairable, principal end items reclaimed from the MK 75 Gun Mount's major assemblies shall be classified SCC "F" (Unserviceable Repairable) per DoD regulations. Once repaired, the principal end item shall be reclassified to SCC "A" per DoD regulations.

3.8.1.2.3 All unserviceable principal end items (which are deemed "not repairable") will be managed and disposed of per DoD 4160.21, Volume 2.

3.8.2 Interferences. Any removed equipment not specifically identified as being replaced shall be retained for reinstallation.

3.8.3 Lighting and Alarms. The Contractor must temporarily remove lighting fixtures, fire and smoke alarms, and junction boxes obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-331-005, 905-WMEC-331-007, 905-WMEC-436-003 and 905-WMEC-153-002 as guidance.

3.8.4 Wireways. The Contractor must temporarily remove wireways obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-321-006 and 905-WMEC-153-002 as guidance.

3.8.5 Ventilation. The Contractor must temporarily remove HVAC duct work obstructing temporary access opening and removal of the MK 13 MOD 0 Air Dryer located under the Fan Room (02 Level) . Use Coast Guard Drawings 905-WMEC-514-006, 901-WMEC- 514-017 and 905-WMEC-153-002 as guidance.

3.8.6 Announcing System Loud Speaker. The Contractor must temporarily remove loud speaker obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-433-001, and 905-WMEC-153-002 as guidance.

3.8.7 AN/SLQ-32 Waveguide & Dry Air. The Contractor must temporarily remove the SLQ-32 waveguide and dry air system obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-441-001, 905-WMEC-401-006 and 905-WMEC-153-002 as guidance.

3.8.8 HVAC. The Contractor must remove the Fan coil Unit and chilled water piping within the Hoist Equipment room (1-26-1-E). The chilled water piping serving the cooling coil in Anchor Windlass RM & Bosun's Workshop (1-12-0-Q) should be modified. This work includes, but is not limited to, piping, valves, fittings, hose, hangers & insulation/lagging. Use Coast Guard Drawings 905-WMEC-514-033 as guidance.

3.8.9 Communications equipment. The Contractor must temporarily remove the audio amplifier AM 3729/SR, VHF speaker #5 and IFF Alarm obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-440-003 and 905-WMEC-455-001 as guidance.

3.9 Temporary equipment removal. See Table 1: Inspection Hold Points.

3.9.1 4010W-53 KU band antenna. The Contractor must remove the KU Band Antenna and foundation located on the 03 Level, temporarily removing it as an interference to allow access cuts in deck above CSC (02-63-0-C). Use Coast Guard Drawings 901-WMEC-441-008 and 905-WMEC-153-002 as guidance.

3.9.2 Antenna, Near Vertical Incidence Skywave (NVIS). The Contractor must remove the NVIS antenna located on the 03 Level, temporarily removing it as an interference to allow access cuts in deck above CSC (02-63-0-C). Use Coast Guard Drawings 905-WMEC-440-003, 905-WMEC-480-002 and 905-WMEC-153-002 as guidance.

3.10 Temporary access cuts.

3.10.1 The Contractor must remove all interferences to the temporary hull access opening and equipment removal.

3.10.2 The Contractor must cut the temporary access opening to remove the MK 92 FCS equipment. Temporary access opening is located in the 03 level above CSC (02-63-0-C). Use Std Spec 1001_STD, MLCA Std Spec 1991_STD, and 905-WMEC-153-002 as guidance. Submit CIR.

3.11 Cables. Contractor must cut cables 3-4 inches above the equipment connector leaving connector in place.

3.12 MK 92 Radar.

3.12.1 MK 53 MOD 0 Combined Antenna System. The Contractor must remove the MK 92 RADAR and modify the foundation from the 03 level. DO NOT cut the cables connecting the junction box to the Combined Antenna System (CAS) located just below the antenna unit. Safely secure the junction box to the CAS unit during removal as not to place stress on the cables. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.12.2 MK 13 MOD 1 Waveguide Air Dryer. The Contractor must remove the waveguide air dryer and tubing located in the Fan Room (02-45-0-Q). Use Coast Guard Drawings 905-WMEC-441-006, 905-WMEC-480-002, 905-WMEC-514-006, 905-WMEC-514-017 and 905-WMEC-184-016 as guidance.

3.12.3 Waveguide. The Contractor must remove the waveguide from the 03 Level to CSC (02-63-0-C). Save the spiral waveguide under the MK 53 CAS and all cabinet interconnecting waveguide runs. Waveguide running from equipment in the Command Support Center (02-63-0-C) to and from the MK 53 CAS are class custom fit and not required to keep. Use Coast Guard Drawing 905-WMEC-480-002 as guidance.

3.12.4 MK 142 MOD 0 Safety Switch. The Contractor must remove safety switch. Use Coast Guard Drawing 905-WMEC-480-002 as guidance.

3.13 MK 92 FIRE CONTROL EQUIPMENT

3.13.1 MK 69 MOD 0 Radar Receiver-Transmitter. The Contractor must remove the Radar Cabinet and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.2 Power Monitor Coupler Assembly. The Contractor must remove the Power Monitor Coupler Assembly. Use Coast Guard Drawings 905-WMEC-480-002 as guidance.

3.13.3 MK 160 MOD 0 Servo Control Cabinet. The Contractor must remove the Servo Control Cabinet and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.4 AN/UYK-7(V) Digital Computer. The Contractor must remove the Digital Computer and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.5 Input/output Console. The Contractor must remove the I/O Console and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.6 Fire Control Switchboard. The Contractor must remove the Fire Control Switchboard and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.7 MK 106 MOD 0 Weapons Control Console. The Contractor must remove the Weapons Control Console and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-184-016 as guidance.

3.13.8 MK 602 MOD 0 Radar Test Set. The Contractor must remove the Radar Test Set. Use Coast Guard Drawings 905-WMEC-480-002 as guidance.

3.13.9 MK 75 GWS Equipment.

3.13.9.1 MK 378 MOD 0 Gun Control Panel. The Contractor must remove the Gun Control Panel and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-713-004 as guidance.

3.13.9.2 MK 390 MOD 0 Water Control Panel. The Contractor must remove the Water Control Panel and foundation. Use Coast Guard Drawings 905-WMEC-480-002, 905-WMEC-256-014, 905-WMEC-533-010 and 905-WMEC-713-004 as guidance.

3.13.9.3 MK 56 MOD 0 Anti-Icing MDS. The Contractor must remove the Anti-Icing MDS and foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 905-WMEC-713-004 as guidance.

3.13.9.4 ASW piping. The Contractor must remove portions of the ASW piping local to the equipment. This work includes, but is not limited to, piping, valves, fittings, hose, hangers & insulation/lagging. Use Coast Guard Drawing 905-WMEC-256-014 as guidance

3.13.9.5 Fresh water piping. The Contractor must remove portions of the fresh water piping local to the equipment. This work includes, but is not limited to, piping, valves, fittings, hose, hangers & insulation/lagging. Use Coast Guard Drawing 905-WMEC-533-010 as guidance.

3.13.10 MK 75, 76MM, 62 caliber gun. The Contractor must remove the MK75 GUN, Slip Ring and inspect the foundation. Use Coast Guard Drawings 905-WMEC-480-002 and 901-WMEC-187-002 as guidance. Submit CIR.

3.13.11 Ventilation. The Contractor must remove portions of the ventilation system in the magazine 1-26-0-M and Hoist Equipment Room (1-26-1-E). The Contractor must remove the supply ventilation system in the magazine 1-26-0-M. This work includes, but is not limited to, removing ventilation ducts, heaters, supports, insulation, and dampers. Use Coast Guard Drawing 905-WMEC-512-018 as guidance.

3.13.12 Sprinkler system. The Contractor must remove portions of the sprinkler system within the magazine (1-26-0-M), Hoist Equipment Room (1-26-1-E), and Passageway (1-26-2-L), continuing to the cutout valve located adjacent to the passageway. This work includes, but is not limited to removing piping, hangers, sprinkler heads. Use Coast Guard Drawing 905-WMEC-522-004 as guidance.

3.13.13 Magazine 1-26-0-M. The Contractor must remove the false deck in Magazine 1-26-0-M. The Contractor must remove the scuttle from the Hoist Equipment Room (1-26-1-E). The Contractor must remove the doors to the Magazine to be lowered, and the Contractor must remove the associated steps. Use Coast Guard Drawings 905-WMEC-187-003 and 905-WMEC-713-004.

3.14 Installations. The Contractor must See Table 1: Inspection Hold Point

3.14.1 MK 38 MOD 3.

3.14.1.1 The Contractor must install a foundation adapter on the existing MK 75 foundation. This work includes, but is not limited to, fabricating and installing the adapter. Use Coast Guard Drawing 901-WMEC-187-002 as guidance.

3.14.1.2 The Contractor must install the MK 38 MOD 3 MGS on the foundation adapter. Use Coast Guard Drawings FL-711-037-1 and 905-WMEC-481-003 as guidance.

3.14.1.3 The Contractor must mount or install the MK 38 equipment in the Pilothouse (03-48-0-C) including but not limited to the Commander Safety Panel in its designated location. Use Coast Guard Drawing 905-WMEC-481-001 and 905-WMEC-481-003 as guidance.

3.14.1.4 The Contractor must install the MK 38 equipment in the Command Support Center (02-63-0-C). The Contractor is not supposed to install the MCP, MFD, or MCU in the ROC table that is contracted out to ISEA.. Use Coast Guard Drawings 905-WMEC-481-001, 905-WMEC-481-003 and 905-WMEC-184-015 as guidance.

3.14.1.5 The Contractor must relocate the sound-powered phone (SPP) Jack in the Command Support Center (02-63-0-C). The Contractor must relocate the SPP Jack in the Gun Control Booth (1-26-1-C) and install a junction box. Use Coast Guard Drawings 905-WMEC-481-001 and 905-WMEC-184-015 as guidance.

3.14.1.6 The Contractor must install MK 38 equipment in the Passageway (1-26-2-L) including but not limited to the charger assembly and junction box. See Coast Guard Drawings 905-WMEC-481-001 and 905-WMEC-184-015 as guidance.

3.14.2 Ventilation. The Contractor must install new ventilation ducts within the magazine (1-26-0-M) and Hoist Equipment Room (1-26-1-E). This work includes, but is not limited to, installing new ducting supports, insulation and dampers. Use Coast Guard Drawing 905-WMEC-512-018 as guidance.

3.14.3 Sprinkler system installation. The Contractor must install new sprinkler system components within the magazine (1-26-0-M) and Hoist Equipment Room (1-26-1-E). This work includes, but is not limited to installing new piping, hangers, sprinkler heads. Use Coast Guard Drawing 905-WMEC-522-004 as guidance.

3.14.4 Ammunition Hoist.

3.14.4.1 The Contractor must install new compressed air to the MK 38 GWS. Use Coast Guard Drawing 905-WMEC-551-009 as guidance.

3.14.4.2 The Contractor must install a new maintenance hoist in the Hoist Equipment Room (1-26-1-E). Use Coast Guard Drawings 905-WMEC-187-003 and 905-WMEC-712-003 as guidance. The system must be weight tested.

3.14.5 Ready service locker. The Contractor must install a new ready service locker. Use Coast Guard Drawing FL-713-002 as guidance.

3.14.6 Maintenance platform. The Contractor must install a new maintenance platform to service the MK 38. Use Coast Guard Drawing 905-WMEC-179-001 as guidance.

3.15 Replacing temporary access cut. The Contractor must reinstall the temporary access opening to remove located in the 03 level above CSC (02-63-0-C). Use Std Spec 1001_STD, MLCA Std Spec 1991_STD, and 905-WMEC-153-002 as guidance.

3.16 Equipment Reinstallation.

3.16.1 The Contractor must reinstall the KU Band Antenna and foundation on the 03 level that was temporarily removed. Use Coast Guard Drawings 901-WMEC-441-008 and 905-WMEC-153-002 as guidance.

3.16.2 The Contractor must reinstall the NVIS antenna located on the 03 level that was temporarily removed. Use Coast Guard Drawings 905-WMEC-440-003, 905-WMEC-480-002 and 905-WMEC-153-002 as guidance.

3.17 Touch-up Preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces, in accordance with COMDTINST M10360.3C, Coatings and Color Manual.

3.18 Insulation. The Contractor must replace any disturbed insulation.

3.19 Interferences. The Contractor must replace all interferences to include but not limited to lighting and alarms, wire ways, duct work, loud speaker and AN/SLQ-32 wave guide. Use Coast Guard Drawings 905-WMEC-321-006, 905-WMEC-331-007, 905-WMEC-433-001, 901-WMEC-441-008, 905-WMEC-401-006, 905-WMEC-436-003, 905-WMEC-514-006, 905-WMEC- 514-017 and 905-WMEC-153-002 as guidance.

3.20 Operational test. After completion of work, the Contractor must witness an operational test (by Coast Guard personnel) of all mechanical and electrical items or shipboard devices that have been disturbed, used, repaired, altered or installed to prove that they are in satisfactory operational condition.

3.21 Labeling. All Hoses, electrical cables, controllers, and other equipment must be properly labeled IAW approved references.

3.22 Hydrostatic Testing. All new and modified piping must be hydrostatic tested in accordance with test pressures listed in their respective Coast Guard Drawings.

3.23 Upon completion of work the contractor shall use Table 3 as a checklist to meet NOSSA compliance:

TABLE 3: NOSSA COMPLIANCE CHECKLIST

ITEM #	LOCATION	SYSTEM/CATEGORY	DESCRIPTION	RESPONSIBILITY	NOTES	DRAWINGS
1	FORECASTLE	DECK	RED DANGER AREA CIRCLE PAINTED IAW DRAWING	CGY		905-WMEC_634_4 901-WMEC_634_4
2	FORECASTLE	RSL	FABRICATED AND INSTALLED IAW DRAWINGS	CGY	Non-metallic washers must be used on either side of dogs.	905-WMEC_180_2 901-WMEC_180_1
3	FORECASTLE	RSL	HIGH SECURITY LOCK INSTALLED	CREW		
4	FORECASTLE	RSL	SAFETY PLACARDS INSTALLED	CREW		
5	FORECASTLE	RSL	TEMP GAGE, CARD AND CARD HOLDER INSTALLED	CREW		
6	MAGAZINE	DECK	NON-SKID DECK PADS INSTALLED	CGY		905-WMEC_713_4 901-WMEC_713_8
7	MAGAZINE	ALARMS	FIRE DETECTION SENSORS AND ALARMS OPERATIONAL AND IN GOOD WORKING CONDITION	CGY	Need to ensure detectors are not coated with paint or damaged.	
8	MAGAZINE	ALARMS	FLOODING DETECTION SENSORS AND ALARMS OPERATIONAL AND IN GOOD WORKING CONDITION	CGY		
9	MAGAZINE	ALARMS	HIGH TEMP SENSORS (3EA, 105°F) OPERATIONAL	CGY	Need to ensure detectors are not	

			AND IN GOOD WORKING CONDITION		coated with paint or damaged. SPENCER has one that either needs to be cleaned up or replaced (STBD-most detector).	
10	MAGAZINE	AMMO HANDLING	AMMO HOIST INSTRUCTION PLACARD INSTALLED	CGY		905-WMEC_712_3 901-WMEC_712_3
11	MAGAZINE	AMMO HANDLING	HOLD-OPEN CAPABILITY INSTALLED ON MAGAZINE WTD AND AFT GCP WTD	CGY	The old hold back devices will need to be adjusted to meet the new door height so that it can be latched open.	
12	MAGAZINE	PHYSICAL SECURITY	MAGAZINE WTH TO MAIN DECK NEEDS LOCKING PINS INSTALLED (INSIDE)	CGY	Had to add these on HARLAN to prevent someone from getting into the magazine by popping out the pins on the main deck.	905-WMEC_712_3 901-WMEC_712_3
13	MAGAZINE	PHYSICAL SECURITY	MAGAZINE WTD NEEDS LOCKING PIN INSTALLED (INSIDE)	CGY	Needed to pin the magazine door from the inside so that GMs can work in	905-WMEC_713_4 901-WMEC_713_8

					the armory where they can't visually monitor the magazine WTD.	
14	MAGAZINE	PHYSICAL SECURITY	HIGH SECURITY HASP INSTALLED ON MAGAZINE WTD	CGY	Hasp & Lock Info	
15	MAGAZINE	PHYSICAL SECURITY	HIGH SECURITY LOCK INSTALLED ON MAGAZINE WTD	CREW	Hasp & Lock Info	
16	MAGAZINE	MISC	MAGAZINE TEMP GAGE INSTALLED NEAR WTD ENTRANCE	CGY		
17	MAGAZINE	MISC	MAGAZINE TEMP CARD AND HOLDER INSTALLED NEAR TEMP GAGE	CGY		
18	MAGAZINE	PLACARDS	AMMUNITION FAR SIDE PLACARDS INSTALLED IN ADJACENT COMPARTMENTS	CREW	Source of Supply	
19	MAGAZINE	PLACARDS	GENERAL SAFETY PRECAUTION PLACARD INSTALLED	CREW	Source of Supply	
20	MAGAZINE	MISC	DECK DRAIN INSTALLED AND OPERATIONAL	CGY		905-WMEC_514_33 901-WMEC_514_54
21	MAGAZINE	PHYSICAL SECURITY	INTERNAL SECURITY HINGE BRACKETS INSTALLED	CGY		

			ON MAGAZINE WTD			
22	MAGAZINE	MISC	NEW (NEALS) EMERGENCY LIGHTING FIXTURES INSTALLED	CGY	If these were not purchased and installed, at a minimum, two battery operated battle lanterns must be installed. One facing magazine WTD and the other facing the armory WTS.	
23	MAGAZINE	MISC	ELECTRICAL JUNCTION BOXES AND FIXTURES THROUGHOUT MAGAZINE ARE WATERTIGHT AND NOT MISSING SCREWS	CREW OR CGY?	We had several last- minute things to do with these boxes/fix- tures on HARLAN. Probably pre- existing.	
24	MAGAZINE	MISC	CABLE PENETRATION S INTACT/PROPE- RLY SEALED	CGY		
25	MAGAZINE	AMMO STORAGE	BATTEN STORAGE SYSTEM PROPERLY INSTALLED	CGY	Need to ensure all areas of the grating are secured so that there is no movement . Had to revisit several times on	905- WMEC_712 _3 901- WMEC_712 _3

					HARLAN.	
26	MAGAZINE	MAG SPRINKLING	SPRINKLER PIPING BRAZED VICE MECHANICALLY CONNECTED	CGY		905-WMEC_522_4 901-WMEC_522_3
27	MAGAZINE	MAG SPRINKLING	SPRINKLER HEADS ARE UNOBSTRUCTED	CGY	We had to lower a few on HARLAN due to obstruction by lighting fixtures and piping.	905-WMEC_522_4 901-WMEC_522_3
28	MAGAZINE	MAG SPRINKLING	SPRINKLER HEADS INSTALLED THROUGH OVERHEAD GRATING OF BATTEN STORAGE	CGY	CGY sent pictures	905-WMEC_522_4 901-WMEC_522_3
29	MAGAZINE	MAG SPRINKLING	FIREMAIN GAGE HAS ADJUSTABLE RED POINTER	CREW OR CGY?		
30	MAGAZINE	MAG SPRINKLING	MAGAZINE SPRINKLER HAS 3-WAY VALVE LOCATED OUTSIDE COMPARTMENT WITH LOCKING BAR ACROSS HANDLE TO PREVENT ACCIDENTIAL ACTIVATION	CREW OR CGY?		
31	MAGAZINE	MAG SPRINKLING	MAGAZINE SPRINKLER FIREMAIN GAUGE(S) CALIBRATED	CREW OR CGY?	Calibration Frequency Message	
32	MAGAZINE	MAG SPRINKLING	MAGAZINE SPRINKLER HAS SPRING TO POSITIVELY	CREW		905-WMEC_522_4 901-WMEC_522

			HOLD IN DETENT			_3
33	MAGAZIN E	MAG SPRINKLING	MAGAZINE SPRINKLER VALVES PROPERLY LABELED TO INCLUDE OPERATING INSTRUCTION S	CREW		
34		PLACARDS	PYRO LOCKER - SAFETY PLACARDS INSTALLED	CREW	Source of Supply	
35		PYRO LOCKER	TEMP GAGE, CARD AND HOLDER INSTALLED	CREW	-	

4. NOTES

4.1 Table 3 lists inspections and/or activities to support quality assurance efforts. Upon completion of each task, submit a CFR and attach CGY-252 and / or drawing redline mark-ups as suitable.

TABLE 3. INSPECTION HOLD TABLE

TASK NO.	SECTION	SHOPS	ACTIVITY	SCHEDULE	CG YARD ENGINEER WITNESS
1	3.9.1	30	Perform functionality test	Prior to removal	Y
2	3.9.2	30	Perform functionality test	Prior to removal	Y
3	3.13.7	30	Perform functionality test	Prior to removal	Y
4	3.13.8	30	Perform functionality test	Prior to removal	Y
5	3.13.9	30	Perform functionality test	Prior to removal	Y
6	3.13.10	30	Perform functionality test	Prior to removal	Y
7	3.13.10	10, 64	Inspect existing foundation for structural integrity and agreement with drawings.	Upon removal of MK 75. Submit CIR.	Y

TASK NO.	SECTION	SHOPS	ACTIVITY	SCHEDULE	CG YARD ENGINEER WITNESS
8	3.14.1.1	10, 64	Verify new foundation installations for MK 38 and its equipment match drawings	Upon installation of new foundations	Y
9	3.14.2	21, 65	Verify installation of ventilation system matches drawing	Prior to insulating	Y
10	3.14.2	21, 65	Final inspection of ventilation system for completeness	Prior to operational test	Y
11	3.14.3	21, 65	Verify installation of sprinkling system matches drawing	Prior to hydrostatic testing	Y
12	3.14.3	21, 65	Final inspection of sprinkling system for completeness	Prior to operational test	Y
13	3.14.4.1	21, 65	Verify installation of compressed air system matches drawing	Prior to hydrostatic testing	Y
14	3.14.4.1	21, 65	Final inspection of compressed air system for completeness	Prior to operational test	Y
15	3.14.4.2	21, 65	Test the lifting capacity of the ammo hoist	Prior to certification	Y
16	3.14.5	10, 64	Verify new ready service locker matches drawing	Upon installation of new equipment	Y
17	3.14.6	10, 64	Verify new MK 38 maintenance platform matches drawing	Upon installation of new equipment	Y

WORK ITEM 43: Shipboard Equipment and Systems, Lay Up

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to lay-up and restore various shipboard equipment for a 270 WMEC during a long term major maintenance availability.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	Battery Assembly (MK39 Gyro UPS)	NSN: 6140-01-513-8204 PN: 1982847	1	\$11,861.24
N	Battery Set (Speed Control UPS)	NSN: 6140-01-530-6548 PN: 223-14535-2	1	\$837.15
N	Battery Assembly (MK27 Gyro UPS)	NSN: XFC066832 PN: 8D	1	

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

***Government-furnished property, which is to be supplied by either the vessel or the C4IT Service Center.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901 WMEC_233_XXX, BLANKING PLATES FOR LAYUP OF ALCO 251 MDE

COAST GUARD PUBLICATIONS

Coast Guard Technical Publication (TP) 4647, FM/ALCO 251 Engine – Operation/Maintenance
Surface Forces Logistics Center Standard Specification 0000 (SFLC Std Spec 0000), 2022,
General Requirements

OTHER REFERENCES

ASTM International (ASTM) D882, 2009, Standard Test Method for Tensile Properties of Thin Plastic Sheeting
ASTM International (ASTM) D1709, 2008, Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method
ASTM International (ASTM) D1922, 2008, Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by the Pendulum Method

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

- None

None.

3.1.2 Tech Rep. The Contractor must provide the required materials and services of a qualified Tech Rep, who is familiar with the ALCO Model 251 engines to accomplish the following tasks – on site:

- Assist in proper engine lay-up and restoration of an engine that is idle for an extended period (greater than 6 months).
- Coast Guard Yard will serve as the Tech Rep.
- Diesel engines must be barred over once a month for the duration of the availability, in the presence of PRO Baltimore inspector.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

3.2 Equipment lay-up particulars. The Contractor must accomplish the tasks in TABLE 1 below, as applicable, in additional to tasks specified in the follow-on sub-paragraphs.

TABLE 1: EQUIPMENT LAY-UP

SYSTEM/EQUIPMENT	LAY-UP PROCEDURES
Batteries	Renew MK39 Gyrocompass UPS battery, Digital Speed Control UPS battery, Fire Alarm panel batteries, and lead acid battery banks in the engine room, aft steering, and comms room at conclusion of MMA.
MPCMS Batteries	Renew all MPCMS batteries.
Reduction Gears	Barred over quarterly two complete shaft revolutions
CPP System	Power will be provided to system for operation of CPP sump heaters and pumps
Exhaust Stacks	The Contractor must take the following protective measures to prevent contamination of non-affected shipboard equipment: Plug, blank, wrap, cover, seal the exhaust stacks using water repellent materials, and prevent entry of contaminants to the MDEs. The protective measure must be suitable to withstand extreme weather conditions such as high winds, rain and snow.
Chill Water Piping System	Open all chill water valves to ensure system ready. Remove chill water pumps and install temporary flushing rig. Fill system with anti-freeze additive and cycle to ensure a complete mixture. If CW piping is to be cropped out for interferences the isolation valves shall be closed for coils to maintain antifreeze in the coil.

Countermeasure Washdown	Open all valves to access all (5) sections, drain water, and apply low pressure air to push all remaining water out of the system. Leave valves open.
Firemain System	Open all valves, drain water, and apply low pressure air to push all remaining water out of the system. Leave valves open.
JP5 Tanks	Commercially clean tanks. Completely drain and dry tanks.
Potable Water Tanks	Commercially clean tanks. Completely drain and dry tanks.
CB-OTH-IV	Assist crew with offloading cutter boat onto trailer provided by SBPL. (SBPL will take possession of OTH-IV for CCBP)
20' CB-L	Assist crew with offloading cutter boat onto trailer provided by SBPL. Provide suitable storage location for the duration of MMA. Shrink wrap the cutter boat once cutter crew completes lay-up procedure.
MK39 MOD3A Gyrocompass	Protect IAW MPC E-C-11821 if conducting welding in the vicinity of the gyrocompass equipment.
MK27 Gyrocompass	Protect IAW 210-WMEC MPC E-C-11902 and E-C-11903 if conducting welding in the vicinity of the gyrocompass equipment.

3.2.1 Shrink wrapping particulars. The Contractor must provide 8-mil marine grade, heat shrink film having a minimum of 3%, ultra-violet inhibitor content that conforms to ASTM D882, D1709, and D1922. Contractor must wrap deck equipment.

3.2.1.1 Pad/cover all sharp protrusions and surfaces that could puncture the covering.

3.2.1.2 Where necessary, fabricate and install ridge poles to support the film in a manner to discourage precipitation pooling or build-up.

3.2.1.3 Seal all seams, pleats, openings, protrusions, gussets and reinforcements in the manner prescribed by the manufacturer for this type enclosure.

3.2.1.4 Inspect the protective covering's integrity at the beginning of each shift during which surface preparation and/or painting will be accomplished.

3.2.1.4.1 Ensure equipment and machinery have not been infiltrated by contaminants.

3.2.1.4.2 If contamination or surface damage has occurred, notify the COR immediately in writing and reseal to prevent further contamination or surface damage.

3.2.1.5 Upon work completion, remove protective material and inspect for the presence of contamination due to improper protection. Clean all contaminated equipment and spaces to original condition of cleanliness.

3.3 Support services. The Contractor must provide access to high volume low pressure compressed air to support winterization efforts conducted by the cutter crew. A regulator must be provided to adjust discharge pressures as low as 10 PSI.

3.4 System restorations. Upon completion of work, the Contractor must accomplish all tasks to return the systems and equipment listed in Table I to an operational readiness condition. This must include, but not limited to, disconnection of temporary services, restoring ship connections, removal of all temporary protective measures (including drainage of preventative fluids), and system fluid refilling. Submit a CFR if additional actions are required beyond the scope of the initial lay-up actions.

3.5 Operational test – post repairs. After completion of work, the Contractor must thoroughly test, in the presence of the Coast Guard Inspector and demonstrate the affected shipboard systems and equipment to be in satisfactory operating condition. Submit a CFR.

4. NOTES

4.1 This section not applicable.

DRAFT

WORK ITEM 44: Ship Service Diesel Generator And Emergency Generator, Replace

1. SCOPE

1.1 Intent. This work item describes the requirements for the Contractor to replace the existing Ships Service Diesel Generator (SSDG) Sets and Emergency Generator (EDG) Sets with new CAT C18 generator sets. This work item includes upgrades to the Ships Service Switchboard (SWBD), Emergency Switchboard, generator foundations, and generator support systems.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	PN	QTY	ESTIMATED COST (\$/UNIT)
N	Caterpillar C18	PN: CAT 497-9131	3	933,637.00
N	CAT C18 Air Starter	PN: CAT 105-5081 IR 7C3372	2	3,617.72
N	CAT C18 Electric Starter	PN: CAT 6V-5539	1	259.72
N	SSDG Silencer, Est. Cost Incl. w/ Item 1.		2	
N	EDG Silencer, Est. Cost Incl. w/ Item 1		1	
N	Ships Service Switchboard Retrofit kits, Est. Cost Incl. w/ Item 1		2	
N	Emergency Switchboard Retrofit Kits, Est. Cost Incl. w/ Item 1		1	
Y	EDG Raw Water Pump MEPCO	PN: ML11418 RP06-20-100-34-3	1	\$15,688.00

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 905-WMEC-101-001, STRUCTURAL PENETRATIONS A AND D

Coast Guard Drawing 905-WMEC-111-008, SLEP CAT C18 SSDG AND EDG ACCESS CUTS & COFFERDAM

Coast Guard Drawing 905-WMEC-130-001, SLEP TEMP ACCESS CUTS FOR MDE, SSDG & EDG REMOVAL

Coast Guard Drawing 905-WMEC-169-002, SLEP TEMPORARY HELO HANGAR REMOVAL

Coast Guard Drawing 905-WMEC-183-001, DIESEL GEN FDNS

Coast Guard Drawing 905-WMEC-183-002, ELECTRICAL FOUNDATIONS ENGINE ROOM

Coast Guard Drawing 905-WMEC-183-011, SLEP CAT C18 SSDG AND EDG FOUNDATIONS

Coast Guard Drawing 905-WMEC-183-012, SLEP CAT C18 SSDG AND EDG ANCILLARY EQUIPMENT FOUNDATIONS

Coast Guard Drawing 905-WMEC-201-001, ENGINE ROOM ARR

Coast Guard Drawing 905-WMEC-202-011, MPCMS UPGRADE ENGINEERING CONTROL CENTER CONSOLE ELECTRICAL MODS

Coast Guard Drawing 905-WMEC-233-007, BLANKING PLATS FOR LAYUP OF ALCO 251 MDE

Coast Guard Drawing 905-WMEC-252-012, MPCMS INTCON WRG DIAG

Coast Guard Drawing 905-WMEC-256-002, ENG RM SW COOLING SYS A&D

Coast Guard Drawing 905-WMEC-256-005, MISC EQPT SW COOLING A&D

Coast Guard Drawing 905-WMEC-256-007, SSDG MODIFICATION - AUX SEA WATER PIPING MOD

Coast Guard Drawing 905-WMEC-256-008, SSDG JACKET WATER COOLING SYSTEM AND MODS

Coast Guard Drawing 905-WMEC-256-011, MN AND AUX SW COOLING SYS DIAG

Coast Guard Drawing 905-WMEC-256-013, SLEP EDG SEA WATER COOLING PUMP REPLACEMENT

Coast Guard Drawing 905-WMEC-259-001, DIESEL EXH

Coast Guard Drawing 905-WMEC-259-002, DIESEL EXHAUST SYSTEM A and D

Coast Guard Drawing 905-WMEC-259-003, SSDG EXHAUST SYSTEM MODS

Coast Guard Drawing 905-WMEC-259-004, SLEP EXHAUST SYSTEM DIAGRAM

Coast Guard Drawing 905-WMEC-260-001, EMER. F.O. SHT DN ARR AND DETS

Coast Guard Drawing 905-WMEC-261-001, F. O. SERV SYS DIAG

Coast Guard Drawing 905-WMEC-261-002, F. O. SERV SYS A AND D

Coast Guard Drawing 905-WMEC-261-004, MODIFICATIONS TO SSDG FUEL OIL SERVICE SYSTEM FILTER & PIPING

Coast Guard Drawing 905-WMEC-262-008, LUBE OIL SYS DIAG

Coast Guard Drawing 905-WMEC-262-002, LUBE OIL SERVICE SYSTEM A&D

Coast Guard Drawing 905-WMEC-264-003, LUBE OIL SPINNER II INSTALL

Coast Guard Drawing 905-WMEC-310-001, S/S DSL GEN INSTL AND ARR

Coast Guard Drawing 905-WMEC-310-002, S/S DSL GEN LO SYS DIAG

Coast Guard Drawing 905-WMEC-310-008, EMER DSL GEN A&D

Coast Guard Drawing 905-WMEC-310-009, EMER DSL GEN LO SYS DIAG

Coast Guard Drawing 905-WMEC-310-0015, SHIP SERVICE AND EMERGENCY DIESEL FUEL OIL SYSTEM DIAGRAM

Coast Guard Drawing 905-WMEC-310-0016, SLEP SSDG ARRANGEMENT

Coast Guard Drawing 905-WMEC-310-017, SLEP EDG ARRANGEMENT

Coast Guard Drawing 905-WMEC-310-018, SLEP CAT C18 SSDG & EDG CGMCS WIRING MODIFICATIONS

Coast Guard Drawing 905-WMEC-311-004, ELECTRIC PLANT LOAD ANALYSIS (PRE-SLEP)

Coast Guard Drawing 905-WMEC-320-014, AS-BUILT ELECTRICAL ONE-LINE DIAGRAM

Coast Guard Drawing 905-WMEC-320-030, EMER SWBD GEN & BUS TIE & SSG'S BRKR REPL

Coast Guard Drawing 905-WMEC-320-025, SSDG MODIFICATION - POWER MODIFICATION

Coast Guard Drawing 905-WMEC-320-053, MPCMS ECCC INTERNAL WIRING/POWER DISTRIBUTION DETAILS

Coast Guard Drawing 905-WMEC-320-057, SLEP EMERGENCY SWBD & EDG INSTALLATION DETAILS

Coast Guard Drawing 905-WMEC-320-058, SLEP SHIP SERVICE SWBD and SSDG INSTALLATION DETAILS

Coast Guard Drawing 905-WMEC-320-059, SLEP CAT C18 LUBE OIL TANK ELECTRICAL WIRING DIAGRAM

Coast Guard Drawing 905-WMEC-331-004, LIGHTING SYSTEM FIRST PLATFORM

Coast Guard Drawing 905-WMEC-342-001, SLEP SSDG SUPPORT SYSTEM MODS

Coast Guard Drawing 905-WMEC-342-002, SLEP EDG SUPPORT SYSTEM MODS

Coast Guard Drawing 905-WMEC-342-003, SLEP SSDG and EDG EXHAUST MODIFICATIONS

Coast Guard Drawing 905-WMEC-342-004, SLEP CAT C18 SSDG & EDG FUEL OIL MODIFICATIONS

Coast Guard Drawing 905-WMEC-342-005, SLEP CAT C18 SSDG STARTING AIR MODIFICATIONS

Coast Guard Drawing 905-WMEC-342-006, SLEP CAT C18 SSDG LUBE OIL MODIFICATIONS

Coast Guard Drawing 905-WMEC-436-003, FIRE and SMOKE DETECTION & QUARTERDECK MONITORING SYSTEM BLK & IWD

Coast Guard Drawing 905-WMEC-475-004, DEGAUSSING SYSTEM CONN BOX WIRING DIA and DET

Coast Guard Drawing 901-WMEC-504-003, SSDG TEMPERATURE GAUGES

Coast Guard Drawing 905-WMEC-505-10, SLEP SSDG PROTOTYPE MISC INTERFERENCES KEY PLAN

Coast Guard Drawing 905-WMEC-505-011, SLEP EDG MISC INTERFERENCES KEY PLAN

Coast Guard Drawing 905-WMEC-505-012, SLEP MDE AUXILARY SYSTEMS INTERFERENCE REMOVALS KEY PLAN

Coast Guard Drawing 905-WMEC-507-001, VALVE LABEL PLATE LIST

Coast Guard Drawing 905-WMEC-507-002, MACHRY LABEL PLATES

Coast Guard Drawing 905-WMEC-512-014, VENT MODS TO STEERING GEAR ROOM 3-228-O-E

Coast Guard Drawing 905-WMEC-513-005 VENT, ER & ECC – PLAN SECT & ELEV A&D

Coast Guard Drawing 905-WMEC-517-001, HEAT RECOVERY/SS DSL GEN JW SYS DIAG

Coast Guard Drawing 905-WMEC-517-002, WASTE HEAT RECOVERY /SHIP SERVICE DIESEL GENERATOR JACKET WATER - A & D

Coast Guard Drawing 905-WMEC-521-008, WASHDOWN COUNTERMEASURE (WDCM) SYSTEM DIAGRAM

Coast Guard Drawing 905-WMEC-521-009, WASHDOWN COUNTERMEASURE SYSTEM A & D MN, DK, & BELOW

Coast Guard Drawing 905-WMEC-551-001, DIAGRAM COMPRESSED AIR SYSTEM DIAGRAM

Coast Guard Drawing 905-WMEC-551-002, COMPRESSED AIR MACH SP - A AND D

Coast Guard Drawing 905-WMEC-611-006 SLEP SSDG PADEYE INSTALLATION DETAILS

Coast Guard Drawing 905-WMEC-622-004, SLEP CAT C18 SSDG DECK GRATING MODS

Coast Guard Drawing 905-WMEC-622-005 SLEP CAT C18 SSDG HANDRAILS

Coast Guard Publications

Coast Guard Commandant Instruction (COMDTINST) M10360.3C, Coatings and Color Manual

Coast Guard Commandant Instruction (COMDTINST) M10360.3, Appendix B (Cutter and Boats Interior Painting System)

Coast Guard Commandant Instruction (COMDTINST) M9000.6F, Naval Engineering Manual

Coast Guard Commandant Instruction (COMDTINST) M4500.5 (series), Property Management Manual

Coast Guard Commandant Instruction (COMDTINST) M5100.47 (series), Safety and Environmental Health Manual

Coast Guard Commandant Instruction (COMDTINST) 9077.1 (series), Equipment Tag-Out Procedure Manual

Coast Guard Tech Pub 2820 (Series), Ship Service Diesel Generator Set

Coast Guard Tech Pub 2821 (Series), Emergency Diesel Generator Set

Operational Test for Degaussing System

Operational Test for MDE Exhaust

SFLC 083-8.4 Markings, Packaging, Delivery, and Inspection/Acceptance Requirements

Surface Forces Logistics Center Standard Specification 0000, (SFLC Std Spec 0000) 2022 Edition, General Requirements

Surface Forces Logistics Center Standard Specification 0740, (SFLC Std Spec 0740) 2022 Edition, Welding and Allied Processes

Surface Forces Logistics Center Standard Specification 3041, (SFLC Std Spec 3041) 2022 Edition, Shipboard Electrical Cable Test

Surface Forces Logistics Center Standard Specification 3042, (SFLC Std Spec 3042) 2022 Edition, Shipboard Electrical Cable Removal, Relocation, Splice, Repair and Installation

Surface Forces Logistics Center Standard Specification 6310, (SFLC Std Spec 6310) 2022 Edition, Requirements for Preservation of Ship Structures

OTHER REFERENCES

270' WMEC SLEP Shipboard Integration Test Plan
Department of Defense Standard Practice, DOD-STD-2003, Electric Plant Installation Standard Methods for Surface Ships and Submarines
DoD 4160.21, Volume 2, Defense Materiel Disposition: Property Disposal and Reclamation
IEEE 45 - IEEE Recommended Practice for Electrical Installation on Shipboard (IEEE Std 45, 2002)
Institute of Electrical and Electronics Engineers Standard, IEEE-45, Recommended Practice for Electrical Installations on Shipboard
MIL-DTL-15024, General Specification for Plates, Tags, and Bands for Identification of Equipment
MIL-DTL-24643, General Specification for Low Smoke Electric Cables and Cords for Shipboard Use
MIL-STD-1310 (SERIES), Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety
MIL-STD-1399, Interface Standard for Shipboard Systems
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 1: Cable)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 2: Equipment)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 3: Penetrations)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 4: Cableways)
NAVSEA Drawing 803-1385866, Rev E, Penetration, Bulkhead and Deck
NAVSEA Drawing 804-5773931, Rev A, Acoustic and Thermal Insulation for Compartments Installation Details
NAVSEA S9086-SN-STM-010, "Naval Ships Technical Manual Chapter 541, Ship Fuel and Fuel Systems" Oct 2005
NAVSEA S9AA0-AB-GOS-010 "General Specifications for Overhaul (GSO) of Surface Ships", April 2016
OSHA 1926.501, DUTY TO HAVE FALL PROTECTION
OSHA 1910.29, FALL PROTECTION SYSTEMS AND FALLING OBJECT PROTECTION-CRITERIA AND PRACTICES
Title 46, Code of Federal Regulations, Subchapter J – Electrical Engineering

3. REQUIREMENTS

3.1 General.

3.1.1 CIR. The Contractor must submit a CIR for the inspections listed in the following paragraph(s):

- 3.2.1 Temporary Access Cut SSDG

- 3.3.4.1 SSDG Foundation
- 3.5.4.3 Temporary Access Cut EDG
- 3.5.5.1 EDG Foundation

3.1.2 Technical Representative. The Government will provide the services of a Technical Representative for the Caterpillar C18 GenSet. The Technical Representative will provide onsite technical support for system testing, commissioning, certification and onboard training of the ship's personnel for the new system.

3.1.2.1 Ensure the Tech Rep is an OEM Certified Representative for the system/equipment stated above and demonstrated on their résumé.

3.1.2.2 Thirty days prior to tech representative services the Contractor must contact COR.

3.1.3 Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must be aware that interferences in way of work include, but are not limited to the below listed. Handle all interferences in accordance with SFLC Std Spec 0000, (Interferences).

- Piping
- Fluid removal (see Table 1)
- Sheathing and Lagging
- Deck plating and structural framework
- Main Diesel Engine (MDE) air boxes and structural supports
- Insulation
- Duct work
- Lighting
- Fire/Smoke detectors
- Wiring
- Hot water tank in 2-82-0-E and piping
- Panels 3-117-1, 3-117-3, 3-117-2, 3-117-4
- Loudspeakers on stanchion at Frame 117
- Handrails around aft end of MDEs (port and starboard)
- Supply/exhaust in Aft Steering, including fan and ducting below Main Deck and gooseneck on Fantail
- Access panels under ship service switchboards

3.1.5 Personnel qualification.

3.1.5.1 Electricians: The Contractor must ensure that all electrical/electronic equipment removals and installations are overseen by an Electrical/ Electronics Work Leader, Journeyman, or Supervisor.

Electrical workers must be knowledgeable of electrical standards and practices implemented by the US Coast Guard in accordance with SFLC STD Spec 0000.

3.1.5.2 Machinery: The Contractor must ensure that all mechanical and piping equipment removals, installations and alignments are overseen by a Pipefitter/Machinist Work Leader, Journeyman, or Supervisor.

3.1.5.3 Structural: The Contractor must ensure that all structural work is overseen by a structural Work Leader, Inspector, or Supervisor. Structural workers must be knowledgeable of structural standards and practices implemented by the US Coast Guard in accordance with SFLC STD Spec 0740.

3.1.6 Related work items. The Contractor shall accomplish this work item in conjunction with Work Item 79, Main Diesel Engines, Replace and Work Item 78 Coast Guard Machinery Control System, Update; which is a separate work item in this specification package.

3.1.7 Tag-out. The Contractor must secure and tag out all equipment affected by this work item in accordance with SFLC Std Spec 0000, General Requirements, paragraph 2.

3.1.8 Gas freeing. When entering a tank, compartment, confined space, or conducting hot work, the Contractor must refer to and comply with SFLC Std Spec 0000, General Requirements, paragraph 3.3.1.2.1.

3.1.9 Welding and brazing. The Contractor must perform all welding, brazing, and non-destructive testing (NDT) in accordance with SFLC Std Spec 0740, Welding and Allied Processes.

3.1.10 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces, in accordance with SFLC Std Spec 6310, Requirements for Preservation of Ship Structures.

3.1.11 Environmental compliance. Contractor must dispose of all system fluids, cleaning and flushing fluids, and wastes in accordance with SFLC Std Spec 0000, Appendix A, Requirements for Environmental Protection.

3.1.12 Electrical requirements. The Contractor must accomplish all electrical work in accordance with SFLC Std Spec 3042 and test cables in accordance with SFLC Std Spec 3041.

3.1.13 Temporary hull access opening. The Contractor must gain access to the engine room (3-103-0-E) and aft steering compartment (3-228-0-E) to remove two (2) SSDG's and one (1) EDG, in accordance with SFLC STD SPEC 8636, Temporary Hull Access.

3.1.14 Staging. Staging will be required in the engine room (3-103-0-E) to mount lifting padeyes in the overhead, remove temporary interferences, remove exhaust, run cables, install exhaust, and re-install temporary interferences.

3.1.15 Temporary decking. Remove, retain and store deck plates as needed for access to piping and mechanical systems. Label original location. Create temporary decking to ensure personnel may safely transverse working areas. Temporary decking shall be no less than 3/4" plywood cut to fit tightly to deck angles when deck support angles are present. Larger openings spanning missing deck support angles or covering areas of missing machinery shall be no less than 2"x12" dimensional lumber of suitable length to span the opening.

3.1.15.1 The Contractor must install and maintain temporary handrails around temporary access cuts and around fall hazards created by large equipment removal IAW OSHA1926.501 and OSHA1910.29.

3.1.16 Joint inspections. The Contractor must abide by the joint inspections laid out in Table 2: Joint Inspection Hold Points throughout the work item.

3.1.17 Equipment lay-up. The Contractor must drain the MTI GenSets in accordance with shipping standards.

3.2 SSDG temporary access cuts.

3.2.1 Removal route. The Contractor must remove two (2) SSDGs and ancillary equipment. The Contractor must cut the temporary access opening to remove the SSDG's and associated equipment. Temporary access opening is located port side engine room (3-103-0-E) above the waterline. Use Std Spec 1001_STD, MLCA Std Spec 1991_STD, and 901-WMEC 111 as guidance. Submit CIR

3.2.2 Interferences. The Contractor must remove all interferences in the vicinity of the temporary hull access opening and in the vicinity of the transit path of the SSDGs. Use Coast Guard Drawing 905-WMEC-505-010 and 905-WMEC-505-012 as guidance.

3.2.3 Inspection. The Contractor must perform a visual inspection of exposed areas after interferences are removed. Submit CFR if damage or excessive corrosion is found.

3.2.4 Electrical interferences.

3.2.4.1 The Contractor must temporarily remove lighting fixtures, fire and smoke alarms, and junction boxes obstructing temporary access opening. Use Coast Guard Drawings 905-WMEC-331-004, 905-WMEC-436-003, and 905-WMEC-111-008 as guidance.

3.2.4.2 The Contractor must remove the existing SSDG gauge panels and all associated cabling. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.2.4.3 The Contractor must remove the existing SSDG starting panels and all associated cabling. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.2.5 Mechanical interferences.

3.2.5.1 The Contractor must remove portions of the auxiliary seawater piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance. All disconnected piping must be capped to prevent leakage and contamination.

3.2.5.2 The Contractor must remove portions of the fuel oil piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, and hangers and shielding. Use Coast Guard Drawing 905-WMEC-342-004 as guidance. All disconnected piping must be capped to prevent leakage and contamination.

3.2.5.3 The Contractor must remove portions of the jacket water piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, gauges, heat exchangers, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance.

3.2.5.4 The Contractor must remove portions of the exhaust local to the equipment. This work includes, but is not limited to, exhaust piping, silencers, expansion joints, foundations, and insulation/lagging. Inspect the remaining exhaust and clean. Use Coast Guard Drawing 905-WMEC-342-003 as guidance. Submit CFR.

3.2.5.5 The Contractor must remove portions of the ship's service compressed air system local to the SSDG's. This work includes, but is not limited to, piping, hoses, valves, fittings and hangers. Use Coast Guard Drawing 905-WMEC-342-005 as guidance.

3.2.5.6 The Contractor must remove portions of the lube oil piping system and two (2) lube oil purifiers. Use Coast Guard Drawings 905-WMEC-264-003 and 905-WMEC-342-006 as guidance.

3.2.5.7 Duct Ventilation ducting in the vicinity of both Port and Starboard SSDG must be removed for future modification. Use Coast Guard Drawing 905-WMEC-513-005 as guidance.

3.3 SSDG removals. The Contractor must remove two (2) Ship Service Diesel Generators (SSDG) through temporary access gained through the port side shell in the engine room (3-103-0-E). Equipment in way of the lifting straps, such as, but not limited to, the cooling tank and piping and filter boxes must be removed. In addition, the starboard and port SSDGs must be decoupled and separated from their skid. The starboard SSDG will traverse the engine room forward of the MDEs. This requires additional interferences to be removed. Use Coast Guard Drawings 901-WMEC_611_XXX, 901-MEC_111_XXX, 901-WMEC_505_XX1, and 901-WMEC_310_XX1 as guidance. Upon removal, inspect existing foundation for structural integrity and agreement with drawings. Submit CIR.

3.3.1 Disposition. The SSDGs shall be separated such that the generator end and the diesel engine shall each be crated separately and returned to SFLC warehouse in accordance with SFLC 083-8.4 Markings, Packaging, Delivery, and Inspection/Acceptance Requirements. The forward end of the rotor must be supported. Contactor must build crate for generator, SFLC MEC Supply will provide metal containers for engines. The contractor shall coordinate the return of the EDG components by requesting a "K" condition due-ins from the MECPL Item Manager, Ms. Martine Duvall, in accordance with Table 1 below.

TABLE 1: SSDG/EDG OLD CONFIGURATION GFE

PARAGRAPH	RETURN TO THE WAREHOUSE:	NSN	REPAIR CODE	CONDITION:	QTY:
3.3.1	Caterpillar D398TA diesel engine (SSDG)	2815-01-671-9891	R	K	2
3.3.1	Kato AC generator (starboard side SSDG)	6115-01-121-8046	R	K	1
3.3.1	Kato AC generator (port side SSDG)	6115-01-528-7750	R	K	1
3.5.1	Kato AC generator (EDG)	6115-01-121-8047	R	K	1
3.5.1	Caterpillar D348TA diesel engine (EDG)	2815-01-585-9265	R	K	1

3.5.1	EDG Raw Water Pump/Motor	4330-01-118-5085	R	K	1
3.6.3._	DECS-150 Voltage Regulator	6110-01-323-1965	R	K	3

3.3.2 Mechanical equipment.

3.3.2.1 The Contractor must remove the port and starboard MDE exhaust from between the outlet side of transition flowerpot to the inlet side of the silencer. Remove portions of the foundation supporting the port and starboard silencers. Provide temporary bracing. MDE exhaust must be blanked after removed, use Drawing 905-WMEC-233-007 as reference. The Contractor must remove and retain blanket exhaust insulation for re-install. Inspect the remaining exhaust and clean.

3.3.2.2 The Contractor must remove AFFF piping beside the foundations of the port and starboard SSDGs as necessary to accommodate future installation work. Use Coast Guard Drawing 905-WMEC-505-010 as guidance.

3.3.3 Disconnects.

3.3.3.1 The Contractor must close system valves and drain the engine fluid systems including: fuel oil supply and return, lube oil, jacket water and sea water. Dispose of fluids in accordance with all applicable Federal, state and local regulations. Use Coast Guard Tech Pub 2820 as guidance.

3.3.3.2 The Contractor must disconnect all ancillary system piping flexible connections, and all instrumentation and control cabling and gauge tubing to allow for engine removal. Connected systems include fuel oil, lube oil, jacket water, sea water, diesel exhaust, and emergency shutdown. All disconnected piping must be capped to prevent leakage and contamination.

3.3.3.3 The Contractor must disconnect all remaining mechanical connected services.

3.3.3.4 The Contractor must verify calibration of all retained gauges and transducers monitoring the C18 and systems disturbed by this work item. Submit CFR if equipment is not suitable condition to reinstall.

3.3.4 Structural foundation.

3.3.4.1 The Contractor must remove portions of the SSDG foundations. Use Coast Guard Drawing 905-WMEC-183-011 as guidance.

3.3.4.2 The Contractor must remove portions of ancillary foundations. Use Coast Guard Drawing 905-WMEC-183-012 as guidance.

3.3.4.3 The Contractor must fabricate transportation stands as needed to facilitate movement of equipment from the cutter to workshop areas.

3.4 Auxiliary seawater EDG pump removals.

3.4.1 Disposition. The contractor shall coordinate the return of the EDG ASW pump by requesting a "K" condition due-in from MECPL manager, Ms. Martine Duvall. EDG ASW pump shall be crated and

returned to SFLC warehouse in accordance with SFLC 083-8.4 Markings, Packaging, Delivery, and Inspection/Acceptance Requirements.

3.4.2 Seawater piping. The Contractor must remove portions of the auxiliary seawater piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance. All disconnected piping must be capped to prevent leakage and contamination.

3.4.3 EDG ASW pump. The Contractor must remove the existing EDG ASW pump. Use Coast Guard Drawing 905-WMEC-256-011 as guidance. The EDG ASW pump shall be crated separately and returned to SFLC warehouse in accordance with SFLC 083-8.4 Markings, Packaging, Delivery, and Inspection/Acceptance Requirements. The contractor shall coordinate the return of the EDG ASW Pump by requesting a “K” condition due-ins from the MECPL Item Manager, Ms. Martine Duvall.

3.4.4 EDG ASW pump structural foundation. Fabricate EDG pump supports. Use 905-WMEC-256-013 and SFLC-STD-SPEC-0740 as guidance.

3.5 EDG removals.

3.5.1 Disposition. The EDG shall be separated such that the generator end and the diesel engine shall each be crated separately and returned to SFLC warehouse in accordance with SFLC 083-8.4 Markings, Packaging, Delivery, and Inspection/Acceptance Requirements. The forward end of the rotor must be supported. Contractor must build crate for generator, SFLC MEC Supply will provide metal containers for engines. The contractor shall coordinate the return of the EDG components by requesting a “K” condition due-ins from the MECPL Item Manager, Ms. Martine Duvall.

3.5.2 EDG interferences.

3.5.2.1 The Contractor must remove at least six (6) light fixtures in the overhead of the aft steering compartment (3-228-0-E); and a section of washdown countermeasure piping in the overhead of the aft steering compartment 3-228-0-E. Use Coast Guard Drawings 905-WMEC-331-004 and 905-WMEC-521-008 as guidance.

3.5.2.2 The Contractor must remove supply/exhaust ventilation in Aft Steering, including fan and ducting below Main Deck. Use Coast Guard Drawing 905-WMEC-512-014 as guidance.

3.5.3 EDG disconnects.

3.5.3.1 The Contractor must close system valves and drain the engine fluid systems including: fuel oil supply and return, lube oil, jacket water and sea water. Dispose of fluids in accordance with all applicable Federal, state and local regulations. All disconnected piping must be capped to prevent leakage and contamination. Use Coast Guard Tech Pub 2821 as guidance.

3.5.3.2 The Contractor must disconnect all ancillary system piping flexible connections, and all instrumentation and control cabling and gauge tubing to allow for engine removal. Connected systems include fuel oil, lube oil, jacket water, sea water, diesel exhaust, and emergency shutdown. All disconnected piping must be capped to prevent leakage and contamination.

3.5.3.3 The Contractor must remove the existing EDG local operating station and all associated cabling. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.5.3.4 The Contractor must remove the existing EDG starting panel and all associated cabling. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.5.3.5 The Contractor must remove portions of the auxiliary seawater piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance. All disconnected piping must be capped to prevent leakage and contamination.

3.5.3.6 The Contractor must remove portions of the fuel oil piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, hangers and sheathing. Use Coast Guard Drawing 905-WMEC-342-004 as guidance. All disconnected piping must be capped to prevent leakage and contamination.

3.5.3.7 The Contractor must remove portions of the jacket water piping system local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, gauges, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-002 as guidance.

3.5.3.8 The Contractor must remove portions of the exhaust local to the equipment. This work includes, but is not limited to, exhaust piping, silencers, expansion joints, foundations, and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-003 as guidance.

3.5.3.9 Monitoring equipment The Contractor must verify calibration of all retained gauges and transducers monitoring the C18 and systems disturbed by this work item. Submit CFR if equipment is not in suitable condition to reinstall.

3.5.4 EDG temporary access cuts.

3.5.4.1 The Contractor must remove all interferences to the temporary hull access opening.

3.5.4.2 The Contractor must cut the temporary access opening in the steering gear compartment 3-228-0-E overhead (main deck) to remove the EDG and associated equipment. Use SFLC-STD-SPEC-8636, and 905-WMEC-111-8 as guidance. Upon removal inspect existing foundation for structural integrity and agreement with drawings. Submit CIR.

3.5.4.3 The Contractor must remove one (1) Emergency Diesel Generator (EDG) through temporary access gained in the overhead of the steering gear room 3-228-0-E (flight deck). Use Coast Guard Drawings 905-WMEC-342-02, 905-WMEC-505-011, 905-WMEC-111-008, and 905-WMEC-611-006 as guidance.

3.5.5 Structural foundation.

3.5.5.1 Upon removal inspect existing foundation for structural integrity and agreement with drawings. Use SFLC Std Spec 0740 and applicable Coast Guard Drawing as guidance. Submit CIR.

3.5.5.2 The Contractor must remove portions of the EDG foundation. Use Coast Guard Drawing 905-WMEC-183-011 as guidance.

3.5.5.3 The Contractor must remove portions of existing ancillary foundations. Use Coast Guard Drawing 905-WMEC-183-012 as guidance.

3.6 Ships service and emergency switchboard (SWBD).

3.6.1 Structural foundation. No SWBD foundation modifications required.

3.6.2 Temporary removals. The access panels under the ship service SWBD must be temporarily removed. Submit CFR if equipment is not in suitable condition to reinstall.

3.6.3 SWBD removals.

3.6.3.1 The Contractor must provide temporary power to cutter's transformers or power panels with cam lock connectors via power distribution units and shore tie cables totaling 800 amps.

3.6.3.2 The Contractor must tag out the ships service SSDG SWBD and migrate critical services to the Contractor-provided temporary power service described in Section 3.6.3.1. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.6.3.3 The Contractor must remove all associated SSDG gauges, pushbuttons, and wiring from ship service SWBD enclosure doors. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.6.3.4 The Contractor must retain existing enclosures, bus bars, generator circuit breakers, and bus tie circuit breakers for reuse. The existing ship service SSDG SWBD doors must be removed and used to fabricate new doors. The circuit breakers must be tested for reuse. Submit CFR.

3.6.3.5 The Contractor must remove all switchgear from the ship service SSDG SWBD. Use Coast Guard Drawing 905-WMEC-320-58 as guidance.

3.6.3.6 The Contractor must remove select distribution circuit breakers from the ship service SSDG SWBD. Use Coast Guard Drawing 905-WMEC-320-58 as guidance.

3.6.3.7 The Contractor must remove the Emergency SWBD enclosure doors (but used to fabricate new doors), and all associated gauges, pushbuttons, and wiring. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.6.3.8 The Contractor must retain existing EDG enclosures, bus bars, generator circuit breakers, and bus tie circuit breakers for reuse. The circuit breakers must be tested for reuse. Submit CFR.

3.6.3.9 The Contractor must remove all switchgear from the Emergency SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.6.3.10 The Contractor must remove select distribution circuit breakers from the Emergency SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.7 SSDG installations.

3.7.1 Structural foundation.

3.7.1.1 The Contractor must install new SSDG foundations and ancillary foundations. Use Coast Guard Drawing 905-WMEC-183-011 and 905-WMEC-183-012 as guidance. Submit CFR.

3.7.1.2 The deck grating (i.e. deck plates and their support structure) must be modified in way of the SSDG foundation modifications. Use Coast Guard Drawing 905-WMEC-622-004 as guidance.

3.7.1.3 The handrails must be modified after the SSDGs are installed. Use Coast Guard Drawing 905-WMEC-622-005 as guidance.

3.7.1.4 The Contractor must reuse the existing foundation for the starboard air receiver tank. Rotate and bolt back down to plate steel foundation.

3.7.2 Mechanical equipment and piping.

3.7.2.1 The Contractor must install two (2) Ship Service Diesel Generators (SSDGs) which must be installed through temporary access gained through the port sideshell in the engine room (3-103-0-E). The SSDGs must be decoupled and detached from their skids and lifted separately into the engine room. Transportation stands must be fabricated to facilitate movement of the genset components from the workshop to the cutter. Existing temporary accesses used for removals will also be used for installations. Use Coast Guard Drawings 901-WMEC_310_XX1, 901-WMEC_342_XX1, 901-WMEC_183_8, 901-WMEC_310_XX1, and 901-WMEC_611_XXX as guidance.

3.7.2.2 The Contractor must install auxiliary seawater piping local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, gauges, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance. Submit CFR.

3.7.2.3 The Contractor must install fuel oil piping local to the equipment. This work includes, but is not limited to, piping, pumps, hoses, filters, valves, fittings, gauges, hangers and shielding. Use Coast Guard Drawing 905-WMEC-342-004 as guidance. All affected fuel oil piping must be hydrostatic tested in accordance with Coast Guard Drawing 905-WMEC-342-004 and flushed in accordance with NSTM 541 and GSO 541 before being connected to equipment. Submit CFR.

3.7.2.4 The Contractor must install exhaust local to the equipment. This work includes, but is not limited to, exhaust piping, silencers, expansion joints, foundations, and insulation/lagging. The Contractor must verify disturbed exhaust flange torque after first heat cycle. Use Coast Guard Drawing 905-WMEC-342-003 as guidance. Submit CFR.

3.7.2.5 The Contractor must install portions of the ship's service compressed air system local to the SSDG's. This work includes, but is not limited to, piping, hoses, valves, fittings and hangers. Additionally, Contractor must rotate the starboard receiver tank 90 degrees clockwise. Use Coast Guard Drawing 905-WMEC-342-005 as guidance. Submit CFR.

3.7.2.6 The Contractor must install portions of the lube oil system local to the SSDG's. This work includes, but is not limited to, piping, hoses, valves, fittings and hangers. Use Coast Guard Drawing 905-WMEC-342-006 as guidance. Submit CFR.

3.7.2.7 The heat exchanger for the starboard SSDG must be rotated 180 degrees so that the inlets and outlets are on the inboard face of the heat exchanger. Use Coast Guard Drawing 905-WMEC-342-002 as guidance. Submit CFR.

3.7.3 Electrical equipment.

3.7.3.1 The Contractor must install new SSDG control panels. Use Coast Guard Drawing 905-WMEC-3200-58 as guidance.

3.7.3.2 The Contractor must install new SSDG Voltage Regulator Cabinets. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.7.3.3 The Contractor must install new power feed cables from the SSDG's to the ship service SWBD. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.7.3.4 The Contractor must adjust the trip settings on the SSDG circuit breakers. Use Coast Guard Drawing 905-WMEC-320-058 as guidance.

3.7.3.5 The Contractor must verify SSDG electrical installation of control panels, voltage regulator panels, and main power feed cables match the referenced drawings. Submit CFR.

3.8 Replacing temporary access cut and other temporary removals.

3.8.1 Replace access cut. In conjunction with Work Item 04, Main Diesel Engines, Replace, The Contractor must reinstall the temporary access created above the MDEs. In the event WI: 04 is not an active work item, the contractor must reinstall the temporary access located port side engine room.

3.8.2 Lighting and alarms. The Contractor must reinstall lighting fixtures, fire and smoke alarms, and junction boxes that were removed to accommodate temporary access opening. Use Coast Guard Drawings 905-WMEC-331-004, 905-WMEC-436-003, and 905-WMEC-111-008 as guidance.

3.8.3 AFFF piping. The Contractor must modify the AFFF piping in way of the SSDG foundations.

3.8.4 Ducting. The Contractor must shorten the ventilation ducting interfering with the Port SSDG. Use Coast Guard Drawing 905-WMEC-513-005 as guidance.

3.9 EDG installations.

3.9.1 EDG structural foundation. The Contractor must install new EDG foundation and foundations for miscellaneous equipment including, but not limited to fan controller, batteries, and EDG voltage regulator. Use Coast Guard Drawing 905-WMEC-183-011 and 905-WMEC-183-012 as guidance. Submit CFR.

3.9.2 EDG. The Contractor must install one (1) Emergency Diesel Generator (EDG) which must be installed through temporary access gained through main deck, overhead of the steering gear compartment (3-228-0-E). The EDG will be oriented so that the generator is forward and the engine is aft. Use Coast Guard Drawings 905-WMEC-310-011, 905-WMEC-342-002, 905-WMEC-183-011.

3.9.3 Seawater piping. The Contractor must install auxiliary seawater piping local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, gauges, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-002 as guidance. Submit CFR.

3.9.4 Fuel oil piping. The Contractor must install fuel oil piping local to the equipment. This work includes, but is not limited to, piping, pumps, hoses, filters, valves, fittings, gauges, and hangers and shielding. Use Coast Guard Drawing 905-WMEC-342-004 as guidance. All fuel oil piping must be hydrostatic tested in accordance with Coast Guard Drawing 905-WMEC-342-004 and flushed in accordance with NSTM 541 and GSO 541 before being connected to equipment. Submit CFR.

3.9.5 Exhaust. The Contractor must install exhaust local to the equipment. This work includes, but is not limited to, exhaust piping, silencers, expansion joints, foundations, and insulation/lagging. The Contractor must verify all freshly joined exhaust flange torque after first heat cycle. Use Coast Guard Drawing 905-WMEC-342-003 as guidance. Submit CFR.

3.9.6 Heat exchanger. The Contractor must rotate the heat exchanger on the generator so that the inlet and outlet are on the inboard side of the unit. Use Coast Guard Drawing 905-WMEC-310-012 and 905-WMEC-342-002 as guidance.

3.9.7 Jacket water piping. The Contractor must install jacket water piping local to the equipment. This work includes, but is not limited to, piping, hoses, pumps, heat exchangers, valves, fittings, gauges, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-002 as guidance.

3.9.8 Duct. The Contractor must modify ventilation ducting in way of the EDG exhaust. Use Coast Guard Drawing 905-WMEC-342-003 and 905-WMEC-512-014 as guidance.

3.9.9 Electrical equipment.

3.9.9.1 The Contractor must install new EDG control panel. On the starboard side, the Contractor must move the existing task light forward to be centered over the new unit location. Use Coast Guard Drawing 905-WMEC-320-57 as guidance.

3.9.9.2 The Contractor must install new EDG Voltage Regulator cabinet. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.9.9.3 The Contractor must adjust the trip settings on the EDG circuit breaker. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.9.9.4 The Contractor must connect 24VDC starter batteries to EDG. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.9.9.5 The Contractor must verify calibration of all new gauges and transducers monitoring the C18 and systems disturbed by this work item.

3.10 Replacing temporary access cut.

3.10.1 Structural. The Contractor must reinstall the temporary access opening in the steering gear compartment 3-228-0-E overhead (fantail). Use SFLC-STD-SPEC-8636 and 905-WMEC-111-008 as guidance.

3.10.2 Lighting and alarms. The Contractor must reinstall lighting fixtures, fire and smoke alarms, and junction boxes that were removed to accommodate temporary access opening. Use Coast Guard Drawings 905-WMEC-331-004, 905-WMEC-436-003, and 905-WMEC-111-008 as guidance.

3.10.3 Mechanical systems. The Contractor must reinstall wash-down countermeasure piping previously removed to accommodate the temporary access opening. Use 905-WMEC-521-008 as guidance.

3.11 EDG ASW pump.

3.11.1 Foundation. The Contractor must modify the existing EDG Auxiliary Sea Water Pump foundation in JP-5 Pump Room 4-186-0-J. Use Coast Guard Drawing 905-WMEC-256-0011 as guidance.

3.11.2 Seawater piping. The Contractor must replace retained portions of the auxiliary seawater piping system local to the equipment and install new portions. This work includes, but is not limited to, piping, hoses, pumps, valves, fittings, hangers and insulation/lagging. Use Coast Guard Drawing 905-WMEC-342-001 as guidance.

3.11.3 EDG ASW pump. The Contractor must install the EDG ASW pump. Use 905-WMEC-256-011 as guidance. Submit CFR.

3.11.4 Electrical. The Contractor must replace overloads in ASW pump motor controller. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12 SWBD installation.

3.12.1 Electrical equipment.

3.12.1.1 The Contractor must install three (3) new ship service SWBD retrofit kits. Use Coast Guard Drawing 905-WMEC-320-057 and 905-WMEC-320-058 as guidance.

3.12.1.2 The Contractor must install new power feed cables from the Emergency SWBD to the Ship Service SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12.1.3 The Contractor must install new power feed and control cables from the EDG to the Emergency SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12.1.4 The Contractor must install select new distribution circuit breakers into the ship service SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12.1.5 The Contractor must fabricate new EDG SWBD enclosure doors to suit new SWBD retrofit kits. Use Coast Guard Drawing 905-WMEC-320-057 as guidance and template from existing.

3.12.1.6 The Contractor must install new emergency SWBD retrofit kits. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12.1.7 The Contractor must install select new distribution circuit breakers into the EDG SWBD. Use Coast Guard Drawing 905-WMEC-320-057 as guidance.

3.12.1.8 The Contractor must fabricate new ship SSDG SWBD enclosure doors to suit new SWBD retrofit kits. Use Coast Guard Drawing 905-WMEC-320-058 as guidance and template from existing.

3.12.1.9 The Contractor must tag out the equipment used to provide temporary power.

3.12.1.10 The Contractor must migrate the services powered by the temporary power equipment back to the ships service SWBD.

3.12.1.11 The Contractor must verify calibration of all new gauges and transducers monitoring the C18 as well as systems disturbed by this work item.

3.12.1.12 The Contractor must test the function of the upgraded SWBDs in accordance with Coast Guard Drawing 905-WMEC-320-057 and 905-WMEC-320-058 prior to commissioning the system. Submit CFR.

3.13 Hydrostatic testing. All new and modified piping must be hydrostatic tested in accordance with test pressures listed in their respective Coast Guard Drawings. Submit CFR.

3.14 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

3.15 Insulation. The Contractor must replace disturbed insulation.

3.16 Fluids. The Contractor must procure all OEM approved fluids for and fill three (3) Caterpillar C18s at the direction of the Technical Representative. Fluids include but are not limited to oil and coolant.

3.17 Operational test. After completion of work, the Contractor must witness an operational test (by Coast Guard personnel) of all mechanical and electrical items or shipboard devices that have been disturbed, used, repaired, altered, or installed to prove that they are in satisfactory operational condition. See Inspection Hold Point Table in 4. Notes for work to be completed prior to operational testing. Use 270' WMEC SLEP Shipboard Integration Test Plan as additional guidance. Submit CFR.

3.18 Labeling. All Hoses, electrical cables, controllers, and other equipment must be labeled IAW approved references.

4. NOTES

4.1 Inspection hold point. Table 2 lists activities engineering and shop personnel will jointly inspect to support quality assurance efforts. Upon assembly of each system or upon uncovering each area as applicable, shop shall schedule a walkthrough with an engineering representative to verify Coast Guard Drawings match vessel configuration. Red-lines of all discrepancies will be recorded for drawing revisions as necessary.

TABLE 2. JOINT INSPECTION HOLD TABLE

TASK NO.	SECTION	SHOPS	ACTIVITY	SCHEDULE
1	3.2.5.7	10, 64	Inspect existing foundation for structural integrity and agreement with drawings.	Upon removal of each SSDG.
2	3.5.4.2	10, 64	Inspect existing foundation for structural integrity and agreement with drawings.	Upon removal of EDG.
3	3.7.2.2	21, 65	Verify installation of seawater system matches drawing and is complete	Prior to hydrostatic testing
4	3.7.2.3	21, 65	Verify installation of fuel oil system matches drawing and is complete	Prior to hydrostatic testing

TASK NO.	SECTION	SHOPS	ACTIVITY	SCHEDULE
5	3.7.2.4	21, 65	Verify installation of exhaust system matches drawing and is complete	Prior to hydrostatic testing
6	3.7.2.5	21, 65	Verify installation of compressed air system matches drawing and is complete	Prior to hydrostatic testing
7	3.7.2.6	21, 65	Verify installation of lube oil system matches drawing and is complete	Prior to hydrostatic testing
8	3.7.1.1	10, 64	Verify new foundation installations for SSDGs and their equipment match drawing	Upon installation of new foundations
9	3.7.3.5	30, 66	Verify SSDG electrical installation matches drawing.	Upon installation of the SSDG control panels, voltage regulator panels, and main power feed cables.
10	3.9.1.1	10, 64	Verify new foundation installations for EDG and its equipment match drawing	Upon completion of tank modifications prior to installation of deck plating
11	3.9.2.2	21, 65	Verify installation of EDG seawater system matches drawing and is complete	Prior to hydrostatic testing
12	3.9.2.3	21, 65	Verify installation of EDG fuel oil system matches drawing and is complete	Prior to hydrostatic testing
13	3.9.2.4	21, 65	Verify installation of EDG exhaust system matches drawing and is complete	Prior to hydrostatic testing
14	3.11.3	21, 65	Verify installation of EDG ASW pump matches drawing	Prior to hydrostatic testing
15	3.9.3	30, 66	Verify EDG electrical installation matches drawing	Upon installation of EDG control panel, voltage regulator panel, and main power feed cables
16	3.10.3	21, 65	Final inspection of temporarily countermeasure wash down piping for completeness	Prior to operational test

4.2 Welding Isolation. Prior to any hot work in the area of new generators ensure that welding isolation of generators is completed.

ESWBS_AUT_MMY_CLASS
 ESWBS_AUT_MMY_Hull#
 ESWBS_MMY_FLT
 ESWBS_MMY_BOAT_FLT
 REC_ESWBS_<<RECMergeField inserted here>>
 ECNNN_ESWBS_AUT_MMY_CLASS
 TCTO_LLNNNN_ESWBS_AUT_MMY_CLASS
 S_ESWBS_AUT_MMY_SWIT
 SYNNN_ESWBS_AUT_MMY_HULL#

WORK ITEM 45: Stern Tube, Replace

1. SCOPE

1.1 Intent. This Work Instruction describes the requirements for the Contractor to replace the stern tubes on the 270' WMEC class cutters by removing the existing stern tubes and fabricating and installing the new stern tubes.

1.2 Government-furnished property.

MTI	ITEM DESCRIPTION	NSN/PN	QTY	ESTIMATED COST (\$/UNIT)
N	** STERN TUBE SPLIT SEAL	NSN: 5330-01-359-5555	4 ea.	\$7,763.86
N	Overhaul kit	NSN: 5330-01-527-0959	2 ea.	\$12,894.00
N	Stern tube bearing	NSN: 3120-01-189-1949	2 ea.	\$12,700.00
N	**Stern tube seal face change out kit	NSN: 5330-01-527-0696	2 kt.	8,954.00
N	Automatic Air Eliminator	PN: AE36 NSN: 4820-01-690-3058	2 ea.	543.00

*Government-loaned property, which must be returned to the vessel upon completion of the availability.

**New or refurbished equipment that the Government may provide for installation in place of existing equipment.

***Government-furnished property, which is to be supplied by either the vessel or the C4IT Service Center.

2. REFERENCES

COAST GUARD DRAWINGS

Coast Guard Drawing 901-WMEC-161-002, Rev. C, "STERN TUBE & BOSS CASTINGS"

Coast Guard Drawing 901-WMEC-241-001, Rev. 9, "STBD RDCN GEAR INSTL DWG"

Coast Guard Drawing 901-WMEC-241-002, Rev. 11, "STBD RDCN GEAR – A&D"

Coast Guard Drawing 901-WMEC-241-004, Rev. 10, "PORT RDCN GEAR INSTL DWG"

Coast Guard Drawing 901-WMEC-241-005, Rev. 10, "PORT RDCN GEAR A&D"

Coast Guard Drawing 901-WMEC-241-006, Rev. 1, "AFT PINION SEAL RETAIN PLATE AND SHIM ASSM"

Coast Guard Drawing 901-WMEC-241-007, Rev. A, "RDCN GEAR L.O. SERVICE SYSTEM DIAGRAM"

Coast Guard Drawing 905 WMEC 243-001, Rev B, Shafting Arrangement

Coast Guard Drawing 905 WMEC 243-002, Rev F, Stern Tubes Strut Brg and Seal – A&D

Coast Guard Drawing 901-WMEC-243-003, Rev. F, "SHAFTING REMOVAL ARRANGEMENT"

Coast Guard Drawing 901-WMEC-243-004, Rev. G, "SHAFTING DETAILS"

Coast Guard Drawing 901-WMEC-243-005, Rev. D, "SHAFTING ALIGNMENT ANALYSIS AND PROCS"

Coast Guard Drawing 905 WMEC 243-006, Rev B, General Arrangement 12.125" Dia. Shaft Type MA Water Lub., Inboard Stern Tube Seal

Coast Guard Drawing 905 WMEC 243-010, Rev A, General Arrangement of 10.500" Dia. Bulkhead Stuffing Box

Coast Guard Drawing 901-WMEC-243-007, Rev. D, "SHAFT COUPLING ASSEMBLIES, CPP"

Coast Guard Drawing 901-WMEC-243-008, Rev. A, "HUB & RUDDER REMOVAL PADEYE A&D"

Coast Guard Drawing 901-WMEC-244-001, Rev. B, "THRUST BNG ASSEMBLY"

Coast Guard Drawing 901-WMEC-245-001, Rev. E, "CPCH PROP HYD SYS DIAG (902-904)"

Coast Guard Drawing 901-WMEC-245-002, Rev. F, "CPCH PROP HYD SYS – A&D"

Coast Guard Drawing 901-WMEC-245-004, Rev. E, "CPP BLADE & PALM"

Coast Guard Drawing 901-WMEC-245-005, Rev. -, "C.P. PROPELLER"

Coast Guard Drawing 901-WMEC-245-006, Rev. -, "CPCH PROP DRILL JIGS"

Coast Guard Drawing 901-WMEC-245-007, Rev. B, "CPCH PROP SPECIAL TOOLS"

Coast Guard Drawing 901-WMEC-245-008, Rev. C, "CPCH PROP ACTUATOR UNIT ASSY"

Coast Guard Drawing 901-WMEC-245-009, Rev. L, "CPCH PROP HYD SYS DIAG"

Coast Guard Drawing 901-WMEC-245-010, Rev. F, "CPCH PROP HYD SYS – A&D (902-904)"

Coast Guard Drawing 901-WMEC-245-011, Rev. -, "CPP BLADE & PALM"

Coast Guard Drawing 901-WMEC-245-012, Rev. -, "CONTROL MODULATOR FOR CPP"

Coast Guard Drawing 901-WMEC-245-013, Rev. B, "PITCH ACTUATOR SYSTEM ENGAGEMENT MECHANISMS, LINKS SHAFT CLAMPS AND BRACKETS"

Coast Guard Drawing 901-WMEC-245-014, Rev. A, "CPP SYSTEM PITCH SETTERS FINAL ASSEMBLY"

Coast Guard Drawing 901-WMEC-245-015, Rev. B, "CPP SYSTEM PITCH SETTERS MAIN BOARDS, SUB-BOARDS SCALES AND MECHANICAL STOPS"

Coast Guard Drawing 901-WMEC-245-017, Rev. B, "MACHINERY AND ENGINEERING CONTROL CENTER"

Coast Guard Drawing 901-WMEC-245-018, Rev. A, "MACHINERY AND ENGINEERING CONTROL CENTER CPP ELECTRICAL INSTALLATION MPCMS UPGRADE"

Coast Guard Drawing 901-WMEC-245-019, Rev. A, "HYDRAULIC OIL PURIFIER (HOP), REMOVAL"

Coast Guard Drawing 901 WMEC 256-012, Rev -, Shaft Seal Vent Line Modification

COAST GUARD PUBLICATIONS

Coast Guard Commandant Instruction (COMDTINST) M9000.6F, Naval Engineering Manual
Coast Guard Technical Publication 2699, (TP: 2699), 2018 Edition, Controllable Pitch Propeller System 270' WMEC (A) Class Cutters
Coast Guard Technical Publication 2817, (TP: 2817), 2018 Edition, Controllable Pitch Propeller System
Coast Guard Technical Publication (TP) 5454, Nov 2013, Technical Manual for Type MA Mechanical Stern Tube Seal
Surface Forces Logistics Center Standard Specification 0000, (SFLC Std Spec 0000) 2022 Edition, General Requirements
Surface Forces Logistics Center Standard Specification 0740, (SFLC Std Spec 0740) 2022 Edition, Welding and Allied Processes
Surface Forces Logistics Center Standard Specification 3041, (SFLC Std Spec 3041) 2022 Edition, Shipboard Electrical Cable Test
Surface Forces Logistics Center Standard Specification 3042, (SFLC Std Spec 3042) 2022 Edition, Shipboard Electrical Cable Removal, Relocation, Splice, Repair and Installation
Surface Forces Logistics Center Standard Specification 6310, (SFLC Std Spec 6310) 2022 Edition, Requirements for Preservation of Ship Structures
Surface Forces Logistics Center Standard Specification 6341, (SFLC Std Spec 6341) 2022 Edition, Install Interior Deck Covering Systems

OTHER REFERENCES

Environmental Protection Agency (EPA) Technician Certification, Type IV (Universal Certification)
MIL-DTL-15024, General Specification for Plates, Tags, and Bands for Identification of Equipment
MIL-DTL-24643, General Specification for Low Smoke Electric Cables and Cords for Shipboard Use
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 1: Cable)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 2: Equipment)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 3: Penetrations)
MIL-STD-2003, Electric Plant Installation Standard Methods (Section 4: Cableways)
NAVSEA Drawing 803-1385866, Rev E, Penetration, Bulkhead and Deck
NAVSEA Drawing 804-5773931, Rev A, Acoustic & Thermal Insulation for Compartments Installation Details

3. REQUIREMENTS

3.1 General.

3.1.1 CIR.

3.2 Inspection and repair requirements

3.1.2 Tech Rep. The Government shall provide the services of a Technical Representative.

3.1.3

Protective measures. The Contractor must furnish and install all protective measures in accordance with SFLC Std Spec 0000, paragraph 3.3.3 (Vessel component, space, and equipment protection).

3.1.4 Interferences. The Contractor must handle all interferences in accordance with SFLC Std Spec 0000, (Interferences). Known interferences include, but are not limited to the following:

- Equipment Foundations
- Insulation
- Fluids (See Table 1)
- Piping
- Sheathing and Lagging
- Shafts
- Rudders

Deck plating and framing

Shelves/supply storage

Safety grating

Propulsion shaft

Cooling water piping

Air line

3.1.5 Gas freeing. The Contractor must, when entering a tank, compartment, confined space, or when conducting hot work, refer to and comply with SFLC Standard Specification 0000, General Requirements, paragraph 3.3.1.2.1.

3.1.6 Preliminary Inspection. The Contractor must perform a visual inspection of exposed areas after interferences are removed. Notify Contracting Officer (KO) if damage or excessive corrosion is found. Submit CFR.

3.1.7 Welding and brazing. The Contractor must perform all welding, brazing, and non-destructive testing (NDT) in accordance with SFLC Standard Specification 0740, Welding and Allied Processes.

3.1.8 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

3.1.9 Equipment/component disposal. The Contractor must dispose of all removed equipment and/or components not designated for reinstallation in accordance with all Federal, State, and Local requirements.

3.2 Removals.

3.2.1 Prior to Docking. The Contractor must perform a strain gage alignment to establish a baseline in accordance with Appendix 5 of 901-WMEC-243-005 or 901-WMEC-243-005.

3.2.2 Preliminary optical alignment. The Contractor must perform preliminary optical alignment in accordance with NSTM 243.

3.2.3 Bearing removal. The Contractor must remove bearings from strut bores and stern tube bores.

3.2.4 Stern tube removal.

3.2.4.1 The Contractor must place targets on the forward end of the reduction gear output (bull gear) shaft and at the stern tube bearing location. This will give the longest reliable alignment axis. These targets must not be disturbed until the stern tube installation is completed.

3.2.4.2 The Contractor must establish the existing location of the forward end of the stern tube. The aft end will be set by the casting, which will remain in place.

3.2.4.3 The Contractor must cut the following welds:

- fillet weld between the forward end of the stern tube casting and the stern tube
- inside of the tube just forward of the bevel weld until it releases the stern tube, being careful to avoid damaging the casting.
- welds joining the stern tube or ring to Bulkhead 186 and grind flush.
- welds around the stern tube in the JP-5 tank at Frame 190 and grind flush.

3.2.4.4 The Contractor must remove the stern tube through Dry Stores.

3.2.4.5 The Contractor must grind the remainder of the bevel weld inside the front of the stern tube casting to regain the original shape of the counter bore in which the stern tube registers. The Contractor must take care to not oversize this hole, as it is what positions the stern tube in the casting.

3.3 Stern tube fabrications.

3.3.1 Replacement stern tube. The Contractor must fabricate the replacement stern tube to the original A-class specs (USCG DWG No. 901-WMEC-161-002). This will result in improved bulkhead stiffness on B-class cutters. Note that the bulkhead penetration will need to be enlarged in B-class cutters. The Contractor must place the tube in a fixture and weld the stern tube flange on so that its face is perpendicular to the axis of the stern tube and the bolt pattern is concentric to the bore. The flange shall have all machining completed prior to installation aboard the cutter.

3.3.1.1 Preservation requirements. The Contractor must prepare and coat the interior surfaces of the stern tubes, using the system specified in Table 1. Stern tube coating systems must be installed in accordance with manufacturer's recommendations.

TABLE 1 – STERN TUBE COATING SYSTEM

SURFACES TO BE PRESERVED	SURFACE PREPARATION / (ANCHOR PROFILE IN MILS)	COATING SYSTEM	DFT (MILS)	NOTES
Stern Tube Interior	SSPC-SP 10/NACE NO. 2 using grit conforming to MIL-A-22262 / (3.0-4.0)	1) One Full Coat Belzona 1341	8.0-10.0	1
		2) One Full Coat Belzona 1341	8.0-10.0	

1. Follow all manufacturer's instructions for application.

3.3.1.2 In-process quality control measures. The Contractor must abide by all the safety, preservation, and quality control requirements specified in SFLC Std Spec 0000, paragraph 3.2.4.2 (In-process QC measures for “critical-coated surfaces”).

NOTE

Surfaces being preserved are considered “critical-coated surfaces”.

3.3.2 Stern tube installation.

3.3.2.1 The Contractor must install five jacking pads (see **Error! Reference source not found.**) on the forward end of Bulkhead 186. The Contractor must weld them in on one side closest to the stern tube flange making sure to space them far enough away to be able to cut them out once the stern tube has been welded in.

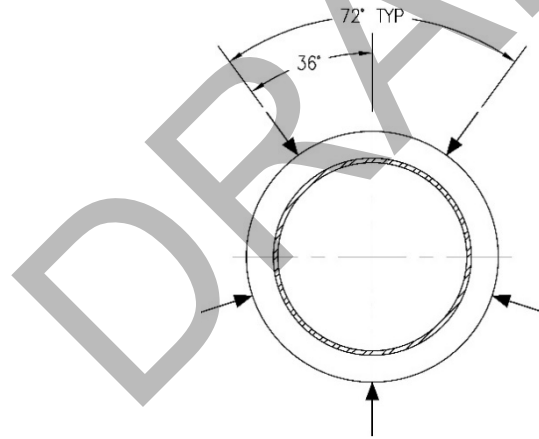


FIGURE 1. JACKING BOLT POSITIONS ON BHD 186, LOOKING AFT

3.3.2.2 The Contractor must install stern tube until it is seated in the recess of the casting. When the original stern tube was removed, the recess should not have been made oversized. Install jacking bolts on the forward end.

3.3.2.3 The Contractor must set up the Optical Alignment Scope on the forward end of the reduction gear and put the target on the aft end of the stern tube bearing.

3.3.2.4 The Contractor must set the Optical Alignment Scope to zero using a glass target in the aft end of the reduction gear.

3.3.2.5 The Contractor must remove the glass target from the aft end of the reduction gear.

3.3.2.6 The Contractor must install stern tube target plate on the stern tube flange. The Contractor shall install metal white target in target plate.

3.3.3 Stern tube modification.

3.3.3.1 The Contractor must adjust stern tube flange into position using the jacking bolts, paying attention to the vertical offsets for cold alignment called out in USCG DWG Nos. 901-WMEC-243-005 and 905-WMEC-243-005. The Contractor shall ensure that it remains seated in the casting.

3.3.3.2 The Contractor must start welding the flange with 1" long tack welds in a star pattern. The second tack weld must be 180° out from the first. The second pair of tack welds should be 90° out from the first. Each successive pair of tack welds should split the difference between the previous sets of tack welds. The Contractor shall allow the flange to cool between tack welds to avoid building up too much heat before the weld is set.

3.3.3.3 The Contractor must check the zero after each pair of tack welds; if it pulls away from zero the tack welds must be cut back position out. The Contractor shall repeat this process until a full pass has been made. At this point, the location has been set and it can be welded normally.

3.3.3.4 The Contractor must remove the target from the stern tube casting.

3.3.3.5 The Contractor must weld the aft end of the stern tube to the casting in short increments to avoid putting too much heat into the Chockfast which has not been removed.

3.3.3.6 The Contractor must weld the stern tube to Frame 190 paying close attention to heat generation. Excessive heat can cause warping.

3.4 Inspection and repair requirements. The Contractor must overhaul the port and starboard stern tube seal assemblies, in accordance with the referenced TP 5454, Chapter 4, Paragraph 4.9 (Complete Overhaul of Seal) and Chapter 8 (Installation), and as specified below. Measure seat wear in accordance with TP 5454, Chapter 4, Section 4.7. Visually inspect the shaft seal mounting faces (including bulkhead mounting flange), and exposed adjacent surfaces for evidence of corrosion or deterioration. Submit a CIR, including recommended repairs, if any.

3.4.1 Seat machining. Machine the seal seats in accordance with TP 5454, Chapter 6, Section 6.4. If a Change Request is released and approved by the KO, build-up/repair the shaft seal bulkhead mounting flange using a commercially available metallic polymer (e.g. Belzona or equivalent).

3.4.2 Seal compression setting. Set the stern tube seal compression during the installation in accordance with TP 5454, Chapter 6, Section 6.2 (Setting Seal Compression).

3.4.3 Alignment of seat. Align each seat to run true with its associated shaft in accordance with TP 5454, Chapter 6, Paragraph 6.3, and as follows:

3.4.3.1 Attach a dial indicator so as to indicate either the face of the seat, aft side of the face carrier, or the flat side of the forward main clamp ring near the butt joint, to measure relative misalignment of the seat in relation to the stationary face insert, as the shaft rotates.

3.4.3.2 Engage turning gear to rotate the shaft.

3.4.3.3 Adjust the alignment bolts and drive bolts to have a total indicator reading of 0.002" or less while the shaft is rotating. Take readings for three complete revolutions of the shaft.

3.4.3.4 Once alignment is complete, lock the alignment bolts with their locking nuts while the shaft is rotating and the indicator is still installed. Recheck alignment, after locking, in the presence of the Coast Guard Inspector. Submit CFR.

3.4.3.5 Disengage turning gear and install splash guard onto the face carrier.

3.4.4 Automatic Air Eliminator, Renew. The Contractor must rip out and dispose of the port and starboard shaft seal air eliminators and renew with government furnished Automatic Air Eliminators, as shown on Coast Guard Drawing 901 WMEC 256-012. Renew all disturbed seals, gaskets and fasteners.

3.5 Tests. The Contractor must perform the following tests to the port and starboard stern tube seal assemblies:

3.5.1 Stern tube seal leak test. Pressurize the stern tube seal with clean fresh water through the flanged water connections at the bottom of the mounting ring to 20 psig. Inspect and correct for any leakage. Submit CFR.

3.5.2 Inflatable seal air test. Pressurize the inflatable seal to 60 psi using Nitrogen, in lieu of Carbon Dioxide.

3.5.2.1 Visually inspect to ensure all pipe joints are tight.

3.5.2.2 With seal inflated to 75-90 psi, shut off all supply valves, and monitor air leakage. Pressure must not drop more than 5 psi in 15 minutes. Submit CFR. Correct any discrepancy noted.

WARNING

Do not rotate shaft while testing inflatable seal!

3.5.3 Inflatable seal water leakage test. After the vessel is waterborne, operate inflatable seal at normal pressure in accordance with TP 5454, Chapter 2, Section 2.3 (use Nitrogen, in lieu of Carbon Dioxide).

3.5.3.1 Slowly open the seal drain valve and drain the seal cavity. Once seal cavity is drained, measure the seal leakage which must not exceed one pint per minute. Submit CFR. Take corrective action if the leakage is exceeding one pint per minute.

3.5.3.2 Align the seal water flush lines for normal operation.

3.6 Stern tube split seal change-out. The Contractor must be aware that the Government reserves the right to furnish a new stern tube split seal for installation in-place of the one removed. When the Government exercises this right, the Contractor must dispose of the removed stern tube split seals as designated by the Coast Guard Inspector.

WARNING

Do not drain any fluids including fresh water, into any space, bilge, or exterior location.

3.7 Pipe flushing. The Contractor must flush all new and disturbed piping with clean fresh water until all debris is removed but no longer than five minutes. Ensure flushing fluid is directed to move scale and foreign debris away from installed machinery to prevent possible damage upon operational testing. Submit a CFR documenting date and time of flushing process and level of pipe cleanliness.

3.7.1 Dispose of flushing fluid in accordance with all applicable Federal, state, and local regulations.

NOTE

Coast Guard personnel will operate all shipboard machinery and equipment.

3.8 Touch-up preservation. The Contractor must prepare and coat all new and disturbed surfaces to match existing adjacent surfaces in accordance with SFLC Std Spec 6310, (Touch-ups and minor coating repairs.)

3.9 Hydrostatic Testing.

3.4.1 Hydrostatically test JP-5 tank. After modifications to affected systems are complete, the Contractor shall hydrostatically test (in the presence of Coast Guard personnel) the JP-5 tank to a static head of seven (7) feet above main deck in accordance with 901-WMEC-192-001 or 905-WMEC-192-001. No leaks are permitted.

3.10 Bearing alignment.

3.10.1 Optical alignment. The Contractor shall install bearings and perform the optical alignment in accordance with NSTM 243.

3.11 Reinstallation and testing.

3.11.1 Shafting reinstallation. Reinstallation is covered under Work Item: Propulsion Shafting; Remove, Inspect, and Reinstall.

3.11.2 Shafting alignment. The Contractor shall perform a strain gage alignment of the shafting in accordance with Coast Guard Drawing 901-WMEC-243-005.

3.11.3 Operational testing – post repairs. Post Op test will be completed in Work Item: Propulsion Shafting; Remove, Inspect, and Reinstall.

4. NOTES

4.1 On-board Spare Parts. Turn over on-board spare parts kit to ship's crew upon completion of the availability.

4.2 References. Not all Drawings or references were completed at the release of Rev 0. Contact CG Yard Engineering for guidance.

4.3 Tech Services Check Sheet. Submit Tech Services Check Sheet prior to Sea Trials (provided on next page).

4.4 Face seal only overhaul kit is available if CIR results of paragraph 3.2 indicate that a complete overhaul of the seal assembly is not required. Only one of the kits (major overhaul kit or face change out kit) should be used and the unused kit returned to stock.

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